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OF LONDON**

Advanced statistics: statistical inference

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Undergraduate study in
**Economics, Management,
Finance and the Social Sciences**

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THE LONDON SCHOOL
OF ECONOMICS AND
POLITICAL SCIENCE ■

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Chapter 1

Introduction

1.1 Route map to the subject guide

This subject guide provides you with a framework for covering the syllabus of the **ST2134 Advanced statistics: statistical inference** half course.

To ‘infer’ means to make general statements on the basis of specific observations. From an early age, people are experts at performing inference. It is such a fundamental part of our intelligence that we do it even subconsciously. We learn to classify objects on the basis of a very limited set of examples. From just a few simple pictures, a child will learn to infer that anything displaying certain characteristics of a long neck and long, thin legs is a giraffe. Of course, the immediacy with which people draw inferences can lead to mistaken inferences – some harmless, or even amusing. However, sometimes misplaced inferences can feed our prejudices and lead us to completely incorrect conclusions about other people’s natures or intentions.

In statistical inference, we go from the specific to the general by way of a mathematical model. The mathematical models which we use draw on probability distributions described in the companion half course **ST2133 Advanced statistics: distribution theory**, which ideally you should study first.

Our specific observations come from a dataset, i.e. a collection of numbers, or at least information which can be expressed numerically (such as by coding). Methods for using probabilistic models to make general statements on the basis of an observed dataset is the central topic of this course.

1.2 Aims and learning outcomes

The principal aim of this course is to provide a thorough theoretical grounding in statistical inference. The half course teaches fundamental material which is required for specialised courses in statistics, actuarial science and econometrics.

At the end of this half course and having completed the recommended reading and activities you should be able to:

- explain the principles of data reduction
- judge the quality of estimators
- choose appropriate methods of statistical inference to tackle real problems.

1.3 How to use this subject guide

The subject guide is intended to help you interpret the syllabus. Detailed exposition is provided for each of the areas along with some suggestions for reading. Precisely how you use the subject guide is clearly a matter of personal preference. A recommended approach is as follows.

- Read each chapter in full reasonably quickly. Do not worry too much if some of it does not make sense initially.
- Go through each chapter section by section by reading, making notes, completing activities and checking that you understand every aspect of the section.
- Go through the learning outcomes and make sure that you are comfortable with all of the points.
- Complete the end-of-chapter sample examination questions, referring to the subject guide as necessary.

The topics in **ST2134 Advanced statistics: statistical inference** build on each other incrementally and sequentially. You should be aware that, although sections are fairly self-contained, examination questions may contain material from many different parts of the subject guide. Finally, remember that you will be assessed on your ability to answer questions. Practice is the *only* way to make sure you can do this successfully.

1.4 Syllabus

The syllabus of **ST2134 Advanced statistics: statistical inference** is as follows.

- **Data reduction:** Sufficiency; minimal sufficiency. Likelihood.
- **Point estimation:** Bias, mean squared error, consistency. Rao–Blackwell theorem. Central limit theorem. Method of moments estimators. Order statistics. Minimum variance unbiased estimators, Cramér–Rao lower bound. Properties of maximum likelihood estimators.
- **Interval estimation:** Coverage probability and length. Pivotal functions.
- **Hypothesis testing:** Most powerful tests. Neyman–Pearson lemma. Likelihood ratio test.

The subject guide is comprised of the following chapters.

- Chapter 1 is this introduction.
- Chapter 2 introduces the fundamental concepts in statistical inference including populations, samples and statistics, sufficiency and likelihood.
- Chapter 3 discusses point estimation, measures of the quality of point estimators and techniques used to generate point estimators.

- Chapter 4 deals with interval estimation and related ideas.
- Chapter 5 discusses hypothesis testing and provides a detailed explanation of likelihood ratio tests.

Basic notation

We often use the symbol ■ to denote the end of a proof, where we have finished explaining why a particular result is true. This is just to make it clear where the proof ends and the following text begins.

Calculators

A calculator may be used when answering questions on the examination paper for **ST2134 Advanced statistics: statistical inference**. It must comply in all respects with the specification given in the Regulations. You should also refer to the admission notice you will receive when entering the examination and the ‘Notice on permitted materials’.

1.4.1 Recommended reading

In principle, this subject guide acts as your essential reading and is sufficient to perform well in the final examination – provided you practise many problems! However, textbooks can be beneficial since they provide:

- an alternative (and in some instances more detailed) perspective on material in the subject guide
- additional illustrative examples
- more exercises to practise.

In looking at textbooks, you will encounter much which will be familiar from looking at the subject guide. However, you will also come across topics which are not covered and questions requiring a deeper level of understanding than is provided here. Remember, this subject guide is the *definitive* syllabus and that the level of examination questions will be similar to the level of questions given at the end of each chapter.

If you do wish to consult a textbook for this half course, we recommend:

- Casella, G. and R.L. Berger, *Statistical Inference*. (Duxbury, 2008) second edition [ISBN 9788131503942].

Indicative reading in this subject guide refers to the above edition of the textbook. Newer editions may have been published by the time you study this half course. You can use a more recent edition by using the detailed chapter and section headings and the index to identify relevant readings. Also, check the virtual learning environment (VLE) regularly for updated guidance on readings.

1.4.2 Further reading

Please note that you are free to read around the subject area in any text, paper or online resource. You can support your learning by reading as widely as possible and by thinking about how these principles apply in the real world. To help you read extensively, you have free access to the virtual learning environment (VLE) and University of London Online Library (see below).

Other useful textbooks for this course include:

- Bartoszynski, R. and M. Niewiadomska-Bugaj, *Probability and Statistical Inference*. (Wiley, 1996) [ISBN 9780471310730].
- Hogg, R.V. and E.A. Tanis, *Probability and Statistical Inference*. (Pearson, 2014) ninth edition [ISBN 9781292062358].
- Larsen R.J. and M.L. Marx, *Introduction to Mathematical Statistics and Its Applications*. (Pearson, 2013) fifth edition [ISBN 9781292023557].
- Miller, I. and M. Miller, *John E. Freund's Mathematical Statistics with Applications*. (Pearson, 2012) eighth edition [ISBN 9780321904409].

1.4.3 Online study resources (the Online Library and the VLE)

In addition to the subject guide and the recommended reading, it is crucial that you take advantage of the study resources which are available online for this course, including the virtual learning environment (VLE) and the Online Library.

You can access the VLE, the Online Library and your University of London email account via the Student Portal at:

<http://my.london.ac.uk>

You should have received your login details for the Student Portal with your official offer, which was emailed to the address that you gave on your application form. You have probably already logged in to the Student Portal in order to register! As soon as you registered, you will automatically have been granted access to the VLE, Online Library and your fully functional University of London email account.

If you forget your login details, please click on the 'Forgotten your password' link on the login page.

The VLE

The VLE, which complements this subject guide, has been designed to enhance your learning experience, providing additional support and a sense of community. It forms an important part of your study experience with the University of London and you should access it regularly.

The VLE provides a range of resources for EMFSS courses:

- **Course materials:** Subject guides and other course materials available for download. In some courses, the content of the subject guide is transferred into the

VLE and additional resources and activities are integrated with the text.

- **Readings:** Direct links, wherever possible, to essential readings in the Online Library, including journal articles and ebooks.
- **Video content:** Including introductions to courses and topics within courses, interviews, lessons and debates.
- **Screencasts:** Videos of PowerPoint presentations, animated podcasts and on-screen worked examples.
- **External material:** Links out to carefully selected third-party resources.
- **Self-test activities:** Multiple-choice, numerical and algebraic quizzes to check your understanding.
- **Collaborative activities:** Work with fellow students to build a body of knowledge.
- **Discussion forums:** A space where you can share your thoughts and questions with fellow students. Many forums will be supported by a ‘course moderator’, a subject expert employed by LSE to facilitate the discussion and clarify difficult topics.
- **Past examination papers:** We provide up to three years of past examinations alongside *Examiners’ commentaries* that provide guidance on how to approach the questions.
- **Study skills:** Expert advice on getting started with your studies, preparing for examinations and developing your digital literacy skills.

Some of these resources are available for certain courses only, but we are expanding our provision all the time and you should check the VLE regularly for updates.

Note: Students registered for Laws courses also receive access to the dedicated Laws VLE.

Making use of the Online Library

The Online Library (<http://onlinelibrary.london.ac.uk>) contains a huge array of journal articles and other resources to help you read widely and extensively.

To access the majority of resources via the Online Library you will either need to use your University of London Student Portal login details, or you will be required to register and use an Athens login.

The easiest way to locate relevant content and journal articles in the Online Library is to use the **Summon** search engine.

If you are having trouble finding an article listed in a reading list, try removing any punctuation from the title, such as single quotation marks, question marks and colons.

For further advice, please use the online help pages (<https://onlinelibrary.london.ac.uk/resources/summon>) or contact the Online Library team: onlinelibrary@shl.london.ac.uk

Additional material

There is a lot of computer-based teaching material available freely over the web. A fairly comprehensive list can be found in the 'Books & Manuals' section of:

<http://statpages.info>

Unless otherwise stated, all websites in this subject guide were accessed in August 2020. We cannot guarantee, however, that they will stay current and you may need to perform an internet search to find the relevant pages.

1.5 Examination advice

Important: the information and advice given here are based on the examination structure used at the time this subject guide was written. Please note that subject guides may be used for several years. Because of this we strongly advise you to always check both the current Programme regulations for relevant information about the examination, and the VLE where you should be advised of any forthcoming changes. You should also carefully check the rubric/instructions on the paper you actually sit and follow those instructions.

The examination paper for this half course is two hours long. There will be *four* questions. You should answer all four questions. The examiners attempt to ensure that the examination provides broad coverage of the syllabus. Some questions may cover more than one topic. A Sample examination paper appears in Appendix D, along with solutions in Appendix E.

There are three reasons why there is no choice of questions in the examination paper:

- i. to make sure you study all parts of the course thoroughly
- ii. to avoid the risk that candidates make bad choices when selecting questions
- iii. to assess you on your ability to perform statistical inference, not your ability to choose examination questions.

The format of the paper is as follows.

- **Section A** (40%): One question divided into several parts. A useful mark allocation for each part will be provided. This question will broadly cover the material in the syllabus.
- **Section B** (60%): Three questions each worth (approximately) 20%. Each question will attempt to test the depth of your knowledge of the material in the syllabus.

You should divide your time in proportion to the marks available for each subquestion. Roughly, this works out as follows: five minutes to read the paper, forty-five minutes on Section A, twenty minutes on each question in Section B, leaving ten minutes for checking.

Remember, it is important to check the VLE for:

- up-to-date information on examination and assessment arrangements for this course
- where available, past examination papers and *Examiners' commentaries* for the course which give advice on how each question might best be answered.

Chapter 2

Fundamentals of statistical inference

2.1 Recommended reading

Casella, G. and R.L. Berger, *Statistical Inference*. Chapter 5, Section 5.1 and Chapter 6, Sections 6.1–6.3.

2.2 Learning outcomes

On completion of this chapter, you should be able to:

- refer to the term population in the context of statistical inference
- explain the difference between a model (represented by the sample) and data (the observed sample)
- distinguish between point estimators and point estimates
- discuss how statistics play a fundamental role in data reduction
- define a sufficient statistic and a minimal sufficient statistic
- determine whether a statistic is sufficient using the factorisation theorem
- define the terms ‘likelihood’, ‘log-likelihood’, ‘score’ and ‘Fisher information’
- exploit the properties of random samples to write simpler forms of likelihood, log-likelihood, score and Fisher information
- calculate the score function and Fisher information for standard distributions
- state two distinct forms of Fisher information and understand the association between them.

2.3 Introduction

This chapter extends the discussion of statistical inference introduced in **ST104b Statistics 2**, so it is advisable to review that material first before proceeding.

Here we present a framework for approaching problems of statistical inference and define some fundamental concepts and terminology. At first sight, some of the distinctions presented in this chapter may seem very subtle, and even pedantic.

However, a detailed understanding of these subtleties and nuances is essential to master the material in the course. In particular, understanding the difference between our *model* (represented in terms of random variables) and the *data* we observe (a collection of numerical values) is important in terms of statistical inference.

This chapter presents the ideas of point estimation, interval estimation and hypothesis testing, which form the basis of the following chapters. We introduce the concepts of *sufficiency* and *likelihood*. Sufficiency provides a formal expression of the intuitive notion that we should use all of the *relevant* information from a random sample. Likelihood is closely connected with probability mass/density functions (with many common probability distributions covered in **ST2133 Advanced statistics: distribution theory**) and likelihood arises frequently as the basis of statistical inference methods.

2.4 Statistical inference: the basics

Before we get into details, let us begin with the ‘big picture’. First, some definitions.

- Research: trying to answer questions about the world in a systematic (scientific) way.
- Empirical research: doing research by first collecting relevant information (*data*) about the world.

Research may be about almost any topic: economics, management, finance, physics, biology, medicine etc. Research in this sense is not just what universities do. Governments, businesses, and all of us as individuals do it too. Statistics is used in essentially the same way for all of these.

Example 2.1 It all starts with a question.

- Can labour regulation hinder economic performance?
- Understanding the gender pay gap: what has competition got to do with it?
- Children and online risk: powerless victims or resourceful participants?
- Refugee protection as a collective action problem: is the European Union (EU) shirking its responsibilities?
- Do directors perform for pay?
- Heeding the push from below: how do social movements persuade the rich to listen to the poor?
- Does devolution lead to regional inequalities in welfare activity?
- The childhood origins of adult socio-economic disadvantage: do cohort and gender matter?
- Parental care as unpaid family labour: how do spouses share?

Key stages of the empirical research process

We can think of the empirical research process as having five key stages.

1. Formulating the *research question*.
2. *Research design*: deciding what kinds of data to collect, how and from where.
3. *Collecting* the data.
4. *Analysis* of the data to answer the research question.
5. *Reporting* the answer and how it was obtained.

The main job of statistics is the *analysis* of data, although it also informs other stages of the research process. Statistics are used when the data are **quantitative**, i.e. in the form of numbers.

Statistical analysis of quantitative data has the following features.

- It can cope with large volumes of data, in which case the first task is to provide an understandable summary of the data. This is the job of **descriptive statistics**.
- It can deal with situations where the observed data are regarded as only a part (a *sample*) from all the data which could have been obtained (the *population*). There is then *uncertainty* in the conclusions. Measuring this uncertainty is the job of **statistical inference**.

We now consider an example of how statistics can be used to help answer a research question.

Example 2.2 CCTV, crime and fear of crime.

Our research question is what is the effect of closed-circuit television (CCTV) surveillance on:

- the number of recorded crimes?
- the fear of crime felt by individuals?

We illustrate this using part of the following study.

- Gill and Spriggs (2005): Assessing the impact of CCTV. *Home Office Research Study 292*.

The research design of the study comprised the following.

- **Target area**: a housing estate in northern England.
- **Control area**: a second, comparable housing estate.

- **Intervention:** CCTV cameras installed in the target area but not in the control area.
- **Compare** measures of crime and the fear of crime in the target and control areas in the 12 months before and 12 months after the intervention.

The data and data collection were as follows.

- **Level of crime:** the number of crimes recorded by the police, in the 12 months before and 12 months after the intervention.
- **Fear of crime:** a *survey* of residents of the areas.
 - Respondents: random samples of residents in each of the areas.
 - In each area, one sample before the intervention date and one about 12 months after.
 - Sample sizes:

	Before	After
Target area	172	168
Control area	215	242

- Question considered here: ‘In general, how much, if at all, do you worry that you or other people in your household will be victims of crime?’ (from 1 = ‘all the time’ to 5 = ‘never’).

Statistical analysis of the data.

% of respondents who worry ‘sometimes’, ‘often’ or ‘all the time’:

Target area			Control area			Confidence interval	
[a]	[b]		[c]	[d]			
Before	After	Change	Before	After	Change	RES	
26	23	−3	53	46	−7	0.98	0.55–1.74

- It is possible to calculate various statistics, for example the *Relative Effect Size* $RES = ([d]/[c])/([b]/[a]) = 0.98$ is a summary measure which compares the changes in the two areas.
- $RES < 1$, which means that the observed change in the reported fear of crime has been a bit *less* good in the target area.
- However, there is *uncertainty because of sampling*: only 168 and 242 individuals were actually interviewed at each time in each area, respectively.
- The *confidence interval* for RES includes 1, which means that changes in the self-reported fear of crime in the two areas are ‘not statistically significantly different’ from each other.

The number of (any kind of) recorded crimes:

Target area			Control area			Confidence interval	
[a]	[b]		[c]	[d]			
Before	After	Change	Before	After	Change	RES	
112	101	−11	73	88	15	1.34	0.79–1.89

- Now the $RES > 1$, which means that the observed change in the number of crimes has been worse in the control area than in the target area.
- However, the numbers of crimes in each area are fairly small, which means that these estimates of the changes in crime rates are fairly uncertain.
- The confidence interval for RES again includes 1, which means that the changes in crime rates in the two areas are not statistically significantly different from each other.

In summary, this study did *not* support the claim that the introduction of CCTV reduces crime or the fear of crime. This serves as an interesting example of how statistics can be employed in the social sciences to investigate research questions.

2.5 Revision of fundamentals

This section is provided as a revision of common statistical operators.

2.5.1 Properties of the summation operator

Let X_i and Y_i , for $i = 1, 2, \dots, n$, be sets of n numbers. Let a denote a *constant*, i.e. a number with the same value for all i .

All of the following results follow simply from the properties of addition (if you are still not convinced, try them with $n = 3$).

$$1. \quad \sum_{i=1}^n a = n \times a = na.$$

Proof:

$$\sum_{i=1}^n a = \overbrace{(a + a + \dots + a)}^{n \text{ times}} = n \times a = na.$$

■

$$2. \quad \sum_{i=1}^n aX_i = a \sum_{i=1}^n X_i.$$

Proof:

$$\begin{aligned} \sum_{i=1}^n aX_i &= (aX_1 + aX_2 + \dots + aX_n) \\ &= a(X_1 + X_2 + \dots + X_n) \\ &= a \sum_{i=1}^n X_i. \end{aligned}$$

■

$$3. \sum_{i=1}^n (X_i + Y_i) = \sum_{i=1}^n X_i + \sum_{i=1}^n Y_i.$$

Proof: Rearranging the elements of the summation, we get:

$$\begin{aligned} \sum_{i=1}^n (X_i + Y_i) &= ((X_1 + Y_1) + (X_2 + Y_2) + \cdots + (X_n + Y_n)) \\ &= ((X_1 + X_2 + \cdots + X_n) + (Y_1 + Y_2 + \cdots + Y_n)) \\ &= (X_1 + X_2 + \cdots + X_n) + (Y_1 + Y_2 + \cdots + Y_n) \\ &= \sum_{i=1}^n X_i + \sum_{i=1}^n Y_i. \end{aligned}$$

■

2.5.2 Extension: double (triple etc.) summation

Sometimes sets of numbers may be indexed with two (or even more) subscripts, for example as X_{ij} , for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$.

Summation over both indices is written as:

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^m X_{ij} &= \sum_{i=1}^n (X_{i1} + X_{i2} + \cdots + X_{im}) \\ &= (X_{11} + X_{12} + \cdots + X_{1m}) + (X_{21} + X_{22} + \cdots + X_{2m}) \\ &\quad + \cdots + (X_{n1} + X_{n2} + \cdots + X_{nm}). \end{aligned}$$

The order of summation can be changed, that is:

$$\sum_{i=1}^n \sum_{j=1}^m X_{ij} = \sum_{j=1}^m \sum_{i=1}^n X_{ij}.$$

2.5.3 Product notation

The analogous notation for the *product* of a set of numbers is:

$$\prod_{i=1}^n X_i = X_1 \times X_2 \times \cdots \times X_n.$$

The following results follow from the properties of multiplication.

1. $\prod_{i=1}^n aX_i = a^n \prod_{i=1}^n X_i.$
2. $\prod_{i=1}^n a = a^n.$
3. $\prod_{i=1}^n X_i Y_i = \left(\prod_{i=1}^n X_i \right) \left(\prod_{i=1}^n Y_i \right).$

2.5.4 The sum of deviations from the sample mean is zero

The mean is ‘in the middle’ of the observations $X_1, X_2, \dots, X_n$, in the sense that positive and negative values of the *deviations* $X_i - \bar{X}$ cancel out, when summed over all the observations, that is:

$$\sum_{i=1}^n (X_i - \bar{X}) = 0.$$

Proof: (The proof uses the definition of $\bar{X}$ and the properties of summation introduced earlier. Note that $\bar{X}$ is a constant in the summation, because it has the same value for all i .)

$$\sum_{i=1}^n (X_i - \bar{X}) = \sum_{i=1}^n X_i - \sum_{i=1}^n \bar{X} = \sum_{i=1}^n X_i - n\bar{X} = \sum_{i=1}^n X_i - n \frac{\sum_{i=1}^n X_i}{n} = \sum_{i=1}^n X_i - \sum_{i=1}^n X_i = 0.$$

■

2.5.5 The sample mean minimises the sum of squared deviations

The smallest possible value of the sum of squared deviations $\sum_{i=1}^n (X_i - C)^2$, for any constant C , is obtained when $C = \bar{X}$.

Proof:

$$\begin{aligned} \sum_{i=1}^n (X_i - C)^2 &= \sum_{i=1}^n (X_i - \overbrace{\bar{X} + \bar{X} - C}^{=0})^2 \\ &= \sum_{i=1}^n ((X_i - \bar{X}) + (\bar{X} - C))^2 \\ &= \sum_{i=1}^n ((X_i - \bar{X})^2 + 2(X_i - \bar{X})(\bar{X} - C) + (\bar{X} - C)^2) \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 + \sum_{i=1}^n 2(X_i - \bar{X})(\bar{X} - C) + \sum_{i=1}^n (\bar{X} - C)^2 \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 + 2(\bar{X} - C) \overbrace{\sum_{i=1}^n (X_i - \bar{X})}^{=0} + n(\bar{X} - C)^2 \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - C)^2 \\ &\geq \sum_{i=1}^n (X_i - \bar{X})^2 \end{aligned}$$

since $n(\bar{X} - C)^2 \geq 0$ for any choice of C . Equality is obtained only when $C = \bar{X}$, so that $n(\bar{X} - C)^2 = 0$.

■

2.5.6 An alternative formula for the sample variance

The sum of squares can also be expressed as:

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2.$$

Proof: We have:

$$\begin{aligned} \sum_{i=1}^n (X_i - \bar{X})^2 &= \sum_{i=1}^n (X_i^2 - 2X_i\bar{X} + \bar{X}^2) \\ &= \sum_{i=1}^n X_i^2 - 2\bar{X} \sum_{i=1}^n X_i + \sum_{i=1}^n \bar{X}^2 \\ &= \sum_{i=1}^n X_i^2 - 2\bar{X} \underbrace{\sum_{i=1}^n X_i}_{=n\bar{X}} + \underbrace{\sum_{i=1}^n \bar{X}^2}_{=n\bar{X}^2} \\ &= \sum_{i=1}^n X_i^2 - n\bar{X}^2. \end{aligned}$$

Therefore, the sample variance can be calculated as:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right)$$

and the sample standard deviation is $S = \sqrt{S^2}$. This formula is most convenient for calculations done by hand when summary statistics such as $\sum_{i=1}^n X_i$ and $\sum_{i=1}^n X_i^2$ are provided. ■

2.5.7 Sample moments

Moments and central moments were covered in **ST2133 Advanced statistics: distribution theory**. Here we consider their sample counterparts. Let us define, for a variable X and for each $r = 1, 2, \dots$, the following:

- the r th sample moment about zero is:

$$M_r = \frac{1}{n} \sum_{i=1}^n X_i^r$$

- the r th central sample moment is:

$$M'_r = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^r.$$

In other words, these are sample averages of the powers X_i^r and $(X_i - \bar{X})^r$, respectively. Clearly:

$$\bar{X} = M_1 \quad \text{and} \quad S^2 = \frac{n}{n-1} M'_2 = \frac{1}{n-1} (nM_2 - n(M_1)^2).$$

Moments of powers 3 and 4 are used in two more summary statistics which are described next. These are used much less often than measures of central tendency and dispersion.

2.5.8 Sample skewness

A measure of the **skewness** of the distribution of a variable X is:

$$G_1 = \frac{M'_3}{S^3} = \frac{\sum_{i=1}^n (X_i - \bar{X})^3 / n}{\left(\sum_{i=1}^n (X_i - \bar{X})^2 / (n-1) \right)^{3/2}}.$$

For this measure, $G_1 = 0$ for a symmetric distribution, $G_1 > 0$ for a positively-skewed distribution, and $G_1 < 0$ for a negatively-skewed distribution.

2.5.9 Sample kurtosis

Kurtosis refers to yet another characteristic of a sample distribution. This has to do with the relative sizes of the ‘peak’ and tails of the distribution (think about shapes of histograms).

- A distribution with high kurtosis (i.e. *leptokurtic*) has a sharp peak and a high proportion of observations in the tails far from the peak.
- A distribution with low kurtosis (i.e. *platykurtic*) is ‘flat’, with no pronounced peak with most of the observations spread evenly around the middle and weak tails.

A sample measure of kurtosis is:

$$G_2 = \frac{M'_4}{(M'_2)^2} - 3 = \frac{\sum_{i=1}^n (X_i - \bar{X})^4 / n}{\left(\sum_{i=1}^n (X_i - \bar{X})^2 / n \right)^2} - 3.$$

$G_2 > 0$ for leptokurtic distributions and $G_2 < 0$ for platykurtic distributions, and $G_2 = 0$ for the normal distribution. Some software packages define a measure of kurtosis without the -3 , i.e. ‘excess kurtosis’.

2.5.10 Calculation of sample quantiles

This is how computer software calculates general sample quantiles (or how you can do so by hand, if you ever needed to).

Suppose we need to calculate the c th sample quantile, q_c , where $0 < c < 100$. Let $R = (n+1)c/100$, and define r as the integer part of R and $f = R - r$ as the fractional part (if R is an integer, $r = R$ and $f = 0$). It follows that:

$$q_c = X_{(r)} + f(X_{(r+1)} - X_{(r)}) = (1-f)X_{(r)} + fX_{(r+1)}.$$

For example, if $n = 10$:

- for q_{50} (the median): $R = 5.5$, $r = 5$, $f = 0.5$, and so we have:

$$q_{50} = X_{(5)} + 0.5(X_{(6)} - X_{(5)}) = 0.5(X_{(5)} + X_{(6)})$$

as before

- for q_{25} (the first quartile): $R = 2.75$, $r = 2$, $f = 0.75$, and so:

$$q_{25} = X_{(2)} + 0.75(X_{(3)} - X_{(2)}) = 0.25X_{(2)} + 0.75X_{(3)}.$$

Example 2.3 Let X_1 , X_2 and X_3 be real numbers with $\bar{X} = (X_1 + X_2 + X_3)/3$. Show that:

(a) $\sum_{i=1}^3 (X_i - \bar{X}) = 0$

(b) $\sum_{i=1}^3 \sum_{j=1}^3 (X_i - \bar{X})(X_j - \bar{X}) = 0$

(c) $\sum_{i=1}^3 \sum_{i \neq j}^3 (X_i - \bar{X})(X_j - \bar{X}) = -\sum_{i=1}^3 (X_i - \bar{X})^2$.

Hint: there are three terms in the expression of (a), nine terms in (b) and six terms in (c). Write out the terms, and try and find ways to simplify them which avoid the need for a lot of messy algebra!

Solution

- (a) We have:

$$\sum_{i=1}^3 (X_i - \bar{X}) = (X_1 - \bar{X}) + (X_2 - \bar{X}) + (X_3 - \bar{X}).$$

However:

$$3\bar{X} = X_1 + X_2 + X_3$$

hence:

$$\sum_{i=1}^3 (X_i - \bar{X}) = 3\bar{X} - 3\bar{X} = 0.$$

- (b) We have:

$$\begin{aligned} \sum_{i=1}^3 \sum_{j=1}^3 (X_i - \bar{X})(X_j - \bar{X}) &= (X_1 - \bar{X})((X_1 - \bar{X}) + (X_2 - \bar{X}) + (X_3 - \bar{X})) \\ &\quad + (X_2 - \bar{X})((X_1 - \bar{X}) + (X_2 - \bar{X}) + (X_3 - \bar{X})) \\ &\quad + (X_3 - \bar{X})((X_1 - \bar{X}) + (X_2 - \bar{X}) + (X_3 - \bar{X})) \\ &= ((X_1 - \bar{X}) + (X_2 - \bar{X}) + (X_3 - \bar{X}))^2 \end{aligned}$$

and $3\bar{X} = X_1 + X_2 + X_3$ as above, so:

$$\sum_{i=1}^3 \sum_{j=1}^3 (X_i - \bar{X})(X_j - \bar{X}) = 0^2 = 0.$$

(c) We have:

$$\sum_{i=1}^3 \sum_{j=1}^3 (X_i - \bar{X})(X_j - \bar{X}) = \sum_{i=1}^3 \sum_{j=1, j \neq i}^3 (X_i - \bar{X})(X_j - \bar{X}) + \sum_{i=1}^3 (X_i - \bar{X})^2$$

We have written the nine terms in the left-hand expression as the sum of the six terms for which $i \neq j$, and the three terms for which $i = j$.

However, we showed in (b) that the left-hand expression is in fact 0, so:

$$0 = \sum_{i=1}^3 \sum_{j=1, j \neq i}^3 (X_i - \bar{X})(X_j - \bar{X}) + \sum_{i=1}^3 (X_i - \bar{X})^2$$

from which the result follows.

Example 2.4 For constants a and b , show that:

- (a) $\bar{y} = a\bar{x} + b$, where $y_i = ax_i + b$ for $i = 1, 2, \dots, n$
- (b) $\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2$
- (c) $\text{s.d.}_y = |a| \text{s.d.}_x$, where s.d._y is the standard deviation of y etc.

Solution

(a) We have:

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n} = \frac{\sum_{i=1}^n (ax_i + b)}{n} = \frac{a \sum_{i=1}^n x_i + nb}{n} = a\bar{x} + b.$$

- (b) Multiply out the square within the summation sign and then evaluate the three expressions, remembering that $\bar{x}$ is a constant with respect to summation and can be taken outside the summation sign as a common factor, i.e. we have:

$$\begin{aligned} \sum_{i=1}^n (x_i - \bar{x})^2 &= \sum_{i=1}^n (x_i^2 - 2x_i\bar{x} + \bar{x}^2) \\ &= \sum_{i=1}^n x_i^2 - 2\bar{x} \sum_{i=1}^n x_i + \sum_{i=1}^n \bar{x}^2 \\ &= \sum_{i=1}^n x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 \end{aligned}$$

hence the result. Recall that $\sum_{i=1}^n x_i = n\bar{x}$.

- (c) It is probably best to work with variances to avoid the square roots. The

variance of y values, say s_y^2 , is given by:

$$\begin{aligned} s_y^2 &= \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2 \\ &= \frac{1}{n-1} \sum_{i=1}^n (ax_i + b - (a\bar{x} + b))^2 \\ &= a^2 \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= a^2 s_x^2. \end{aligned}$$

The result follows on taking the square root, observing that the standard deviation cannot be a negative quantity.

Example 2.5 Use the results in Example 2.4 to answer the following.

- (a) What are the mean and standard deviation of the set:

$$\{x_1 + k, x_2 + k, \dots, x_n + k\}$$

where k is a constant?

- (b) What are the mean and standard deviation of the set $\{cx_1, cx_2, \dots, cx_n\}$ where c is a constant?

Solution

- (a) Adding a constant k to each value of a dataset adds k to the mean and leaves the standard deviation unchanged. This corresponds to a transformation $y_i = ax_i + b$ with $a = 1$ and $b = k$. Apply (a) and (c) in Example 2.4 with these values.
- (b) Multiplying each value of a dataset by a constant c multiplies the mean by c and also the standard deviation by $|c|$. This corresponds to a transformation $y_i = cx_i$ with $a = c$ and $b = 0$. Apply (a) and (c) in Example 2.4 with these values.

Activity 2.1 Show that:

$$\sum_{i=1}^n \sum_{j=1}^n (x_i - x_j)^2 = 2n \left[\sum_{i=1}^n (x_i - \bar{x})^2 \right].$$

Note solutions to all activities can be found in Appendix B.

2.6 Populations, samples and parameters

In the univariate setting, we consider a real-world random variable X , with assumed probability distribution F_X . An *observed sample* of size n is a collection of real numbers:

$$\mathbf{x} = \{x_1, x_2, \dots, x_n\} = (x_1, x_2, \dots, x_n)^T$$

drawn from F_X which can be viewed as one *realisation* of a collection of random variables, the *sample*, denoted as:

$$\mathbf{X} = \{X_1, X_2, \dots, X_n\} = (X_1, X_2, \dots, X_n)^T.$$

We will assume that the sample is a sequence of independent and identically distributed random variables drawn from F_X , referred to as a *random sample*.

Random sample versus observed sample

A **random sample** is a sequence of random variables:

$$\mathbf{X} = \{X_1, X_2, \dots, X_n\}$$

from a population with distribution function F_X if $\{X_1, X_2, \dots, X_n\}$ are mutually independent, and:

$$X_i \sim F_X \quad \text{for } i = 1, 2, \dots, n.$$

An **observed sample** is a realised collection of observations:

$$\mathbf{x} = \{x_1, x_2, \dots, x_n\}$$

on $\{X_1, X_2, \dots, X_n\}$.

The assumptions we make about $\{X_1, X_2, \dots, X_n\}$ form our *model*, i.e. our approach to represent the data-generating process mathematically. We need a model for the entire population because the research questions we ask concern the entire population, not just the data we have. Having a model for the population tells us that the remaining population is not that much different from our data or, in other words, that the data are *representative* of the population. We need a *random* model because the process of drawing a sample from a population is a bit like the process of generating random variables. A different sample would produce different values. Therefore, the population from which we draw a random sample is represented as a probability distribution.

So a random sample of size n implies a particular probability model described by the population $F_{X_i}(x_i; \theta)$, that is by the distribution function of each X_i . Notice that it depends on one or more parameters θ (we will use $\boldsymbol{\theta}$ to denote a vector of scalar parameters), and if we know θ then the model would be completely specified. However, θ is in general unknown and it is the object we are interested in estimating. For this reason we highlight the dependence on θ when writing the distribution function, $F_{X_i}(x_i; \theta)$, the density function (for continuous random variables), $f_{X_i}(x_i; \theta)$, and mass function (for discrete random variables), $p_{X_i}(x_i; \theta)$.

Joint density function

Let X be a continuous random variable. Since $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ is assumed to be a sequence of independent and identically distributed random variables, then the **joint density function** is given by:

$$\begin{aligned} f_{\mathbf{X}}(\mathbf{x}; \theta) &= f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n; \theta) = f_X(x_1; \theta) f_X(x_2; \theta) \cdots f_X(x_n; \theta) \\ &= \prod_{i=1}^n f_X(x_i; \theta) \end{aligned}$$

where θ is constant because the random variables are identically distributed, and we take the product because the random variables are mutually independent.

Example 2.6 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from an exponential distribution with parameter $\lambda > 0$. The joint density function of the random sample $\mathbf{X}$ is:

$$f_{\mathbf{X}}(\mathbf{x}; \lambda) = f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n; \lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum_{i=1}^n x_i}.$$

Example 2.7 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a $N(\mu, \sigma^2)$ population. The joint density function of the random sample $\mathbf{X}$ is:

$$\begin{aligned} f_{\mathbf{X}}(\mathbf{x}; \mu, \sigma^2) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) \\ &= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right). \end{aligned}$$

The joint density function is also equal to:

$$f_{\mathbf{X}}(\mathbf{x}; \mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{n(\bar{x} - \mu)^2 + (n-1)s^2}{2\sigma^2}\right).$$

To prove this, it suffices to show that:

$$\sum_{i=1}^n (x_i - \mu)^2 = n(\bar{x} - \mu)^2 + \sum_{i=1}^n (x_i - \bar{x})^2$$

since $s^2 = \sum_{i=1}^n (x_i - \bar{x})^2 / (n-1)$. Note that:

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n (x_i^2 - 2\mu x_i + \mu^2) = \sum_{i=1}^n x_i^2 - 2\mu n\bar{x} + n\mu^2$$

and:

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n (x_i^2 - 2\bar{x}x_i + \bar{x}^2) = \sum_{i=1}^n x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2.$$

Subtracting the second of these equations from the first yields:

$$\sum_{i=1}^n (x_i - \mu)^2 - \sum_{i=1}^n (x_i - \bar{x})^2 = n\bar{x}^2 - 2\mu n\bar{x} + n\mu^2 = n(\bar{x} - \mu)^2.$$

Hence:

$$\sum_{i=1}^n (x_i - \mu)^2 = n(\bar{x} - \mu)^2 + \sum_{i=1}^n (x_i - \bar{x})^2.$$

Joint mass function

Let X be a discrete random variable. Since $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ is assumed to be a sequence of independent and identically distributed random variables, then the **joint mass function** is given by:

$$\begin{aligned} p_{\mathbf{X}}(\mathbf{x}; \theta) &= p_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n; \theta) = p_X(x_1; \theta) p_X(x_2; \theta) \cdots p_X(x_n; \theta) \\ &= \prod_{i=1}^n p_X(x_i; \theta). \end{aligned}$$

Example 2.8 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a Poisson distribution with parameter $\lambda > 0$. The joint mass function is:

$$p_{\mathbf{X}}(\mathbf{x}; \lambda) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n; \lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}.$$

2.6.1 Sampling with replacement versus without replacement

When selecting members of a population, we have a choice of whether to sample with or without *replacement*.

- **Sampling with replacement:** A sample is selected at random from the population. After each member is selected and observed, it is replaced such that every member of the population has an equal probability of selection which remains unchanged throughout the sampling process. Population members *can* be selected more than once.
- **Sampling without replacement:** Once selected and observed, selected population members are not returned to the sampling population. Population members *cannot* be selected more than once.

A (true) random sample is when sampling is performed *with* replacement. In this case the population from which selections are made never changes, hence $\{X_1, X_2, \dots, X_n\}$

comprises identically distributed random variables since $X_i \sim F_X$ for $i = 1, 2, \dots, n$. The random variables are also mutually independent since the first selected population member does not affect which members are subsequently selected, for example:

$$P(X_2 \leq x_2 | X_1 \leq x_1) = P(X_2 \leq x_2) = F_X(x_2).$$

However, when sampling *without* replacement, after the first selection all subsequent selections are made from an ever-smaller population, whose composition depends on which members were previously selected. Hence the random variables in the sample are not identically distributed, nor are they independent. However, if the sample size n is small relative to the population size N , such that $n \ll N$ (where ‘ $\ll$ ’ means ‘much less than’) then in practice the difference between sampling with and without replacement is negligible such that a random sample is a sufficiently good approximating model even when sampling without replacement.

2.6.2 Parameters

The population distribution F_X has certain attributes which interest us. These attributes are represented by a vector of r *parameters*, denoted $\boldsymbol{\theta}$, where $r \geq 1$, where:

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_r)^T$$

such that $\theta_i \in \mathbb{R}$ for $i = 1, 2, \dots, r$. The **parameter space** is $\Theta \subseteq \mathbb{R}^r$. Examples of parameters of interest include the population mean, $\mu_X = E(X)$, and the population variance, $\sigma_X^2 = \text{Var}(X)$.

In problems of statistical inference, we will assume the parametric family $F_X(x; \boldsymbol{\theta})$ is known, but the vector of parameters $\boldsymbol{\theta}$ is unknown. Statistical inference will then involve the estimation and/or testing of the values of $\boldsymbol{\theta}$ using the observed sample $\mathbf{x}$ drawn from $F_X(x; \boldsymbol{\theta})$. The connection between a population and a sample is shown in Figure 2.1, where F_X is parameterised by the parameter $\theta \in \mathbb{R}$.

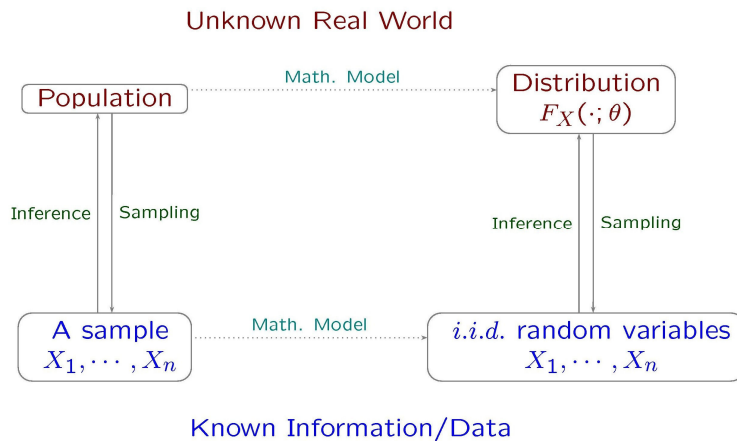


Figure 2.1: Representation of the connection between a population and a sample.

Example 2.9 As seen in **ST2133 Advanced statistics: distribution theory**, examples of parameters associated with common probability distributions include:

- π for the Bernoulli, binomial and geometric distributions, i.e. $X \sim \text{Bernoulli}(\pi)$, $X \sim \text{Bin}(n, \pi)$ and $X \sim \text{Geo}(\pi)$
- λ for the Poisson and exponential distributions, i.e. $X \sim \text{Pois}(\lambda)$ and $X \sim \text{Exp}(\lambda)$
- a and b for the continuous uniform distribution, i.e. $X \sim \text{Uniform}[a, b]$
- μ and σ^2 for the normal distribution, i.e. $X \sim N(\mu, \sigma^2)$
- α and β for the gamma and beta distributions, i.e. $X \sim \text{Gamma}(\alpha, \beta)$ and $X \sim B(\alpha, \beta)$.

We typically assume an appropriate theoretical probability distribution as an approximating model of X such that the ‘stylised facts’ (real-world characteristics such as support, skewness etc.) of X are adequately captured by F_X .

2.7 Inference, data reduction and statistics

Given an observed sample $\mathbf{x}$, *inference* requires us to *infer* characteristics of F_X . The raw data provided by $\mathbf{x}$ must be ‘refined’ by calculating various descriptive statistics (also called summary statistics) of interest. In **ST104a Statistics 1**, common descriptive statistics such as the sample mean, $\bar{x}$, and sample variance, s^2 , were introduced. Both of these statistics are examples of **data reduction** in that they reduce n real numbers $\{x_1, x_2, \dots, x_n\}$ into a single number. While there is a benefit of data reduction (simplicity by reducing the dimensionality of the raw data), it comes at a cost in terms of a loss of *information*.

Example 2.10 Consider the observed sample:

$$\mathbf{x} = \{10, 8, 13, 9, 11, 14, 6, 4, 12, 7, 5\}.$$

This yields a sample mean of:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{99}{11} = 9$$

and a sample variance of:

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{10} \times 110 = 11.$$

Suppose a second observed sample is:

$$\mathbf{y} = \{8, 8, 8, 8, 8, 8, 8, 19, 8, 8, 8\}.$$

This yields a sample mean of:

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i = \frac{99}{11} = 9$$

and a sample variance of:

$$s_y^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2 = \frac{1}{10} \times 110 = 11.$$

So the observed samples $\mathbf{x}$ and $\mathbf{y}$ have the same mean and variance, but $\mathbf{x}$ and $\mathbf{y}$ are clearly very different. Data reduction is achieved as we can summarise $n = 11$ values into two summary attributes (the mean and variance), but we have lost some information – knowledge of these summary statistics alone would not allow us to recover the raw values contained in $\mathbf{x}$ and $\mathbf{y}$, respectively.

A ‘good’ inferential procedure should provide an acceptable compromise between data reduction and loss of information. We seek an adequate summary of the data while not discarding too much relevant information. We will consider the concept of *sufficiency* later in this chapter.

The rest of the chapters in this course cover point estimation, interval estimation and hypothesis testing, respectively. All three topics were covered in **ST104b Statistics 2**, although in this course we study these at a deeper level. All inferential procedures related to these topics use *statistics*, each of which has a distribution known as a *sampling distribution*.

Statistic and sampling distribution

For a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ of size n drawn from a population, let $T(\mathbf{x}) = T(x_1, x_2, \dots, x_n)$ be a real-valued function (or $\mathbf{T}(\mathbf{x})$ a vector-valued function) with domain including the sample space of $\{X_1, X_2, \dots, X_n\}$. This function is a function of the sample and known constants only; it is never a function of unknown parameters.

A **statistic** is the random variable:

$$Y = T(\mathbf{X}) = T(X_1, X_2, \dots, X_n)$$

or random vector:

$$\mathbf{Y} = \mathbf{T}(\mathbf{X}).$$

The **observed value of a statistic** given an observed sample $\mathbf{x}$ is $y = T(\mathbf{x})$, and can be thought of as one possible realisation of the statistic $Y = T(\mathbf{X})$.

The probability distribution of a statistic Y is the **sampling distribution** of Y .

Useful statistics have dimensions which are smaller than that of the sample. We will tend to focus on scalar statistics, i.e. statistics which reduce $\mathbf{x}$ to a single value; that is, n values are reduced to 1. Point estimation of parameters uses scalar statistics heavily.

2.7.1 Point estimation

Suppose a probability distribution F_X is parameterised by the scalar parameter θ , such that θ is an unknown number. Any estimate of θ based on a random sample of size n can be represented by a function $T : \mathbb{R}^n \rightarrow \mathbb{R}$. Applied to an observed sample $\mathbf{x}$ it produces a *point estimate* $T(\mathbf{x})$, which is a real number. In order to examine the statistical properties of the point estimation method, we apply the function to a random sample, i.e. $T(\mathbf{X})$, which is the *point estimator*.

Point estimator and point estimate

A scalar statistic, $T(\mathbf{X})$, may be used as a **point estimator** of a scalar parameter θ . An observed value of this statistic, $T(\mathbf{x})$, is a **point estimate**.

Chapter 3 covers point estimation in detail, extending the material covered in **ST104b Statistics 2**. Example 2.11 considers some common point estimators of parameters.

Example 2.11 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample with an observed sample of $\mathbf{x} = \{x_1, x_2, \dots, x_n\}$. The so-called *order statistics*, denoted as $x_{(1)}, x_{(2)}, \dots, x_{(n)}$, represent the observed values when placed in ascending order.

(a) The sample mean:

$$T(\mathbf{X}) = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

is the (usual) estimator of the population mean, μ , with observed value:

$$T(\mathbf{x}) = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

(b) The sample variance:

$$T(\mathbf{X}) = S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

is the (usual) estimator of the population variance, σ^2 , with observed value:

$$T(\mathbf{x}) = s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

(c) The sample minimum, or first order statistic:

$$T(\mathbf{X}) = X_{(1)} = \min_{1 \leq i \leq n} X_i$$

is the estimator of the population minimum, with observed value:

$$T(\mathbf{x}) = x_{(1)} = \min_{1 \leq i \leq n} x_i.$$

(d) The sample maximum, or n th order statistic:

$$T(\mathbf{X}) = X_{(n)} = \max_{1 \leq i \leq n} X_i$$

is the estimator of the population maximum, with observed value:

$$T(\mathbf{x}) = x_{(n)} = \max_{1 \leq i \leq n} x_i.$$

(e) The sample median:

$$T(\mathbf{X}) = \begin{cases} X_{((n+1)/2)} & \text{for odd } n \\ (X_{(n/2)} + X_{(n/2+1)})/2 & \text{for even } n \end{cases}$$

is the estimator of the population median, with observed value:

$$T(\mathbf{x}) = \begin{cases} x_{((n+1)/2)} & \text{for odd } n \\ (x_{(n/2)} + x_{(n/2+1)})/2 & \text{for even } n. \end{cases}$$

Point estimator of a vector of parameters

Often we have more than one parameter of interest, which we group together into a vector:

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_r)^T$$

which can be considered either as a single vector parameter or as a vector of scalar parameters. For notational brevity, the transpose T is often dropped, hence we may write $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_r)$. The statistic used as a point estimator of $\boldsymbol{\theta}$ is an $r \times 1$ random vector with a corresponding point estimate which is an $r \times 1$ vector of real numbers, i.e. a point in $\mathbb{R}^r$.

Example 2.12 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, where μ and σ^2 are unknown. We have $\boldsymbol{\theta} = (\mu, \sigma^2)$. A point estimator of $\boldsymbol{\theta}$ would be:

$$\hat{\boldsymbol{\theta}} = (\bar{X}, S^2).$$

2.7.2 Interval estimation

A point estimate provides a single value estimate of a parameter of interest based on an observed sample $\mathbf{x}$. While a random sample is free of selection bias, there is no guarantee that the observed sample will be completely representative of the population distribution from which it is drawn, hence there is a possibility of *sampling error* (also known as *estimation error*). Interval estimation takes into account the likely sampling error by providing an interval estimate in which we believe the true parameter value lies.

Interval estimator

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from F_X parameterised by θ . Let $Y_1 = T_1(\mathbf{X})$ and $Y_2 = T_2(\mathbf{X})$ be statistics with $Y_1 \leq Y_2$. The random interval:

$$[Y_1, Y_2]$$

is an **interval estimator** of θ . If the observed sample is $\mathbf{x}$, then the observed values of the statistics are $y_1 = T_1(\mathbf{x})$ and $y_2 = T_2(\mathbf{x})$, respectively. The interval $[y_1, y_2]$ is hence an interval estimate of θ .

Chapter 4 covers interval estimation in detail, extending the material covered in **ST104b Statistics 2**. Recall we seek an interval estimate which is as narrow as possible while having a large probability that it covers the true parameter value. The balance between *interval length* and *coverage probability* will be discussed in Chapter 4. Example 2.13 considers a common application of interval estimation.

Example 2.13 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, where μ is unknown (but σ^2 is known). We seek an interval estimator of μ . The sample mean and its sampling distribution are:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

Standardising, we have:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

For $Z \sim N(0, 1)$ it is well-known that:

$$P(-1.96 \leq Z \leq 1.96) = 0.95$$

hence:

$$P\left(-1.96 \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq 1.96\right) = 0.95.$$

Rearranging, we have:

$$P\left(\bar{X} - 1.96 \times \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + 1.96 \times \frac{\sigma}{\sqrt{n}}\right) = 0.95.$$

Therefore, a sensible interval estimator of μ is:

$$\left[\bar{X} - 1.96 \times \frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96 \times \frac{\sigma}{\sqrt{n}}\right]$$

with a corresponding interval estimate of:

$$\left[\bar{x} - 1.96 \times \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \times \frac{\sigma}{\sqrt{n}}\right]$$

which has a 95% confidence level.

2.7.3 Hypothesis testing

Hypothesis testing was first introduced in **ST104a Statistics 1** from an applied perspective, with theoretical aspects considered in **ST104b Statistics 2**. Using a random sample we test whether there the observed sample $\mathbf{x}$ provides sufficient *evidence* to justify rejecting the null hypothesis, denoted H_0 , which is a statement about the population.

An appropriate *test statistic* is constructed, with its distribution under H_0 known. Conditional on H_0 being true, we consider the extremity of the observed value of the test statistic in terms of the test statistic distribution. Recall inferential errors can be made in the form of Type I errors (false positives) and Type II errors (false negatives). We have to balance the probability of these errors, although we typically control for the probability of a Type I error through the *significance level*, and then we seek to minimise the probability of a Type II error by opting for a hypothesis testing procedure which maximises the *power* of the test. Chapter 5 covers hypothesis testing in detail, extending the material covered in **ST104b Statistics 2**. Example 2.14 is a reminder of a z -test of a population mean.

Example 2.14 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, where μ is unknown (but σ^2 is known). We wish to test:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu \neq \mu_0.$$

Under H_0 , the sample mean and its sampling distribution are:

$$\bar{X} \sim N\left(\mu_0, \frac{\sigma^2}{n}\right).$$

Standardising, we have:

$$\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$$

hence:

$$P\left(-1.96 \leq \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \leq 1.96\right) = 0.95.$$

Using this fact, we may construct a rejection region, R , for the test:

$$R = \{T(\mathbf{x}) \in \mathbb{R} : T(\mathbf{x}) < -1.96 \text{ or } T(\mathbf{x}) > 1.96\}$$

where:

$$T(\mathbf{x}) = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}.$$

Hence we reject H_0 if:

$$T(\mathbf{x}) < -1.96 \quad \text{or} \quad T(\mathbf{x}) > 1.96.$$

By construction, the probability of a Type I error (the significance level) is $\alpha = 0.05$, or 5%.

2.7.4 Assumptions in statistical inference

In order to make statistical inferences, it is necessary to make assumptions about the relationship between the population and the sample. These assumptions could be quite weak, such as assuming that all the random variables in the sample are identically distributed. In general, stronger assumptions allow us to make stronger statements about the population. The risk is that a model with strong assumptions is less likely to be a reasonable model for a real-world situation. In practice, it is necessary to balance a model's ability to make statements about the population and the danger of introducing unrealistic assumptions which would invalidate our inferences.

2.8 Sufficiency

Each univariate statistic reduces an observed sample $\mathbf{x}$ into a single number, hence statistics serve a useful purpose in terms of data reduction. However, we have previously noted that inevitably there is some loss of information whenever we undertake data reduction. A good statistical inferential procedure is based on statistics which do not throw away too much information. The concept of *sufficiency* makes this more precise.

2.8.1 Sufficient statistics

Let $\mathbf{X}$ be a random sample parameterised by $\boldsymbol{\theta}$ (a vector of scalar parameters). Sufficiency considers a statistic which summarises, or captures, all of the *relevant* information about $\boldsymbol{\theta}$ contained in $\mathbf{X}$.

Sufficient statistic

Let $\mathbf{X}$ be a random sample parameterised by $\boldsymbol{\theta}$. A statistic $\mathbf{U} = \mathbf{T}(\mathbf{X})$ is a **sufficient statistic** for $\boldsymbol{\theta}$ if the conditional distribution of $\mathbf{X}$ given $\mathbf{U}$ does not depend on $\boldsymbol{\theta}$.

The above definition of a sufficient statistic makes reference to the (conditional) distribution of $\mathbf{X}$. As such, we can speak of a statistic being sufficient for a family of distributions, i.e. $\{F_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) : \boldsymbol{\theta} \in \boldsymbol{\Theta}\}$. Hence $\mathbf{U}$ being a sufficient statistic equates to saying that the conditional distribution function $F_{\mathbf{X}|\mathbf{U}}(\mathbf{x}|\mathbf{u})$ is not a function of $\boldsymbol{\theta}$. Equivalently, if $f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta})$ is the joint density function of $\mathbf{X}$, then $f_{\mathbf{X}|\mathbf{U}}(\mathbf{x}|\mathbf{u})$ is not a function of $\boldsymbol{\theta}$ (the continuous case). Similarly, if $p_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta})$ is the joint mass function of $\mathbf{X}$, then $p_{\mathbf{X}|\mathbf{U}}(\mathbf{x}|\mathbf{u})$ is not a function of $\boldsymbol{\theta}$ (the discrete case).

The conditional mass and density functions $p_{\mathbf{X}|\mathbf{U}}(\mathbf{x}|\mathbf{u})$ and $f_{\mathbf{X}|\mathbf{U}}(\mathbf{x}|\mathbf{u})$, respectively, may seem strange since we are conditioning on $\mathbf{U}$ which, of course, is a function of the random sample $\mathbf{X}$. The properties of conditioning in this way allow us to derive results for sufficient statistics.

We consider the discrete case such that the conditional mass function can be viewed as a probability. Note that since $\mathbf{U} = \mathbf{T}(\mathbf{X})$, we have $P(\mathbf{X} = \mathbf{x}, \mathbf{U} = \mathbf{u}) = P(\mathbf{X} = \mathbf{x})$, indeed the event $\{\mathbf{X} = \mathbf{x}\}$ is a subset of the event $\{\mathbf{T}(\mathbf{X}) = \mathbf{T}(\mathbf{x})\}$ (consider, for example, the case $\mathbf{T}(\mathbf{X}) = \sum_{i=1}^n X_i$).

Conditioning on a function of the sample

Let $\mathbf{X}$ be a random sample from a discrete distribution, and let $\mathbf{U} = \mathbf{T}(\mathbf{X})$ be a (multivariate) statistic. The conditional mass function of $\mathbf{X}$ given $\mathbf{U}$ is:

$$p_{\mathbf{X}|\mathbf{U}}(\mathbf{x} | \mathbf{u}) = \begin{cases} \frac{p_{\mathbf{X}}(\mathbf{x})}{p_{\mathbf{U}}(\mathbf{u})} & \text{for } \mathbf{u} = \mathbf{T}(\mathbf{x}) \text{ and } p_{\mathbf{U}}(\mathbf{u}) \neq 0 \\ 0 & \text{otherwise.} \end{cases}$$

Proof: We define $\mathbf{u} = \mathbf{T}(\mathbf{x})$, hence:

$$\mathbf{X} = \mathbf{x} \Rightarrow \mathbf{T}(\mathbf{X}) = \mathbf{T}(\mathbf{x}) \Rightarrow \mathbf{U} = \mathbf{u}.$$

Expressed in terms of events, we have:

$$\{\mathbf{X} = \mathbf{x}\} \Rightarrow \{\mathbf{U} = \mathbf{u}\}.$$

Therefore, by standard probability theory, the joint mass function of $\mathbf{X}$ and $\mathbf{U}$ is:

$$p_{\mathbf{X},\mathbf{U}}(\mathbf{x}, \mathbf{u}) = P(\mathbf{X} = \mathbf{x}, \mathbf{U} = \mathbf{u}) = P(\mathbf{X} = \mathbf{x}) = p_{\mathbf{X}}(\mathbf{x}).$$

If $\mathbf{u} \neq \mathbf{T}(\mathbf{x})$, then $\{\mathbf{X} = \mathbf{x}\} \cap \{\mathbf{U} = \mathbf{u}\} = \emptyset$, hence $p_{\mathbf{X}|\mathbf{U}}(\mathbf{x} | \mathbf{u}) = 0$. By definition of conditional probability:

$$p_{\mathbf{X}|\mathbf{U}}(\mathbf{x} | \mathbf{u}) = \frac{p_{\mathbf{X},\mathbf{U}}(\mathbf{x}, \mathbf{u})}{p_{\mathbf{U}}(\mathbf{u})} = \frac{p_{\mathbf{X}}(\mathbf{x})}{p_{\mathbf{U}}(\mathbf{u})}.$$

■

Similarly, in the continuous case, if $f_{\mathbf{X}}(\mathbf{x})$ is the joint density function of the random sample $\mathbf{X}$, and $f_{\mathbf{U}}(\mathbf{u})$ is the density function of $\mathbf{U}$, then $\mathbf{U}$ is a sufficient statistic for θ if:

$$f_{\mathbf{X}|\mathbf{U}}(\mathbf{x} | \mathbf{u}) = \frac{f_{\mathbf{X},\mathbf{U}}(\mathbf{x}, \mathbf{u})}{f_{\mathbf{U}}(\mathbf{u})} = \frac{f_{\mathbf{X}}(\mathbf{x})}{f_{\mathbf{U}}(\mathbf{u})} \quad \text{for } \mathbf{u} = \mathbf{T}(\mathbf{x}) \text{ and } f_{\mathbf{U}}(\mathbf{u}) \neq 0$$

is a constant as a function of θ for all $\mathbf{x}$ (that is, it does not depend on θ), and 0 otherwise.

2.8.2 The sufficiency principle

If we deem $\mathbf{U}$ to be a sufficient statistic for θ then it contains all *relevant* information from a random sample, hence any statistical inference made about θ should only make use of $\mathbf{U}$. This is known as the *sufficiency principle*.

Sufficiency principle

If $\mathbf{U} = \mathbf{T}(\mathbf{X})$ is a sufficient statistic for θ then any inference about θ should be based on the random sample $\mathbf{X}$ only through $\mathbf{T}(\mathbf{X})$.

That is, if $\mathbf{x}$ and $\mathbf{y}$ are two observed samples in the univariate case (i.e. we observe $\mathbf{x} = \{x_1, x_2, \dots, x_n\}$ and $\mathbf{y} = \{y_1, y_2, \dots, y_n\}$ such that $T(\mathbf{x}) = T(\mathbf{y})$, then the inference about θ should be the same regardless of whether $\mathbf{X} = \mathbf{x}$ or $\mathbf{X} = \mathbf{y}$ was observed.

Example 2.15 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a $\text{Pois}(\lambda)$ distribution, and let $U = T(\mathbf{X}) = \sum_{i=1}^n X_i$. It can be shown that $U \sim \text{Pois}(n\lambda)$ (see **ST104b Statistics 2**). We can also write:

$$P(\mathbf{X} = \mathbf{x} | U = u) = \frac{P(\mathbf{X} = \mathbf{x}, U = u)}{P(U = u)}$$

and note that:

$$P(\mathbf{X} = \mathbf{x}, U = u) = \begin{cases} P(\mathbf{X} = \mathbf{x}) & \text{for } U = u \\ 0 & \text{otherwise} \end{cases}$$

so we can then write:

$$\begin{aligned} P(\mathbf{X} = \mathbf{x} | U = u) &= \frac{P(\mathbf{X} = \mathbf{x})}{P(U = u)} = \frac{\prod_{i=1}^n e^{-\lambda} \lambda^{x_i} / x_i!}{e^{-n\lambda} (n\lambda)^u / u!} = \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i} / \prod_{i=1}^n x_i!}{e^{-n\lambda} (n\lambda)^u / u!} \\ &= \frac{\lambda^u / \prod_{i=1}^n x_i!}{n^u \lambda^u / u!} \\ &= \frac{u!}{n^u \prod_{i=1}^n x_i!}. \end{aligned}$$

We conclude that $U = T(\mathbf{X}) = \sum_{i=1}^n X_i$ is a sufficient statistic for λ , since there is no λ in the above conditional probability expression.

Example 2.16 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a $\text{Bernoulli}(\pi)$ distribution, such that:

$$p_X(x) = \begin{cases} \pi^x (1 - \pi)^{1-x} & \text{for } x = 0, 1 \\ 0 & \text{otherwise.} \end{cases}$$

Let $U = T(\mathbf{X}) = \sum_{i=1}^n X_i$. For any given observed sample $\mathbf{x}$, the number of successes (i.e. the number of 1s) is $u = \sum_{i=1}^n x_i$, and the number of failures (i.e. the number of 0s) is $n - u$. By independence, the joint mass function of $\mathbf{X}$ is:

$$p_{\mathbf{X}}(\mathbf{x}) = \pi^u (1 - \pi)^{n-u}$$

where $u = \sum_{i=1}^n x_i$, and 0 otherwise. It can be shown that $U \sim \text{Bin}(n, \pi)$ by the properties of repeated independent and identically distributed Bernoulli trials (see **ST104b Statistics 2**). Hence the mass function of U is:

$$p_U(u) = \binom{n}{u} \pi^u (1 - \pi)^{n-u} \quad \text{for } u = 0, 1, 2, \dots, n$$

and 0 otherwise. Hence:

$$p_{\mathbf{X}|U}(\mathbf{x} | u) = \frac{p_{\mathbf{X}}(\mathbf{x})}{p_U(u)} = \frac{\pi^u(1 - \pi)^{n-u}}{\binom{n}{u}\pi^u(1 - \pi)^{n-u}} = \frac{1}{\binom{n}{u}} \quad \text{for } u = 0, 1, 2, \dots, n$$

where $u = \sum_{i=1}^n x_i$, and 0 otherwise. For a sense check, this seems reasonable because if there are n trials and we know that there are u successes, then there are $\binom{n}{u}$ ways of achieving these u successes, with each one being equally likely. Since:

$$p_{\mathbf{X}|U}(\mathbf{x} | u) = \frac{1}{\binom{n}{u}}$$

does not depend on π , we conclude that $U = T(\mathbf{X}) = \sum_{i=1}^n X_i$ is a sufficient statistic for π .

Note that according to the sufficiency principle, when making inferences about π if different observed samples gave us the same value of U then we would make the same inference about π . Consider, for example, the point estimator:

$$\hat{\pi} = \frac{U}{n}$$

which is the *sample proportion* of successes in the n trials. $\hat{\pi}$ is consistent with the sufficiency principle as seen by the following two observed samples:

$$\mathbf{x} = \{0, 1, 1, 0, 1, 0, 1, 1, 1, 0\}$$

and:

$$\mathbf{y} = \{1, 1, 0, 0, 1, 1, 1, 0, 0, 1\}.$$

Despite $\mathbf{x} \neq \mathbf{y}$, nonetheless $\sum_i x_i = \sum_i y_i = 6$, so the observed value u is the same for $\mathbf{x}$ and $\mathbf{y}$. Our point estimate of π would be:

$$\hat{\pi} = \frac{u}{n} = \frac{6}{10} = 0.6$$

regardless of whether we observed $\mathbf{x}$ or $\mathbf{y}$ in practice.

2.8.3 Factorisation theorem

While the concept of a sufficient statistic may be appealing in terms of capturing all relevant information in a sample about a parameter, it does not seem obvious how we would determine what the sufficient statistic is in the first place.

Using the definition, so far we must ‘guess’ what the statistic $\mathbf{U} = \mathbf{T}(\mathbf{X})$ should be, then find the mass or density function of $\mathbf{U}$ and check to see if the ratio of the mass or density functions does not depend on $\boldsymbol{\theta}$. A good ‘guess’ requires a lot of intuition, and the checking of the ratio of functions can be tiresome.

Fortunately, the *factorisation theorem* provides a useful condition such that a statistic is sufficient.

Factorisation theorem

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample with joint density function $f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta})$. The statistic $\mathbf{U} = \mathbf{T}(\mathbf{X})$ is sufficient for the vector of parameters $\boldsymbol{\theta}$ if and only if we can find functions g and h such that:

$$f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) = g(\mathbf{T}(\mathbf{x}), \boldsymbol{\theta}) h(\mathbf{x})$$

for all $\mathbf{x} \in \mathbb{R}^n$ and $\boldsymbol{\theta} \in \Theta$. Note $\mathbf{T}(\mathbf{X})$ and $\boldsymbol{\theta}$ can be scalars, denoted $T(\mathbf{X})$ and θ , respectively.

In the discrete case with joint mass function $p_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta})$, we have:

$$p_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) = g(\mathbf{T}(\mathbf{x}), \boldsymbol{\theta}) h(\mathbf{x})$$

for all $\mathbf{x} \in \mathbb{R}^n$ and $\boldsymbol{\theta} \in \Theta$.

We give the proof for discrete distributions only. The proof for the continuous case is quite technical and beyond the scope of this course.

Proof: We begin with the ‘only if’ part of the theorem. Let $\mathbf{U} = \mathbf{T}(\mathbf{X})$ be a sufficient statistic for $\boldsymbol{\theta}$. Given the definition of a sufficient statistic, the conditional distribution of $\mathbf{X}$ given $\mathbf{U}$ does not depend on $\boldsymbol{\theta}$. Hence:

$$p_{\mathbf{X}|\mathbf{U}}(\mathbf{x} | \mathbf{u}) = h(\mathbf{x})$$

for some function h . Since the conditional mass function of $\mathbf{X}$ given $\mathbf{U}$ is:

$$p_{\mathbf{X}|\mathbf{U}}(\mathbf{x} | \mathbf{u}) = \frac{p_{\mathbf{X}}(\mathbf{x})}{p_{\mathbf{U}}(\mathbf{u})}$$

then:

$$p_{\mathbf{X}}(\mathbf{x}) = p_{\mathbf{X}|\mathbf{U}}(\mathbf{x} | \mathbf{u}) p_{\mathbf{U}}(\mathbf{u}) = h(\mathbf{x}) p_{\mathbf{U}}(\mathbf{u})$$

where $\mathbf{u} = \mathbf{T}(\mathbf{x})$. Making the dependence on the parameters explicit, we have:

$$p_{\mathbf{U}}(\mathbf{u}; \boldsymbol{\theta}) = p_{\mathbf{U}}(\mathbf{T}(\mathbf{x}); \boldsymbol{\theta}) = g(\mathbf{T}(\mathbf{x}), \boldsymbol{\theta})$$

for some function g . Substituting yields:

$$p_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) = g(\mathbf{T}(\mathbf{x}), \boldsymbol{\theta}) h(\mathbf{x}).$$

Now we consider the ‘if’ part of the theorem. Suppose we can find functions g and h such that:

$$p_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) = g(\mathbf{T}(\mathbf{x}), \boldsymbol{\theta}) h(\mathbf{x}).$$

We have:

$$\begin{aligned}
 p_{\mathbf{U}}(\mathbf{u}) &= \sum_{\mathbf{x}} p_{\mathbf{X}, \mathbf{U}}(\mathbf{x}, \mathbf{u}) && \text{(properties of discrete distributions)} \\
 &= \sum_{\mathbf{x}: T(\mathbf{x})=\mathbf{u}} p_{\mathbf{X}}(\mathbf{x}) && \text{(since } p_{\mathbf{X}, \mathbf{U}}(\mathbf{x}, \mathbf{u}) = p_{\mathbf{X}}(\mathbf{x}) \text{ when } \mathbf{u} = T(\mathbf{x})) \\
 &= \sum_{\mathbf{x}: T(\mathbf{x})=\mathbf{u}} g(T(\mathbf{x}), \boldsymbol{\theta}) h(\mathbf{x}) && \text{(factorisation theorem)} \\
 &= g(T(\mathbf{x}), \boldsymbol{\theta}) \sum_{\mathbf{x}: T(\mathbf{x})=\mathbf{u}} h(\mathbf{x}).
 \end{aligned}$$

Now if $\mathbf{u} = T(\mathbf{x})$, then:

$$p_{\mathbf{X}|\mathbf{U}}(\mathbf{x} | \mathbf{u}) = \frac{p_{\mathbf{X}}(\mathbf{x})}{p_{\mathbf{U}}(\mathbf{u})} = \frac{g(T(\mathbf{x}), \boldsymbol{\theta}) h(\mathbf{x})}{g(T(\mathbf{x}), \boldsymbol{\theta}) \sum_{\mathbf{x}: T(\mathbf{x})=\mathbf{u}} h(\mathbf{x})} = \frac{h(\mathbf{x})}{\sum_{\mathbf{x}: T(\mathbf{x})=\mathbf{u}} h(\mathbf{x})}.$$

Therefore, in full:

$$p_{\mathbf{X}|\mathbf{U}}(\mathbf{x} | \mathbf{u}) = \begin{cases} \frac{h(\mathbf{x})}{\sum_{\mathbf{x}: T(\mathbf{x})=\mathbf{u}} h(\mathbf{x})} & \text{for } \mathbf{u} = T(\mathbf{x}) \\ 0 & \text{otherwise.} \end{cases}$$

Hence the distribution of $\mathbf{X}$ conditional on $\mathbf{U}$ does not depend on $\boldsymbol{\theta}$, and so $\mathbf{U}$ is a sufficient statistic for $\boldsymbol{\theta}$. ■

Example 2.17 In Example 2.16 we saw that $U = \sum_{i=1}^n X_i$ is a sufficient statistic for π when sampling from a Bernoulli(π) distribution. We now confirm this using the factorisation theorem. Since:

$$p_{\mathbf{X}}(\mathbf{x}) = \pi^u (1 - \pi)^{n-u}$$

where $u = \sum_{i=1}^n x_i$, and 0 otherwise, this factorises as:

$$p_{\mathbf{X}}(\mathbf{x}) = g\left(\sum_{i=1}^n x_i, \pi\right) h(\mathbf{x})$$

where, trivially:

$$g\left(\sum_{i=1}^n x_i, \pi\right) = g(T(\mathbf{x}), \pi) = g(u, \pi) = \pi^u (1 - \pi)^{n-u} \quad \text{and} \quad h(\mathbf{x}) = 1.$$

Hence $U = \sum_{i=1}^n X_i$ is a sufficient statistic for π .

Example 2.18 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, with μ unknown and σ^2 known. The sample mean is $\bar{X} = \sum_{i=1}^n X_i/n$. The joint density function may be written as:

$$f_{\mathbf{X}}(\mathbf{x}; \mu) = \prod_{i=1}^n f_X(x_i; \mu) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right).$$

The joint density function can be factorised by noting that:

$$\sum_{i=1}^n (x_i - \mu)^2 = n(\bar{x} - \mu)^2 + \sum_{i=1}^n (x_i - \bar{x})^2$$

as seen in Example 2.7. Hence we may express the joint density function as:

$$f_{\mathbf{X}}(\mathbf{x}; \mu) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp \left(-\frac{n}{2\sigma^2} (\bar{x} - \mu)^2 \right) \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \bar{x})^2 \right).$$

This factorisation involves a term which depends on μ and the observed sample mean $\bar{x}$ (a statistic), such that we may write:

$$g(\bar{x}, \mu) = g(T(\mathbf{x}), \mu) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp \left(-\frac{n}{2\sigma^2} (\bar{x} - \mu)^2 \right)$$

and another term which depends on $\mathbf{x}$ only:

$$h(\mathbf{x}) = \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \bar{x})^2 \right).$$

Hence $U = \bar{X}$ is a sufficient statistic for μ .

Example 2.19 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, with μ and σ^2 unknown. Define $\boldsymbol{\theta}$ to be the vector of these unknown parameters, i.e. $\boldsymbol{\theta} = (\mu, \sigma^2)$. The joint density function may be factorised as:

$$\begin{aligned} f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) &= f_{\mathbf{X}}(\mathbf{x}; \mu, \sigma^2) = \prod_{i=1}^n f_X(x_i; \mu, \sigma^2) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp \left(-\frac{1}{2\sigma^2} \left(\sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i + n\mu^2 \right) \right) \\ &= g(\mathbf{T}(\mathbf{x}), \boldsymbol{\theta}) h(\mathbf{x}). \end{aligned}$$

By defining:

$$\mathbf{U} = \mathbf{h}(\mathbf{X}) = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right)$$

then $\mathbf{U}$ is sufficient for (μ, σ^2) . Noting that a one-to-one function of a sufficient statistic is sufficient (covered in the next subsection), since:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{and} \quad S^2 = \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - \frac{1}{n} \left(\sum_{i=1}^n X_i \right)^2 \right)$$

then $(\bar{X}, S^2)$ is also sufficient for (μ, σ^2) . This can also be seen from the following factorisation of the joint density function:

$$f_{\mathbf{X}}(\mathbf{x}; \mu, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp \left(-\frac{1}{2\sigma^2} n(\bar{x} - \mu)^2 \right) \exp \left(-\frac{1}{2\sigma^2} (n-1)s^2 \right).$$

Example 2.20 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from the continuous Uniform $[0, \theta]$ distribution. Note unlike the previous examples the support of the distribution depends on θ .

Let $I_{[0, \theta]}(x_i)$ denote an indicator function which equals 1 if $x_i \in [0, \theta]$, and 0 otherwise. The joint density function may then be written as:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = \prod_{i=1}^n f_X(x_i; \theta) I_{[0, \theta]}(x_i) = \prod_{i=1}^n \frac{1}{\theta} I_{[0, \theta]}(x_i) = \frac{1}{\theta^n} \prod_{i=1}^n I_{[0, \theta]}(x_i).$$

Note that the product of indicator functions can only be equal to 1 if $x_i \in [0, \theta]$ for $i = 1, 2, \dots, n$, and is equal to 0 otherwise. The statement that $x_i \in [0, \theta]$ for $i = 1, 2, \dots, n$ is equivalent to $\min_{1 \leq i \leq n} x_i \geq 0$ and $\max_{1 \leq i \leq n} x_i \leq \theta$, i.e. the minimum observed value is at least 0 and the maximum observed value is at most θ . Hence:

$$\prod_{i=1}^n I_{[0, \theta]}(x_i) = I_{[0, \infty)} \left(\min_{1 \leq i \leq n} x_i \right) I_{(-\infty, \theta]} \left(\max_{1 \leq i \leq n} x_i \right).$$

The joint density function can now be factorised as:

$$\begin{aligned} f_{\mathbf{X}}(\mathbf{x}; \theta) &= \left(\frac{1}{\theta^n} I_{(-\infty, \theta]} \left(\max_{1 \leq i \leq n} x_i \right) \right) I_{[0, \infty)} \left(\min_{1 \leq i \leq n} x_i \right) \\ &= g \left(\max_{1 \leq i \leq n} x_i, \theta \right) h(\mathbf{x}). \end{aligned}$$

Hence by the factorisation theorem, $\max_{1 \leq i \leq n} X_i = X_{(n)}$ is a sufficient statistic for θ , where recall $X_{(n)}$ is the n th order statistic. So if we know the lower bound of a continuous uniform distribution (here it is 0), then the sample maximum is a sufficient statistic for the upper bound.

Example 2.21 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from the following distributions. Find a sufficient statistic for each of the unknown parameters.

- (a) A Bin(m, π) distribution with π unknown and m known.

- (b) A $N(0, \sigma^2)$ distribution.
- (c) A $\text{Uniform}(a, b)$ distribution.
- (d) An $\text{Exp}(1)$ location family, with density function:

$$f_X(x; \theta) = \begin{cases} e^{-(x-\theta)} & \text{for } x \geq \theta \\ 0 & \text{otherwise.} \end{cases}$$

Solution

In each of these cases we use the factorisation theorem. For the first two the support does not involve the parameters. However, for the last two the support does involve the parameters, so we must proceed carefully in order to draw the correct conclusions from the factorisation theorem. In these two cases, indicator functions provide a convenient mechanism for representing the support, as per Example 2.20.

- (a) The joint density function of the sample may be written as:

$$f_{\mathbf{X}}(\mathbf{x}; \pi) = \prod_{i=1}^n \binom{m}{x_i} \pi^{x_i} (1 - \pi)^{m-x_i} = \pi^{\sum_{i=1}^n x_i} (1 - \pi)^{mn - \sum_{i=1}^n x_i} \prod_{i=1}^n \binom{m}{x_i}.$$

The joint density function factorises into a function:

$$g\left(\sum_{i=1}^n x_i, \pi\right) = \pi^{\sum_{i=1}^n x_i} (1 - \pi)^{mn - \sum_{i=1}^n x_i}$$

and a function of the data alone:

$$h(\mathbf{x}) = \prod_{i=1}^n \binom{m}{x_i}.$$

This means that the one-dimensional statistic $T(\mathbf{X}) = \sum_{i=1}^n X_i$ is sufficient for π .

- (b) For $-\infty < x_i < \infty$, the joint density function is:

$$f_{\mathbf{X}}(\mathbf{x}; \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x_i^2/2\sigma^2} = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2\right).$$

Here we can take $g\left(\sum_{i=1}^n x_i^2, \sigma^2\right)$ to be the joint density function, and $h(\mathbf{x}) = 1$.

It follows by the factorisation theorem that $\sum_{i=1}^n X_i^2$ is a sufficient statistic for σ^2 .

- (c) We use an indicator function to represent the support, hence we can write the joint density function as:

$$f_{\mathbf{X}}(\mathbf{x}; a, b) = \prod_{i=1}^n \frac{1}{b-a} I_{[a,b]}(x_i) = \left(\frac{1}{b-a}\right)^n \prod_{i=1}^n I_{[a,b]}(x_i).$$

A product of indicator functions only takes the value 1 if all of the component indicators take the value 1. Therefore:

$$\prod_{i=1}^n I_{[a,b]}(x_i) = \begin{cases} 1 & \text{if all } x_i \in [a, b] \\ 0 & \text{otherwise.} \end{cases}$$

In order for all x_i s to be in the interval $[a, b]$ we require that the minimum is greater than or equal to a and that the maximum is less than or equal to b . We conclude that:

$$f_{\mathbf{X}}(\mathbf{x}; a, b) = \left(\frac{1}{b-a} \right)^n I_{[a,\infty)} \left(\min_{1 \leq i \leq n} x_i \right) I_{(-\infty, b]} \left(\max_{1 \leq i \leq n} x_i \right).$$

We can take:

$$g \left(\min_{1 \leq i \leq n} x_i, \max_{1 \leq i \leq n} x_i; a, b \right) = \left(\frac{1}{b-a} \right)^n I_{[a,\infty)} \left(\min_{1 \leq i \leq n} x_i \right) I_{(-\infty, b]} \left(\max_{1 \leq i \leq n} x_i \right)$$

and $h(\mathbf{x}) = 1$. By the factorisation theorem, the two-dimensional statistic:

$$\left(\min_{1 \leq i \leq n} X_i, \max_{1 \leq i \leq n} X_i \right)$$

is a sufficient statistic for (a, b) .

(d) The joint density function is:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = \prod_{i=1}^n e^{-(x_i - \theta)} I_{[\theta, \infty)}(x_i) = e^{n\theta} I_{[\theta, \infty)} \left(\min_{1 \leq i \leq n} x_i \right) e^{-n\bar{x}}.$$

We can take:

$$g \left(\min_{1 \leq i \leq n} x_i, \theta \right) = e^{n\theta} I_{[\theta, \infty)} \left(\min_{1 \leq i \leq n} x_i \right)$$

and $h(\mathbf{x}) = e^{-n\bar{x}}$ to show that by the factorisation theorem $\min_{1 \leq i \leq n} X_i$ is a sufficient statistic for θ .

Activity 2.2 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample. Use the factorisation theorem to find a simple sufficient statistic for the unknown parameter in each of the following cases:

- (a) $X \sim \text{Pois}(\lambda)$
- (b) $X \sim \text{Geometric}(\pi)$.

2.9 Minimal sufficiency

Trivially, a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ is itself a sufficient statistic for any parameter. Denoting the joint density function of the sample $\mathbf{X}$ by $f_{\mathbf{X}}(\mathbf{x}; \theta)$, we note

that:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = f_{\mathbf{X}}(\mathbf{x}; \theta) \times 1 = g(T(\mathbf{x}), \theta) h(\mathbf{x})$$

where:

$$T(\mathbf{X}) = \mathbf{X}, \quad g(T(\mathbf{x}), \theta) = f_{\mathbf{X}}(\mathbf{x}; \theta) \quad \text{and} \quad h(\mathbf{x}) = 1.$$

Hence every sample is itself a sufficient statistic. In essence, the distribution of $\mathbf{X}$ on $\mathbf{X} = \mathbf{x}$ is the degenerate distribution (see **ST2133 Advanced statistics: distribution theory**) with all probability mass concentrated at $\mathbf{x}$.

While there exist many sufficient statistics, ideally we would like the *simplest* possible sufficient statistic such that to use it as a point estimator we would require some degree of data reduction, i.e. its dimension should be smaller than the dimension of $\mathbf{X}$. So we seek a sufficient statistic which achieves maximum data reduction without loss of information about the parameters. This motivates *minimal sufficiency*, although first we consider the concept of functions of statistics.

Data reduction involves calculating statistics, which are known functions of a sample. Clearly, the output from a function cannot contain *more* information than the argument. For example, for $n > 1$, the sample mean:

$$\bar{X} = T(\mathbf{X}) = \frac{1}{n} \sum_{i=1}^n X_i$$

contains less information than $\mathbf{X}$ does. Recall how in Example 2.10 the observed samples $\mathbf{x}$ and $\mathbf{y}$ had the same sample mean, but only knowledge of $\bar{x} = \bar{y} = 9$ results in a loss of information as we would be unable to reconstitute all of the exact observations in $\mathbf{x}$ and $\mathbf{y}$ from just the sample mean.

Statistics can be functions of other statistics. For example, consider the two-dimensional statistic:

$$\mathbf{V} = (V_1, V_2) = \left(X_1, \sum_{i=2}^n X_i \right).$$

Immediately we see that $\bar{X}$ is a function, say b , of $\mathbf{V}$, since:

$$\bar{X} = b(\mathbf{V}) = \frac{V_1 + V_2}{n}.$$

Hence $\bar{X}$ contains less information about $\mathbf{X}$ than $\mathbf{V}$ does.

If a statistic $\mathbf{U}$ is a function of another statistic $\mathbf{V}$, then $\mathbf{U}$ contains no more information than $\mathbf{V}$. We look at those sufficient statistics which are still sufficient even though they are functions of other statistics. Also, every statistic that is a one-to-one function of a sufficient statistic is itself a sufficient statistic.

Functions of statistics and sufficiency

Suppose $\mathbf{V}$ is a statistic. If $\mathbf{U}$ is a function of $\mathbf{V}$ only, i.e. $\mathbf{U} = \mathbf{b}(\mathbf{V})$ for some function $\mathbf{b}$, then:

- i. $\mathbf{U}$ is a statistic
- ii. if $\mathbf{U}$ is a sufficient statistic, then $\mathbf{V}$ is a sufficient statistic
- iii. if $\mathbf{V}$ is not a sufficient statistic, then $\mathbf{U}$ is not a sufficient statistic
- iv. if $\mathbf{b}$ is a one-to-one function and $\mathbf{V}$ is a sufficient statistic, then $\mathbf{U}$ is a sufficient statistic.

Proof:

- i. $\mathbf{V}$ is a statistic, i.e. $\mathbf{V}$ is a function of data and known constants only. We could write $\mathbf{V} = \mathbf{T}(\mathbf{X})$ for some function $\mathbf{T}$. By definition $\mathbf{U} = \mathbf{b}(\mathbf{T}(\mathbf{X}))$ and hence $\mathbf{U}$ is a function of data and known constants alone. We conclude that $\mathbf{U}$ is a statistic.
- ii. We start by using the factorisation theorem. Since $\mathbf{U}$ is a sufficient statistic, then:

$$f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) = g_1(\mathbf{b}(\mathbf{T}(\mathbf{x})), \boldsymbol{\theta}) h(\mathbf{x}).$$

Since $\mathbf{x}$ only enters in g_1 via $\mathbf{T}(\mathbf{x})$, we can find another function g_2 such that:

$$g_1(\mathbf{b}(\mathbf{T}(\mathbf{x})), \boldsymbol{\theta}) = g_2(\mathbf{T}(\mathbf{x}), \boldsymbol{\theta})$$

and hence:

$$f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) = g_2(\mathbf{T}(\mathbf{x}), \boldsymbol{\theta}) h(\mathbf{x}).$$

We conclude that $\mathbf{T}(\mathbf{X})$ is sufficient. The intuition here is that since $\mathbf{U}$ is a function of $\mathbf{V}$, we know that $\mathbf{V}$ must contain at least as much information from the sample as $\mathbf{U}$. Therefore, if $\mathbf{U}$ is a sufficient statistic, then $\mathbf{V}$ must also be a sufficient statistic.

- iii. This is logically equivalent to part ii. The intuition is also the same; as $\mathbf{U}$ contains less information about the sample than $\mathbf{V}$, if $\mathbf{V}$ is not sufficient then $\mathbf{U}$ cannot be.
- iv. If $\mathbf{b}$ is a one-to-one function then $\mathbf{b}^{-1}$ is also a function. We can write $\mathbf{V} = \mathbf{b}^{-1}(\mathbf{U})$, then appeal to the result in part ii.

■

So if $\mathbf{V}$ is sufficient, then it is certain that $\mathbf{U} = \mathbf{b}(\mathbf{V})$ does not contain *more* information than $\mathbf{V}$ because it is a function of $\mathbf{V}$ alone. It is possible, of course, that some information is actually lost. However, if $\mathbf{U}$ is sufficient for all parameters of interest, we have not lost any *relevant* information moving from $\mathbf{V}$ to $\mathbf{U}$. We now define a *minimal sufficient statistic*.

Minimal sufficient statistic

A sufficient statistic $\mathbf{U} = \mathbf{T}(\mathbf{X})$ is a **minimal sufficient statistic** if, for any other sufficient statistic $\mathbf{V} = \mathbf{T}'(\mathbf{X})$, $\mathbf{U}$ is a function of $\mathbf{V}$.

Some facts about minimal sufficient statistics are the following.

- If a sufficient statistic has dimension 1, i.e. if it is scalar, it must be a minimal sufficient statistic.
- Minimal sufficient statistics are not unique. However, if two statistics are minimally sufficient, they must have the same dimension.
- A one-to-one function of a minimal sufficient statistic is also a minimal sufficient statistic.
- The dimension of a minimal sufficient statistic is not always the same as the dimension of the parameter of interest.

Consider the partition generated by the statistic $\mathbf{U} = \mathbf{T}(\mathbf{X})$ such that $\mathbf{T} : P \rightarrow Q$ for some sets P and Q . As with any function, $\mathbf{T}$ defines a partition on its domain P . We say $\mathbf{x}$ and $\mathbf{y}$ belong to the same element of the partition if $\mathbf{T}(\mathbf{x}) = \mathbf{T}(\mathbf{y})$. The elements of this partition can be indexed by the range Q , so for $\mathbf{q} \in Q$ we have:

$$A_{\mathbf{q}} = \{\mathbf{x} \in P : \mathbf{T}(\mathbf{x}) = \mathbf{q}\}.$$

A non-examinable proof can be found in Appendix A.

A minimal sufficient statistic is a sufficient statistic for which the associated partition is as coarse as possible. For example, consider another sufficient statistic $\mathbf{V} = \mathbf{R}(\mathbf{X})$. If the partition associated with $\mathbf{U}$ is as coarse as possible, any two points giving the same value of $\mathbf{V}$ must also give the same value of $\mathbf{U}$, that is:

$$\mathbf{R}(\mathbf{x}) = \mathbf{R}(\mathbf{y}) \quad \Rightarrow \quad \mathbf{T}(\mathbf{x}) = \mathbf{T}(\mathbf{y}).$$

Hence we can find a function $\mathbf{b}$ such that $\mathbf{T}(\mathbf{x}) = \mathbf{b}(\mathbf{R}(\mathbf{x}))$ for all $\mathbf{x} \in P$. We conclude that $\mathbf{U} = \mathbf{b}(\mathbf{V})$, and $\mathbf{U}$ is minimal sufficient.

Activity 2.3 Consider the functions $R : S \rightarrow P$ and $T : S \rightarrow Q$ for arbitrary sets P , S and Q . Prove that if:

$$R(x) = R(y) \quad \Rightarrow \quad T(x) = T(y)$$

then we can find a function $b : P \rightarrow Q$ such that:

$$T(x) = b(R(x)).$$

The idea of partitioning using sufficient statistics forms the basis of the proof of the following proposition.

Characterisation of minimal sufficiency

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta})$. If we can find a function $\mathbf{T}$ such that:

$$\mathbf{T}(\mathbf{x}) = \mathbf{T}(\mathbf{y}) \Leftrightarrow f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) = k(\mathbf{x}, \mathbf{y}) f_{\mathbf{X}}(\mathbf{y}; \boldsymbol{\theta}) \Leftrightarrow \frac{f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta})}{f_{\mathbf{X}}(\mathbf{y}; \boldsymbol{\theta})} = k(\mathbf{x}, \mathbf{y})$$

then $\mathbf{T}(\mathbf{X})$ is a minimal sufficient statistic for $\boldsymbol{\theta}$. Here $k(\mathbf{x}, \mathbf{y})$ does not depend on $\boldsymbol{\theta}$.

A non-examinable proof can be found in Appendix A.

Example 2.22 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$ with μ and σ^2 unknown. We check whether the sufficient statistic:

$$\left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right)$$

is minimally sufficient for $\boldsymbol{\theta} = (\mu, \sigma^2)$. Consider the observed samples $\mathbf{x}$ and $\mathbf{y}$. We have (see Activity 2.4):

$$\frac{f_{\mathbf{X}}(\mathbf{x}; \mu, \sigma^2)}{f_{\mathbf{X}}(\mathbf{y}; \mu, \sigma^2)} = \exp \left(-\frac{1}{2\sigma^2} \left(\left(\sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i \right) - \left(\sum_{i=1}^n y_i^2 - 2\mu \sum_{i=1}^n y_i \right) \right) \right).$$

This ratio is a constant as a function of μ and σ^2 if and only if:

$$\sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i = \sum_{i=1}^n y_i^2 - 2\mu \sum_{i=1}^n y_i.$$

Viewing these expressions as polynomials in μ and comparing coefficients, we conclude that:

$$f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) = k(\mathbf{x}, \mathbf{y}) f_{\mathbf{X}}(\mathbf{y}; \boldsymbol{\theta}) \Leftrightarrow \left(\sum_{i=1}^n x_i, \sum_{i=1}^n x_i^2 \right) = \left(\sum_{i=1}^n y_i, \sum_{i=1}^n y_i^2 \right).$$

Hence:

$$\left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right)$$

is minimally sufficient for $\boldsymbol{\theta} = (\mu, \sigma^2)$.

Activity 2.4 Suppose that $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ is a random sample from $N(\mu, \sigma^2)$ with μ and σ^2 unknown. Consider two observed samples $\mathbf{x}$ and $\mathbf{y}$. Show that:

$$\frac{f_{\mathbf{X}}(\mathbf{x}; \mu, \sigma^2)}{f_{\mathbf{X}}(\mathbf{y}; \mu, \sigma^2)} = \exp \left(-\frac{1}{2\sigma^2} \left(\left(\sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i \right) - \left(\sum_{i=1}^n y_i^2 - 2\mu \sum_{i=1}^n y_i \right) \right) \right).$$

2.10 Likelihood

In **ST104b Statistics 2**, you were introduced to the concept of *likelihood* when the point estimation technique of maximum likelihood estimation was discussed. As a reminder, we use the notation:

$$f_{\mathbf{X}}(\mathbf{x}; \theta)$$

to represent a joint density function of a sample (and similarly $p_{\mathbf{X}}(\mathbf{x}; \theta)$ for a joint mass function) making the dependency on the parameter θ explicit through use of the ‘; θ ’ term, where in practice the value of θ is unknown. The joint distribution describes the probability distribution of the random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ with θ being an unknown parameter (or $\boldsymbol{\theta}$ for the vector parameter case).

In contrast, the *likelihood function* (discussed in this section) is a function of θ while $\mathbf{x}$ is treated as a given set of observations. While the joint distribution and likelihood have the same functional form, in the likelihood context we have a mechanism for investigating what happens across a range of parameter values, i.e. for all $\theta \in \Theta$, where Θ is the parameter space.

2.10.1 Likelihood function and log-likelihood function

We first define the likelihood function.

Likelihood function

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample with joint density function $f_{\mathbf{X}}(\mathbf{x}; \theta)$, for a scalar parameter θ . Given $\mathbf{X} = \mathbf{x}$ is observed, the function of θ defined by the joint density function of $\mathbf{X} = \mathbf{x}$ denoted:

$$L_{\mathbf{X}}(\theta; \mathbf{x}) = f_{\mathbf{X}}(\mathbf{x}; \theta)$$

is called the **likelihood function**. Replace $f_{\mathbf{X}}(\mathbf{x}; \theta)$ with $p_{\mathbf{X}}(\mathbf{x}; \theta)$ for the discrete case.

While the functional form of the likelihood $L_{\mathbf{X}}$ is the same as $f_{\mathbf{X}}$, note the reversal of arguments such that the likelihood is viewed as a function of θ rather than $\mathbf{x}$. If the dependence on $\mathbf{x}$ is not important, we could simply express the likelihood as $L_{\mathbf{X}}(\theta)$.

So the distribution of $\mathbf{X}$ is thought of as a function of $\mathbf{x}$ dependent on an unknown parameter value θ , whereas the likelihood function is thought of as a function of θ for a given observed sample $\mathbf{x}$.

The likelihood, as a function of θ , is *not* a density function because it does not integrate to 1 over θ . Instead, the likelihood value $L_{\mathbf{X}}(\theta; \mathbf{x})$ is a measure of the plausibility (or *likelihood*, hence the name!) of a particular value of θ given the observed sample $\mathbf{x}$.

Example 2.23 Suppose for θ_1 and θ_2 we have:

$$L_{\mathbf{X}}(\theta_1; \mathbf{x}) > L_{\mathbf{X}}(\theta_2; \mathbf{x})$$

then the observed sample $\mathbf{x}$ is more likely to have occurred if $\theta = \theta_1$ than if $\theta = \theta_2$. In other words θ_1 is a more plausible value than θ_2 .

Example 2.24 Consider a continuous random variable X with density function $f_X(x; \theta)$. For small ϵ we have:

$$\frac{P_\theta(x - \epsilon < X < x + \epsilon)}{2\epsilon} \simeq f_X(x; \theta) = L_X(\theta; x).$$

Therefore, if we compare the probabilities for different values of θ , we have:

$$\frac{P_{\theta_1}(x - \epsilon < X < x + \epsilon)}{P_{\theta_2}(x - \epsilon < X < x + \epsilon)} \simeq \frac{L_X(\theta_1; x)}{L_X(\theta_2; x)}$$

and the value of θ which gives the higher likelihood is more likely to be associated with the observed sample since this gives a higher probability. (This construction forms the basis of likelihood ratio tests in Chapter 5.)

Later chapters will make use of the likelihood function in various ways, such as a deeper treatment of maximum likelihood estimation in Chapter 3 than covered in **ST104b Statistics 2**, and in likelihood ratio tests (see Example 2.24). For now, we state some essential definitions and prove important results which will be useful in our later work.

Sometimes it is more convenient to work with the log-likelihood function, $l_{\mathbf{X}}(\theta; \mathbf{x})$ which is just the log of the likelihood function.

Log-likelihood function

If $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ is a random sample with likelihood function $L_{\mathbf{X}}(\theta; \mathbf{x})$, then the **log-likelihood function**, denoted $l_{\mathbf{X}}$, is:

$$l_{\mathbf{X}}(\theta; \mathbf{x}) \equiv \log L_{\mathbf{X}}(\theta; \mathbf{x})$$

provided $L_{\mathbf{X}}(\theta; \mathbf{x}) \neq 0$.

Because a logarithmic transformation is a monotone transformation, then the location of any maxima and minima are unaffected by the transformation. In applications of maximum likelihood estimation (when the support of $\mathbf{X}$ does not depend on θ) then applying calculus to maximise the log-likelihood function is often easier than maximising the likelihood function directly. Hence the log-likelihood function is often easier to work with in practice.

Constructing the likelihood and log-likelihood functions for random samples

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample, i.e. a sequence of n independent and identically distributed random variables each with a marginal density function $f_X(x_i; \theta)$, or marginal mass function $p_X(x_i; \theta)$, for $i = 1, 2, \dots, n$. The likelihood function and log-likelihood function, respectively, for a single observation are:

$$L_X(\theta; x) \quad \text{and} \quad l_X(\theta; x).$$

Exploiting the independence of the X_i s, the joint mass/density function is the product of the marginal distributions. Because mass/density and likelihood functions have the same functional form, the likelihood function of a random sample is:

$$L_{\mathbf{X}}(\theta; \mathbf{x}) = \prod_{i=1}^n L_X(\theta; x_i) = \begin{cases} \prod_{i=1}^n p_X(x_i; \theta) & \text{for the discrete case} \\ \prod_{i=1}^n f_X(x_i; \theta) & \text{for the continuous case.} \end{cases}$$

Taking logs produces the log-likelihood function:

$$l_{\mathbf{X}}(\theta; \mathbf{x}) = \sum_{i=1}^n l_X(\theta; x_i)$$

where recall ‘the log of the product is the sum of the logs’.

Example 2.25 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from an exponential distribution with parameter $\lambda > 0$. Hence the marginal density function is:

$$f_X(x_i; \lambda) = \begin{cases} \lambda e^{-\lambda x_i} & \text{for } x_i \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

The likelihood function is, due to independence:

$$L_{\mathbf{X}}(\lambda; \mathbf{x}) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n \exp \left(-\lambda \sum_{i=1}^n x_i \right)$$

for $\lambda > 0$, and the log-likelihood function is:

$$l_{\mathbf{X}}(\lambda; \mathbf{x}) = n \log \lambda - \lambda \sum_{i=1}^n x_i$$

again, for $\lambda > 0$.

Vector parameter case

Sometimes the assumed probability distribution for the population has more than one parameter. In such instances the likelihood can equivalently be thought of as a function of either a single vector parameter or as a function of several scalar parameters.

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample parameterised by $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_r)^T$. The likelihood function can be written as:

$$L_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x}) = L_{\mathbf{X}}(\theta_1, \theta_2, \dots, \theta_r; \mathbf{x}) = \prod_{i=1}^n L_X(\theta_1, \theta_2, \dots, \theta_r; x_i)$$

while the log-likelihood function can be written as:

$$l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x}) = l_{\mathbf{X}}(\theta_1, \theta_2, \dots, \theta_r; \mathbf{x}) = \sum_{i=1}^n l_X(\theta_1, \theta_2, \dots, \theta_r; x_i).$$

Example 2.26 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$. Here we let $\boldsymbol{\theta}$ denote the vector parameter such that $\boldsymbol{\theta} = (\mu, \sigma^2)$. The marginal density function is:

$$f_X(x_i; \boldsymbol{\theta}) = f_X(x_i; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) \quad \text{for } -\infty < x_i < \infty.$$

The likelihood function is, due to independence:

$$\begin{aligned} L_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x}) &= L_{\mathbf{X}}(\mu, \sigma^2; \mathbf{x}) = \prod_{i=1}^n f_X(x_i; \mu, \sigma^2) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right) \end{aligned}$$

for $\mu \in \mathbb{R}$ and $\sigma^2 > 0$, and the log-likelihood function is:

$$l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x}) = l_{\mathbf{X}}(\mu, \sigma^2; \mathbf{x}) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Example 2.27 Suppose that from a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ we have the observed sample $\mathbf{x} = \{x_1, x_2, \dots, x_n\}$. Find the likelihood and the log-likelihood functions for the following distributions.

- (a) Poisson with parameter λ .
- (b) Geometric with parameter π .
- (c) Binomial with parameters m and π , where m is known.
- (d) Gamma with parameters α and β .

Solution

- (a) The likelihood function is:

$$L_{\mathbf{X}}(\lambda; \mathbf{x}) = \prod_{i=1}^n p_X(x_i; \lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}$$

and the log-likelihood function is:

$$l_{\mathbf{X}}(\lambda; \mathbf{x}) = -n\lambda + \sum_{i=1}^n x_i \log \lambda - \sum_{i=1}^n \log(x_i!).$$

- (b) The likelihood function is:

$$L_{\mathbf{X}}(\pi; \mathbf{x}) = \prod_{i=1}^n p_X(x_i; \pi) = \prod_{i=1}^n (1 - \pi)^{x_i - 1} \pi = (1 - \pi)^{\sum_{i=1}^n x_i - n} \pi^n$$

and the log-likelihood function is:

$$l_{\mathbf{X}}(\pi; \mathbf{x}) = \left(\sum_{i=1}^n x_i - n \right) \log(1 - \pi) + n \log(\pi).$$

(c) The likelihood function is:

$$\begin{aligned} L_{\mathbf{X}}(\pi; \mathbf{x}) &= \prod_{i=1}^n p_X(x_i; \pi) = \prod_{i=1}^n \left(\binom{m}{x_i} \pi^{x_i} (1 - \pi)^{m-x_i} \right) \\ &= \left(\prod_{i=1}^n \binom{m}{x_i} \right) \pi^{\sum_{i=1}^n x_i} (1 - \pi)^{nm - \sum_{i=1}^n x_i} \end{aligned}$$

and the log-likelihood function is:

$$l_{\mathbf{X}}(\pi; \mathbf{x}) = \sum_{i=1}^n \log \binom{m}{x_i} + \sum_{i=1}^n x_i \log \pi + \left(nm - \sum_{i=1}^n x_i \right) \log(1 - \pi).$$

(d) The likelihood function is:

$$\begin{aligned} L_{\mathbf{X}}(\alpha, \beta; \mathbf{x}) &= \prod_{i=1}^n f_X(x_i; \alpha, \beta) = \prod_{i=1}^n \frac{1}{\Gamma(\alpha)} \beta^\alpha x_i^{\alpha-1} e^{-\beta x_i} \\ &= \Gamma(\alpha)^{-n} \beta^{n\alpha} \left(\prod_{i=1}^n x_i^{\alpha-1} \right) \exp \left(-\beta \sum_{i=1}^n x_i \right) \end{aligned}$$

and the log-likelihood function is:

$$l_{\mathbf{X}}(\alpha, \beta; \mathbf{x}) = -n \log(\Gamma(\alpha)) + n\alpha \log \beta + (\alpha - 1) \sum_{i=1}^n \log x_i - \beta \sum_{i=1}^n x_i.$$

2.10.2 Score function and Fisher information

When estimating parameters using maximum likelihood estimation for probability distributions whose supports are not defined by unknown parameters,¹ we will use calculus to maximise the log-likelihood function which requires us to obtain derivatives.

As seen in **ST104b Statistics 2**, maximising the log-likelihood function (rather than the likelihood function) is more convenient, and since taking logs is a monotone transformation it does not affect the maximum solution.

We now define the *score function* to be the first derivative of the log-likelihood function. This will be a useful construct for us throughout the course.

¹An example of a distribution with its support defined by its parameters is the continuous uniform distribution, i.e. when $X \sim \text{Uniform}[a, b]$, where $a \leq x \leq b$.

Score function

The **score function** associated with the log-likelihood function $l_{\mathbf{X}}(\theta; \mathbf{x})$ is:

$$s_{\mathbf{X}}(\theta; \mathbf{x}) = \frac{\partial}{\partial \theta} l_{\mathbf{X}}(\theta; \mathbf{x}).$$

Applying properties of logs and the chain rule:

$$s_{\mathbf{X}}(\theta; \mathbf{x}) = \frac{1}{L_{\mathbf{X}}(\theta; \mathbf{x})} \frac{\partial}{\partial \theta} L_{\mathbf{X}}(\theta; \mathbf{x}).$$

The above definition of the score function makes use of an observed sample $\mathbf{x}$. Alternatively, we may apply the score function to the random sample represented as $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$, i.e. a sequence of independent and identically distributed random variables, denoted $s_{\mathbf{X}}(\theta; \mathbf{X})$. The expectation of this score function is zero. We proceed assuming $\mathbf{X}$ is a random sample of a continuous distribution, although similar proofs apply for discrete distributions.

Property of the score function

Under sufficient regularity conditions to permit differentiation under integral signs:

$$E(s_{\mathbf{X}}(\theta; \mathbf{X})) = 0.$$

Proof: For the continuous case (for the discrete case replace the integrals with sums):

$$\begin{aligned} E(s_{\mathbf{X}}(\theta; \mathbf{X})) &= \int_{\mathbb{R}^n} s_{\mathbf{X}}(\theta; \mathbf{x}) f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x} \\ &= \int_{\mathbb{R}^n} \frac{1}{L_{\mathbf{X}}(\theta; \mathbf{x})} \frac{\partial}{\partial \theta} L_{\mathbf{X}}(\theta; \mathbf{x}) f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x} \\ &= \int_{\mathbb{R}^n} \frac{\partial}{\partial \theta} f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x} && (\text{since } L_{\mathbf{X}}(\theta; \mathbf{x}) = f_{\mathbf{X}}(\mathbf{x}; \theta)) \\ &= \frac{\partial}{\partial \theta} \underbrace{\int_{\mathbb{R}^n} f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x}}_{=1} \\ &= \frac{d}{d\theta} 1 \\ &= 0 \end{aligned}$$

because the last integral is equal to one, since the density function is normalised. ■

Although the score function is usually viewed as a function of θ , the expectation is taken with respect to $\mathbf{X}$, actually with respect to the distribution of $\mathbf{X}$, which depends on θ . This may be interpreted as follows. If the experiment was repeated many times, the score function would on average equal zero. That is, if we start at the true value of the parameter, *on average* over many experiments the likelihood does not change if we make an infinitesimal change to the parameter.

For an observed sample $\mathbf{x}$ we could plot $L_{\mathbf{X}}(\theta; \mathbf{x})$ (or $l_{\mathbf{X}}(\theta; \mathbf{x})$) on the y -axis and values of θ on the x -axis. In most cases, if we plot the likelihood function against θ we get a curve with a peak at the maximum. The sharper the peak is, the more information about θ exists in the sample. The following are two extreme scenarios.

- i. $L_{\mathbf{X}}(\theta; \mathbf{x})$ has a sharp peak indicating that a small subset of values of $\theta \in \Theta$ are much more plausible than others.
- ii. $L_{\mathbf{X}}(\theta; \mathbf{x})$ is quite flat indicating that a large subset of values of $\theta \in \Theta$ are approximately equally plausible.

In case i. the likelihood provides a lot of information, allowing us to reduce the possible values of θ to a small subset. In case ii. the likelihood is of little-to-no assistance to us when attempting to make inferences about θ . Hence the usefulness of the likelihood depends on how rapidly it changes with θ . When the likelihood has a sharp peak then the absolute values of the first derivatives are large. This concept is formalised with *Fisher information* by looking at the square (rather than the absolute value) of the derivative of the log-likelihood and taking expectations across sample values.

Fisher information

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a probability distribution parameterised by θ with log-likelihood function $l_{\mathbf{X}}(\theta; \mathbf{x})$. The **(total) Fisher information** about θ in the random sample $\mathbf{X}$ is:

$$\mathcal{I}_{\mathbf{X}}(\theta) = E((s_{\mathbf{X}}(\theta; \mathbf{X}))^2) = E\left(\left(\frac{\partial}{\partial \theta} l_{\mathbf{X}}(\theta; \mathbf{X})\right)^2\right).$$

The word ‘total’ may be used when looking at information contained in the entire sample.

Fisher information may be expressed in different forms. Note $\mathcal{I}_{\mathbf{X}}(\theta)$ is the variance of the score function (when computed at the true value of θ), so the larger it is the more the score function at the true value is affected by minimal changes in the parameter, hence the sharper is the peak, hence the more useful is our information about θ .

Since $E(s_{\mathbf{X}}(\theta; \mathbf{X})) = 0$, we have that:

$$\begin{aligned} \text{Var}(s_{\mathbf{X}}(\theta; \mathbf{X})) &= E((s_{\mathbf{X}}(\theta; \mathbf{X}))^2) - (E(s_{\mathbf{X}}(\theta; \mathbf{X})))^2 \\ &= E((s_{\mathbf{X}}(\theta; \mathbf{X}))^2) - 0^2 \\ &= E((s_{\mathbf{X}}(\theta; \mathbf{X}))^2) \end{aligned}$$

hence:

$$\mathcal{I}_{\mathbf{X}}(\theta) = \text{Var}(s_{\mathbf{X}}(\theta; \mathbf{X})).$$

Another representation of Fisher information, which can be computationally convenient, is the following.

Alternative form of Fisher information

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a probability distribution parameterised by θ with log-likelihood function $l_{\mathbf{X}}(\theta; \mathbf{x})$. Fisher information about θ in the random sample $\mathbf{X}$ can also be expressed as:

$$\mathcal{I}_{\mathbf{X}}(\theta) = -E \left(\frac{\partial^2}{\partial \theta^2} l_{\mathbf{X}}(\theta; \mathbf{X}) \right) = -E \left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}) \right).$$

Proof: We proceed by differentiating $E(s_{\mathbf{X}}(\theta; \mathbf{X}))$ with respect to θ . We have:

$$\begin{aligned} 0 &= \frac{d}{d\theta} E(s_{\mathbf{X}}(\theta; \mathbf{X})) \\ &= \frac{d}{d\theta} \int_{\mathbb{R}^n} s_{\mathbf{X}}(\theta; \mathbf{x}) f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x} \\ &= \int_{\mathbb{R}^n} \frac{d}{d\theta} (s_{\mathbf{X}}(\theta; \mathbf{x}) f_{\mathbf{X}}(\mathbf{x}; \theta)) d\mathbf{x} \\ &= \int_{\mathbb{R}^n} \left(\left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{x}) \right) f_{\mathbf{X}}(\mathbf{x}; \theta) + s_{\mathbf{X}}(\theta; \mathbf{x}) \frac{\partial}{\partial \theta} f_{\mathbf{X}}(\mathbf{x}; \theta) \right) d\mathbf{x} \\ &= \int_{\mathbb{R}^n} \left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{x}) + (s_{\mathbf{X}}(\theta; \mathbf{x}))^2 \right) f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x} \quad \text{since } \frac{\partial}{\partial \theta} f_{\mathbf{X}}(\mathbf{x}; \theta) = s_{\mathbf{X}}(\theta; \mathbf{x}) f_{\mathbf{X}}(\mathbf{x}; \theta) \\ &= E \left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}) + (s_{\mathbf{X}}(\theta; \mathbf{X}))^2 \right) \\ &= E \left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}) \right) + E((s_{\mathbf{X}}(\theta; \mathbf{X}))^2). \end{aligned}$$

Hence:

$$E((s_{\mathbf{X}}(\theta; \mathbf{X}))^2) = -E \left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}) \right)$$

and so:

$$\mathcal{I}_{\mathbf{X}}(\theta) = E((s_{\mathbf{X}}(\theta; \mathbf{X}))^2) = -E \left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}) \right).$$

In this case at the true value of θ the Fisher information is also the negative Hessian of the log-likelihood so it measures the concavity of the log-likelihood. In particular since the Fisher information must always be positive (it is a variance), then the Hessian must be negative which jointly with a zero expectation of the score (the first derivative of the log-likelihood) tells us that the true value of θ is a maximum of the log-likelihood for any realisation $\mathbf{X} = \mathbf{x}$.

Example 2.28 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $\text{Exp}(\lambda)$. Recall that the log-likelihood function (see Example 2.25) is:

$$l_{\mathbf{X}}(\lambda; \mathbf{x}) = n \log \lambda - \lambda \sum_{i=1}^n x_i.$$

We obtain the score function by differentiating $l_{\mathbf{X}}(\lambda; \mathbf{x})$ with respect to λ . Hence:

$$s_{\mathbf{X}}(\lambda; \mathbf{x}) = \frac{n}{\lambda} - \sum_{i=1}^n x_i.$$

In this case, Fisher information is most easily derived using the alternative form:

$$\mathcal{I}_{\mathbf{X}}(\theta) = -E \left(\frac{\partial^2}{\partial \theta^2} l_{\mathbf{X}}(\theta; \mathbf{X}) \right) = -E \left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}) \right).$$

Hence:

$$\mathcal{I}_{\mathbf{X}}(\lambda) = -E \left(-\frac{n}{\lambda^2} \right) = \frac{n}{\lambda^2}.$$

Example 2.29 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, with σ^2 known. The log-likelihood function (see Example 2.26) is:

$$l_{\mathbf{X}}(\mu; \mathbf{x}) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

As σ^2 is known, the only unknown parameter is μ . We obtain the score function by differentiating $l_{\mathbf{X}}(\mu; \mathbf{x})$ with respect to μ . Hence:

$$s_{\mathbf{X}}(\mu; \mathbf{x}) = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu).$$

In this case, Fisher information is most easily derived using:

$$\mathcal{I}_{\mathbf{X}}(\theta) = E((s_{\mathbf{X}}(\theta; \mathbf{X}))^2).$$

Hence:

$$\begin{aligned} \mathcal{I}_{\mathbf{X}}(\mu) &= E \left(\frac{1}{(\sigma^2)^2} \left(\sum_{i=1}^n (X_i - \mu) \right)^2 \right) \\ &= \frac{1}{\sigma^4} \left(\sum_{i=1}^n E((X_i - \mu)^2) \right) \\ &= \frac{1}{\sigma^4} n \text{Var}(X) \\ &= \frac{n}{\sigma^2}. \end{aligned}$$

Note that this expression says that information contained in the sample about μ increases with the sample size n (more data means more information, other things equal), but decreases with the population variance σ^2 (greater variability in the population leads to greater sampling variability and hence less precise estimates, other things equal).

Example 2.30 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from the following distributions. Find the score function and Fisher information in each case.

- (a) Poisson with parameter λ .
- (b) Geometric with parameter π .
- (c) Binomial with parameters m and π , where m is known.

Solution

- (a) The score function is the first derivative with respect to λ of the previously found log-likelihood function (see Example 2.27):

$$s_{\mathbf{X}}(\lambda; \mathbf{x}) = \frac{\partial}{\partial \lambda} l_{\mathbf{X}}(\lambda; \mathbf{x}) = -n + \frac{1}{\lambda} \sum_{i=1}^n x_i.$$

The second derivative of the log-likelihood function is:

$$\frac{\partial^2}{\partial \lambda^2} l_{\mathbf{X}}(\lambda; \mathbf{x}) = -\frac{1}{\lambda^2} \sum_{i=1}^n x_i.$$

Hence Fisher information is (noting that $E(X_i) = \lambda$ for $i = 1, 2, \dots, n$):

$$\mathcal{I}_{\mathbf{X}}(\lambda) = -E \left(-\frac{1}{\lambda^2} \sum_{i=1}^n X_i \right) = \frac{1}{\lambda^2} \sum_{i=1}^n E(X_i) = \frac{n\lambda}{\lambda^2} = \frac{n}{\lambda}.$$

- (b) The score function is the first derivative with respect to π of the previously found log-likelihood function (see Example 2.27):

$$s_{\mathbf{X}}(\pi; \mathbf{x}) = -\frac{\sum_{i=1}^n x_i - n}{1 - \pi} + \frac{n}{\pi}.$$

The second derivative of the log-likelihood function is:

$$\frac{\partial^2}{\partial \pi^2} l_{\mathbf{X}}(\pi; \mathbf{x}) = -\frac{\sum_{i=1}^n x_i - n}{(1 - \pi)^2} - \frac{n}{\pi^2}.$$

Hence Fisher information is (noting that $E(X_i) = 1/\pi$ for $i = 1, 2, \dots, n$):

$$\begin{aligned} \mathcal{I}_{\mathbf{X}}(\pi) &= -E \left(-\frac{\sum_{i=1}^n X_i - n}{(1 - \pi)^2} - \frac{n}{\pi^2} \right) \\ &= \frac{\sum_{i=1}^n E(X_i) - n}{(1 - \pi)^2} + \frac{n}{\pi^2} \\ &= \frac{n(1 - \pi)/\pi}{(1 - \pi)^2} + \frac{n}{\pi^2} \\ &= \frac{n}{\pi^2(1 - \pi)}. \end{aligned}$$

- (c) The score function is the first derivative with respect to π of the previously found log-likelihood function (see Example 2.27):

$$s_{\mathbf{X}}(\pi; \mathbf{x}) = \frac{\partial}{\partial \pi} l_{\mathbf{X}}(\pi; \mathbf{x}) = \frac{\sum_{i=1}^n x_i}{\pi} - \frac{nm - \sum_{i=1}^n x_i}{1 - \pi}.$$

The second derivative of the log-likelihood function is:

$$\frac{\partial^2}{\partial \pi^2} l_{\mathbf{X}}(\pi; \mathbf{x}) = -\frac{\sum_{i=1}^n x_i}{\pi^2} - \frac{nm - \sum_{i=1}^n x_i}{(1 - \pi)^2}.$$

Hence Fisher information is (noting that $E(X_i) = m\pi$ for $i = 1, 2, \dots, n$):

$$\begin{aligned} \mathcal{I}_{\mathbf{X}}(\pi) &= -E \left(-\frac{\sum_{i=1}^n X_i}{\pi^2} - \frac{nm - \sum_{i=1}^n X_i}{(1 - \pi)^2} \right) \\ &= \frac{\sum_{i=1}^n E(X_i)}{\pi^2} + \frac{nm - \sum_{i=1}^n E(X_i)}{(1 - \pi)^2} \\ &= \frac{nm\pi}{\pi^2} + \frac{nm - nm\pi}{(1 - \pi)^2} \\ &= n \left(\frac{m}{\pi} + \frac{m}{1 - \pi} \right) \\ &= \frac{nm}{\pi(1 - \pi)}. \end{aligned}$$

Simplification for random samples

For a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$, we have seen that the likelihood function equals the product of the individual likelihoods, and the log-likelihood function is the sum of the individual log-likelihoods. The score function and Fisher information have similar results.

Let $s_X(\theta; x)$ and $\mathcal{I}_X(\theta)$ denote the score function and Fisher information associated with a single observation, respectively. Since:

$$l_{\mathbf{X}}(\theta; \mathbf{x}) = \sum_{i=1}^n l_X(\theta; x_i)$$

we can differentiate to obtain the score function:

$$s_{\mathbf{X}}(\theta; \mathbf{x}) = \sum_{i=1}^n s_X(\theta; x_i).$$

Proof: The log-likelihood function is:

$$l_{\mathbf{X}}(\theta; \mathbf{x}) = \log \left(\prod_{i=1}^n f_X(x_i; \theta) \right) = \sum_{i=1}^n l_X(\theta; x_i).$$

Hence:

$$s_{\mathbf{X}}(\theta; \mathbf{x}) = \frac{\partial l_{\mathbf{X}}(\theta; \mathbf{x})}{\partial \theta} = \frac{\partial}{\partial \theta} \sum_{i=1}^n l_X(\theta; x_i) = \sum_{i=1}^n \frac{\partial}{\partial \theta} l_X(\theta; x_i) = \sum_{i=1}^n s_X(\theta; x_i).$$

■

We also have that:

$$\mathcal{I}_{\mathbf{X}}(\theta) = n\mathcal{I}_X(\theta)$$

which says that for a random sample the total Fisher information is the Fisher information for a single observation multiplied by the sample size.

Proof: We exploit the result for the relationship between individual score functions and the overall score function, as well as the independence of the X_i s. We have:

$$\mathcal{I}_{\mathbf{X}}(\theta) = \text{Var}(s_{\mathbf{X}}(\theta; \mathbf{X})) = \text{Var} \left(\sum_{i=1}^n s_X(\theta; X_i) \right) = \sum_{i=1}^n \text{Var}(s_X(\theta; X_i)) = n\mathcal{I}_X(\theta).$$

■

Activity 2.5 Consider again Example 2.28. Derive the Fisher information using the original definition of Fisher information. Show that you arrive at the same result starting from the information associated with a single observation.

Activity 2.6 Consider again Example 2.29. Derive the Fisher information using the alternative form of Fisher information. Show that you arrive at the same result starting from the information associated with a single observation.

Activity 2.7 Consider a distribution parameterised by θ where the log-likelihood for a single observation can be written as:

$$l_X(\theta; x) = a(\theta) t(x) + b(\theta) + c(x)$$

for some functions a , t , b and c meeting regularity conditions. Using the properties of the score function, find:

- (a) $E(t(X))$
- (b) $\mathcal{I}_X(\theta)$ (by differentiating twice)
- (c) $\text{Var}(t(X))$ (using part (b))

in terms of a and b and their respective derivatives. Apply these results to the binomial distribution where the number of trials, n , is fixed and the parameter of interest is the probability of success, π .

Vector parameter case

If the population probability distribution is parameterised by more than one parameter, then we have a vector of parameters $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_r)^T$ (sometimes with the transpose T , omitted for brevity). In this case, the score function is an $r \times 1$ vector and Fisher information is an $r \times r$ matrix.

Score vector function and Fisher information matrix

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample parameterised by $\boldsymbol{\theta}$ and with log-likelihood function $l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x})$. The **score vector function** is:

$$s_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x}) = \nabla_{\boldsymbol{\theta}} l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x}) = \left(\frac{\partial}{\partial \theta_1} l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x}), \frac{\partial}{\partial \theta_2} l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x}), \dots, \frac{\partial}{\partial \theta_r} l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x}) \right)^T$$

and the **Fisher information matrix** is the matrix:

$$\mathcal{I}_{\mathbf{X}}(\boldsymbol{\theta}) = E(s_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X}) s_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X})^T).$$

The (i, j) th element of the Fisher information matrix is:

$$(\mathcal{I}_{\mathbf{X}}(\boldsymbol{\theta}))_{ij} = -E \left(\frac{\partial^2}{\partial \theta_i \partial \theta_j} l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X}) \right) = E \left(\frac{\partial}{\partial \theta_i} l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X}) \frac{\partial}{\partial \theta_j} l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X}) \right).$$

Example 2.31 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, with μ and σ^2 unknown. Recall that the log-likelihood function (see Example 2.26) is:

$$l_{\mathbf{X}}(\mu, \sigma^2; \mathbf{x}) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

As μ and σ^2 are unknown, we have the vector parameter case with $r = 2$. We obtain the score vector function elements by partially differentiating $l_{\mathbf{X}}(\mu, \sigma^2; \mathbf{x})$ with respect to μ and σ^2 , respectively. Hence:

$$\frac{\partial}{\partial \mu} l_{\mathbf{X}}(\mu, \sigma^2; \mathbf{x}) = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

and:

$$\frac{\partial}{\partial \sigma^2} l_{\mathbf{X}}(\mu, \sigma^2; \mathbf{x}) = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \mu)^2$$

such that the score vector function is:

$$s_{\mathbf{X}}(\mu, \sigma^2; \mathbf{x}) = \begin{pmatrix} \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) \\ -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \mu)^2 \end{pmatrix}.$$

The Fisher information matrix is of dimension 2×2 , so has four elements, which are:

$$\mathcal{I}_{\mathbf{X}}(\mu, \sigma^2)_{11} = -E \left(\frac{\partial^2}{\partial \mu^2} l_{\mathbf{X}}(\mu, \sigma^2; \mathbf{X}) \right) = -E \left(-\frac{n}{\sigma^2} \right) = \frac{n}{\sigma^2}$$

also:

$$\begin{aligned}\mathcal{I}_{\mathbf{X}}(\mu, \sigma^2)_{22} &= -\mathbb{E} \left(\frac{\partial^2}{\partial (\sigma^2)^2} l_{\mathbf{X}}(\mu, \sigma^2; \mathbf{X}) \right) = -\mathbb{E} \left(\frac{n}{2\sigma^4} - \frac{1}{\sigma^6} \sum_{i=1}^n (X_i - \mu)^2 \right) \\ &= -\mathbb{E} \left(-\frac{n}{2\sigma^4} \right) \\ &= \frac{n}{2\sigma^4}\end{aligned}$$

and:

$$\mathcal{I}_{\mathbf{X}}(\mu, \sigma^2)_{12} = -\mathbb{E} \left(\frac{\partial^2}{\partial \mu \partial \sigma^2} l_{\mathbf{X}}(\mu, \sigma^2; \mathbf{X}) \right) = -\mathbb{E} \left(-\frac{1}{\sigma^4} \sum_{i=1}^n (X_i - \mu) \right) = 0$$

since:

$$\mathbb{E} \left(\sum_{i=1}^n (X_i - \mu) \right) = \mathbb{E} \left(\sum_{i=1}^n X_i \right) - \mathbb{E} \left(\sum_{i=1}^n \mu \right) = \sum_{i=1}^n \mathbb{E}(X_i) - n\mu = n\mu - n\mu = 0.$$

Note that $\mathcal{I}_{\mathbf{X}}(\mu, \sigma^2)_{12} = \mathcal{I}_{\mathbf{X}}(\mu, \sigma^2)_{21}$ since the order of differentiation does not matter. Therefore, the Fisher information matrix is:

$$\mathcal{I}_{\mathbf{X}}(\mu, \sigma^2) = \begin{pmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{pmatrix}.$$

Activity 2.8 Find the Fisher information matrix for (π_1, π_2) associated with one observation, (x_1, x_2) , from the *trinomial* distribution with mass function:

$$p_{X_1, X_2}(x_1, x_2; \pi_1, \pi_2) = \frac{n!}{x_1! x_2! (n - x_1 - x_2)!} \pi_1^{x_1} \pi_2^{x_2} (1 - \pi_1 - \pi_2)^{n - x_1 - x_2}.$$

Properties of the score vector function and Fisher information matrix

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample parameterised by $\boldsymbol{\theta}$ and with log-likelihood function $l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x})$. The following relationships hold.

- i. $\mathbb{E}(s_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X})) = \mathbf{0}$.
- ii. $\mathcal{I}_{\mathbf{X}}(\boldsymbol{\theta}) = \text{Var}(s_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X}))$, where $\text{Var}(\cdot)$ denotes a covariance matrix.
- iii. $\mathcal{I}_{\mathbf{X}}(\boldsymbol{\theta}) = -\mathbb{E}(\nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}}^T l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X})) = -\mathbb{E}(\nabla_{\boldsymbol{\theta}} s_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X}))$ or else:

$$(\mathcal{I}_{\mathbf{X}}(\boldsymbol{\theta}))_{i,j} = -\mathbb{E} \left(\frac{\partial^2}{\partial \theta_i \partial \theta_j} l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X}) \right).$$

Activity 2.9 Prove the above properties of the score vector function and Fisher information matrix.

2.11 A reminder of your learning outcomes

On completion of this chapter, you should be able to:

- refer to the term population in the context of statistical inference
- explain the difference between a model (represented by the sample) and data (the observed sample)
- distinguish between point estimators and point estimates
- discuss how statistics play a fundamental role in data reduction
- define a sufficient statistic and a minimal sufficient statistic
- determine whether a statistic is sufficient using the factorisation theorem
- define the terms ‘likelihood’, ‘log-likelihood’, ‘score’ and ‘Fisher information’
- exploit the properties of random samples to write simpler forms of likelihood, log-likelihood, score and Fisher information
- calculate the score function and Fisher information for standard distributions
- state two distinct forms of Fisher information and understand the association between them.

2.12 Sample examination questions

This chapter provides the foundation for the course. The ideas discussed in this chapter are of fundamental importance in answering examination questions. However, they do not arise independently but rather in the context of point estimation, interval estimation and hypothesis testing. For this reason, no sample examination questions are given for this chapter. You will find the material discussed occurs repeatedly in the sample examination questions for the remaining chapters.

Chapter 3

Point estimation

3.1 Recommended reading

Casella, G. and R.L. Berger, *Statistical Inference*. Chapter 7, Sections 7.1, 7.2.1, 7.2.2, 7.3.1, 7.3.2 and 7.3.3.

3.2 Learning outcomes

On completion of this chapter, you should be able to:

- calculate the bias, variance and mean squared error of the sample mean
- establish whether an estimator is consistent
- apply conditioning on a sufficient statistic to improve an estimator
- prove the Rao–Blackwell theorem
- demonstrate that the sample mean is unbiased, and derive its variance
- prove the central limit theorem in the special case where the moment generating function is well-defined
- discuss practical uses and limitations of the central limit theorem
- demonstrate that the sample variance is an unbiased estimator
- demonstrate the ‘trick’ of splitting $(X_i - \bar{X})$ into independent terms
- derive the distributions of the sample mean and sample variance for a normal population
- derive method of moments estimators of parameters for specified probability distributions
- construct sample quantities using order statistics
- derive the distribution of the sample minimum and sample maximum for a random sample from specified probability distributions
- apply the joint density function of order statistics in simple cases
- prove the Cramér–Rao inequality

- employ the Cramér–Rao inequality to determine the lower bound of the variance of unbiased estimators of parameters of specified probability distributions
- determine whether an estimator is minimum variance unbiased for the parameters for specified probability distributions
- explain the basic principles of maximum likelihood estimation
- describe the large sample properties of maximum likelihood estimators
- analytically derive maximum likelihood estimators in simple situations
- derive the updating step for the method of scoring to numerically approximate the maximum likelihood estimates of parameters for specified probability distributions.

3.3 Introduction

Suppose that a real-world phenomenon may be described by a probability model defined through the random variable X with distribution function $F_X(x; \theta)$. Suppose also that a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ is drawn from this distribution. We want to use the information in the observed sample $\mathbf{x}$ to get a best guess of θ . In other words we want a *point estimate* of θ . A function of $\mathbf{X}$ which gives a point estimate of θ is a *point estimator* of θ . If we use the observed random sample $\mathbf{x}$, we get the estimate of θ which is a particular value of the estimator.

In this chapter we review key statistical properties of estimators (bias, variance and mean squared error) which were first introduced in **ST104b Statistics 2**, and we also consider how sufficiency can be used to improve estimators. We will discuss method of moments estimation, which is a useful point estimation technique when we wish to estimate population moments of distributions. However, we are also sometimes interested in estimating population quantiles, such as the median, which can be estimated with sample quantiles.

When we wish to restrict ourselves to unbiased estimators of parameters, then we will seek the most *efficient* estimator, i.e. the one with the smallest variance. We will show how to establish a lower bound on the variance of unbiased estimators. Finally, we will consider maximum likelihood estimation, including a discussion of the statistical properties of maximum likelihood estimators.

3.4 Statistical properties of point estimators

All point estimators are statistics, i.e. known functions of a sample. When estimating a scalar parameter θ , we will denote its point estimator by $\hat{\theta}$, as explained in **ST104b Statistics 2**. Emphasising that $\hat{\theta}$ is a function of a random sample $\mathbf{X}$, with its value determined by an observed sample $\mathbf{x}$, we can distinguish between a point estimator and a point estimate as follows.

Notation for a point estimator versus a point estimate

Consider a random sample $\mathbf{X}$ parameterised by θ , and an observed sample $\mathbf{x}$.

A point estimator of θ is denoted $\hat{\theta}(\mathbf{X})$, and a point estimate of θ is denoted $\hat{\theta}(\mathbf{x})$.

To avoid excessive notation, where the context is clear, $\hat{\theta}$ will be used to denote a point estimator or a point estimate as appropriate.

Since there are many possible observed samples which may be drawn from the population, the *observed value* of $\hat{\theta}$ will vary from sample to sample. Being a function of the sample, an estimator is itself a random variable with a corresponding probability distribution, which recall we refer to as its *sampling distribution*.

Statistical properties of $\hat{\theta}$ determine how useful it is as an estimator of θ . A ‘good’ estimator should have most of its probability mass concentrated around θ . In general, this ensures that $\hat{\theta}$ has a high probability of returning a value close to θ .

3.4.1 Bias, variance and mean squared error

Our preferred measure of central tendency of the sampling distribution of an estimator is its mean, i.e. the expected value of an estimator, which is $E(\hat{\theta})$. The mean is used to determine the *bias* of an estimator.

Bias of an estimator

Let $\hat{\theta}$ be an estimator of θ . The **bias** of the estimator is:

$$\text{Bias}_{\theta}(\hat{\theta}) = E_{\theta}(\hat{\theta}) - \theta.$$

If $E_{\theta}(\hat{\theta}) = \theta$, then it is an *unbiased* estimator of θ .

The θ subscript on $\text{Bias}_{\theta}(\hat{\theta})$ indicates that the estimator is being considered to estimate θ . This θ subscript is also used on the expectation and probability operators in this context because the *distribution* of a statistic may depend on unknown parameters. We could write:

$$\hat{\theta}(\mathbf{X}) \sim F_{\hat{\theta}(\mathbf{X})}(\hat{\theta}(\mathbf{x}); \theta) = P_{\theta}(\hat{\theta}(\mathbf{X}) < \hat{\theta}(\mathbf{x}))$$

hence:

$$E_{\theta}(\hat{\theta}(\mathbf{X})) = \begin{cases} \sum_{\hat{\theta}(\mathbf{x})} \hat{\theta}(\mathbf{x}) p_{\hat{\theta}(\mathbf{X})}(\hat{\theta}(\mathbf{x}); \theta) & \text{for the discrete case} \\ \int_{-\infty}^{\infty} \hat{\theta}(\mathbf{x}) f_{\hat{\theta}(\mathbf{X})}(\hat{\theta}(\mathbf{x}); \theta) d\hat{\theta}(\mathbf{x}) & \text{for the continuous case} \end{cases}$$

where $p_{\hat{\theta}(\mathbf{X})}(\hat{\theta}(\mathbf{x}); \theta)$ and $f_{\hat{\theta}(\mathbf{X})}(\hat{\theta}(\mathbf{x}); \theta)$ are the mass and density functions, respectively, associated with $F_{\hat{\theta}(\mathbf{X})}(\hat{\theta}(\mathbf{x}); \theta)$.

To avoid excessive notation, the subscript will be dropped on the expectation, variance and probability operators when the relationship between parameters and the sampling distribution of the estimators is obvious.

3. Point estimation

While having $E(\hat{\theta}) = \theta$ (i.e. an unbiased estimator) is desirable, other things equal, the problem is that other things may not be equal! We are also concerned about the variability of point estimates. As such, we also consider the variance of an estimator, denoted:

$$\text{Var}(\hat{\theta}).$$

Ideally, we would like this variance to be as small as possible such that an estimate is *precise*.

Example 3.1 Consider two different estimators of a parameter θ , denoted $\hat{\theta}_1$ and $\hat{\theta}_2$. Suppose the bias and variance attributes for these two estimators are:

Estimator	$\hat{\theta}_1$	$\hat{\theta}_2$
Bias	0	$\neq 0$
Variance	Small	Large

Unambiguously we prefer $\hat{\theta}_1$ since it is unbiased (relative to the biased estimator $\hat{\theta}_2$) and it has a smaller variance.

Suppose instead these two estimators had the following properties:

Estimator	$\hat{\theta}_1$	$\hat{\theta}_2$
Bias	0	$\neq 0$
Variance	Large	Small

Now it is not obvious which estimator is better. While $\hat{\theta}_1$ is unbiased, $\hat{\theta}_2$ has a smaller variance.

Example 3.1 demonstrates that we may face a trade-off between bias and variance of an estimator. An estimator which has a large probability attached to values which are much larger and much smaller than θ is undesirable, yet it may still be an unbiased estimator. One approach to deal with this problem is to consider the *mean squared error* which is the expected squared distance between an estimator and the true parameter value.

Mean squared error

The **mean squared error (MSE)** of an estimator $\hat{\theta}$ is the function of θ defined by:

$$\begin{aligned} \text{MSE}_{\theta}(\hat{\theta}) &= E_{\theta}((\hat{\theta} - \theta)^2) \\ &= \text{Var}_{\theta}(\hat{\theta}) + (\text{Bias}_{\theta}(\hat{\theta}))^2. \end{aligned}$$

An estimator $\hat{\theta}_1$ is uniformly better than $\hat{\theta}_2$ if it has a smaller mean squared error for all θ .

Proof: We have (dropping the θ subscript):

$$\begin{aligned}
 \text{MSE}_\theta(\hat{\theta}) &= E((\hat{\theta} - \theta)^2) \\
 &= E((\underbrace{\hat{\theta} - E(\hat{\theta})}_{=0} + E(\hat{\theta}) - \theta)^2) \\
 &= E((\hat{\theta} - E(\hat{\theta}))^2 + (E(\hat{\theta}) - \theta)^2 + 2(\hat{\theta} - E(\hat{\theta}))(E(\hat{\theta}) - \theta)) \\
 &= E((\hat{\theta} - E(\hat{\theta}))^2) + E((\text{Bias}(\hat{\theta}))^2) + 2 \underbrace{(E(\hat{\theta}) - E(\hat{\theta}))}_{=0} (E(\hat{\theta}) - E(\theta)) \\
 &= \text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2.
 \end{aligned}$$

An estimator with a small bias and small variance is appealing. A small variance means that probability mass is concentrated around the centre of the distribution, yet given that we also have a small bias then the centre of the distribution is close to the true parameter value. Therefore, an estimator with a small mean squared error will have a sampling distribution with probability mass concentrated near θ . In practice, a small bias and a small variance are competing demands, as demonstrated in the second case of Example 3.1. ■

3.4.2 Consistency

In **ST104b Statistics 2** an estimator was called *consistent* if its mean squared error tended to zero as the sample size $n \rightarrow \infty$, hence consistency is an *asymptotic* property. The idea being that if we had an infinite sample size then we would expect our estimator to be perfect such that the point estimate equals the true parameter value, i.e. the true parameter value is obtained with probability 1.

Here we provide a slightly more precise definition of consistency. First we define a sequence of estimators. Up to this point we have denoted the sample size as n without representing the dependence of an estimator on n explicitly. Suppose we did make the dependence on n explicit for the sample mean, i.e. we now write the estimator as:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

hence:

$$\bar{X}_1 = X_1, \quad \bar{X}_2 = \frac{X_1 + X_2}{2}, \quad \bar{X}_3 = \frac{X_1 + X_2 + X_3}{3} \quad \text{etc.}$$

For a generic estimator we could use the notation $\hat{\theta}_n$ where the n subscript indicates the sample size on which the estimator is based. This allows us to define $\{\hat{\theta}_n : n = 1, 2, \dots\}$ to be a sequence of estimators, or $\{\hat{\theta}_n\}$ for short. We now define consistency.

Consistency

The sequence $\{\hat{\theta}_n : n = 1, 2, \dots\}$ is a **consistent** sequence of estimators of θ if it converges in probability to θ , i.e. for every $\epsilon > 0$ and $\theta \in \Theta$ we have:

$$\lim_{n \rightarrow \infty} P_{\theta}(|\hat{\theta}_n - \theta| < \epsilon) = 1$$

or equivalently:

$$\lim_{n \rightarrow \infty} P_{\theta}(|\hat{\theta}_n - \theta| \geq \epsilon) = 0.$$

This definition of consistency says that, for any $\delta > 0$, we can find a positive integer $N \in \mathbb{N}^+$ such that for any $n > N$ we have:

$$1 - P_{\theta}(|\hat{\theta}_n - \theta| < \epsilon) < \delta \quad \Leftrightarrow \quad P_{\theta}(|\hat{\theta}_n - \theta| \geq \epsilon) < \delta.$$

This can be thought of as follows. Suppose $\{\hat{\theta}_n\}$ is a consistent sequence of estimators of θ . Hence for any open interval around θ , such as $(\theta - \epsilon, \theta + \epsilon)$, we can find a number $N \in \mathbb{N}^+$ such that whenever $n > N$ the probability of obtaining a point estimate outside of the interval is arbitrarily small; meaning we can reduce this probability simply by increasing the value of N .

We now consider the relationship between mean squared error and consistency.

Relationship between mean squared error and consistency

Let $\{\hat{\theta}_n\}$ be a sequence of statistics and θ be an unknown parameter. If, for all $\theta \in \Theta$, we have:

$$\lim_{n \rightarrow \infty} \text{MSE}_{\theta}(\hat{\theta}_n) = 0$$

then $\{\hat{\theta}_n\}$ is a consistent sequence of estimators of θ .

Proof: If $\lim_{n \rightarrow \infty} \text{MSE}_{\theta}(\hat{\theta}_n) = 0$ holds for all $\theta \in \Theta$, then by definition:

$$\lim_{n \rightarrow \infty} E_{\theta}((\hat{\theta}_n - \theta)^2) = 0 \quad \text{for all } \theta \in \Theta.$$

Applying the Chebyshev inequality (see **ST2133 Advanced statistics: distribution theory**) to the probability in our definition of consistency gives:

$$P_{\theta}(|\hat{\theta}_n - \theta| \geq \epsilon) \leq \frac{E_{\theta}((\hat{\theta}_n - \theta)^2)}{\epsilon^2}$$

where the numerator of the right-hand side converges to zero for all $\theta \in \Theta$. The left-hand side is a non-negative number (by the first axiom of probability) which is bounded above by something which converges to zero. Hence:

$$\lim_{n \rightarrow \infty} P_{\theta}(|\hat{\theta}_n - \theta| \geq \epsilon) = 0$$

and so $\{\hat{\theta}_n\}$ is a consistent sequence of estimators of θ . ■

We now consider the relationship between bias, variance and consistency.

Relationship between bias, variance and consistency

Let $\{\hat{\theta}_n\}$ be a sequence of statistics and θ be an unknown parameter. If, for all $\theta \in \Theta$, we have:

$$\lim_{n \rightarrow \infty} \text{Bias}_\theta(\hat{\theta}_n) = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \text{Var}_\theta(\hat{\theta}_n) = 0$$

then $\{\hat{\theta}_n\}$ is a consistent sequence of estimators of θ .

3

Example 3.2 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, 1)$. Consider the statistics:

- (a) $\hat{\mu} = X_1 + (X_2 - X_3)^2 - 2$ as an estimator of μ
- (b) $\hat{\mu}^2 = \bar{X}^2 - 1/n$ as an estimator of μ^2 .

For each estimator, we determine whether they are (i) unbiased, and (ii) consistent.

We assume that if $X_n \xrightarrow{p} x$ and g is a continuous function then $g(X_n) \xrightarrow{p} g(x)$, and we will use the fact that:

$$E(X^2) = \text{Var}(X) + (E(X))^2 = 1 + \mu^2.$$

- (a) Working directly to establish unbiasedness, we have:

$$\begin{aligned} E(\hat{\mu}) &= E(X_1 + (X_2 - X_3)^2 - 2) \\ &= E(X_1) + E((X_2 - X_3)^2) - E(2) \\ &= \mu + E(X_2^2 - 2X_2X_3 + X_3^2) - 2 \\ &= \mu + 2(1 + \mu^2) - 2\mu^2 - 2 \\ &= \mu \end{aligned}$$

since $E(X_2X_3) = E(X_2)E(X_3)$, due to independence. This estimator cannot be consistent as its value does not depend on n .

- (b) Again, proceeding directly, we have unbiasedness since:

$$\begin{aligned} E(\hat{\mu}^2) &= E\left(\bar{X}^2 - \frac{1}{n}\right) \\ &= \text{Var}(\bar{X}) + (E(\bar{X}))^2 - E\left(\frac{1}{n}\right) \\ &= \frac{1}{n} + \mu^2 - \frac{1}{n} \\ &= \mu^2 \end{aligned}$$

since $\text{Var}(\bar{X}) = \sigma^2/n$ and $E(\bar{X}) = \mu$. We know that $\bar{X} \xrightarrow{p} \mu$, hence $\bar{X}^2 \xrightarrow{p} \mu^2$. Therefore, since $1/n \rightarrow 0$, we have that the estimator is consistent.

Activity 3.1 Let $\{X_1, X_2, \dots, X_n\}$ be a sample from a population with mean μ and variance σ^2 . The sample is *not* random since $\text{Cov}(X_i, X_j) = \rho\sigma^2$ for all $i \neq j$. Let:

$$\hat{\mu} = \sum_{i=1}^n a_i X_i.$$

Provide the condition on the constants $a_1, a_2, \dots, a_n$ for $\hat{\mu}$ to be an unbiased estimator of μ . Under this condition, calculate the mean squared error of $\hat{\mu}$.

Activity 3.2 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a Bernoulli(π) distribution. Define $S = \sum_{i=1}^n X_i$. Show that:

$$\frac{S(S-1)}{n(n-1)}$$

is an unbiased estimator of π^2 . Is this estimator consistent?

Activity 3.3 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, 1)$. Define Y_n to be a discrete random variable with support $\{0, n\}$ and mass function:

$$p_Y(y) = \begin{cases} 1 - 1/n & \text{for } y = 0 \\ 1/n & \text{for } y = n. \end{cases}$$

Show that the intuitively unappealing estimator:

$$\hat{\mu}_n = \bar{X} + Y_n$$

is consistent for μ . What is the limit of the mean squared error of this estimator as $n \rightarrow \infty$?

3.5 Application of sufficiency in point estimation

Treating mean squared error as a measure of the quality of an estimator, we seek an estimator with the smallest possible mean squared error. The sufficiency principle (covered in Chapter 2) suggests that statistical inference should be based on sufficient statistics. Here we show how to combine minimal mean squared error and sufficiency to derive a preferred estimator.

Suppose $\hat{\theta}$ is an estimator of θ . We can find a function of a sufficient statistic for θ which is at least as good as $\hat{\theta}$ in terms of mean squared error. We then have the best of both worlds – satisfying the sufficiency principle without losing mean squared error performance. The result, known as the *Rao–Blackwell theorem*, depends on a lemma about conditioning on a sufficient statistic, which we introduce first.

Conditioning on a sufficient statistic

Let $\mathbf{X}$ be a random sample parameterised by $\boldsymbol{\theta}$. Let $\mathbf{Y}(\mathbf{X})$ be a statistic and $\mathbf{S}(\mathbf{X})$ be a sufficient statistic for $\boldsymbol{\theta}$. If we define $\mathbf{T}(\mathbf{X}) = E(\mathbf{Y}(\mathbf{X}) | \mathbf{S}(\mathbf{X}))$, then $\mathbf{T}(\mathbf{X})$ is a statistic, i.e. $\mathbf{T}(\mathbf{X})$ does not depend on $\boldsymbol{\theta}$.

Proof: Since $\mathbf{S}(\mathbf{X})$ is a sufficient statistic, by definition we know that the conditional density function $f_{\mathbf{X}|\mathbf{S}(\mathbf{X})}(\mathbf{x} | \mathbf{s}(\mathbf{x}))$ does not depend on $\boldsymbol{\theta}$. The statistic $\mathbf{Y}(\mathbf{X})$ is some function of $\mathbf{X}$, hence:

$$E(\mathbf{Y}(\mathbf{X}) | \mathbf{S}(\mathbf{X}) = \mathbf{s}(\mathbf{x})) = \int_{\mathbb{R}^n} \mathbf{y}(\mathbf{x}) f_{\mathbf{X}|\mathbf{S}(\mathbf{X})}(\mathbf{x} | \mathbf{s}(\mathbf{x})) d\mathbf{x}$$

does not depend on $\boldsymbol{\theta}$, and so $\mathbf{T}(\mathbf{X}) = E(\mathbf{Y}(\mathbf{X}) | \mathbf{S}(\mathbf{X}))$ is a proper statistic. ■

We can now prove the Rao–Blackwell theorem, which shows that for any point estimator $\hat{\boldsymbol{\theta}}$ we can find a point estimator which is a function of a sufficient statistic and is at least as good as $\hat{\boldsymbol{\theta}}$ in terms of mean squared error.

Rao–Blackwell theorem

Let $\mathbf{X}$ be a random sample parameterised by $\boldsymbol{\theta}$. Let $\mathbf{Y}(\mathbf{X})$ be a statistic of the same dimension as $\boldsymbol{\theta}$, and let $\mathbf{S}(\mathbf{X})$ be a sufficient statistic for $\boldsymbol{\theta}$. If we define $\mathbf{T}(\mathbf{X}) = E(\mathbf{Y}(\mathbf{X}) | \mathbf{S}(\mathbf{X}))$, then $\mathbf{T}(\mathbf{X})$ is a statistic and:

$$\text{MSE}_{\boldsymbol{\theta}}(\mathbf{T}(\mathbf{X})) \leq \text{MSE}_{\boldsymbol{\theta}}(\mathbf{Y}(\mathbf{X})).$$

Proof: $\mathbf{T}(\mathbf{X})$ is a sufficient statistic due to the sufficiency of $\mathbf{S}(\mathbf{X})$. For the mean squared error result, we begin by considering the variance of $\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta}$, conditional on $\mathbf{S}(\mathbf{X})$, i.e. we consider:

$$\text{Var}((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta}) | \mathbf{S}(\mathbf{X})) = E((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta})^2 | \mathbf{S}(\mathbf{X})) - (E((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta}) | \mathbf{S}(\mathbf{X})))^2.$$

Since $\text{Var}((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta}) | \mathbf{S}(\mathbf{X})) \geq 0$ (it is a variance, hence is non-negative), we have:

$$E((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta})^2 | \mathbf{S}(\mathbf{X})) \geq (E((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta}) | \mathbf{S}(\mathbf{X})))^2.$$

We can now prove the result directly:

$$\begin{aligned} \text{MSE}_{\boldsymbol{\theta}}(\mathbf{Y}(\mathbf{X})) &= E((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta})^2) && \text{(by definition)} \\ &= E(E((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta})^2 | \mathbf{S}(\mathbf{X}))) && \text{(law of iterated expectations)} \\ &\geq E(E((\mathbf{Y}(\mathbf{X}) - \boldsymbol{\theta}) | \mathbf{S}(\mathbf{X}))^2) \\ &= E((E(\mathbf{Y}(\mathbf{X}) | \mathbf{S}(\mathbf{X})) - \boldsymbol{\theta})^2) && \text{(properties of expectation)} \\ &= E((\mathbf{T}(\mathbf{X}) - \boldsymbol{\theta})^2) \\ &= \text{MSE}_{\boldsymbol{\theta}}(\mathbf{T}(\mathbf{X})). \end{aligned}$$

■

So the Rao–Blackwell theorem allows us to find an improved estimator in terms of mean squared error (or at least not a worse mean squared error) by taking conditional expectations when conditioning on a sufficient statistic. Note that the bias is unaffected, since:

$$E(\mathbf{T}(\mathbf{X})) = E(E(\mathbf{Y}(\mathbf{X}) | \mathbf{S}(\mathbf{X}))) = E(\mathbf{Y}(\mathbf{X}))$$

by iterated expectations.

The Rao–Blackwell theorem is a remarkable result. Starting off with an unbiased estimator, simply by taking conditional expectations conditioning on a sufficient statistic, we end up with another unbiased estimator with a variance (and hence mean squared error) at least as small as the initial unbiased estimator.

Example 3.3 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a distribution with mean μ and variance σ^2 . Suppose that $S(\mathbf{X}) = \sum_{i=1}^n X_i$ is a sufficient statistic for μ . We consider $\hat{\mu}_1 = X_1$ as an initial estimator of μ and seek a better one.

For the estimator $\hat{\mu}_1$, we have:

$$E(\hat{\mu}_1) = E(X_1) = \mu \quad \text{and} \quad \text{Var}(\hat{\mu}_1) = \text{Var}(X_1) = \sigma^2.$$

Since the X_i s are identically distributed, for all $i = 1, 2, \dots, n$, we have:

$$E(X_i | S(\mathbf{X})) = E(X_1 | S(\mathbf{X})).$$

The Rao–Blackwell theorem states that the following estimator is better:

$$\begin{aligned} \hat{\mu}_2 &= E(\hat{\mu}_1 | S(\mathbf{X})) = E(X_1 | S(\mathbf{X})) = \frac{1}{n} \sum_{i=1}^n E(X_i | S(\mathbf{X})) \\ &= \frac{1}{n} \sum_{i=1}^n E(X_i | S(\mathbf{X})) \\ &= \frac{1}{n} E\left(\sum_{i=1}^n X_i \mid S(\mathbf{X})\right) \\ &= \frac{1}{n} E\left(\sum_{i=1}^n X_i \mid \sum_{i=1}^n X_i\right) \\ &= \frac{1}{n} \sum_{i=1}^n X_i \\ &= \bar{X}. \end{aligned}$$

Example 3.4 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, 1)$. Let $Y(\mathbf{X}) = X_1$ be the initial estimator of μ . Since $E(Y(\mathbf{X})) = E(X_1) = \mu$, this is an unbiased estimator, and:

$$\text{MSE}(Y(\mathbf{X})) = \text{Var}(Y(\mathbf{X})) = \text{Var}(X_1) = \sigma^2 = 1.$$

A sufficient statistic for μ is $S(\mathbf{X}) = \sum_{i=1}^n X_i$. It can be shown (see Activity 3.4) that $(Y(\mathbf{X}), S(\mathbf{X}))^T$ is jointly normally distributed with:

$$E((Y(\mathbf{X}), S(\mathbf{X}))^T) = \begin{pmatrix} \mu \\ n\mu \end{pmatrix}$$

and:

$$\text{Var}((Y(\mathbf{X}), S(\mathbf{X}))^T) = \begin{pmatrix} 1 & 1 \\ 1 & n \end{pmatrix}.$$

Using bivariate normal distribution theory:

$$\begin{aligned} E(Y(\mathbf{X}) | S(\mathbf{X})) &= E(Y(\mathbf{X})) + \text{Cov}(Y(\mathbf{X}), S(\mathbf{X}))(\text{Var}(S(\mathbf{X})))^{-1}(S(\mathbf{X}) - E(S(\mathbf{X}))) \\ &= \mu + 1 \times n^{-1} \times (S(\mathbf{X}) - n\mu) \\ &= \frac{1}{n} S(\mathbf{X}) \\ &= \frac{1}{n} \sum_{i=1}^n X_i \\ &= \bar{X}. \end{aligned}$$

Hence the Rao–Blackwell theorem leads us to the sample mean $\bar{X}$ (unsurprisingly) to be a better estimator of μ than X_1 . Since:

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

then:

$$\text{MSE}_\mu(\bar{X}) \leq \text{MSE}_\mu(X_1).$$

Activity 3.4 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, 1)$. Let $U(\mathbf{X}) = X_1$ and $S(\mathbf{X}) = \sum_{i=1}^n X_i$. Show that $(U(\mathbf{X}), S(\mathbf{X}))^T$ is jointly normally distributed.

Activity 3.5 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from the Bernoulli(π) distribution. Use the Rao–Blackwell theorem to find an estimator of π which is an improvement on X_1 .

3.6 Sample moments

As seen in **ST104b Statistics 2** when method of moments estimation was introduced, *sample moments* provide natural estimators of *population moments*. As revised in Chapter 2, we define the r th sample moment about zero and r th central sample moment as follows.

Sample moments and central sample moments

For a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$, for each $r = 1, 2, \dots$ we define the r th **sample moment** (about zero) to be:

$$M_r = \frac{1}{n} \sum_{i=1}^n X_i^r$$

and we define the r th **central sample moment** to be:

$$M'_r = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^r.$$

These can be used as estimators of the corresponding population moments, which were covered in **ST2133 Advanced statistics: distribution theory**, such that the r th population moment is:

$$\mu_r = E(X^r)$$

and the r th central population moment is:

$$\mu'_r = E((X - E(X))^r) = E((X - \mu_1)^r).$$

Note the special cases of the population mean:

$$E(X) = \mu = \mu_1$$

and the population variance:

$$\text{Var}(X) = \sigma^2 = \mu'_2 = \mu_2 - (\mu_1)^2.$$

The sample moments are analogous to taking expectations by taking sample averages (summing and dividing by n). Sample moments are statistics (known functions of data), hence are random variables.

The sample mean, $\bar{X}$, is the first sample moment:

$$\bar{X} = M_1 = \frac{1}{n} \sum_{i=1}^n X_i.$$

The sample variance (with the $n - 1$ divisor) is not the second central sample moment, but:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{n}{n-1} M'_2.$$

Higher-order sample moments measure properties such as skewness and kurtosis of sample distributions.

We now state the following result for central sample moments.

Central sample moment property

When determining the properties of central sample moments, we may assume a population mean of zero.

Proof: Let $\mathbf{Y} = \{Y_1, Y_2, \dots, Y_n\}$ represent a transformation of the random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ by defining:

$$Y_i = X_i - \frac{1}{n} \sum_{j=1}^n X_j = X_i - \bar{X} \quad \text{for } i = 1, 2, \dots, n.$$

Hence:

$$E(Y_i) = E(X_i - \bar{X}) = E(X_i) - E(\bar{X}) = \mu - \mu = 0 \quad \text{for } i = 1, 2, \dots, n.$$

Denoting the r th central sample moment of $\mathbf{Y}$ by $M'_r(\mathbf{Y})$, for $r \geq 1$ we have:

$$\begin{aligned} M'_r(\mathbf{Y}) &= \sum_{i=1}^n \left(Y_i - \frac{1}{n} \sum_{j=1}^n Y_j \right)^r \\ &= \sum_{i=1}^n \left(X_i - \frac{1}{n} \sum_{j=1}^n X_j - \frac{1}{n} \sum_{j=1}^n \left(X_i - \frac{1}{n} \sum_{k=1}^n X_k \right) \right)^r \\ &= \sum_{i=1}^n \left(X_i - \frac{1}{n} \sum_{j=1}^n X_j \right)^r \\ &= M'_r(\mathbf{X}) \end{aligned}$$

where $M'_r(\mathbf{X})$ is the r th central sample moment of $\mathbf{X}$. Hence any statements about the central sample moments of $\mathbf{Y}$ (with population mean zero) hold equally for $\mathbf{X}$. ■

3.6.1 Sample mean

The sample mean and its statistical properties (mean and variance) have been covered extensively in **ST104b Statistics 2**. Given the widespread use of the sample mean in statistical inference, these results are provided again here for completeness before considering the asymptotic (i.e. large sample) properties of the sample mean in the next subsection.

Sample mean

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from the probability distribution F_X . The **sample mean** is:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

We routinely use the sample mean, $\bar{X}$, as the estimator of the population mean, $E(X) = \mu$, i.e. we could write $\hat{\mu} = \bar{X}$. We now consider the expected value and variance of the sample mean, for which *neither* property requires any distributional assumption about the parametric form of F_X .

Statistical properties of the sample mean

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a population distribution with mean μ and finite variance, i.e. $\sigma^2 < \infty$.

- i. The sample mean is an unbiased estimator of μ , i.e. $E(\bar{X}) = \mu$.
- ii. The variance of the sample mean is $\text{Var}(\bar{X}) = \sigma^2/n$.

These properties hold regardless of the sample size n .

Proof:

- i. Since the X_i s are identically distributed, we have:

$$\begin{aligned}
 E(\bar{X}) &= E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \\
 &= \frac{1}{n} \sum_{i=1}^n E(X_i) && \text{(properties of expectation)} \\
 &= \frac{1}{n} \sum_{i=1}^n \mu && \text{(as identically distributed)} \\
 &= \mu.
 \end{aligned}$$

- ii. Since the X_i s are independent and identically distributed, we have:

$$\begin{aligned}
 \text{Var}(\bar{X}) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \\
 &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) && \text{(as independent)} \\
 &= \frac{1}{n^2} \sum_{i=1}^n \sigma^2 && \text{(as identically distributed)} \\
 &= \frac{\sigma^2}{n}.
 \end{aligned}$$

■

Example 3.5 We find the sampling distribution of $\bar{X}$ in the following cases.

- (a) A random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ is drawn from $N(\mu, \sigma^2)$, where:

$$M_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right).$$

Given the moment generating function of a $N(\mu, \sigma^2)$ random variable, the moment generating function of $\bar{X}$, noting that the X_i s are independent and identically distributed, is:

$$\begin{aligned}
 M_{\bar{X}}(t) &= E \left(\exp \left(t \frac{1}{n} \sum_{i=1}^n X_i \right) \right) = E \left(\prod_{i=1}^n \exp \left(\frac{t}{n} X_i \right) \right) \\
 &= \prod_{i=1}^n E \left(\exp \left(\frac{t}{n} X_i \right) \right) \\
 &= \left(M_X \left(\frac{t}{n} \right) \right)^n \\
 &= \left(\exp \left(\mu \frac{t}{n} + \frac{\sigma^2 (t^2/n^2)}{2} \right) \right)^n \\
 &= \exp \left(\frac{\mu n t}{n} + n \frac{\sigma^2 t^2 / n^2}{2} \right) \\
 &= \exp \left(\mu t + \frac{(\sigma^2/n) t^2}{2} \right)
 \end{aligned}$$

which is the moment generating function of a $N(\mu, \sigma^2/n)$ distribution. Therefore:

$$\bar{X} \sim N \left(\mu, \frac{\sigma^2}{n} \right).$$

- (b) A random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ is drawn from $\text{Gamma}(\alpha, \beta)$, where:

$$M_X(t) = \left(1 - \frac{t}{\beta} \right)^{-\alpha}.$$

Given the moment generating function of a $\text{Gamma}(\alpha, \beta)$ random variable, the moment generating function of $\bar{X}$ is:

$$\begin{aligned}
 M_{\bar{X}}(t) &= E \left(\exp \left(t \frac{1}{n} \sum_{i=1}^n X_i \right) \right) = E \left(\prod_{i=1}^n \exp \left(\frac{t}{n} X_i \right) \right) \\
 &= \prod_{i=1}^n E \left(\exp \left(\frac{t}{n} X_i \right) \right) \\
 &= \left(M_X \left(\frac{t}{n} \right) \right)^n \\
 &= \left(\left(1 - \frac{t/n}{\beta} \right)^{-\alpha} \right)^n \\
 &= \left(1 - \frac{t}{n\beta} \right)^{-n\alpha}
 \end{aligned}$$

which is the moment generating function of a $\text{Gamma}(n\alpha, n\beta)$ distribution. Therefore:

$$\bar{X} \sim \text{Gamma}(n\alpha, n\beta).$$

3.6.2 Asymptotic properties of the sample mean

Following our discussion of consistency, we now make explicit the dependence of the sample mean on the sample size n by writing $\bar{X}_n$. We now consider the *weak law of large numbers*.

Weak law of large numbers

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a population distribution with finite mean, i.e. $\mu < \infty$. By the **weak law of large numbers**, then:

$$\bar{X}_n \xrightarrow{p} \mu \quad \text{as } n \rightarrow \infty$$

where $\bar{X}_n$ is the sample mean.

Proof: Since $\bar{X}_n$ is an unbiased estimator of μ , then $\text{MSE}_\mu(\bar{X}_n) = \text{Var}_\mu(\bar{X}_n)$. Hence:

$$\lim_{n \rightarrow \infty} \text{MSE}_\mu(\bar{X}_n) = \lim_{n \rightarrow \infty} \frac{\sigma^2}{n} = 0$$

for $\sigma^2 < \infty$. Hence $\bar{X}_n$ is a consistent estimator of μ such that $\bar{X}_n \xrightarrow{p} \mu$ as $n \rightarrow \infty$. ■

This weak law of large numbers is a slightly stronger version of the convergence in distribution of the sample mean to μ , known as the *law of large numbers*.

Law of large numbers

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a population distribution with finite mean, i.e. $\mu < \infty$. By the **law of large numbers**, then:

$$\bar{X}_n \xrightarrow{d} \mu \quad \text{as } n \rightarrow \infty$$

where $\bar{X}_n$ is the sample mean.

Activity 3.6 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample and $\bar{X}_n$ be the sample mean. Also, let $E(X) = \mu$ and suppose the moment generating function M_X is well-defined. Using cumulant generating functions, prove that:

$$\bar{X}_n \xrightarrow{d} \mu \quad \text{as } n \rightarrow \infty.$$

With slightly stronger assumptions, we come to the *strong law of large numbers* which tells us that the sample mean converges almost surely to μ .

Strong law of large numbers

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a population distribution with finite mean, i.e. $\mu < \infty$, and $E(|X|) < \infty$. By the **strong law of large numbers**, then:

$$\bar{X}_n \xrightarrow{a.s.} \mu \quad \text{as } n \rightarrow \infty$$

where $\bar{X}_n$ is the sample mean.

The weak law and strong law tell us that, under relatively mild conditions, the sample mean converges to a degenerate distribution with all of the probability mass at μ as $n \rightarrow \infty$. However, in practice we have finite sample sizes such that we seek a sufficiently good approximation of the sampling distribution of $\bar{X}_n$ for finite n . The *central limit theorem* provides such a result.

Central limit theorem

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a population distribution with mean μ and finite variance, i.e. $\sigma^2 < \infty$. The **central limit theorem** says that:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1) \quad \text{as } n \rightarrow \infty$$

where $\bar{X}_n$ is the sample mean.

A non-examinable proof can be found in Appendix A.

The central limit theorem is a remarkable result allowing us to make statistical inferences (such as in interval estimation and hypothesis testing) about the sample mean using the normal distribution regardless of the population distribution F_X from which a random sample is drawn.

The rate of convergence of the sampling distribution of $\bar{X}_n$ to normality depends on the shape of the underlying distribution F_X . If $F_X \sim N(\mu, \sigma^2)$ then:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

exactly regardless of the sample size n . (Recall that a linear combination of normal random variables is also normally distributed, hence if $X_i \sim N(\mu, \sigma^2)$, then

$\bar{X} = \sum_{i=1}^n X_i/n$ is such a linear combination.)

However, if we are sampling, say, from a *heavily skewed* and *discrete* distribution (which differs from the *symmetric* and *continuous* normal distribution) a large sample size would be needed to ensure sufficiently adequate convergence. The central limit theorem itself does not provide any indication about how good a normal approximation would be in finite samples.

The following examples are from **ST104b Statistics 2**, but are reproduced here to demonstrate the rate of convergence when sampling from the exponential distribution

(which, although positively skewed, is at least continuous) and the Bernoulli distribution (which is heavily skewed and discrete).

Example 3.6 In the first case, we simulate random samples of sizes:

$$n = 1, 5, 10, 30, 100 \text{ and } 1,000$$

from the $\text{Exponential}(0.25)$ distribution, for which:

$$E(X) = \mu = \frac{1}{\lambda} = \frac{1}{0.25} = 4 \quad \text{and} \quad \text{Var}(X) = \sigma^2 = \frac{1}{\lambda^2} = 16.$$

This is clearly a skewed distribution, as shown by the histogram for $n = 1$ in Figure 3.1.

10,000 independent random samples of each size were generated. Histograms of the values of $\bar{x}$ from these observed samples are shown in Figure 3.1. Each plot also shows the probability density function (pdf) of the approximating normal distribution:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) = N\left(4, \frac{16}{n}\right).$$

The normal approximation is reasonably good already for $n = 30$, very good for $n = 100$, and practically perfect for $n = 1,000$.

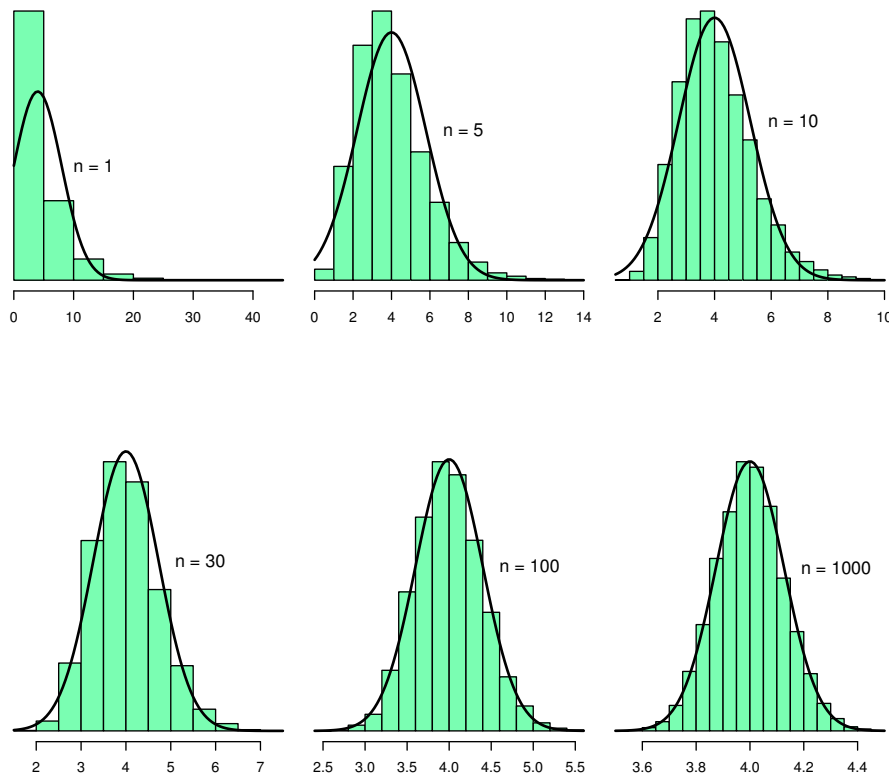


Figure 3.1: Sampling distributions of $\bar{X}$ for various n when sampling from the $\text{Exponential}(0.25)$ distribution.

Example 3.7 In the second case, we simulate 10,000 independent random samples of sizes:

$$n = 1, 10, 30, 50, 100 \text{ and } 1,000$$

from the Bernoulli(0.2) distribution, for which:

$$E(X) = \mu = \pi = 0.2 \quad \text{and} \quad \text{Var}(X) = \sigma^2 = \pi(1 - \pi) = 0.16.$$

Here the distribution of X_i itself is not even continuous, and has only two possible values, 0 and 1. Nevertheless, the sampling distribution of $\bar{X}$ can be very well-approximated by the normal distribution, when n is large enough. Each plot also shows the pdf of the approximating normal distribution:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) = N\left(0.2, \frac{0.16}{n}\right).$$

Note that since here $X_i = 1$ or $X_i = 0$ for all i , $\bar{X} = \sum_{i=1}^n X_i/n = m/n$, where m is the number of observations for which $X_i = 1$. In other words, $\bar{X}$ is the *sample proportion* of the event $X = 1$.

The normal approximation is clearly very bad for small n , but reasonably good already for $n = 50$, as shown by the histograms in Figure 3.2.

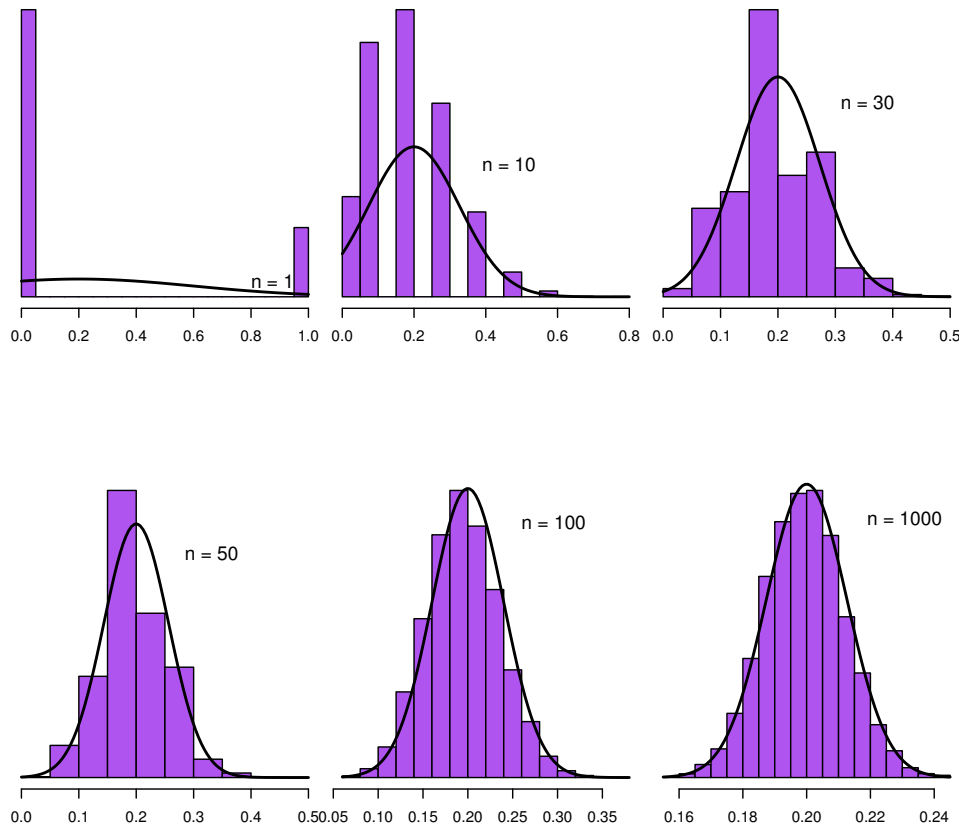


Figure 3.2: Sampling distributions of $\bar{X}$ for various n when sampling from the Bernoulli(0.2) distribution.

3.6.3 Sample variance

For any random sample $\mathbf{X}$ with $n \geq 2$ we can generate a point estimator of the population variance σ^2 .

Sample variance

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample with $n \geq 2$ from F_X . The **sample variance** is:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Properties of the summation operator allow us to write alternative representations of the sample variance.

Four representations of the sample variance

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample with $n \geq 2$ from F_X . The sample mean is:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

and the sample variance is:

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

The following are four representations of the sample variance.

i. For $n \geq 2$:

$$(n-1)S_n^2 = \sum_{i=1}^n X_i(X_i - \bar{X}_n).$$

ii. For $n \geq 2$:

$$(n-1)S_n^2 = \sum_{i=1}^n X_i^2 - n\bar{X}_n^2.$$

iii. For $n \geq 2$:

$$(n-1)S_n^2 = \left(\sum_{i=2}^n (X_i - \bar{X}_n) \right)^2 + \sum_{i=2}^n (X_i - \bar{X}_n)^2.$$

iv. For $n \geq 3$:

$$(n-1)S_n^2 = (n-2)S_{n-1}^2 + \frac{n-1}{n}(X_n - \bar{X}_{n-1})^2.$$

Non-examinable proofs can be found in Appendix A.

We now consider the statistical properties of the sample variance.

Statistical properties of the sample variance

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample with $n \geq 2$ from a population with mean μ and finite variance, i.e. $\sigma^2 < \infty$. The sample variance:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

has the following properties.

- i. S^2 is an unbiased estimator of σ^2 , i.e. $E(S^2) = \sigma^2$.

Provided F_X has a finite fourth central moment, i.e. $\mu'_4 < \infty$, then the following properties also hold.

- ii. The mean squared error of S^2 as an estimator of σ^2 is:

$$\text{MSE}_{\sigma^2}(S^2) = \text{Var}(S^2) = \frac{1}{n} \left(\mu'_4 - \frac{n-3}{n-1} (\mu'_2)^2 \right) = \frac{1}{n} \left(\mu'_4 - \frac{n-3}{n-1} \sigma^4 \right).$$

- iii. S^2 is a consistent estimator of σ^2 .

Proof:

- i. Noting that $E(X_i) = \mu$ and $E(X_i^2) = \text{Var}(X) + (E(X))^2 = \sigma^2 + \mu^2$ for all $i = 1, 2, \dots, n$, and that:

$$E(\bar{X}^2) = \text{Var}(\bar{X}) + (E(\bar{X}))^2 = \frac{\sigma^2}{n} + \mu^2$$

we have:

$$\begin{aligned} E(S^2) &= E \left(\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \right) \\ &= \frac{1}{n-1} E \left(\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right) \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n E(X_i^2) - n E(\bar{X}^2) \right) \\ &= \frac{1}{n-1} \left(n(\sigma^2 + \mu^2) - n \left(\frac{\sigma^2}{n} + \mu^2 \right) \right) \\ &= \frac{1}{n-1} (n-1)\sigma^2 \\ &= \sigma^2. \end{aligned}$$

Note an alternative *non-examinable* proof can be found in Appendix A.

- ii. A non-examinable proof can be found in Appendix A.

iii. From property ii. we have:

$$\lim_{n \rightarrow \infty} \text{MSE}_{\sigma^2}(S^2) = 0$$

hence S^2 is a consistent estimator of σ^2 . ■

Example 3.8 We compare the mean squared errors of S^2 , $(n-1)S^2/n$, and $\hat{\sigma}^2 = 1$ as estimators of σ^2 in the presence of a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ from $N(\mu, \sigma^2)$. Noting that $E(S^2) = \sigma^2$, then S^2 is an unbiased estimator. We also have:

$$E\left(\frac{n-1}{n}S^2\right) = \frac{n-1}{n}E(S^2) = \frac{n-1}{n}\sigma^2 \neq \sigma^2$$

hence $(n-1)S^2/n$ is a biased estimator, with bias:

$$\text{Bias}\left(\frac{n-1}{n}S^2\right) = E\left(\frac{n-1}{n}S^2\right) - \sigma^2 = \frac{n-1}{n}\sigma^2 - \sigma^2 = -\frac{\sigma^2}{n}.$$

Noting that the fourth central population moment of the normal distribution is $\mu'_4 = 3\sigma^4$, the mean squared error of S^2 is:

$$\text{MSE}(S^2) = \text{Var}(S^2) = \frac{1}{n}\left(\mu'_4 - \frac{n-3}{n-1}\sigma^4\right) = \frac{1}{n}\left(3\sigma^4 - \frac{n-3}{n-1}\sigma^4\right) = \frac{2}{n-1}\sigma^4.$$

The variance of $(n-1)S^2/n$ is:

$$\text{Var}\left(\frac{n-1}{n}S^2\right) = \frac{(n-1)^2}{n^2}\text{Var}(S^2) = \frac{(n-1)^2}{n^2}\frac{2}{n-1}\sigma^4 = \frac{2(n-1)}{n^2}\sigma^4$$

hence:

$$\begin{aligned}\text{MSE}\left(\frac{n-1}{n}S^2\right) &= \text{Var}\left(\frac{n-1}{n}S^2\right) + \left(\text{Bias}\left(\frac{n-1}{n}S^2\right)\right)^2 \\ &= \frac{2(n-1)}{n^2}\sigma^4 + \left(-\frac{\sigma^2}{n}\right)^2 \\ &= \frac{2n-1}{n^2}\sigma^4.\end{aligned}$$

We have that:

$$\text{MSE}(S^2) = \frac{2}{n-1}\sigma^4 > \frac{2n-1}{n^2}\sigma^4 = \text{MSE}\left(\frac{n-1}{n}S^2\right)$$

for $n = 2, 3, \dots$. Hence $(n-1)S^2/n$ is uniformly better than S^2 . However, it is not *uniformly* better than $\hat{\sigma}^2 = 1$, which has a mean squared error of zero when $\sigma^2 = 1$.

Having derived the expected value and variance for both of $\bar{X}$ and S^2 , we now note the association between these two statistics without proof.

Association between the sample mean and sample variance

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample with $n \geq 2$ from a population with mean μ and finite variance, i.e. $\sigma^2 < \infty$.

- i. $\bar{X}$ is uncorrelated with $(X_i - \bar{X})$ for $i = 1, 2, \dots, n$.

Provided F_X has a finite third central moment, i.e. $\mu'_3 < \infty$, then the following property also holds.

- ii. The covariance between the sample mean and sample variance is a function of the third central moment:

$$\text{Cov}(\bar{X}, S^2) = \frac{1}{n} \mu'_3.$$

Non-examinable proofs can be found in Appendix A.

3.6.4 Sample mean and sample variance when sampling from a normal population

Here we consider a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ from $N(\mu, \sigma^2)$ and the sampling distributions of $\bar{X}$ and S^2 , as well as the independence of these two statistics.

Independence of sample mean and variance for normal population

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$. In this case, the following properties hold.

- i. The sample mean $\bar{X}$ and terms of the form $(X_i - \bar{X})$ are independent, for $i = 1, 2, \dots, n$, i.e. $\bar{X} \perp\!\!\!\perp (X_i - \bar{X})$.
- ii. The sample mean and sample variance are independent, i.e. $\bar{X} \perp\!\!\!\perp S^2$.

Proof: For a normal distribution, $\mu'_3 = 0$. Uncorrelated normal random variables are independent. Hence the results follow from the association results in the previous subsection. ■

In order to determine the sampling distribution of the sample variance, recall the definition of the *chi-squared distribution* introduced in **ST104b Statistics 2**.

Chi-squared distribution

Let $\mathbf{Z} = \{Z_1, Z_2, \dots, Z_n\}$ be a random sample from $N(0, 1)$, i.e. the standard normal distribution. Since $Z_i^2 \sim \chi_1^2$, then:

$$\sum_{i=1}^n Z_i^2 \sim \chi_n^2.$$

We now derive the sampling distributions of the sample mean and sample variance when sampling from the standard normal distribution, i.e. from $N(0, 1)$.

Distribution of sample mean and variance for standard normal case

Let $\mathbf{Z} = \{Z_1, Z_2, \dots, Z_n\}$ be a random sample from $N(0, 1)$. Let the sample mean and sample variance be denoted, respectively, by:

$$\bar{Z} = \frac{1}{n} \sum_{i=1}^n Z_i \quad \text{and} \quad S_Z^2 = \frac{1}{n-1} \sum_{i=1}^n (Z_i - \bar{Z})^2.$$

The sampling distributions of these statistics are as follows.

- i. The sample mean is normally distributed:

$$\bar{Z} \sim N\left(0, \frac{1}{n}\right).$$

- ii. $(n-1)$ times the sample variance is chi-squared distributed:

$$(n-1)S_Z^2 \sim \chi_{n-1}^2.$$

Proof:

- i. We consider the moment generating function of $\bar{Z}$ and appeal to the one-to-one mapping between a moment generating function and probability distribution. We have:

$$\begin{aligned} M_{\bar{Z}}(t) &= E(\exp(t\bar{Z})) = E\left(\exp\left(\frac{t}{n}(Z_1 + Z_2 + \dots + Z_n)\right)\right) \\ &= E\left(\prod_{i=1}^n \exp\left(\frac{t}{n}Z_i\right)\right) \\ &= \prod_{i=1}^n E\left(\exp\left(\frac{t}{n}Z_i\right)\right) \quad (\text{by independence}) \\ &= \left(M_Z\left(\frac{t}{n}\right)\right)^n \quad (\text{as identically distributed}) \\ &= \left(\exp\left(\frac{1}{2}\left(\frac{t}{n}\right)^2\right)\right)^n \quad (\text{as } Z \sim N(0, 1)) \\ &= \exp\left(\frac{1}{n} \frac{t^2}{2}\right) \end{aligned}$$

which is the moment generating function of $N(0, 1/n)$, hence $\bar{Z} \sim N(0, 1/n)$.

- ii. We use various properties of the normal and chi-squared distributions. We consider

an inductive argument on:

$$(n-1)S_n^2 = \sum_{i=1}^n (Z_i - \bar{Z}_n)^2 \quad \text{where } \bar{Z}_n = \frac{1}{n} \sum_{i=1}^n Z_i.$$

Assume that:

$$(n-2)S_{n-1}^2 \sim \chi_{n-1}^2.$$

Using representation 'iv.' of the sample variance, we have:

$$(n-1)S_n^2 = (n-2)S_{n-1}^2 + \frac{n-1}{n} (Z_n - \bar{Z}_{n-1})^2.$$

Hence:

$$Z_n \sim N(0, 1) \quad \text{and} \quad \bar{Z}_{n-1} \sim N\left(0, \frac{1}{n-1}\right) \Rightarrow (Z_n - \bar{Z}_{n-1}) \sim N\left(0, \frac{n}{n-1}\right)$$

and so:

$$\frac{n-1}{n} (Z_n - \bar{Z}_{n-1})^2 \sim \chi_1^2.$$

We note that $(Z_n - \bar{Z}_{n-1})^2$ and S_{n-1}^2 are functions of sets of random variables which are independent of each other. Since functions of independent random variables are also independent, we have that:

$$(Z_n - \bar{Z}_{n-1})^2 \perp\!\!\!\perp S_{n-1}^2.$$

Given the additive property of chi-squared distributions, if this holds for S_{n-1}^2 it also holds for S_n^2 . Hence we just need to establish the result for S_2^2 . We have:

$$S_2^2 = \frac{1}{2} (Z_2 - \bar{Z}_1)^2 = \frac{1}{2} (Z_2 - Z_1)^2$$

hence:

$$Z_2 - Z_1 \sim N(0, 2) \Rightarrow \frac{Z_2 - Z_1}{\sqrt{2}} \sim N(0, 1) \Rightarrow S_2^2 \sim \chi_1^2.$$

Therefore, the result is true for S_2^2 . Hence we have established that:

$$(n-2)S_{n-1}^2 \sim \chi_{n-2}^2 \Rightarrow (n-1)S_n^2 \sim \chi_{n-1}^2$$

so, by induction, the result holds for $n = 2, 3, \dots$

■

Activity 3.7 Suppose that $X_1 \sim N(\mu_1, \sigma_1^2)$, $X_2 \sim N(\mu_2, \sigma_2^2)$ and $X_1 \perp\!\!\!\perp X_2$. Show that:

$$X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

Activity 3.8 Let $X \sim \chi_k^2$. Show that the moment generating function of X is:

$$M_X(t) = (1 - 2t)^{-k/2}.$$

Activity 3.9 Let $X_1 \sim \chi_m^2$, $X_2 \sim \chi_n^2$ and $X_1 \perp\!\!\!\perp X_2$. Show that:

$$X_1 + X_2 \sim \chi_{m+n}^2.$$

We now consider a random sample from a general normal distribution, i.e. from $N(\mu, \sigma^2)$.

Distribution of sample mean and variance for general normal case

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$. Let the sample mean and sample variance be denoted, respectively, by:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{and} \quad S_X^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

The sampling distributions of these statistics are as follows.

- i. The sample mean is normally distributed:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

- ii. $(n-1)/\sigma^2$ times the sample variance is chi-squared distributed:

$$\frac{(n-1)S_X^2}{\sigma^2} \sim \chi_{n-1}^2.$$

Proof:

- i. Since $Z_i = (X_i - \mu)/\sigma \sim N(0, 1)$, the result then follows directly from the fact that:

$$\bar{Z} = \frac{\bar{X} - \mu}{\sigma} \sim N\left(0, \frac{1}{n}\right) \Rightarrow \bar{X} - \mu \sim N\left(0, \frac{\sigma^2}{n}\right) \Rightarrow \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

- ii. Since:

$$(Z_i - \bar{Z})^2 = \frac{(X_i - \bar{X})^2}{\sigma^2}$$

we have:

$$S_Z^2 = \frac{1}{n-1} \sum_{i=1}^n (Z_i - \bar{Z})^2 = \frac{1}{\sigma^2} \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{\sigma^2} S_X^2.$$

Therefore:

$$(n-1)S_Z^2 \sim \chi_{n-1}^2 \Rightarrow \frac{(n-1)S_X^2}{\sigma^2} \sim \chi_{n-1}^2.$$

■

3.6.5 Method of moments estimation

The point estimation technique of *method of moments estimation* was introduced in **ST104b Statistics 2**. While the technique remains unchanged, in this course you may have to derive estimators for more difficult distributions. We review the technique here for completeness with slightly modified notation.

Remember that population moments are functions of the parameters of the distribution, while sample moments are statistics (functions of the sample). By equating population moments with their corresponding sample moments we can then solve the resulting set of simultaneous equations to derive the method of moments estimators.

Method of moments estimation

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a population distribution F_X parameterised by $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_r)^T$. Let:

$$\mu_r = \mu_r(\boldsymbol{\theta}) = E(X^r)$$

denote the r th population moment. μ_r depends on the unknown parameter(s) $\boldsymbol{\theta}$, as everything else about the distribution F_X is known.

Denote the r th sample moment by:

$$M_r = M_r(\mathbf{X}) = \frac{1}{n} \sum_{i=1}^n X_i^r.$$

The **method of moments estimators (MMEs)** of $\theta_1, \theta_2, \dots, \theta_r$ are obtained by solving:

$$\mu_i(\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_r) = M_i(\mathbf{X}) \quad \text{for } i = 1, 2, \dots, r$$

for expressions of $\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_r$ in terms of $\mathbf{X}$.

The idea is that M_r should be close to μ_r for large enough values of n , i.e. we expect:

$$M_r \rightarrow \mu_k \quad \text{as } n \rightarrow \infty.$$

Method of moments estimation is easy to implement, but method of moments estimators can have undesirable statistical properties (such as producing biased estimators), as well as possibly poor performance in practice for small sample sizes (the larger the sample size, the better the convergence of sample moments to their corresponding population moments).

Example 3.9 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, with μ and σ^2 unknown parameters, hence $\boldsymbol{\theta} = (\mu, \sigma^2)$, i.e. $r = 2$. Method of moments estimators are found by solving:

$$\mu_1(\hat{\mu}, \hat{\sigma}^2) = M_1(\mathbf{X}) \quad \text{and} \quad \mu_2(\hat{\mu}, \hat{\sigma}^2) = M_2(\mathbf{X}).$$

This produces:

$$\hat{\mu} = \bar{X}$$

and:

$$\hat{\sigma}^2 + \hat{\mu}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 \Rightarrow \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{n-1}{n} S^2.$$

Note we have:

$$\begin{aligned} E(\hat{\sigma}^2) &= E\left(\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2\right) \\ &= \frac{1}{n} \sum_{i=1}^n E(X_i^2) - E(\bar{X}^2) \\ &= E(X^2) - E(\bar{X}^2) \\ &= \sigma^2 + \mu^2 - \left(\frac{\sigma^2}{n} + \mu^2\right) \\ &= \frac{n-1}{n} \sigma^2. \end{aligned}$$

Since:

$$E(\hat{\sigma}^2) - \sigma^2 = \frac{n-1}{n} \sigma^2 - \sigma^2 = -\frac{\sigma^2}{n} < 0$$

$\hat{\sigma}^2$ is a negatively-biased estimator of σ^2 .

Activity 3.10 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, with μ and σ^2 unknown parameters. Find the bias, variance and mean squared error of the method of moments estimator of the variance:

$$\hat{\sigma}^2 = \frac{n-1}{n} S^2.$$

You may use the fact that the fourth central population moment of the normal distribution is $\mu'_4 = 3\sigma^4$.

Example 3.10 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a $\text{Bin}(m, \pi)$ distribution, with m and π unknown parameters, hence $\boldsymbol{\theta} = (m, \pi)$, i.e. $r = 2$. Method of moments estimators are found by solving:

$$\mu_1(\hat{m}, \hat{\pi}) = M_1(\mathbf{X}) \quad \text{and} \quad \mu_2(\hat{m}, \hat{\pi}) = M_2(\mathbf{X})$$

such that:

$$E(X) = m\pi \quad \text{and} \quad E(X^2) = \text{Var}(X) + (E(X))^2 = m\pi(1 - \pi) + m^2\pi^2.$$

This produces:

$$\hat{m}\hat{\pi} = \bar{X} \quad \text{and} \quad \hat{m}\hat{\pi}(1 - \hat{\pi}) + \hat{m}^2\hat{\pi}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2.$$

Solving for $\hat{m}$ by substituting $\hat{\pi} = \bar{X}/\hat{m}$, we have:

$$\hat{m} = \frac{\bar{X}^2}{\bar{X} - \sum_{i=1}^n (X_i - \bar{X})^2/n} \quad \text{and} \quad \hat{\pi} = \frac{\bar{X}}{\hat{m}}.$$

Example 3.11 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a $\text{Gamma}(\alpha, \beta)$ distribution with density function:

$$f_X(x; \alpha, \beta) = \frac{1}{\Gamma(\alpha)} \beta^\alpha x^{\alpha-1} e^{-\beta x}$$

for $\alpha, \beta > 0$ and $x \geq 0$, and 0 otherwise.

We find the method of moments estimators of α and β .

For a $\text{Gamma}(\alpha, \beta)$ distribution we have that:

$$E(X) = \frac{\alpha}{\beta} \quad \text{and} \quad \text{Var}(X) = \frac{\alpha}{\beta^2}.$$

Hence:

$$E(X^2) = \text{Var}(X) + (E(X))^2 = \frac{\alpha}{\beta^2} + \left(\frac{\alpha}{\beta}\right)^2.$$

If we equate the first two sample moments with the first two population moments we get the following system:

$$\frac{\hat{\alpha}}{\hat{\beta}} = \bar{X} \quad \Rightarrow \quad \hat{\alpha} = \hat{\beta} \bar{X}$$

and:

$$\frac{\hat{\alpha}}{\hat{\beta}^2} + \left(\frac{\hat{\alpha}}{\hat{\beta}}\right)^2 = \frac{1}{n} \sum_{i=1}^n X_i^2$$

or else:

$$\frac{\bar{X}}{\hat{\beta}} + \bar{X}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 \quad \Rightarrow \quad \frac{\bar{X}}{\hat{\beta}} = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2.$$

Hence:

$$\hat{\beta} = \frac{\bar{X}}{\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2} = \frac{\bar{X}}{(n-1)S^2/n}$$

which also gives:

$$\hat{\alpha} = \frac{\bar{X}^2}{(n-1)S^2/n}.$$

3.7 Sample quantiles and order statistics

So far our focus has been on the (method of moments) estimation of the population mean μ as population variance σ^2 as measures of central tendency and dispersion, respectively. While the most widely-reported attributes of a probability distribution, there are alternatives.

The *median* is another common measure of central tendency, which is particularly useful for heavily skewed distributions (such as for income studies) whose means are

affected by extreme tail event values with non-negligible probabilities of occurrence.

The *interquartile range* (IQR) is another measure of dispersion. Both the median and IQR are calculated from the *quantiles* of a distribution. Quantile-based statistics are in general more *robust* than moment-based statistics, i.e. quantile-based statistics are less sensitive to extreme observations (recall that the sample mean is sensitive to outliers whereas the sample median is not).

Population quantile

Let X be a random variable with distribution function F_X . A **population quantile** of X , known as the α -quantile, is the smallest q_α such that:

$$F_X(q_\alpha) = \alpha.$$

Special quantiles are the following.

- The median of a distribution is $\text{median}(X) = q_{0.50}$.
- The interquartile range of a distribution is $\text{IQR}(X) = q_{0.75} - q_{0.25}$, i.e. the difference between the upper and lower quartiles such that $Q_3 = q_{0.75}$ and $Q_1 = q_{0.25}$.

In the context of statistical inference, we may view quantiles such as the median and interquartile range as population parameters to be estimated. In order to construct estimators of these *population* quantiles (and functions of them) we must define *sample* quantiles, which are easily constructed from *order statistics*, which we define first.

Order statistics

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample. The **order statistics** are the sample values arranged in ascending order. Hence $X_{(i)}$ is the i th smallest value in the sample, for $i = 1, 2, \dots, n$.

Order statistics are the random variables $X_{(1)}, X_{(2)}, \dots, X_{(n)}$ such that:

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}.$$

More precisely, since random variables are functions from the sample space, Ω , to the real line, $\mathbb{R}$, the order statistics are functions such that for every $\omega \in \Omega$ we have:

$$X_{(1)}(\omega) \leq X_{(2)}(\omega) \leq \dots \leq X_{(n)}(\omega).$$

Since the order statistics are random variables, each will have a probability distribution. Over repeated samples, we can then treat the observed order statistics $x_{(i)}$ as realisations of the corresponding order statistic $X_{(i)}$, for $i = 1, 2, \dots, n$.

Let $\mathbf{x}_j = \{x_1, x_2, \dots, x_n\}$ be the j th observed sample, for $j = 1, 2, \dots, m$, where m is the number of observed samples, each with the same sample size n . The observed order statistics are then:

$$x_{(1)}^j, x_{(2)}^j, \dots, x_{(n)}^j \quad \text{for } j = 1, 2, \dots, m.$$

Hence the minimum values across the m samples are:

$$x_{(1)}^1, x_{(1)}^2, \dots, x_{(1)}^m$$

which are realisations of the first order statistic $X_{(1)}$. Similarly, the maximum values across the m samples are:

$$x_{(n)}^1, x_{(n)}^2, \dots, x_{(n)}^m$$

which are realisations of the n th order statistic $X_{(n)}$.

It is important to realise that the distribution of $X_{(i)}$ is *not* the same as F_X . For example, since $X_{(1)}$ is the *smallest* member of a random sample, then:

$$E(X_{(1)}) < E(X)$$

for any non-degenerate distribution. Similarly, since $X_{(n)}$ is the *largest* member of a random sample, then:

$$E(X_{(n)}) > E(X).$$

We will derive the probability distributions of the order statistics shortly.

Since $X_{(i)}$ is the i th smallest value in a random sample, it seems a natural choice of estimator of the point which has i/n of the mass of the distribution *below* it. This motivates our definition of *sample* quantiles. We restrict our definition to:

$$\alpha \in \left(\frac{1}{2n}, 1 - \frac{1}{2n} \right)$$

because this produces helpful symmetry properties while also reflecting the fact that our ability to estimate the extremes of a distribution are limited by the sample size n .

Sample quantile

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from F_X . A **sample quantile**, known as the sample α -quantile, is denoted Q_α . We define Q_α for $\alpha = 0.50$ (i.e. the median) and for $\alpha \neq 0.50$.

The median is:

$$Q_{0.50} = \begin{cases} X_{((n+1)/2)} & \text{for odd } n \\ (X_{(n/2)} + X_{((n/2)+1)})/2 & \text{for even } m. \end{cases}$$

For $\alpha \neq 0.5$ we have:

$$Q_\alpha = \begin{cases} X_{\{n\alpha\}} & \text{for } 1/(2n) < \alpha < 0.50 \\ X_{n+1-\{n(1-\alpha)\}} & \text{for } 0.50 < \alpha < 1 - 1/(2n) \end{cases}$$

where $\{k\}$ denotes k rounded to the nearest integer.

So, as introduced in **ST104a Statistics 1**, if the location of the median falls between two order statistics, i.e. when n is even, we interpolate by taking the average of order statistics either side of the location of the median. Other sample quantiles are obtained by finding the closest order statistic.

Activity 3.11 Show that sample quantiles have the property that:

$$Q_\alpha = X_{(i)} \Rightarrow Q_{1-\alpha} = X_{(n+1-i)}$$

i.e. if Q_α is given by the i th smallest value, then $Q_{1-\alpha}$ is given by the i th largest value.

3.7.1 Sample minimum and sample maximum

The minimum and maximum values of a sample are now discussed.

Sample minimum and sample maximum

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample.

The **sample minimum** is:

$$X_{(1)} = \min_{1 \leq i \leq n} (X_1, X_2, \dots, X_n).$$

The **sample maximum** is:

$$X_{(n)} = \max_{1 \leq i \leq n} (X_1, X_2, \dots, X_n).$$

Since $X_{(1)}$ and $X_{(n)}$ are random variables, we now derive their respective probability distributions, beginning with their distribution functions.

Distribution function of sample minimum and sample maximum

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from F_X .

The **distribution function of the sample minimum** is:

$$F_{X_{(1)}}(x) = P(X_{(1)} \leq x) = 1 - (1 - F_X(x))^n = 1 - (\bar{F}_X(x))^n$$

where $\bar{F}_X = 1 - F_X(x)$ is the survival function.

The **distribution function of the sample maximum** is:

$$F_{X_{(n)}}(x) = P(X_{(n)} \leq x) = (F_X(x))^n.$$

These hold regardless of whether X is discrete or continuous.

Proof: Note that if $X_{(1)} > x$, then $X_i > x$ for all $i = 1, 2, \dots, n$. Hence:

$$\begin{aligned}
 F_{X_{(1)}}(x) &= P(X_{(1)} \leq x) = 1 - P(X_{(1)} > x) \\
 &= 1 - P(X_1 > x, X_2 > x, \dots, X_n > x) \\
 &= 1 - P(X_1 > x) P(X_2 > x) \cdots P(X_n > x) \quad (\text{by independence}) \\
 &= 1 - (P(X > x))^n \quad (\text{as identically distributed}) \\
 &= 1 - (1 - F_X(x))^n \\
 &= 1 - (\bar{F}_X(x))^n.
 \end{aligned}$$

By definition of the sample maximum, if $X_{(n)} \leq x$, then $X_i \leq x$ for all $i = 1, 2, \dots, n$. Hence:

$$\begin{aligned}
 F_{X_{(n)}}(x) &= P(X_{(n)} \leq x) = P(X_1 \leq x, X_2 \leq x, \dots, X_n \leq x) \\
 &= P(X_1 \leq x) P(X_2 \leq x) \cdots P(X_n \leq x) \quad (\text{by independence}) \\
 &= (P(X \leq x))^n \quad (\text{as identically distributed}) \\
 &= (F_X(x))^n.
 \end{aligned}$$

■

The derivations of the mass function and density function differ. We first consider the discrete case. If X is a discrete random variable then its support will be a countable set of real numbers, $\{a_1, a_2, \dots\}$. Without loss of generality, let us assume that the indexing is chosen such that $a_1 \leq a_2 \leq \dots$. The mass function is given by taking differences. For example, for the sample maximum we have:

$$\begin{aligned}
 p_{X_{(n)}}(x) &= F_{X_{(n)}}(a_k) - F_{X_{(n)}}(a_{k-1}) \\
 &= \begin{cases} (F_X(a_k))^n - (F_X(a_{k-1}))^n & \text{for } x = a_k \text{ for some } k \in \mathbb{N}^+ \\ 0 & \text{otherwise.} \end{cases}
 \end{aligned}$$

If X is a continuous random variable, then the density function is simply obtained by differentiating the distribution function. Hence for the sample minimum we have:

$$f_{X_{(1)}}(x) = \frac{d}{dx} F_{X_{(1)}}(x) = n f_X(x) (1 - F_X(x))^{n-1} = n f_X(x) (\bar{F}_X(x))^{n-1}$$

for $x \in \Omega$, and 0 otherwise, where $\bar{F}_X$ is the survival function.

For the sample maximum, we have:

$$f_{X_{(n)}}(x) = \frac{d}{dx} F_{X_{(n)}}(x) = n f_X(x) (F_X(x))^{n-1}$$

for $x \in \Omega$, and 0 otherwise. Note that the supports of $f_{X_{(1)}}(x)$ and $f_{X_{(n)}}(x)$ are the same as the support of X .

Example 3.12 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $\text{Uniform}[0, \theta]$, such that:

$$f_X(x) = \begin{cases} \theta^{-1} & \text{for } 0 \leq x \leq \theta \\ 0 & \text{otherwise.} \end{cases}$$

The associated distribution function is:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt = \int_0^x \frac{1}{\theta} dt = \left[\frac{t}{\theta} \right]_0^x = \frac{x}{\theta}$$

hence, in full:

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ x/\theta & \text{for } 0 \leq x \leq \theta \\ 1 & \text{for } x > \theta. \end{cases}$$

The distribution function of the sample maximum is:

$$F_{X_{(n)}}(x) = (F_X(x))^n = \begin{cases} 0 & \text{for } x < 0 \\ (x/\theta)^n & \text{for } 0 \leq x \leq \theta \\ 1 & \text{for } x > \theta. \end{cases}$$

Differentiating yields the density function (note the use of the chain rule):

$$f_{X_{(n)}}(x) = \frac{n}{\theta} \left(\frac{x}{\theta} \right)^{n-1}$$

for $0 \leq x \leq \theta$, and 0 otherwise.

Example 3.13 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a distribution with density function:

$$f_X(x; \theta) = \begin{cases} 2\theta^{-2}x & \text{for } 0 \leq x \leq \theta \\ 0 & \text{otherwise} \end{cases}$$

where $\theta > 0$ is an unknown parameter. Let $X_{(n)} = \max_{1 \leq i \leq n} X_i$. We find the density function of $X_{(n)}$.

For $x \in [0, \theta]$, we have:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt = \int_0^x 2\theta^{-2}t dt = \left[\frac{t^2}{\theta^2} \right]_0^x = \left(\frac{x}{\theta} \right)^2.$$

Hence:

$$\begin{aligned} f_{X_{(n)}}(x) &= n f_X(x) (F_X(x))^{n-1} = n 2\theta^{-2}x \left(\left(\frac{x}{\theta} \right)^2 \right)^{n-1} \\ &= \frac{2nx^{2n-1}}{\theta^{2n}}. \end{aligned}$$

for $0 \leq x \leq \theta$, and 0 otherwise.

Activity 3.12 Derive the density function of the minimum of a random sample, i.e. $X_{(1)}$, for each of the following distributions.

(a) Pareto distribution:

$$f_X(x) = \frac{a}{(1+x)^{a+1}}$$

for $0 < x < \infty$ and $a > 0$, and 0 otherwise.

(b) Weibull distribution:

$$f_X(x) = c\tau x^{\tau-1} \exp(-cx^\tau)$$

for $0 < x < \infty$, $\tau > 0$ and $c > 0$, and 0 otherwise.

(c) Extreme value distribution:

$$f_X(x) = \lambda e^x \exp(-\lambda e^x)$$

for $-\infty < x < \infty$ and $\lambda > 0$.

3.7.2 Distribution of the i th order statistic

We now consider the distribution of the i th order statistic, for $i = 1, 2, \dots, n$. Note so far we have only considered the special cases of $i = 1$ and $i = n$, representing the sample minimum and sample maximum, respectively.

Distribution function of the i th order statistic

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from F_X . The **distribution function of the i th order statistic** is:

$$F_{X_{(i)}}(x) = \sum_{j=i}^n \binom{n}{j} (F_X(x))^j (1 - F_X(x))^{n-j}$$

Proof: We consider the event that exactly j values in the sample are at most x . Since the X_i s are identically distributed, for each sample value we have $F_X(x) = P(X \leq x)$. As the X_i s are independent, the probability that exactly j are at most x is given by the binomial mass function:

$$\binom{n}{j} (F_X(x))^j (1 - F_X(x))^{n-j}.$$

For the event $X_{(i)} \leq x$, this occurs when at least i individual sample values are at most x . Hence:

$$F_{X_{(i)}}(x) = \sum_{j=i}^n \binom{n}{j} (F_X(x))^j (1 - F_X(x))^{n-j}$$

■

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The density function of $X_{(i)}$ is obtained by differentiating $F_{X_{(i)}}(x)$.

Density function of the i th order statistic

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from F_X . The **density function of the i th order statistic** is:

$$f_{X_{(i)}}(x) = \frac{n!}{(i-1)!(n-i)!} f_X(x) (F_X(x))^{i-1} (1 - F_X(x))^{n-i}.$$

The support of $f_{X_{(i)}}(x)$ is the same as the support of f_X .

A non-examinable proof can be found in Appendix A.

For the discrete case, the mass function $p_{X_{(i)}}(x)$ is derived through differencing. We omit details in this course.

Example 3.14 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $\text{Uniform}[0, \theta]$. From Example 3.12 we have $f_X(x)$ and $F_X(x)$, hence the density function of the i th order statistic is:

$$\begin{aligned} f_{X_{(i)}}(x) &= \frac{n!}{(i-1)!(n-i)!} f_X(x) (F_X(x))^{i-1} (1 - F_X(x))^{n-i} \\ &= \frac{n!}{(i-1)!(n-i)!} \frac{1}{\theta} \left(\frac{x}{\theta}\right)^{i-1} \left(1 - \frac{x}{\theta}\right)^{n-i} \end{aligned}$$

for $0 \leq x \leq \theta$, and 0 otherwise.

Since the order statistics are the sample values arranged in ascending order, they are *not* independent random variables. We now state the joint density function of $X_{(i)}$ and $X_{(j)}$, i.e. the i th and j th order statistics.

Joint density function of the i th and j th order statistics

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from F_X . The **joint density function of the i th and j th order statistics** is:

$$\begin{aligned} f_{X_{(i)}, X_{(j)}}(u, v) &= \frac{n!}{(i-1)!(j-i-1)!(n-j)!} f_X(u) f_X(v) (F_X(u))^{i-1} (F_X(v) - F_X(u))^{j-i-1} (1 - F_X(v))^{n-j}. \end{aligned}$$

The support is the region where $u < v$, $f_X(u) > 0$ and $f_X(v) > 0$.

A non-examinable proof can be found in Appendix A.

Example 3.15 Suppose that X_1, X_2 and X_3 are independent and identically distributed exponential random variables. Define:

$$Y_1 = \frac{X_1}{\sum_{i=1}^3 X_i}, \quad Y_2 = \frac{X_1 + X_2}{\sum_{i=1}^3 X_i} \quad \text{and} \quad Y_3 = \sum_{i=1}^3 X_i.$$

We show that Y_1 and Y_2 have the same joint distribution as the two order statistics for a random sample of size two from the continuous uniform distribution on $[0, 1]$.

Consider a random sample $\mathbf{X} = \{X_1, X_2\}$ from a continuous uniform distribution on $[0, 1]$, for which:

$$f_X(x) = \begin{cases} 1 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } 0 \leq x \leq 1 \\ 1 & \text{for } x > 1. \end{cases}$$

For $n = 2$, the joint density function of the first and second (and hence only) order statistics is (noting that $0! = 1$):

$$\begin{aligned} & f_{X_{(1)}, X_{(2)}}(x_1, x_2) \\ &= \frac{n!}{(i-1)!(j-i-1)!(n-j)!} f_X(x_1) f_X(x_2) (F_X(x_1))^{i-1} (F_X(x_2) - F_X(x_1))^{j-i-1} (1 - F_X(x_2))^{n-j} \\ &= \frac{2!}{(1-1)!(2-1-1)!(2-2)!} \times 1 \times 1 \times (x_1)^{1-1} (x_2 - x_1)^{2-1-1} (1 - x_2)^{2-2} \\ &= \frac{2!}{1 \times 1 \times 1} \times 1 \times 1 \times 1 \times 1 \times 1 \\ &= \begin{cases} 2 & \text{for } 0 \leq x_1 < x_2 \leq 1 \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

This is the joint density function which we want to establish for Y_1 and Y_2 defined above. (Note the rest of this example is an examinable technique with respect to **ST2133 Advanced statistics: distribution theory only**.)

Consider the inverse of the specified transformation:

$$x_1 = y_1 y_3, \quad x_2 = y_2 y_3 - y_1 y_3 \quad \text{and} \quad x_3 = y_3 - y_2 y_3.$$

The Jacobian of this transformation is given by the determinant:

$$\begin{vmatrix} y_3 & -y_3 & 0 \\ 0 & y_3 & -y_3 \\ y_1 & y_2 - y_1 & 1 - y_2 \end{vmatrix} = y_3^2.$$

The original random variables X_1, X_2 and X_3 are exponential so have positive support. Due to independence, the joint density function is:

$$f_{X_1, X_2, X_3}(x_1, x_2, x_3) = \begin{cases} \lambda^3 \exp(-\lambda(x_1 + x_2 + x_3)) & \text{for } x_1, x_2, x_3 \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

The change of variable formula then yields:

$$f_{Y_1, Y_2, Y_3}(y_1, y_2, y_3) = \begin{cases} \lambda^3 \exp(-\lambda y_3) |y_3^2| & \text{for } 0 \leq y_1 < y_2 \leq 1, y_3 > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Integrating out y_3 gives the desired result:

$$\begin{aligned} f_{Y_1, Y_2}(y_1, y_2) &= \int_{-\infty}^{\infty} f_{Y_1, Y_2, Y_3}(y_1, y_2, y_3) dy_3 = \lambda^2 \int_{-\infty}^{\infty} y_3^2 \lambda \exp(-\lambda y_3) dy_3 \\ &= \lambda^2 E(Y^2) \\ &= \lambda^2 \frac{2}{\lambda^2} \\ &= 2 \end{aligned}$$

for $0 \leq y_1 < y_2 \leq 1$, where $Y \sim \text{Exp}(\lambda)$.

3.8 Variance of unbiased estimators

Unbiasedness of estimators is a desirable property, other things equal, since an unbiased estimator correctly estimates the corresponding parameter *on average*, i.e. the expected estimation error is zero. As such, we may wish to *only* consider unbiased estimators in which case they are equal in terms of the expected value being equal to the parameter. Among this class of unbiased estimators we would wish to identify the one with the *smallest* variance such that we choose the **minimum variance unbiased estimator (MVUE)**.

We proceed by proving the *Cramér–Rao lower bound* which represents the minimum variance we can expect for an unbiased estimator. We assume a random sample is drawn from a distribution with a scalar parameter θ .

Cramér–Rao lower bound for any sample

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a sample from a distribution parameterised by θ . Let $T(\mathbf{X})$ be an unbiased estimator of a function of the parameter, denoted $g(\theta)$. Under regularity conditions permitting differentiation under an integral, for any $\theta \in \Theta$ it holds that:

$$\text{Var}(T(\mathbf{X})) \geq \frac{\left(\frac{d}{d\theta} g(\theta)\right)^2}{\mathcal{I}_{\mathbf{X}}(\theta)}$$

where $\mathcal{I}_{\mathbf{X}}(\theta)$ is the total Fisher information about θ in $\mathbf{X}$. The right-hand side is the **Cramér–Rao lower bound** on the variance of such an unbiased estimator.

Proof: We consider the continuous case. We make use of the variance/covariance form of the Cauchy–Schwarz inequality which states that for two random variables X and Y we have:

$$(\text{Cov}(X, Y))^2 \leq \text{Var}(X) \text{Var}(Y).$$

By unbiasedness of $T(\mathbf{X})$, we have:

$$E(T(\mathbf{X})) = g(\theta)$$

hence:

$$g(\theta) = \int_{\mathbb{R}^n} T(\mathbf{x}) f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x}.$$

Differentiating both sides with respect to θ gives us:

$$\frac{d}{d\theta} g(\theta) = \int_{\mathbb{R}^n} T(\mathbf{x}) \frac{\partial}{\partial \theta} f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x} = \int_{\mathbb{R}^n} T(\mathbf{x}) s_{\mathbf{X}}(\theta; \mathbf{x}) f_{\mathbf{X}}(\mathbf{x}; \theta) d\mathbf{x} = E(T(\mathbf{X}) s_{\mathbf{X}}(\theta; \mathbf{X})).$$

Since $E(s_{\mathbf{X}}(\theta; \mathbf{X})) = 0$, we have:

$$\begin{aligned} \text{Cov}(T(\mathbf{X}), s_{\mathbf{X}}(\theta; \mathbf{X})) &= E(T(\mathbf{X}) s_{\mathbf{X}}(\theta; \mathbf{X})) - E(T(\mathbf{X})) E(s_{\mathbf{X}}(\theta; \mathbf{X})) \\ &= E(T(\mathbf{X}) s_{\mathbf{X}}(\theta; \mathbf{X})) \end{aligned}$$

hence we may write:

$$\frac{d}{d\theta} g(\theta) = \text{Cov}(T(\mathbf{X}), s_{\mathbf{X}}(\theta; \mathbf{X})).$$

Applying the Cauchy–Schwarz inequality then yields:

$$\left(\frac{d}{d\theta} g(\theta) \right)^2 \leq \text{Var}(T(\mathbf{X})) \text{Var}(s_{\mathbf{X}}(\theta; \mathbf{X})).$$

Rearranging, we have:

$$\text{Var}(T(\mathbf{X})) \geq \frac{\left(\frac{d}{d\theta} g(\theta) \right)^2}{\text{Var}(s_{\mathbf{X}}(\theta; \mathbf{X}))} = \frac{\left(\frac{d}{d\theta} g(\theta) \right)^2}{\mathcal{I}_{\mathbf{X}}(\theta)}$$

since $\mathcal{I}_{\mathbf{X}}(\theta) = \text{Var}(s_{\mathbf{X}}(\theta; \mathbf{X}))$. ■

Other than some requirements for regularity of the joint distribution, the Cramér–Rao lower bound does not make any assumptions about the nature of the sample. If $\mathbf{X}$ is a *random* sample, then recall that:

$$\mathcal{I}_{\mathbf{X}}(\theta) = n\mathcal{I}_X(\theta)$$

where $\mathcal{I}_{\mathbf{X}}(\theta)$ is the total Fisher information, and $\mathcal{I}_X(\theta)$ is the Fisher information of a single observation. We now have the following corollary.

Cramér–Rao lower bound for a random sample

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a distribution parameterised by θ . Let $T(\mathbf{X})$ be an unbiased estimator of a function of the parameter, denoted $g(\theta)$. Under regularity conditions permitting differentiation under an integral, for any $\theta \in \Theta$ it holds that:

$$\text{Var}(T(\mathbf{X})) \geq \frac{\left(\frac{d}{d\theta} g(\theta) \right)^2}{n\mathcal{I}_X(\theta)}$$

where $\mathcal{I}_X(\theta)$ is the Fisher information about θ in a single observation.

If we are interested in estimating θ , rather than some function of the parameter $g(\theta)$, then the Cramér–Rao lower bound simplifies further. If $\hat{\theta}$ is an unbiased estimator of θ based on a random sample $\mathbf{X}$, then the Cramér–Rao lower bound on the estimator's variance is:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{n\mathcal{I}_X(\theta)}.$$

Example 3.16 We find the Cramér–Rao lower bound for the variance of an unbiased estimator of β for a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ from the Gamma distribution with density function:

$$f_X(x) = \frac{1}{\Gamma(\alpha)} \beta^\alpha x^{\alpha-1} e^{-\beta x}$$

for $x \geq 0$, α is known, where $\beta > 0$, and 0 otherwise.

In order to calculate the Cramér–Rao lower bound we just need information which is based on the log-likelihood function. For a single observation, the log-likelihood function is:

$$l_X(\beta; x) = -\log \Gamma(\alpha) + \alpha \log \beta + (\alpha - 1) \log x - \beta x$$

and the score function is:

$$s_X(\beta; x) = \frac{\partial}{\partial \beta} l_X(\beta; x) = \frac{\alpha}{\beta} - x.$$

Hence Fisher information is:

$$\mathcal{I}_X(\beta) = \text{Var}(s_X(\beta; X)) = \frac{\alpha}{\beta^2}.$$

The Cramér–Rao lower bound is given by:

$$\frac{1}{n\mathcal{I}_X(\beta)} = \frac{\beta^2}{n\alpha}.$$

Example 3.17 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $\text{Pois}(\lambda)$. The likelihood function is:

$$L_{\mathbf{X}}(\lambda; \mathbf{x}) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}$$

hence the log-likelihood function is:

$$l_{\mathbf{X}}(\lambda; \mathbf{x}) = -n\lambda + \sum_{i=1}^n x_i \log \lambda - \sum_{i=1}^n \log x_i!.$$

The score function is:

$$s_{\mathbf{X}}(\lambda; \mathbf{x}) = -n + \frac{1}{\lambda} \sum_{i=1}^n x_i = -n + \frac{n\bar{x}}{\lambda}$$

and:

$$\frac{\partial}{\partial \lambda} s_{\mathbf{X}}(\lambda; \mathbf{x}) = -\frac{n\bar{x}}{\lambda^2}.$$

Hence total Fisher information is:

$$\mathcal{I}_{\mathbf{X}}(\lambda) = -E \left(\frac{\partial}{\partial \lambda} s_{\mathbf{X}}(\lambda; \mathbf{X}) \right) = -E \left(-\frac{n\bar{X}}{\lambda^2} \right) = \frac{n}{\lambda}$$

since $E(\bar{X}) = \mu = \lambda$.

Consider the estimators $\hat{\lambda}_1 = \bar{X}$ and $\hat{\lambda}_2 = S^2$. They are both unbiased estimators of λ since both the mean and the variance of a Poisson distribution are equal to λ , i.e. $\mu = \sigma^2 = \lambda$. Hence:

$$E(\bar{X}) = E(X) = \mu = \lambda \quad \text{and} \quad E(S^2) = \text{Var}(X) = \sigma^2 = \lambda.$$

Since the Cramér–Rao lower bound is:

$$\frac{1}{\mathcal{I}_{\mathbf{X}}(\lambda)} = \frac{\lambda}{n}$$

and since:

$$\text{Var}(\bar{X}) = \frac{\text{Var}(X)}{n} = \frac{\lambda}{n}$$

then $\hat{\lambda}_1$ is unbiased and attains the Cramér–Rao lower bound for λ , it is also a minimum variance unbiased estimator of λ so we cannot obtain an unbiased estimator with a smaller variance. Therefore, we would choose $\bar{X}$ to be our estimator of λ .

Note that the Cramér–Rao lower bound establishes a floor (i.e. a minimum value) for the variance of an unbiased estimator of $g(\theta)$. There is no guarantee that the minimum variance unbiased estimator actually *attains* this floor. If the variance of the estimator equals the Cramér–Rao lower bound then it *must* be a minimum variance unbiased estimator, but there can be instances when the variance does not actually attain the Cramér–Rao lower bound.

We require an extra condition to ensure that the Cramér–Rao lower bound is actually attained, presented next.

Attaining the Cramér–Rao lower bound

If $T(\mathbf{X})$ is an unbiased estimator of $g(\theta)$, then $T(\mathbf{X})$ **attains the Cramér–Rao lower bound** if and only if:

$$s_{\mathbf{X}}(\theta; \mathbf{x}) = b(\theta)(T(\mathbf{x}) - g(\theta))$$

where $b(\theta)$ is a function involving the parameter θ , but not the observed sample $\mathbf{x}$.

Proof: We again make use of the Cauchy–Schwarz inequality:

$$(\text{Cov}(X, Y))^2 \leq \text{Var}(X) \text{Var}(Y)$$

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which achieves *equality* if and only if X and Y are linearly related, i.e. if:

$$Y = \alpha + \beta X \quad \text{or} \quad Y = \beta \left(\frac{\alpha}{\beta} + X \right).$$

We use the fact that:

$$\text{Cov}(T(\mathbf{X}), s_{\mathbf{X}}(\theta; \mathbf{X})) \leq \text{Var}(T(\mathbf{X})) \text{Var}(s_{\mathbf{X}}(\theta; \mathbf{X})).$$

Hence for equality we require:

$$s_{\mathbf{X}}(\theta; \mathbf{X}) = \alpha + \beta T(\mathbf{X}).$$

Since we know that $E(s_{\mathbf{X}}(\theta; \mathbf{X})) = 0$ and $E(T(\mathbf{X})) = g(\theta)$, we have that:

$$\alpha + \beta g(\theta) = 0 \quad \Rightarrow \quad \frac{\alpha}{\beta} = -g(\theta).$$

Setting $\beta = b(\theta)$ to denote the possibility that the constant depends on θ gives the result. ■

Example 3.18 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, with σ^2 known. In Example 2.29 we showed that the total Fisher information about μ contained in $\mathbf{X}$ is:

$$\mathcal{I}_{\mathbf{X}}(\mu) = \frac{n}{\sigma^2}.$$

Since we are estimating μ (rather than a function of μ) using a random sample, if we use $T(\mathbf{X})$ as an estimator of μ such that it is unbiased (i.e. assume $E(T(\mathbf{X})) = \mu$), then according to the Cramér–Rao lower bound:

$$\text{Var}(T(\mathbf{X})) \geq \frac{\sigma^2}{n}.$$

We know this bound is attained if we set $T(\mathbf{X}) = \bar{X}$, since $\text{Var}(\bar{X}) = \sigma^2/n$, and $E(\bar{X}) = \mu$ hence it is an unbiased estimator. Formally, the score function is:

$$s_{\mathbf{X}}(\mu; \mathbf{x}) = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = \frac{n}{\sigma^2} (\bar{x} - \mu)$$

which clearly meets the factorisation criterion:

$$s_{\mathbf{X}}(\mu; \mathbf{x}) = b(\mu)(T(\mathbf{x}) - g(\mu))$$

where:

$$b(\mu) = \frac{n}{\sigma^2}, \quad T(\mathbf{x}) = \bar{x} \quad \text{and} \quad g(\mu) = \mu$$

hence $T(\mathbf{X}) = \bar{X}$ attains the Cramér–Rao lower bound. Note that $b(\mu)$ does not explicitly depend on μ , but it does not have to.

Example 3.19 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $\text{Pois}(\lambda)$. If we use $T(\mathbf{X}) = \bar{X}$ as our estimator of λ , then it is unbiased since:

$$E(\bar{X}) = \lambda.$$

From Example 3.17, the score function is:

$$s_{\mathbf{X}}(\lambda; \mathbf{x}) = -n + \frac{1}{\lambda} \sum_{i=1}^n x_i = -n + \frac{n\bar{x}}{\lambda} = \frac{n}{\lambda} (\bar{x} - \lambda)$$

which clearly meets the factorisation criterion:

$$s_{\mathbf{X}}(\lambda; \mathbf{x}) = b(\lambda)(T(\mathbf{x}) - g(\lambda)).$$

where:

$$b(\lambda) = \frac{n}{\lambda}, \quad T(\mathbf{x}) = \bar{x} \quad \text{and} \quad g(\lambda) = \lambda$$

hence $T(\mathbf{X}) = \bar{X}$ attains the Cramér–Rao lower bound.

Activity 3.13 Find the Cramér–Rao lower bound for the variance of an unbiased estimator of the square of the probability of success of a binomial distribution. Is this bound attainable?

For completeness we now consider the Cramér–Rao lower bound when the population distribution is parameterised by a vector of parameters, $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_r)^T$. The variance of the estimator $\mathbf{T}(\mathbf{X})$ and Fisher information matrix $\mathcal{I}_{\mathbf{X}}(\boldsymbol{\theta})$ are now $r \times r$ matrices.

Cramér–Rao lower bound for the vector parameter case

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a distribution parameterised by $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_r)^T$. If $\mathbf{T}(\mathbf{X})$ is an unbiased estimator of $\boldsymbol{\theta}$, then:

$$\text{Var}(\mathbf{T}(\mathbf{X})) - (\mathcal{I}_{\mathbf{X}}(\boldsymbol{\theta}))^{-1}$$

is a non-negative definite matrix.

3.9 Maximum likelihood estimation

We have previously discussed that the likelihood is a measure of the plausibility of various parameter values given a sample. In *maximum likelihood estimation* (introduced in **ST104b Statistics 2**) we take as the point estimator the value which maximises the likelihood function. We begin with the scalar parameter case.

Maximum likelihood estimate

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample parameterised by θ , and let $L_{\mathbf{X}}$ be the likelihood function. For an observed sample $\mathbf{x}$, the **maximum likelihood estimate** of θ is the value $\hat{\theta}(\mathbf{x})$ which maximises $L_{\mathbf{X}}(\theta; \mathbf{x})$ as a function of θ .

We now present some important comments on maximum likelihood estimation.

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1. We may use the notation $\hat{\theta}(\mathbf{X})$ to denote a maximum likelihood *estimator*, and $\hat{\theta}(\mathbf{x})$ to denote a maximum likelihood *estimate*. Provided the context is clear, simply $\hat{\theta}$ may be used to denote a maximum likelihood estimator or estimate, as appropriate.
2. Denoting the parameter space of θ by Θ , such that $\theta \in \Theta$, the maximum likelihood estimate of θ given an observed sample $\mathbf{x}$ is the value $\hat{\theta}$ such that:

$$L_{\mathbf{X}}(\hat{\theta}; \mathbf{x}) = \sup_{\theta \in \Theta} L_{\mathbf{X}}(\theta; \mathbf{x}).$$

3. By definition, the *maximum* likelihood estimate means there is no other possible value of θ which has a larger likelihood given an observed sample $\mathbf{x}$. Hence:

$$L_{\mathbf{X}}(\theta; \mathbf{x}) \leq L_{\mathbf{X}}(\hat{\theta}; \mathbf{x})$$

for all $\theta \in \Theta$.

4. Since taking logs is a monotone transformation, maximising the log-likelihood function, $l_{\mathbf{X}}(\theta; \mathbf{x})$, produces the same maximum. Therefore, if $\hat{\theta}$ is the maximum likelihood estimate of θ given an observed sample $\mathbf{x}$, then:

$$l_{\mathbf{X}}(\hat{\theta}; \mathbf{x}) = \sup_{\theta \in \Theta} l_{\mathbf{X}}(\theta; \mathbf{x}).$$

5. Provided the likelihood function is differentiable (which will be the case for distributions for which the support does *not* depend on θ), we can obtain the maximum likelihood estimate of θ given an observed sample $\mathbf{x}$ by solving:

$$\left. \frac{\partial}{\partial \theta} L_{\mathbf{X}}(\theta; \mathbf{x}) \right|_{\theta=\hat{\theta}} = 0$$

for $\hat{\theta}$. Equivalently, we could maximise the log-likelihood function, $l_{\mathbf{X}}(\theta; \mathbf{x})$. By exploiting the definition of the score function as the derivative of $l_{\mathbf{X}}(\theta; \mathbf{x})$, in regular cases the maximum likelihood estimate satisfies:

$$s_{\mathbf{X}}(\hat{\theta}; \mathbf{x}) = 0.$$

Example 3.20 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample drawn from an exponential distribution with parameter λ . We derive the maximum likelihood estimator of λ .

From Example 2.28, the score function is:

$$s_{\mathbf{X}}(\lambda; \mathbf{x}) = \frac{n}{\lambda} - \sum_{i=1}^n x_i.$$

Setting $s_{\mathbf{X}}(\hat{\lambda}; \mathbf{x}) = 0$ gives:

$$\frac{n}{\hat{\lambda}} - \sum_{i=1}^n x_i = 0 \quad \Rightarrow \quad \hat{\lambda} = \frac{1}{\bar{X}}.$$

The second derivative evaluated at $\bar{x}$ is:

$$-\frac{n}{\hat{\lambda}^2} = -\frac{n}{\bar{x}^2} < 0$$

indicating that this point corresponds to a maximum. We conclude that the maximum likelihood estimator is $\hat{\lambda} = \bar{X}^{-1}$.

Example 3.21 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample drawn from a Poisson distribution with parameter λ . We derive the maximum likelihood estimator of λ .

From Example 2.30, the score function is:

$$s_{\mathbf{X}}(\lambda; \mathbf{x}) = -n + \frac{1}{\lambda} \sum_{i=1}^n x_i.$$

Setting $s_{\mathbf{X}}(\hat{\lambda}; \mathbf{x}) = 0$ gives:

$$-n + \frac{1}{\hat{\lambda}} \sum_{i=1}^n x_i = 0 \quad \Rightarrow \quad \hat{\lambda} = \bar{X}.$$

The second derivative evaluated at $\bar{x}$ is:

$$-\frac{1}{\hat{\lambda}^2} \sum_{i=1}^n x_i = -\frac{1}{\bar{x}^2} \sum_{i=1}^n x_i < 0$$

indicating that this point corresponds to a maximum. We conclude that the maximum likelihood estimator is $\hat{\lambda} = \bar{X}$.

Example 3.22 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample drawn from a geometric distribution with parameter π . We derive the maximum likelihood estimator of π .

From Example 2.30, the score function is:

$$s_{\mathbf{X}}(\pi; \mathbf{x}) = -\frac{\sum_{i=1}^n x_i - n}{1 - \pi} + \frac{n}{\pi} = \frac{n - \sum_{i=1}^n x_i}{1 - \pi} + \frac{n}{\pi}.$$

Solving $s_{\mathbf{X}}(\hat{\pi}; \mathbf{x}) = 0$ gives us a candidate for the maximum likelihood estimator:

$$\frac{n - \sum_{i=1}^n x_i}{1 - \hat{\pi}} + \frac{n}{\hat{\pi}} = 0 \quad \Rightarrow \quad n - n\hat{\pi} = -n\hat{\pi} + \hat{\pi} \sum_{i=1}^n x_i$$

or:

$$\hat{\pi} = \frac{n}{\sum_{i=1}^n X_i} = \bar{X}^{-1}.$$

The second derivative of the log-likelihood function (derivative of the score function) is equal to:

$$\frac{n - \sum_{i=1}^n x_i}{(1 - \pi)^2} - \frac{n}{\pi^2}$$

which is always negative since $n < \sum_{i=1}^n x_i$, indicating that this point corresponds to a maximum. We conclude that the maximum likelihood estimator is:

$$\hat{\pi} = \frac{n}{\sum_{i=1}^n X_i} = \bar{X}^{-1}.$$

Example 3.23 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample drawn from a $\text{Bin}(m, \pi)$ distribution, with m known. We derive the maximum likelihood estimator of π .

From Example 2.30, the score function is:

$$s_{\mathbf{X}}(\pi; \mathbf{x}) = \frac{\sum_{i=1}^n x_i}{\pi} - \frac{nm - \sum_{i=1}^n x_i}{1 - \pi}.$$

Setting $s_{\mathbf{X}}(\hat{\pi}; \mathbf{x}) = 0$ gives:

$$\frac{nm - \sum_{i=1}^n x_i}{1 - \hat{\pi}} = \frac{\sum_{i=1}^n x_i}{\hat{\pi}} = 0 \quad \Rightarrow \quad nm\hat{\pi} - \hat{\pi} \sum_{i=1}^n x_i = \sum_{i=1}^n x_i - \hat{\pi} \sum_{i=1}^n x_i$$

or:

$$\hat{\pi} = \frac{\sum_{i=1}^n X_i}{nm}.$$

The second derivative of the log-likelihood function (derivative of the score function) is equal to:

$$-\frac{\sum_{i=1}^n x_i}{\pi^2} - \frac{nm - \sum_{i=1}^n x_i}{(1 - \pi)^2}$$

which is always negative since $nm > \sum_{i=1}^n x_i$, indicating that this point corresponds to a maximum. We conclude that the maximum likelihood estimator is:

$$\hat{\pi} = \frac{\sum_{i=1}^n X_i}{nm}.$$

Example 3.24 Consider the maximum likelihood estimator of π for the binomial distribution in Example 3.23. We determine whether it is a minimum variance unbiased estimator.

We know that the maximum likelihood estimator is:

$$\hat{\pi} = \frac{\sum_{i=1}^n X_i}{nm}.$$

We can verify that $\hat{\pi}$ is an unbiased estimator of π . Note if $X_i \sim \text{Bin}(m, \pi)$, then $E(X_i) = m\pi$, for $i = 1, 2, \dots, n$, hence:

$$E(\hat{\pi}) = E\left(\frac{\sum_{i=1}^n X_i}{nm}\right) = \frac{\sum_{i=1}^n E(X_i)}{nm} = \frac{\sum_{i=1}^n m\pi}{nm} = \frac{nm\pi}{nm} = \pi.$$

Note if $X_i \sim \text{Bin}(m, \pi)$, then $\text{Var}(X_i) = m\pi(1 - \pi)$, for $i = 1, 2, \dots, n$, hence the variance of the estimator is, due to independence:

$$\text{Var}(\hat{\pi}) = \text{Var}\left(\frac{\sum_{i=1}^n X_i}{nm}\right) = \frac{\sum_{i=1}^n \text{Var}(X_i)}{n^2 m^2} = \frac{\sum_{i=1}^n m\pi(1 - \pi)}{n^2 m^2} = \frac{nm\pi(1 - \pi)}{n^2 m^2} = \frac{\pi(1 - \pi)}{nm}$$

which is equal to the inverse of the (total) Fisher information (see Example 2.30), which is the Cramér–Rao lower bound. Hence $\hat{\pi}$ is a minimum variance unbiased estimator.

Example 3.25 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, 1)$. We find the maximum likelihood estimator of μ .

The likelihood function is:

$$L_{\mathbf{X}}(\mu; \mathbf{x}) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_i - \mu)^2}{2}\right) = \left(\frac{1}{\sqrt{2\pi}}\right)^n \exp\left(-\frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2\right).$$

The log-likelihood function is equal to:

$$l_{\mathbf{X}}(\mu; \mathbf{x}) = -\frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2 + c = -\frac{1}{2} \sum_{i=1}^n x_i^2 + \mu \sum_{i=1}^n x_i - \frac{1}{2} n\mu^2 + c$$

where $c = n \log(1/\sqrt{2\pi})$ is an arbitrary constant which does not depend on μ .

Setting the score function equal to 0 yields a candidate for the global maximum:

$$\frac{\partial}{\partial \mu} l_{\mathbf{X}}(\hat{\mu}; \mathbf{x}) = 0 \quad \Rightarrow \quad \sum_{i=1}^n x_i - n\hat{\mu} = 0 \quad \Rightarrow \quad \hat{\mu} = \bar{X}.$$

We could check whether it corresponds to a maximum (and not a minimum) if the second derivative of the log-likelihood is negative:

$$\frac{\partial^2}{\partial \mu^2} l_{\mathbf{X}}(\hat{\mu}; \mathbf{x}) = -n < 0.$$

Therefore, the maximum likelihood estimator of μ is $\hat{\mu} = \bar{X}$.

Activity 3.14 Find the maximum likelihood estimate of θ for an observed sample $\mathbf{x}$ of size n from the probability distribution:

$$f_X(x; \theta) = \begin{cases} 2x/\theta^2 & \text{for } 0 \leq x \leq \theta \\ 0 & \text{otherwise.} \end{cases}$$

Vector parameter case

Maximum likelihood estimation can, of course, be applied to the vector parameter case. Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample parameterised by $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_r)^T$

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such that $\theta \in \Theta$. The maximum likelihood estimates $\hat{\theta}$ of the elements of θ are the values which satisfy:

$$L_{\mathbf{X}}(\hat{\theta}; \mathbf{x}) = \sup_{\theta \in \Theta} L_{\mathbf{X}}(\theta; \mathbf{x})$$

or equivalently:

$$l_{\mathbf{X}}(\hat{\theta}; \mathbf{x}) = \sup_{\theta \in \Theta} l_{\mathbf{X}}(\theta; \mathbf{x}).$$

In regular cases the maximum likelihood estimates satisfy:

$$s_{\mathbf{X}}(\hat{\theta}; \mathbf{x}) = \mathbf{0}.$$

Example 3.26 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, with μ and σ^2 unknown parameters. Suppose $\mathbf{x}$ is an observed sample. From Example 2.19, the joint density function is:

$$f_{\mathbf{X}}(\mathbf{x}; \mu, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp \left(-\frac{1}{2\sigma^2} (n(\bar{x} - \mu)^2) \right) \exp \left(-\frac{1}{2\sigma^2} ((n-1)s^2) \right)$$

hence the log-likelihood function is:

$$l_{\mathbf{X}}(\mu, \sigma^2; \mathbf{x}) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{n}{2\sigma^2} (\bar{x} - \mu)^2 - \frac{n-1}{2\sigma^2} s^2$$

where, using standard notation, the observed sample mean and observed sample variance are, respectively:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{and} \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

We equate the partial derivatives of $l_{\mathbf{X}}(\mu, \sigma^2; \mathbf{x})$ with respect to zero to find the maximum likelihood estimators. For μ , we have:

$$\frac{\partial}{\partial \mu} l_{\mathbf{X}}(\mu, \sigma^2; \mathbf{x}) = \frac{n}{\sigma^2} (\bar{x} - \mu)$$

hence:

$$\frac{\partial}{\partial \mu} l_{\mathbf{X}}(\mu, \sigma^2; \mathbf{x}) \Big|_{\mu=\hat{\mu}} = 0 \quad \Rightarrow \quad \hat{\mu} = \bar{X}.$$

This is clearly a maximum since as $L_{\mathbf{X}}(\mu, \sigma^2; \mathbf{x})$ is a quadratic in μ and the coefficient of μ^2 is negative. Substituting into $l_{\mathbf{X}}(\mu, \sigma^2; \mathbf{x})$ produces the *profile log-likelihood* function:

$$l_{\mathbf{X}}(\hat{\mu}, \sigma^2; \mathbf{x}) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{n-1}{2\sigma^2} s^2.$$

Differentiating with respect to σ^2 (note σ^2 , not σ) gives us:

$$\frac{\partial}{\partial \sigma^2} l_{\mathbf{X}}(\hat{\mu}, \sigma^2; \mathbf{x}) = -\frac{n}{2\sigma^2} + \frac{n-1}{2(\sigma^2)^2} s^2$$

hence:

$$\frac{\partial}{\partial \sigma^2} l_{\mathbf{X}}(\hat{\mu}, \sigma^2; \mathbf{x}) \Big|_{\sigma^2=\hat{\sigma}^2} = 0 \quad \Rightarrow \quad \hat{\sigma}^2 = \frac{n-1}{n} S^2.$$

Therefore, the maximum likelihood estimators of μ and σ^2 when sampling from a normal distribution are:

$$\hat{\mu} = \bar{X} \quad \text{and} \quad \hat{\sigma}^2 = \frac{n-1}{n} S^2.$$

Note these are the same as the method of moments estimators, and we have already examined the statistical properties of these estimators (see Example 3.9).

3

3.9.1 Properties of maximum likelihood estimators

We consider the statistical properties of maximum likelihood estimators of a scalar parameter θ , starting with *consistency*.

Consistency of maximum likelihood estimators

Under weak conditions, the maximum likelihood estimator $\hat{\theta}$ is a consistent estimator of θ .

Consistency, while desirable, does not tell us anything about the sampling distribution of maximum likelihood estimators. However, we may use the central limit theorem and the law of large numbers to establish that maximum likelihood estimators are *asymptotically normal*.

Asymptotic normality of maximum likelihood estimators

Under weak conditions, a maximum likelihood estimator $\hat{\theta}$ satisfies:

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \mathcal{I}_{\mathbf{X}}^{-1}(\theta)).$$

A non-examinable partial proof can be found in Appendix A.

A corollary of this result is that for large sample sizes we may approximate the sampling distribution of maximum likelihood estimators with the normal distribution.

Large sample sampling distributions of maximum likelihood estimators

For large sample sizes, the sampling distribution of a maximum likelihood estimator $\hat{\theta}$ is approximately normally distributed. We have:

$$\hat{\theta} \sim N(\theta, \mathcal{I}_{\mathbf{X}}^{-1}(\theta))$$

approximately, for large n .

3.9.2 Invariance of maximum likelihood estimators

Instead of θ , we may have interest in some function of θ , say $g(\theta)$. Given an estimator $\hat{\theta}$ of θ , a **plug-in estimator** would mean estimating $g(\theta)$ with $g(\hat{\theta})$.

In general, statistical properties of plug-in estimators differ from those of the original estimator, as the following example demonstrates.

Example 3.27 The sample variance is:

$$S^2 = \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right).$$

This means that $(n-1)S^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2$, hence:

$$E((n-1)S^2) = (n-1)E(S^2) = E\left(\sum_{i=1}^n X_i^2 - n\bar{X}^2\right) = nE(X_i^2) - nE(\bar{X}^2).$$

For a random sample, the X_i s are identically distributed, hence $E(X_i^2) = E(X^2)$ for $i = 1, 2, \dots, n$. Since:

$$\text{Var}(X) = \sigma^2 = E(X^2) - \mu^2$$

then:

$$E(X^2) = \sigma^2 + \mu^2 \quad \text{and} \quad E(\bar{X}^2) = \frac{\sigma^2}{n} + \mu^2.$$

Therefore:

$$(n-1)E(S^2) = n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right) = (n-1)\sigma^2$$

so $E(S^2) = \sigma^2$, which means that S^2 is an unbiased estimator of σ^2 .

The standard formula for $\text{Var}(X)$, applied to S , states that:

$$E(S^2) = \text{Var}(S) + (E(S))^2$$

which means that:

$$E(S) = \sqrt{E(S^2) - \text{Var}(S)} = \sqrt{\sigma^2 - \text{Var}(S)} < \sigma = \sqrt{\sigma^2}$$

since all variances are strictly positive. It follows that S is a biased estimator of σ (with its average value lower than the true value σ) even though S^2 is an unbiased estimator of σ^2 . Therefore, unbiasedness of estimators is *not* an invariant property.

Activity 3.15 A random sample of n Bernoulli trials with success probability π results in R successes. Derive an unbiased estimator of $\pi(1-\pi)$.

A useful property of maximum likelihood estimators is that plug-in estimators are also maximum likelihood estimators, i.e. if $\hat{\theta}$ is the maximum likelihood estimator of θ , then $g(\hat{\theta})$ is the maximum likelihood estimator of $g(\theta)$.

If the function g is one-to-one this is straightforward to show. Let $\phi = g(\theta)$ for some one-to-one function g . We can reparameterise using the fact that $\theta = g^{-1}(\phi)$. The likelihood function can then be expressed in terms of ϕ such that:

$$L_{\mathbf{X}}(\theta; \mathbf{x}) = L_{\mathbf{X}}(g^{-1}(\phi); \mathbf{x}).$$

This attains its maximum at $\hat{\theta}$, hence $g^{-1}(\hat{\phi}) = \hat{\theta}$, which implies that $\hat{\phi} = g(\hat{\theta})$.

However, in general we are interested in functions for g which are not one-to-one. In this case, we need to clearly define what is meant by the maximum likelihood estimator of $\phi = g(\theta)$ which leads us to the *induced likelihood function*.

Induced likelihood function

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample parameterised by θ . Suppose $\phi = g(\theta)$ where $g : \Theta \rightarrow \Phi$. The **induced likelihood function** of ϕ given an observed sample $\mathbf{x}$ is:

$$L'_{\mathbf{X}}(\phi; \mathbf{x}) = \sup_{\{\theta: g(\theta)=\phi\}} L_{\mathbf{X}}(\theta; \mathbf{x})$$

for all $\phi \in \Phi$.

Note that here Θ is the parameter space of θ , and Φ is the domain of the function g . It is then natural to define the maximum likelihood estimator $\hat{\phi}$ as the value which maximises $L'_{\mathbf{X}}(\phi; \mathbf{x})$ as a function of ϕ . Using this framework we can now prove the general *invariance property* of maximum likelihood estimators.

Invariance of maximum likelihood estimators

If $\hat{\theta}$ is the maximum likelihood estimator of θ then, for any function $g(\theta)$, the maximum likelihood estimator of $g(\theta)$ is $g(\hat{\theta})$.

Proof: Given the definition of the induced likelihood function, we just need to show that:

$$L'_{\mathbf{X}}(g(\hat{\theta}); \mathbf{x}) = L'_{\mathbf{X}}(\hat{\phi}; \mathbf{x}).$$

It is clear from the definition that the induced likelihood function has the same maximum as the likelihood function. Hence:

$$\begin{aligned} L'_{\mathbf{X}}(\hat{\phi}; \mathbf{x}) &= \sup_{\phi \in \Phi} L'_{\mathbf{X}}(\phi; \mathbf{x}) && \text{(by definition of } \hat{\phi}) \\ &= \sup_{\phi \in \Phi} \sup_{\{\theta: g(\theta)=\phi\}} L_{\mathbf{X}}(\theta; \mathbf{x}) && \text{(by definition of induced likelihood)} \\ &= \sup_{\theta \in \Theta} L_{\mathbf{X}}(\theta; \mathbf{x}) \\ &= L_{\mathbf{X}}(\hat{\theta}; \mathbf{x}). \end{aligned}$$

If we restrict our attention to the set of parameters $\{\theta : g(\theta) = g(\hat{\theta})\}$, then $\hat{\theta}$ is in this set, hence:

$$L_{\mathbf{X}}(\hat{\theta}; \mathbf{x}) = \sup_{\theta \in \Theta} L_{\mathbf{X}}(\theta; \mathbf{x}) = \sup_{\{\theta: g(\theta)=g(\hat{\theta})\}} L_{\mathbf{X}}(\theta; \mathbf{x}) = L'_{\mathbf{X}}(g(\hat{\theta}); \mathbf{x})$$

by definition of the induced likelihood function. We conclude that:

$$L'_{\mathbf{X}}(g(\hat{\theta}); \mathbf{x}) = L'_{\mathbf{X}}(\hat{\phi}; \mathbf{x}).$$

■

Example 3.28 In Example 3.25 we derived the maximum likelihood estimator of μ when sampling from $N(\mu, 1)$ to be $\hat{\mu} = \bar{X}$. By the invariance property of maximum likelihood estimators, the maximum likelihood estimator of μ^2 is $\hat{\mu}^2 = \bar{X}^2$.

Example 3.29 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from the gamma distribution with density function:

$$f_X(x; \alpha, \beta) = \begin{cases} \frac{1}{\Gamma(\alpha)} \beta^\alpha x^{\alpha-1} e^{-\beta x} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where $\alpha > 0$ is known, and $\beta > 0$ is an unknown parameter.

- Find the maximum likelihood estimators of β and $1/\beta$.
- Do the maximum likelihood estimators of β and $1/\beta$ attain the Cramér–Rao lower bound for their variances?
- Find the Fisher information of β .

Solution

- The likelihood function is:

$$L_{\mathbf{X}}(\beta; \mathbf{x}) = (\Gamma(\alpha))^{-n} \beta^{n\alpha} \left(\prod_{i=1}^n x_i^{\alpha-1} \right) \exp \left(-\beta \sum_{i=1}^n x_i \right).$$

The log-likelihood function is:

$$l_{\mathbf{X}}(\beta; \mathbf{x}) = n\alpha \log \beta - \beta \sum_{i=1}^n x_i + c$$

where c is independent of β . Hence the score function is:

$$s_{\mathbf{X}}(\beta; \mathbf{x}) = \frac{\partial}{\partial \beta} l_{\mathbf{X}}(\beta; \mathbf{x}) = \frac{n\alpha}{\beta} - \sum_{i=1}^n x_i.$$

Given that for any β we have:

$$\frac{\partial}{\partial \beta} s_{\mathbf{X}}(\beta; \mathbf{x}) = -\frac{n\alpha}{\beta^2} < 0$$

we may set $s_{\mathbf{X}}(\hat{\beta}; \mathbf{x}) = 0$ to get the maximum likelihood estimator:

$$\frac{n\alpha}{\hat{\beta}} - n\bar{X} = 0 \quad \Rightarrow \quad \hat{\beta} = \frac{\alpha}{\bar{X}}.$$

By the invariance property of maximum likelihood estimators, the maximum likelihood estimator of $1/\beta$ is:

$$\frac{1}{\hat{\beta}} = \frac{\bar{X}}{\alpha}.$$

- (b) The score function $s_{\mathbf{X}}(\beta; \mathbf{x})$ may be factorised as:

$$s_{\mathbf{X}}(\beta; \mathbf{x}) = -n\alpha \left(\frac{\bar{X}}{\alpha} - \frac{1}{\beta} \right)$$

which indicates that:

- if the estimator $\bar{X}/\alpha$ of $1/\beta$ is unbiased, then it attains the Cramér–Rao lower bound
- no unbiased estimator of β attains the Cramér–Rao lower bound.

Since $E(X_i)$ is α/β , we get that the maximum likelihood estimator of $1/\beta$ is unbiased since:

$$E\left(\frac{\bar{X}}{\alpha}\right) = \frac{E(X_i)}{\alpha} = \frac{\alpha/\beta}{\alpha} = \frac{1}{\beta}.$$

Hence the maximum likelihood estimator of β does not attain the Cramér–Rao lower bound, whereas the maximum likelihood estimator of $1/\beta$ does.

- (c) The Fisher information is equal to:

$$\mathcal{I}_{\mathbf{X}}(\beta) = -E\left(\frac{\partial}{\partial \beta} s_{\mathbf{X}}(\beta; \mathbf{X})\right) = -E\left(-\frac{n\alpha}{\beta^2}\right) = \frac{n\alpha}{\beta^2}.$$

There is an important association between likelihood-based statistical inference and sufficiency, which we now state and prove.

Relationship between maximum likelihood and sufficiency

The maximum likelihood estimate $\hat{\theta}$ is a function of every sufficient statistic for θ .

Proof: This is a consequence of the factorisation theorem. If $T(\mathbf{X})$ is a sufficient statistic for θ , then the density function (for the continuous case) can be factorised as:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = g(T(\mathbf{x}), \theta) h(\mathbf{x}).$$

The likelihood function has the same functional form as $f_{\mathbf{X}}$, so the log-likelihood function can be written as:

$$l_{\mathbf{X}}(\theta) = \log g(T(\mathbf{x}), \theta) + \log h(\mathbf{x}).$$

The value of θ which maximises this function is the value which maximises $\log g(T(\mathbf{x}), \theta)$. So the maximum likelihood estimate depends on $\mathbf{x}$ only through $T(\mathbf{x})$. Hence the maximum likelihood estimate is a function of the sufficient statistic $T(\mathbf{X})$. ■

3.9.3 Numerical procedures for likelihood maximisation

In the vector parameter case, where $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_r)^T$, we saw that in regular cases the maximum likelihood estimates satisfy:

$$\mathbf{s}_{\mathbf{X}}(\hat{\boldsymbol{\theta}}; \mathbf{x}) = \mathbf{0}$$

which defines a system of r equations. In most practical cases of interest, there is no closed-form solution to this set of equations. Hence we typically require *numerical methods* to find maximum likelihood estimates. This is the subject of this subsection.

The score vector function features heavily here. We will use $\dot{\mathbf{s}}$ to refer to the matrix given by the first derivative of the transpose of the score vector function, hence:

$$\dot{\mathbf{s}}_{\mathbf{X}}(\boldsymbol{\theta}, \mathbf{x}) = \nabla_{\boldsymbol{\theta}} \mathbf{s}_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x})^T = \nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}}^T l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x}).$$

Suppose that our current estimate of $\boldsymbol{\theta}$ is $\boldsymbol{\theta}^*$. We seek the maximum likelihood estimator $\hat{\boldsymbol{\theta}}$. In regular cases, $\hat{\boldsymbol{\theta}}$ is the solution to $\mathbf{s}_{\mathbf{X}}(\hat{\boldsymbol{\theta}}; \mathbf{x}) = \mathbf{0}$. We use a Taylor series expansion of $\mathbf{s}_{\mathbf{X}}(\hat{\boldsymbol{\theta}}; \mathbf{x})$ around $\boldsymbol{\theta}^*$ to give us an idea of how we could improve on our estimate. Ignoring terms of second order and higher, we have:

$$\mathbf{s}_{\mathbf{X}}(\hat{\boldsymbol{\theta}}; \mathbf{x}) \approx \mathbf{s}_{\mathbf{X}}(\boldsymbol{\theta}^*; \mathbf{x}) + \dot{\mathbf{s}}_{\mathbf{X}}(\boldsymbol{\theta}^*, \mathbf{x})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*).$$

Since $\mathbf{s}_{\mathbf{X}}(\hat{\boldsymbol{\theta}}; \mathbf{x}) = \mathbf{0}$, we can rearrange to give:

$$\hat{\boldsymbol{\theta}} \approx \boldsymbol{\theta}^* - (\dot{\mathbf{s}}_{\mathbf{X}}(\boldsymbol{\theta}^*, \mathbf{x}))^{-1} \mathbf{s}_{\mathbf{X}}(\boldsymbol{\theta}^*, \mathbf{x}). \quad (3.1)$$

Clearly, if $\boldsymbol{\theta}^*$ is far from $\hat{\boldsymbol{\theta}}$, the second-order terms, i.e. terms in $(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*)^2$ will not be small and the approximation given by (3.1) will be poor. In practice, (3.1) is used as part of an *iterative* process known as the **Newton–Raphson method**.

Suppose that our current estimate is $\boldsymbol{\theta}_k^*$, then we define the next estimate to be:

$$\boldsymbol{\theta}_{k+1}^* = \boldsymbol{\theta}_k^* - (\dot{\mathbf{s}}_{\mathbf{X}}(\boldsymbol{\theta}_k^*, \mathbf{x}))^{-1} \mathbf{s}_{\mathbf{X}}(\boldsymbol{\theta}_k^*, \mathbf{x}).$$

The iterations are initialised with some sensible estimate $\boldsymbol{\theta}_0^*$, such as a method of moments estimate as the initial value for the procedure. The iterative steps are then repeated for $k = 1, 2, \dots$ anticipating that the sequence $\boldsymbol{\theta}_1^*, \boldsymbol{\theta}_2^*, \dots$ will eventually *converge* to the true maximum likelihood estimate.

The **method of scoring** is an alternative numerical procedure, which is a modification of the Newton–Raphson method. For this, we exploit the result that:

$$E(-\dot{\mathbf{s}}_{\mathbf{X}}(\boldsymbol{\theta}, \mathbf{x})) = \mathcal{I}_{\mathbf{X}}(\boldsymbol{\theta}).$$

Replacing the term $-\dot{\mathbf{s}}_{\mathbf{X}}(\boldsymbol{\theta}_k^*, \mathbf{x})$ in the Newton–Raphson method by its expected value, the iterations become:

$$\boldsymbol{\theta}_{k+1}^* = \boldsymbol{\theta}_k^* + (\mathcal{I}_{\mathbf{X}}(\boldsymbol{\theta}_k^*))^{-1} \mathbf{s}_{\mathbf{X}}(\boldsymbol{\theta}_k^*, \mathbf{x}).$$

The Fisher information matrix does not depend on the observed sample $\mathbf{x}$ and is often simpler to evaluate than the derivative of the score vector function.

Numerical procedures are not guaranteed to find the maximum likelihood estimate, i.e. we cannot be certain of locating the global maximum of the likelihood function

$L_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x})$. A necessary condition for finding the maximum likelihood estimate is that the iterative procedure converges. Convergence is judged using the *difference* between consecutive estimates. We set some arbitrary tolerance $\epsilon > 0$ and say that the procedure has converged if we can find an i for which:

$$|\boldsymbol{\theta}_i^* - \boldsymbol{\theta}_{i-1}^*| < \epsilon.$$

The maximum likelihood estimate is then taken to be the vector (which may be scalar):

$$\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}_i^*.$$

All numerical procedures suffer from potential drawbacks related to the following important issues.

- i. The procedure may not converge. This is likely to be problematic when dealing with relatively flat likelihood functions.
- ii. Even if the procedure converges, it may not converge to the correct value. For example, we may have found a *local* maximum instead of the *global* maximum.

Inevitably a balance is required. If ϵ is set too small, the procedure may take a long time to achieve convergence, or may not converge at all. If ϵ is set too large, there is a risk that the procedure converges to a value which is not close to the global maximum. Hence the value of ϵ is important and may depend on several factors, such as the accuracy of the computer, the nature of the problem, and past experience.

Comparison of the relative advantages of the Newton–Raphson method and the method of scoring is not straightforward. If both procedures converge, then the Newton–Raphson method usually converges more quickly. However, the method of scoring is often less sensitive to the choice of initial values. In practice, slightly more sophisticated variants of the Newton–Raphson method are used to find the maximum likelihood estimates, although these are beyond the scope of this course.

Activity 3.16 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample of size n from the logistic distribution, with density function:

$$f_X(x; \mu) = \frac{\exp(x - \mu)}{(1 + \exp(x - \mu))^2}.$$

for $\mu \in \mathbb{R}$ and $x \in \mathbb{R}$. Find the log-likelihood function and derive iterations for the method of scoring.

Hint: Write quantities, as much as possible, in terms of the density function and the distribution function for the logistic distribution.

3.10 A reminder of your learning outcomes

On completion of this chapter, you should be able to:

- calculate the bias, variance and mean squared error of the sample mean

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- establish whether an estimator is consistent
- apply conditioning on a sufficient statistic to improve an estimator
- prove the Rao–Blackwell theorem
- demonstrate that the sample mean is unbiased, and derive its variance
- prove the central limit theorem in the special case where the moment generating function is well-defined
- discuss practical uses and limitations of the central limit theorem
- demonstrate that the sample variance is an unbiased estimator
- demonstrate the ‘trick’ of splitting $(X_i - \bar{X})$ into independent terms
- derive the distributions of the sample mean and sample variance for a normal population
- derive method of moments estimators of parameters for specified probability distributions
- construct sample quantities using order statistics
- derive the distribution of the sample minimum and sample maximum for a random sample from specified probability distributions
- apply the joint density function of order statistics in simple cases
- prove the Cramér–Rao inequality
- employ the Cramér–Rao inequality to determine the lower bound of the variance of unbiased estimators of parameters of specified probability distributions
- determine whether an estimator is minimum variance unbiased for the parameters for specified probability distributions
- explain the basic principles of maximum likelihood estimation
- describe the large sample properties of maximum likelihood estimators
- analytically derive maximum likelihood estimators in simple situations
- derive the updating step for the method of scoring to numerically approximate the maximum likelihood estimates of parameters for specified probability distributions.

3.11 Sample examination questions

Solutions can be found in Appendix C.

- Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample drawn from an exponential distribution. The density function of X_i , for $i = 1, 2, \dots, n$, is given by:

$$f_X(x; \lambda) = \begin{cases} \frac{1}{\mu} \exp\left(-\frac{x}{\mu}\right) & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

and $E(X_i) = \mu$.

- Consider the random variable $T(\mathbf{X}) = \min_{1 \leq i \leq n} X_i$. Show that its distribution is exponential with mean μ/n .
 - Find the method of moments estimator of μ .
 - Find the maximum likelihood estimator of μ and the Fisher information for μ .
- Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a $N(0, \theta)$ distribution. In other words, the density function of each X_i is:

$$f_X(x; \theta) = \frac{1}{\sqrt{2\pi\theta}} \exp\left(-\frac{x^2}{2\theta}\right)$$

for $\theta > 0$.

- Find a sufficient statistic of dimension one for θ and derive the log-likelihood function.
 - Find the maximum likelihood estimator of θ . Is it a minimum variance unbiased estimator? Justify your answer.
- Hint: You may want to use the fact that:

$$E(X_i^2) = \text{Var}(X_i) = \theta \quad \text{and} \quad \text{Var}(X_i^2) = 2\theta^2.$$

- Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a distribution with density function:

$$f_X(x; \alpha, \theta) = \begin{cases} \alpha e^{-\alpha(x-\theta)} & \text{for } x \geq \theta \\ 0 & \text{otherwise} \end{cases}$$

where $\alpha > 0$ and $\theta > 0$ are parameters.

- Find a sufficient statistic of dimension one and the maximum likelihood estimator of:
 - α , when θ is known
 - θ , when α is known.
- Let $\mu = 1/\alpha$ and suppose θ is known. Is the maximum likelihood estimator of μ a minimum variance unbiased estimator? Justify your answer.

Chapter 4

Interval estimation

4.1 Recommended reading

Casella, G. and R.L. Berger, *Statistical Inference*. Chapter 9, Sections 9.1, 9.2.1, 9.2.2, 9.2.3, 9.3.1 and 9.3.2.

4.2 Learning outcomes

On completion of this chapter, you should be able to:

- explain the distinction between an interval estimator and an interval estimate
- interpret interval estimates
- define the terms ‘coverage probability’ and ‘coverage coefficient’
- calculate the expected length of an interval estimator
- state pivotal functions using standard cases
- derive confidence intervals using pivotal functions
- compute the confidence coefficient of an interval estimator based on order statistics.

4.3 Introduction

In Chapter 3 our focus was on methods for obtaining point estimators of unknown parameters. Given the possibility of sampling error in random sampling, another important class of statistical inference methods is that of **interval estimators**, which provide a range (i.e. an interval) of possible values for an unknown parameter rather than just a single point estimate. Interval estimates (also known as **confidence intervals**, as seen in **ST104a Statistics 1** and **ST104b Statistics 2**) are widely used, but also often misinterpreted. Interval estimators can be viewed as a specific case of a confidence set. We proceed with clarifying some terminology.

4.4 Interval estimators and confidence sets

Point estimates provide a single value as a ‘best guess’ of the parameter of interest. Interval estimates provide an interval which we believe contains the true value of the

parameter. Recall that a point *estimator* is a statistic, say $\hat{\theta}(\mathbf{X})$ (i.e. a known function of a random sample $\mathbf{X}$) and hence a random variable. The realised value of a point estimator is computed from an observed sample $\mathbf{x}$ and is a number, called the point *estimate*, say $\hat{\theta}(\mathbf{x})$.

It is an analogous distinction when discussing interval estimation. An *interval estimator* is a random interval, with the endpoints of an interval being statistics (i.e. known functions of a random sample $\mathbf{X}$). For an observed sample $\mathbf{x}$, we compute the realised values of these endpoints and end up with an *interval estimate* which is simply a section of the real line, $\mathbb{R}$.

Interval estimator versus interval estimate

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from F_X parameterised by θ . Let $T_1(\mathbf{X})$ and $T_2(\mathbf{X})$ be statistics such that $T_1(\mathbf{x}) \leq T_2(\mathbf{x})$ for any observed sample $\mathbf{x}$. The random interval:

$$[T_1(\mathbf{X}), T_2(\mathbf{X})]$$

is an **interval estimator** of θ .

If the observed sample is $\mathbf{x}$, then the interval:

$$[T_1(\mathbf{x}), T_2(\mathbf{x})]$$

is an **interval estimate** (or confidence interval) of θ .

Example 4.1 Extending Example 2.13, consider a random sample $\mathbf{X}$ drawn from $N(\mu, 1)$, with μ unknown. The random interval:

$$\left[\bar{X} - 1.96 \times \frac{1}{\sqrt{n}}, \bar{X} + 1.96 \times \frac{1}{\sqrt{n}} \right]$$

is a sensible interval estimator of μ . This interval is constructed using:

$$\sqrt{n}(\bar{X} - \mu) \sim N(0, 1)$$

where the distribution is the standard normal distribution regardless of the value of the unknown parameter μ . Note that this is the standardised version of $\bar{X}$ with sampling distribution:

$$\bar{X} \sim N\left(\mu, \frac{1}{n}\right).$$

Functions which possess the property where the distribution does *not* depend on unknown parameters are known as *pivotal functions* which we consider later in this chapter.

Example 4.2 Continuing Example 4.1, since $\sqrt{n}(\bar{X} - \mu)$ is a pivotal function, the probability that the random interval covers (or spans) the true μ is 0.95 regardless of

what the true value of μ actually is, since:

$$P(-1.96 \leq Z \leq 1.96) = 0.95 \quad \text{for } Z \sim N(0, 1).$$

Hence we assign a *confidence coefficient* of 0.95 to the interval estimator. In percentage point terms, we may refer to a $100(1 - \alpha)\%$ confidence interval, where $\alpha \in [0, 1]$, as discussed in **ST104b Statistics 2**.

If $\alpha = 0.05$, an interval estimator would have a confidence coefficient of $1 - \alpha = 1 - 0.05 = 0.95$, resulting in a $100(1 - \alpha) = 100 \times (1 - 0.05) = 95\%$ confidence level.

Although our examples will mainly involve the construction of confidence intervals, more generally we may look for a *confidence set* (not necessarily an interval) – that is, we may view a confidence interval as a special case of a confidence set. Confidence sets are useful:

- when we are unsure whether the result of a procedure is an interval
- when we have more than one parameter, i.e. a vector of parameters, in which case we may refer to the confidence set as a *confidence region*.

Confidence set

A **confidence set** with confidence coefficient $1 - \alpha$ for a vector parameter $\theta \in \Theta$ is defined as the random set $S(\mathbf{X}) \subseteq \Theta$ where:

$$P(\theta \in S(\mathbf{X})) = 1 - \alpha.$$

Note in the above definition that $S(\mathbf{X})$ is a random variable (because it is a function of the random sample $\mathbf{X}$). For an observed sample $\mathbf{x}$ we would have the observed confidence set $S(\mathbf{x})$.

Given that there are a large (infinite) number of intervals which are valid interval estimators, it is necessary to establish mechanisms to decide what constitutes a ‘good’ interval estimator. We proceed to study this next.

Activity 4.1 The confidence level of a confidence interval is sometimes described as the probability that the true parameter value lies in the interval. Why is this slightly misleading?

Activity 4.2 Consider a random sample from $N(\mu, \sigma^2)$, with σ^2 known. Show that the random set:

$$\left(-\infty, \bar{X} - 0.126 \times \sqrt{\frac{\sigma^2}{n}}\right] \cup \left[\bar{X} + 0.126 \times \sqrt{\frac{\sigma^2}{n}}, \infty\right)$$

is a confidence set for the mean with a 95% confidence level. Why would you not use this in practice?

4.5 Coverage probability and expected length

‘Good’ interval estimators should, intuitively, have a large probability of containing the true parameter value being estimated. While $(-\infty, \infty)$ would be certain to include any real-valued parameter, an interval estimate of infinite length is of no practical use – we would basically be saying that an unknown number is some number!

As seen in **ST104b Statistics 2**, there is a *trade-off* between the coverage probability and interval length. Other things equal, we prefer a large coverage probability, while other things equal we would like the interval length to be short. In practice we have to sacrifice one for the other. A ‘good’ interval would hence be one with a high coverage probability *and* a short length.

As the probability of an interval estimator covering the true parameter value may depend on the parameter, we distinguish between the terms *coverage probability* and *confidence coefficient*.

Coverage probability

The probability that the random interval contains the true parameter θ is called the **coverage probability** and denoted as:

$$P(T_1(\mathbf{X}) \leq \theta \leq T_2(\mathbf{X})).$$

Confidence coefficient

The infimum of all the coverage probabilities (for each θ) is called the **confidence coefficient** (or confidence level, when expressed as a percentage) of the interval, such that:

$$\inf_{\theta \in \Theta} P(T_1(\mathbf{X}) \leq \theta \leq T_2(\mathbf{X})).$$

The random variables in the coverage probability are $T_1(\mathbf{X})$ and $T_2(\mathbf{X})$. The interval may be interpreted as the probability that $T_1(\mathbf{X})$ and $T_2(\mathbf{X})$ *contain* or *span* θ .

If an interval has a confidence coefficient of $1 - \alpha$, for $\alpha \in [0, 1]$, the interpretation is: ‘If the experiment was repeated many times, $100 \times (1 - \alpha)\%$ of the corresponding intervals would contain the true parameter θ .’

The random variables in $P(T_1(\mathbf{X}) \leq \theta \leq T_2(\mathbf{X}))$ are $T_1(\mathbf{X})$ and $T_2(\mathbf{X})$. Hence:

$$\begin{aligned} P(T_1(\mathbf{X}) \leq \theta \leq T_2(\mathbf{X})) &= P(\{T_1(\mathbf{X}) \leq \theta\} \cap \{T_2(\mathbf{X}) \geq \theta\}) \\ &= 1 - P(\{T_1(\mathbf{X}) > \theta\} \cup \{T_2(\mathbf{X}) < \theta\}) \\ &= 1 - (P(T_1(\mathbf{X}) > \theta) + P(T_2(\mathbf{X}) < \theta) \\ &\quad - P(\{T_1(\mathbf{X}) > \theta\} \cap \{T_2(\mathbf{X}) < \theta\})) \\ &= 1 - P(T_1(\mathbf{X}) > \theta) - P(T_2(\mathbf{X}) < \theta) \end{aligned}$$

since $P(\{T_1(\mathbf{X}) > \theta\} \cap \{T_2(\mathbf{X}) < \theta\}) = 0$ as $\{T_1(\mathbf{X}) > \theta\}$ and $\{T_2(\mathbf{X}) < \theta\}$ are mutually exclusive events because $T_1(\mathbf{X}) \leq T_2(\mathbf{X})$.

Example 4.3 Given a random sample $\mathbf{X} = \{X_1, X_2, X_3, X_4\}$ from $N(\mu, 1)$, we compare the sample mean $\bar{X}$, which is a point estimator, with the interval estimator $[\bar{X} - 1, \bar{X} + 1]$.

At first sight with the interval estimator we just lose precision (an interval versus a point), but we actually gained in confidence. Indeed, while $P(\bar{X} = \mu) = 0$ and noting that:

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} = \frac{1}{4}$$

we have:

$$\begin{aligned} P(\bar{X} - 1 \leq \mu \leq \bar{X} + 1) &= P(-1 \leq \bar{X} - \mu \leq 1) \\ &= P\left(-2 \leq \frac{\bar{X} - \mu}{\sqrt{1/4}} \leq 2\right) \\ &= 0.9544 \end{aligned}$$

because $\bar{X} \sim N(\mu, 1/4)$. Therefore, we lose in precision but we now have over a 95% probability of covering the unknown parameter with this interval estimator.

In general, the *length* of an interval estimator will be a random variable. One possible measure of the length of an interval is its *expected length*.

Expected length

Consider an interval estimator $[T_1(\mathbf{X}), T_2(\mathbf{X})]$. The length $T_2(\mathbf{X}) - T_1(\mathbf{X})$ is a random variable. Its **expected length** is:

$$E(T_2(\mathbf{X}) - T_1(\mathbf{X})).$$

A good interval estimator should *minimise* the expected length while *maximising* the coverage probability.

Example 4.4 Suppose that we have a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ from $N(\mu, 1)$ and we require an interval estimator of μ . Let k_1 and k_2 be positive constants. All the intervals below have the same length of $k_1 + k_2$.

- (a) $[-k_1, k_2]$.
- (b) $[X_1 - k_1, X_1 + k_2]$.
- (c) $[\bar{X} - k_1, \bar{X} + k_2]$.

Let us evaluate their respective coverage probabilities and confidence coefficients. Note that $X_i - \mu \sim N(0, 1)$.

- (a) This interval does not depend on the sample. If $\mu \in [-k_1, k_2]$ the coverage probability is 1, otherwise it is 0. Hence the confidence coefficient (which recall is the infimum coverage probability over μ) for this interval is 0.

(b) The coverage probability is:

$$\begin{aligned}
 P(X_1 - k_1 \leq \mu \leq X_1 + k_2) &= 1 - P(X_1 - k_1 > \mu) - P(X_1 + k_2 < \mu) \\
 &= 1 - P(X_1 - \mu > k_1) - P(X_1 - \mu < -k_2) \\
 &= \Phi(k_1) - \Phi(-k_2) \\
 &= \Phi(k_1) + \Phi(k_2) - 1
 \end{aligned}$$

which does not depend on μ . The confidence coefficient (i.e. the infimum coverage probability over μ) is also $\Phi(k_1) + \Phi(k_2) - 1$.

(c) Using the fact that $\sqrt{n}(\bar{X} - \mu) \sim N(0, 1)$ and similar calculations to (b), we get a coverage probability of:

$$\begin{aligned}
 P(\bar{X} - k_1 \leq \mu \leq \bar{X} + k_2) &= 1 - P(\bar{X} - k_1 > \mu) - P(\bar{X} + k_2 < \mu) \\
 &= 1 - P(\bar{X} - \mu > k_1) - P(\bar{X} - \mu < -k_2) \\
 &= \Phi(\sqrt{n}k_1) - \Phi(-\sqrt{n}k_2) \\
 &= \Phi(\sqrt{n}k_1) + \Phi(\sqrt{n}k_2) - 1.
 \end{aligned}$$

We have that:

$$\Phi(\sqrt{n}k_1) + \Phi(\sqrt{n}k_2) - 1 \geq \Phi(k_1) + \Phi(k_2) - 1$$

because the cumulative distribution function of the standard normal distribution, $\Phi(\cdot)$, is a non-decreasing function, $k_1 \geq 0$, $k_2 \geq 0$ and $n \geq 1$. Therefore, the confidence coefficient is equal to the coverage probability.

Clearly, the interval in (a) with a confidence coefficient of 0 is of no practical use. If we have to choose between the intervals in (b) and (c), we would choose (c) as it has a greater confidence coefficient. Note that this is consistent with the principle of sufficiency as $\bar{X}$ is a sufficient statistic for μ .

Example 4.4 controls for the length, i.e. all three intervals have the same length of $k_1 + k_2$. Example 4.5 (below) considers the more usual approach where the confidence coefficient is controlled for (i.e. it is fixed) and we then seek the shortest corresponding interval.

A reminder that we adopt the same notation as used in **ST104a Statistics 1** and **ST104b Statistics 2** such that z_α is the z -value which cuts off $100\alpha\%$ probability in the *upper* tail of the standard normal distribution, *not* the α -quantile. For example, $z_{0.025} = 1.96$.

Example 4.5 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, 1)$ and we want an interval estimator of μ with confidence coefficient $1 - \alpha$. We know that:

$$Z = \sqrt{n}(\bar{X} - \mu) \sim N(0, 1)$$

is a pivotal function. If α_1, α_2 are positive numbers such that $\alpha = \alpha_1 + \alpha_2$, we can

write:

$$\begin{aligned}
 P(z_{1-\alpha_1} \leq \sqrt{n}(\bar{X} - \mu) \leq z_{\alpha_2}) &= 1 - P(Z < z_{1-\alpha_1}) - P(Z > z_{\alpha_2}) \\
 &= 1 - (1 - (1 - \alpha_1)) - \alpha_2 \\
 &= 1 - (\alpha_1 + \alpha_2) \\
 &= 1 - \alpha.
 \end{aligned}$$

By rearrangement we get the interval estimator of μ with confidence coefficient $1 - \alpha$ of (noting that $z_{1-\alpha_1} = -z_{\alpha_1}$):

$$\left[\bar{X} - z_{\alpha_2} \times \frac{1}{\sqrt{n}}, \bar{X} + z_{\alpha_1} \times \frac{1}{\sqrt{n}} \right].$$

The length of this interval is:

$$\left(\bar{X} + z_{\alpha_1} \times \frac{1}{\sqrt{n}} \right) - \left(\bar{X} - z_{\alpha_2} \times \frac{1}{\sqrt{n}} \right) = \frac{1}{\sqrt{n}}(z_{\alpha_1} + z_{\alpha_2})$$

hence if either $\alpha_1 = 0$ or $\alpha_2 = 0$ then the length of the interval is infinite since $z_0 = \infty$. Note that the expected length of the interval is the same as the length since there is no random component.

Using statistical tables we can construct the following table, where we fix $\alpha_1 + \alpha_2 = 0.05$. Hence we have $\sqrt{n} \times \text{length}$ of 95% confidence intervals for the mean of a normal distribution with unit variance for various lower and upper endpoints.

α_1	α_2	z_{α_1}	z_{α_2}	$\sqrt{n} \times \text{length}$ $= z_{\alpha_1} + z_{\alpha_2}$
0	0.05	∞	1.645	∞
0.001	0.049	3.090	1.655	4.745
0.01	0.04	2.326	1.751	4.077
0.02	0.03	2.054	1.881	3.935
0.025	0.025	1.96	1.96	3.920
0.03	0.02	1.881	2.054	3.935
0.04	0.01	1.751	2.326	4.077
0.049	0.001	1.655	3.090	4.745
0.05	0	1.645	∞	∞

It is clearly seen that the shortest interval occurs when $\alpha_1 = \alpha_2 = 0.025$. As we might have expected, for a given confidence coefficient, the shortest confidence intervals for μ of a normal distribution will be *symmetric* about the sample mean.

Why 95%? Let us consider symmetric intervals with confidence coefficients of 0.80, 0.90, 0.95 and 0.99. Using the previous procedure and statistical tables we can construct the following table where we have $\sqrt{n} \times \text{length}$ of intervals for the mean of a normal distribution with unit variance for various confidence levels.

Confidence coefficient	α_1	α_2	z_{α_1}	z_{α_2}	$\sqrt{n} \times \text{length}$ $= z_{\alpha_1} + z_{\alpha_2}$
0.80	0.10	0.10	1.2816	1.2816	2.563
0.90	0.05	0.05	1.645	1.645	3.290
0.95	0.025	0.025	1.96	1.96	3.920
0.99	0.005	0.005	2.576	2.576	5.152

A confidence level of 95% is chosen as an adequate compromise between length and confidence.

Activity 4.3 Consider a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ from $N(\mu, \sigma^2)$, where μ is known and σ^2 is unknown. Two possible interval estimators of σ^2 are:

$$\left[\frac{1}{a_1} \sum_{i=1}^n (X_i - \bar{X})^2, \frac{1}{a_2} \sum_{i=1}^n (X_i - \bar{X})^2 \right] \quad \text{and} \quad \left[\frac{1}{b_1} \sum_{i=1}^n (X_i - \mu)^2, \frac{1}{b_2} \sum_{i=1}^n (X_i - \mu)^2 \right]$$

where a_1, a_2, b_1 and b_2 are constants. For $n = 10$, find the values of a_1, a_2, b_1 and b_2 which give intervals with a 90% confidence level. Compare the expected lengths of the intervals and comment.

4.6 Finding interval estimators from pivotal functions

A way to construct a $100(1 - \alpha)\%$ confidence set for θ is by using a *pivotal function*. Pivotal functions play a fundamental role (dare we say a *pivotal* role!) in the construction of confidence intervals. We proceed with a formal definition.

Pivotal function

Consider a random sample $\mathbf{X}$ from F_X parameterised by θ , where θ is a scalar parameter. Suppose that we are interested in constructing an interval estimator of θ .

Let $g(\mathbf{X}, \theta)$ be a function of $\mathbf{X}$ and θ which does not involve any unknown parameters other than θ .

$g(\mathbf{X}, \theta)$ is a **pivotal function** if its distribution is known and does not depend on θ .

Note that a pivotal function defines a random variable, say $Y = g(\mathbf{X}, \theta)$. By definition, the *distribution* of Y does not depend on θ .

Example 4.6 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, with μ unknown and σ^2 known. We know that:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

and we can use the above to get the following pivotal function for μ since:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

Note that although Z depends on μ , its *distribution* does not change regardless of the value of μ .

Example 4.7 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, with μ *known* and σ^2 *unknown*. We know that:

$$Z_i = \frac{X_i - \mu}{\sigma} \sim N(0, 1)$$

and hence:

$$Z_i^2 = \frac{(X_i - \mu)^2}{\sigma^2} \sim \chi_1^2.$$

The Z_i s are independent because the X_i s are independent. Considering $Y = \sum_{i=1}^n Z_i^2$ gives us the following pivotal function for σ^2 since:

$$Y = \sum_{i=1}^n Z_i^2 = \frac{\sum_{i=1}^n (X_i - \mu)^2}{\sigma^2} \sim \chi_n^2.$$

Note that although Y depends on σ^2 , its *distribution* does not change regardless of the value of σ^2 .

Example 4.8 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$ with both μ and σ^2 *unknown*. Now we *cannot* use:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

as a pivotal function for μ because its distribution also depends on the unknown parameter σ^2 . Instead we use (see **ST104b Statistics 2**):

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

In the same way we *cannot* use $\sum_{i=1}^n Z_i^2$ as a pivotal function for σ^2 since its distribution depends on μ , which is unknown. Instead we use (see **ST104b Statistics 2**):

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

Pivotal functions provide a simple mechanism for generating interval estimators with a given confidence coefficient. We now consider the procedure for constructing an interval from a pivotal function.

Constructing an interval from a pivotal function

Suppose that we have a random sample $\mathbf{X}$ from F_X parameterised by θ . To construct an interval estimator of θ with a $100(1-\alpha)\%$ confidence level using a pivotal function, we use the following procedure.

1. Find a pivotal function $g(\mathbf{X}, \theta)$ based on a reasonable point estimator of θ .
2. Use the distribution of the pivotal function to find values k_1 and k_2 such that:

$$P(k_1 \leq g(\mathbf{X}, \theta) \leq k_2) = 1 - \alpha.$$

3. Manipulate $g(\mathbf{X}, \theta) \geq k_1$ and $g(\mathbf{X}, \theta) \leq k_2$ to make θ the reference point. This yields inequalities of the form:

$$\theta \geq T_1(\mathbf{X}, k_1, k_2) \quad \text{and} \quad \theta \leq T_2(\mathbf{X}, k_1, k_2)$$

for some functions $T_1(\cdot)$ and $T_2(\cdot)$ independent of θ .

4. Construct the following interval:

$$[T_1(\mathbf{X}, k_1, k_2), T_2(\mathbf{X}, k_1, k_2)]$$

which is then an interval estimator of θ with a confidence coefficient of $1 - \alpha$.

Note the endpoints T_1 and T_2 are usually functions of one (and only one) of k_1 or k_2 .

We now consider several examples demonstrating the use of pivotal functions.

Example 4.9 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, with σ^2 known, and we require an interval estimator of μ with a $100(1 - \alpha)\%$ confidence level.

1. We know that:

$$Z = Z(\mathbf{X}, \mu) = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

Therefore, Z is a pivotal function because σ^2 is known.

2. We can write:

$$\begin{aligned} P(z_{1-\alpha/2} \leq Z \leq z_{\alpha/2}) &= 1 - P(Z < z_{1-\alpha/2}) - P(Z > z_{\alpha/2}) \\ &= 1 - \left(1 - \left(1 - \frac{\alpha}{2}\right)\right) - \frac{\alpha}{2} \\ &= 1 - \left(\frac{\alpha}{2} + \frac{\alpha}{2}\right) \\ &= 1 - \alpha. \end{aligned}$$

3. Rearranging the inequalities we get:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \geq z_{1-\alpha/2} \quad \text{and} \quad \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}$$

which we can rewrite as:

$$\mu \geq \bar{X} - z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \quad \text{and} \quad \mu \leq \bar{X} - z_{1-\alpha/2} \times \frac{\sigma}{\sqrt{n}}.$$

4. Note that $z_{\alpha/2} = -z_{1-\alpha/2}$. We get the interval estimator:

$$\left[\bar{X} - z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \right].$$

Suppose we have the observed sample $\mathbf{x}$ with $n = 10$, $\bar{x} = 5.2$, $\sigma^2 = 2.4$ and $\alpha = 0.05$. Since $z_{0.025} = 1.96$, an interval estimate of μ with a confidence coefficient of $1 - \alpha$ is:

$$\left[5.2 - 1.96 \times \sqrt{\frac{2.4}{10}}, 5.2 + 1.96 \times \sqrt{\frac{2.4}{10}} \right]$$

or else $[4.24, 6.16]$.

Example 4.10 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, with σ^2 unknown, and we require an interval estimator of μ with a $100(1 - \alpha)\%$ confidence level.

1. We know (from **ST104b Statistics 2**) that:

$$T = T(\mathbf{X}, \mu) = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

where S is the sample standard deviation. Therefore, T is a pivotal function because S is a function of $\mathbf{X}$, and the distribution of T does not depend on μ .

2. We can write:

$$\begin{aligned} P(t_{1-\alpha/2, n-1} \leq T \leq t_{\alpha/2, n-1}) &= 1 - P(T < t_{1-\alpha/2, n-1}) - P(T > t_{\alpha/2, n-1}) \\ &= 1 - \left(1 - \left(1 - \frac{\alpha}{2}\right)\right) - \frac{\alpha}{2} \\ &= 1 - \left(\frac{\alpha}{2} + \frac{\alpha}{2}\right) \\ &= 1 - \alpha. \end{aligned}$$

3. Rearranging the inequalities we get:

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \geq t_{1-\alpha/2, n-1} \quad \text{and} \quad \frac{\bar{X} - \mu}{S/\sqrt{n}} \leq t_{\alpha/2, n-1}$$

which we can rewrite as:

$$\mu \geq \bar{X} - t_{\alpha/2, n-1} \times \frac{S}{\sqrt{n}} \quad \text{and} \quad \mu \leq \bar{X} - t_{1-\alpha/2, n-1} \times \frac{S}{\sqrt{n}}.$$

4. Note that $t_{\alpha/2, n-1} = -t_{1-\alpha/2, n-1}$. We get the interval estimator:

$$\left[\bar{X} - t_{\alpha/2, n-1} \times \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2, n-1} \times \frac{S}{\sqrt{n}} \right].$$

Suppose we have the observed sample $\mathbf{x}$ with $n = 10$, $\bar{x} = 5.2$, $s^2 = 2.4$ and $\alpha = 0.05$. Since $t_{0.025, 9} = 2.262$, an interval estimate of μ with a confidence coefficient of $1 - \alpha$ is:

$$\left[5.2 - 2.262 \times \sqrt{\frac{2.4}{10}}, 5.2 + 2.262 \times \sqrt{\frac{2.4}{10}} \right]$$

or else $[4.09, 6.31]$.

Compared with the σ^2 known case in Example 4.9, the confidence interval is longer despite the fact that $s^2 = \sigma^2 = 2.4$. The t distribution has fatter tails than the standard normal distribution, hence $t_{\alpha/2, n-1} > z_{\alpha/2}$. However, as $n \rightarrow \infty$ we have that $t_{\alpha/2, n-1} \rightarrow z_{\alpha/2}$ because the t distribution converges to $N(0, 1)$ as the degrees of freedom tend to infinity.

Example 4.11 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from an $\text{Exp}(\lambda)$ distribution. We seek an interval estimator of the parameter λ . We know that if $X \sim \text{Exp}(\lambda)$, then:

$$E(\lambda X) = \lambda E(X) = \lambda \times \frac{1}{\lambda} = 1$$

so we might consider $\lambda \bar{X}$ as a potential source of pivotal functions. In fact we can show that if:

$$Y = g(\mathbf{X}, \lambda) = \lambda \sum_{i=1}^n X_i$$

then Y is a pivotal function.

It is clear that Y does not involve any unknown parameters other than λ . Using moment generating functions (introduced in **ST2133 Advanced statistics: distribution theory**), we can establish that the distribution of Y does not depend on λ .

Since $X \sim \text{Exp}(\lambda)$, the moment generating function of X is:

$$M_X(t) = \left(1 - \frac{t}{\lambda}\right)^{-1}.$$

The moment generating function of Y , noting the independence of the X_i s, is:

$$\begin{aligned} M_Y(t) &= E(e^{tY}) = E\left(\exp\left(t\lambda \sum_{i=1}^n X_i\right)\right) = E\left(\prod_{i=1}^n e^{t\lambda X_i}\right) \\ &= \prod_{i=1}^n E(e^{t\lambda X_i}) \quad (\text{by independence}) \\ &= (M_X(t\lambda))^n \quad (\text{as identically distributed}) \\ &= (1 - t)^{-n}. \end{aligned}$$

Recall that the probability distribution of a random variable is completely characterised by its moment generating function. Since $M_Y(t)$ does not depend on λ , we conclude that the distribution of Y does not depend on λ , and hence Y is a pivotal function.

To proceed to construct an interval estimator of λ we require the parametric form of the distribution of Y . Once more, we will make use of moment generating functions. Note that if $V \sim \chi_k^2$, then:

$$M_V(t) = (1 - 2t)^{-k/2}.$$

Comparing moment generating functions, we see that:

$$2Y \sim \chi_{2n}^2.$$

Exploiting the parametric form of $2Y$ we have:

$$P\left(\chi_{1-\alpha/2, 2n}^2 \leq 2\lambda \sum_{i=1}^n X_i \leq \chi_{\alpha/2, 2n}^2\right) = 1 - \alpha.$$

Rearranging the inequalities we have that:

$$\left[\frac{\chi_{1-\alpha/2, 2n}^2}{2 \sum_{i=1}^n X_i}, \frac{\chi_{\alpha/2, 2n}^2}{2 \sum_{i=1}^n X_i} \right]$$

is an interval estimator of λ with a confidence coefficient of $1 - \alpha$.

Activity 4.4 Suppose $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ is a random sample from $N(\mu, \sigma^2)$. Find a pivotal function for σ^2 for the cases when μ is known and unknown, respectively. In each case derive an interval estimator of σ^2 with a $100(1 - \alpha)\%$ confidence level and provide an expression for the expected length of the interval.

Activity 4.5 Suppose that we have a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ from $N(\mu_X, \sigma_X^2)$ and another random sample $\mathbf{Y} = \{Y_1, Y_2, \dots, Y_m\}$ from $N(\mu_Y, \sigma_Y^2)$, where μ_X and μ_Y are known. Find a pivotal function for the ratio σ_X^2/σ_Y^2 and hence provide an expression for an interval estimator of σ_X^2/σ_Y^2 with a $100(1 - \alpha)\%$ confidence level.

Hint: Recall the definition of the F distribution from **ST104b Statistics 2**.

4.7 Asymptotic pivotal functions from maximum likelihood estimators

Remember that in interval estimation we want to use the information from the random sample $\mathbf{X}$ to provide an interval which we believe contains the true value of the unknown parameter of interest of the probability distribution from which the random sample was drawn. The probability that the random interval contains the true parameter θ is called the coverage probability. The infimum of all the coverage probabilities is called the confidence coefficient (or level) of the interval.

A way to construct an interval is by using a pivotal function, that is a function of $\mathbf{X}$ and θ with a distribution independent of θ . There may exist more than one interval with the same confidence coefficient. One way to choose between them is through their expected length. However, sometimes it may be even hard to find any ‘reasonable’ interval estimator.

Here we consider the construction of *asymptotic* interval estimators based on the asymptotic sampling distributions of maximum likelihood estimators (covered in Chapter 3).

Asymptotic pivotal functions

If $\hat{\theta}$ is a maximum likelihood estimator of θ , then under mild regularity conditions:

$$\hat{\theta} \xrightarrow{d} N(\theta, \mathcal{I}_{\mathbf{X}}^{-1}(\theta)).$$

For large n we have:

$$\sqrt{\mathcal{I}_{\mathbf{X}}(\theta)}(\hat{\theta} - \theta) \sim N(0, 1)$$

approximately.

As $\sqrt{\mathcal{I}_{\mathbf{X}}(\theta)}(\hat{\theta} - \theta)$ converges to a distribution which does not depend on θ , we say that it is **asymptotically pivotal** for θ .

Since $\hat{\theta}$ is a consistent estimator of θ , the quantity $\mathcal{I}_{\mathbf{X}}(\hat{\theta})$ converges in probability to $\mathcal{I}_{\mathbf{X}}(\theta)$, which is total Fisher information. Therefore, in a second level of approximation, we can write for large sample sizes that:

$$\sqrt{\mathcal{I}_{\mathbf{X}}(\hat{\theta})}(\hat{\theta} - \theta) \sim N(0, 1)$$

approximately, and use this function to construct asymptotic confidence intervals for the parameter θ . The approximation improves as $n \rightarrow \infty$.

Example 4.12 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from Bernoulli(π). We know that:

- $Y = \sum_{i=1}^n X_i \sim \text{Bin}(n, \pi)$
- the maximum likelihood estimator of π is $\hat{\pi} = \bar{X}$
- the total Fisher information is:

$$\mathcal{I}_{\mathbf{X}}(\pi) = \frac{n}{\pi(1 - \pi)}.$$

The estimator $\hat{\pi}$ is consistent, unbiased (hence also asymptotically unbiased) and efficient since it attains the Cramér–Rao lower bound. The asymptotic distribution of $\hat{\pi}$ is:

$$\hat{\pi} \sim N\left(\pi, \frac{\pi(1 - \pi)}{n}\right)$$

approximately, for large n . In an extra level of approximation we may use:

$$\hat{\pi} \sim N\left(\pi, \frac{\hat{\pi}(1-\hat{\pi})}{n}\right)$$

approximately, for large n . Let:

$$S_{\hat{\pi}} = \frac{\hat{\pi}(1-\hat{\pi})}{n}$$

then, approximately:

$$\hat{\pi} \sim N(\pi, S_{\hat{\pi}}).$$

Standardising, we have:

$$\frac{\hat{\pi} - \pi}{\sqrt{S_{\hat{\pi}}}} \sim N(0, 1).$$

We can use the above to construct an asymptotic $100(1-\alpha)\%$ confidence interval for π which is:

$$\left[\hat{\pi} - z_{\alpha/2} \times \sqrt{S_{\hat{\pi}}}, \hat{\pi} + z_{\alpha/2} \times \sqrt{S_{\hat{\pi}}} \right]$$

or in full:

$$\left[\hat{\pi} - z_{\alpha/2} \times \sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{n}}, \hat{\pi} + z_{\alpha/2} \times \sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{n}} \right].$$

Note that the above interval may take values outside of $[0, 1]$ while, of course, $\pi \in [0, 1]$.

Activity 4.6 Suppose that $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ is a random sample from $\text{Bernoulli}(\pi)$. Using the normal approximation to the binomial, find a function which is approximately pivotal for π . Hence construct an interval estimator of π .

Activity 4.7 Suppose that $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ is a random sample from $\text{Pois}(\lambda)$. Using the approximate normality of the maximum likelihood estimator, construct an interval estimator of λ .

4.8 Confidence intervals for quantiles

So far, we have looked at how to construct confidence intervals for parameters such as the mean, μ , and variance, σ^2 , of various well-known distributions. Another useful class of population parameters are the *quantiles*. We have seen that the order statistics provide a natural mechanism for statistical inference about population quantiles. As we might expect, interval estimators of quantiles also use order statistics.

We begin by determining the coverage probability associated with an interval based on the first and last order statistics as an interval estimator of a scalar parameter θ .

Interval estimators based on the first and last order statistics

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a distribution with scalar parameter θ . Let F_X be the distribution function of X and let $X_{(i)}$ denote the i th order statistic.

The probability that the random interval $[X_{(1)}, X_{(n)}]$ covers θ is:

$$1 - (1 - F_X(\theta))^n - (F_X(\theta))^n.$$

Proof: Using the properties of order statistics, we know that if $X_{(1)} > \theta$ then we must have $X_i > \theta$ for all $i = 1, 2, \dots, n$. Similarly, if $X_{(n)} \leq \theta$ then $X_i \leq \theta$ for all $i = 1, 2, \dots, n$. Hence:

$$P(X_{(1)} > \theta) = (1 - F_X(\theta))^n \quad \text{and} \quad P(X_{(n)} \leq \theta) = (F_X(\theta))^n.$$

Since the events $\{X_{(1)} > \theta\}$ and $\{X_{(n)} \leq \theta\}$ are mutually exclusive, then:

$$\begin{aligned} P(\{X_{(1)} > \theta\} \cup \{X_{(n)} \leq \theta\}) &= P(X_{(1)} > \theta) + P(X_{(n)} \leq \theta) \\ &= (1 - F_X(\theta))^n + (F_X(\theta))^n. \end{aligned}$$

Applying de Morgan's laws (see **ST104b Statistics 2**), the result follows:

$$\begin{aligned} P(\{X_{(1)} \leq \theta\} \cap \{X_{(n)} > \theta\}) &= 1 - P(\{X_{(1)} > \theta\} \cup \{X_{(n)} \leq \theta\}) \\ &= 1 - (1 - F_X(\theta))^n - (F_X(\theta))^n. \end{aligned}$$

Suppose that the parameter of interest is the β -quantile, i.e. $\theta = q_\beta$, where q_β satisfies $F_X(q_\beta) = \beta$. The coverage probability is then a confidence coefficient. The following corollary discusses.

 $[X_{(1)}, X_{(n)}]$ as an interval estimator of a quantile

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from F_X . Let q_β be the β -quantile of X . The random interval $[X_{(1)}, X_{(n)}]$ is an interval estimator of q_β with confidence coefficient:

$$1 - (1 - \beta)^n - \beta^n.$$

In practice we would like an interval estimator of q_β for any given confidence coefficient. Greater flexibility is gained by considering the interval estimator associated with general order statistics $X_{(i)}$ and $X_{(j)}$, for $i < j$.

 $[X_{(i)}, X_{(j)}]$ as an interval estimator

Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from F_X parameterised by the scalar parameter θ . Let $F_{X_{(i)}}$ be the distribution function of the i th order statistic. If $i < j$, the probability that the random interval $[X_{(i)}, X_{(j)}]$ covers θ is:

$$F_{X_{(j)}}(\theta) - F_{X_{(i)}}(\theta).$$

Proof: By definition:

$$P(X_{(i)} > \theta) = 1 - F_{X_{(i)}}(\theta) \quad \text{and} \quad P(X_{(j)} \leq \theta) = F_{X_{(j)}}(\theta).$$

By mutual exclusivity:

$$P(\{X_{(i)} > \theta\} \cup \{X_{(j)} \leq \theta\}) = 1 - F_{X_{(i)}}(\theta) + F_{X_{(j)}}(\theta).$$

Using de Morgan's laws (see **ST104b Statistics 2**):

$$P(X_{(i)} \leq \theta < X_{(j)}) = P(\{X_{(i)} \leq \theta\} \cap \{X_{(j)} > \theta\}) = F_{X_{(i)}}(\theta) - F_{X_{(j)}}(\theta).$$

Using the distribution function of the i th order statistic, we have:

$$F_{X_{(i)}}(\theta) = \sum_{j=i}^n \binom{n}{j} (F_X(\theta))^j (1 - F_X(\theta))^{n-j}.$$

This calculation involves the sum of binomial probabilities and so can be more conveniently expressed in terms of a binomial distribution function. Suppose we define:

$$Y \sim \text{Bin}(n, F_X(\theta))$$

then:

$$F_{X_{(i)}}(\theta) = P(X_{(i)} \leq \theta) = P(Y \geq i) = 1 - F_Y(i - 1).$$

Hence the probability that the random interval $[X_{(i)}, X_{(j)}]$ covers θ is:

$$F_Y(j - 1) - F_Y(i - 1)$$

where $Y \sim \text{Bin}(n, F_X(\theta))$. Once again, this result is most usefully applied to quantiles. If $\theta = q_\beta$, then $Y \sim \text{Bin}(n, \beta)$.

Example 4.13 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from any distribution. The median is the quantile $q_{0.50}$.

The interval with endpoints given by the sample minimum and sample maximum is an interval estimator for the median with confidence coefficient:

$$1 - \left(\frac{1}{2}\right)^n - \left(\frac{1}{2}\right)^n = 1 - \left(\frac{1}{2}\right)^{n-1} = \frac{2^{n-1} - 1}{2^{n-1}}.$$

For a random sample of size $n = 4$, this confidence coefficient is $7/8 = 0.875$. Clearly, for a more reasonable sample size, the probability that $[X_{(1)}, X_{(n)}]$ covers the median will be very close to 1.

Example 4.14 Suppose that we seek an interval estimator of the median with a confidence coefficient of $1 - \alpha$. We need to find i and j such that:

$$F_Y(j - 1) - F_Y(i - 1) = 1 - \alpha$$

where $Y \sim \text{Bin}(n, 0.50)$. For example, if $n = 100$ then:

$$P(Y \leq 39) = 0.0176 \quad \text{and} \quad P(Y \leq 59) = 0.9716$$

hence $[X_{(40)}, X_{(60)}]$ provides a confidence interval for the median with a confidence level of 95.4%.

Activity 4.8 Suppose that $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ is a random sample from any distribution. Find interval estimators with approximate 95% confidence levels for the upper quartile, i.e. $q_{0.75}$, when $n = 100$.

4.9 A reminder of your learning outcomes

On completion of this chapter, you should be able to:

- explain the distinction between an interval estimator and an interval estimate
- interpret interval estimates
- define the terms ‘coverage probability’ and ‘coverage coefficient’
- calculate the expected length of an interval estimator
- state pivotal functions using standard cases
- derive confidence intervals using pivotal functions
- compute the confidence coefficient of an interval estimator based on order statistics.

4.10 Sample examination questions

Solutions can be found in Appendix C.

1. Suppose that we have a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ from $N(\mu, \sigma^2)$.
 - (a) Find a pivotal function for σ^2 when:
 - i. μ is known
 - ii. μ is unknown.
 - (b) In each case in (a) derive a 90% confidence interval for $n = 10$ and find the expected length.
 - (c) Compare the expected lengths in (b) and comment.
2. Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a distribution with density function:

$$f_X(x; \mu) = \left(\frac{1}{2\pi x^3} \right)^{1/2} \exp \left(\frac{1}{\mu} - \frac{x}{2\mu^2} - \frac{1}{2x} \right), \quad \text{for } x \geq 0, \mu > 0$$

and 0 otherwise.

- (a) Derive the likelihood function and provide a one-dimensional sufficient statistic for μ .

- (b) Find the maximum likelihood estimator of μ .
 - (c) Find an approximate pivotal function for μ and construct an asymptotic interval estimator of μ with a $100(1 - \alpha)\%$ confidence level, where $\alpha \in [0, 1]$.
3. Suppose that we have a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ from a Bernoulli(π) distribution.
- (a) Find an asymptotically unbiased and efficient estimator of the *odds* of success, which are equal to $\pi/(1 - \pi)$.
 - (b) Construct a $100(1 - \alpha)\%$ asymptotic interval estimator of the odds of success.

Chapter 5

Hypothesis testing

5.1 Recommended reading

Casella, G. and R.L. Berger, *Statistical Inference*. Chapter 8, Sections 8.1, 8.2.1, 8.3.1, 8.3.2 and 8.3.4.

5.2 Learning outcomes

On completion of this chapter, you should be able to:

- explain the distinction between simple and composite hypotheses, and the distinction between the size and significance level of a statistical test
- define Type I and Type II errors and describe the association between these errors and the concepts of size and power
- summarise the use of p -values in hypothesis testing
- construct hypothesis tests using pivotal functions
- state and prove the Neyman–Pearson lemma
- construct most powerful tests for simple hypotheses
- determine whether a test is uniformly most powerful
- explain the rationale behind the likelihood ratio test
- describe how the likelihood ratio test is used in the presence of nuisance parameters
- construct likelihood ratio tests for standard cases.

5.3 Introduction

Hypothesis testing featured heavily in both **ST104a Statistics 1** and **ST104b Statistics 2**, such that the basic mechanics of conducting hypothesis tests should be familiar to you. In this course we consider hypothesis testing in greater depth looking at how to deal with composite null hypotheses, most powerful tests and likelihood ratio tests. We begin with a review of the standard hypothesis testing procedure.

5.4 Statistical tests

Suppose that a real-world phenomenon (population) may be described by a probability model defined through the random variable X with associated distribution function $F_X(x; \theta)$. Suppose also that a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ is drawn from that distribution/population. We want to use the information in the observed sample $\mathbf{x}$ to answer statements about the parameter θ , if scalar, or $\boldsymbol{\theta}$ as a vector of parameters.

A statistical hypothesis is a statement about the value of a population parameter, θ . The two complementary hypotheses in a hypothesis testing problem are called the **null hypothesis**, H_0 , and the **alternative hypothesis**, H_1 . Hypotheses can be *simple* or *composite*.

Simple versus composite hypotheses

A **simple hypothesis** takes the form:

$$H_0 : \theta = c$$

where c is a constant, i.e. it states an exact specification of θ .

If a vector of parameters is being tested, then we may write $H_0 : \boldsymbol{\theta} = \mathbf{c}$, where $\mathbf{c}$ is a vector of constants of the same dimension as $\boldsymbol{\theta}$.

A hypothesis which is not simple is called a **composite hypothesis**, for example:

$$H_0 : \theta \leq c.$$

In classical hypothesis testing, the null hypothesis is the hypothesis which is actually tested, i.e. we perform the test *conditional* on H_0 . The alternative hypothesis informs the manner in which the null hypothesis is specified.

The null hypothesis is usually ‘conservative’ meaning it reflects established thinking. Hence under H_0 we typically posit statements reflecting ‘no change’, ‘no difference’ or ‘no effect’. The alternative hypothesis reflects our suspicion about values which θ might take. H_1 is usually a composite hypothesis and can be thought of as reflecting ‘a change’, ‘a difference’ or ‘an effect’.

General hypotheses

For a vector of parameters $\boldsymbol{\theta}$ and parameter space Θ , we test:

$$H_0 : \boldsymbol{\theta} \in \Theta_0 \quad \text{vs.} \quad H_1 : \boldsymbol{\theta} \in \Theta_1$$

where $\Theta_0 \subset \Theta$, $\Theta_1 \subset \Theta$ and $\Theta_0 \cap \Theta_1 = \emptyset$.

Note that Θ_0 and Θ_1 must be mutually exclusive such that it is impossible for H_0 and H_1 to be true simultaneously. However, there is no requirement for Θ_0 and Θ_1 to be collectively exhaustive, i.e. we do not require that $\Theta_0 \cup \Theta_1 = \Theta$.

5.4.1 Decision rules

Classical hypothesis testing involves a binary choice such that we:

$$\text{reject } H_0 \quad \text{or} \quad \text{not reject } H_0.$$

Note a hypothesis testing procedure is incapable of ‘accepting’ a hypothesis – a test can only check to what extent the available evidence is *consistent* or *inconsistent* with the null hypothesis. The evidence used to make the decision is given by the observed sample $\mathbf{x} = \{x_1, x_2, \dots, x_n\}$. A *test statistic*, $T(\mathbf{X})$, is constructed and the computed test statistic *value*, $T(\mathbf{x})$, is used to make the decision of whether or not to reject H_0 .

For now, let us summarise the decision-making process by a single function, $\delta(\mathbf{x})$, such that:

$$\delta(\mathbf{x}) = \begin{cases} 1 & \text{when } H_0 \text{ is rejected} \\ 0 & \text{when } H_0 \text{ is not rejected.} \end{cases}$$

The function δ is known as the *decision rule*. δ partitions the space of possible observed samples into two sets:

$$R = \{\mathbf{x} : \delta(\mathbf{x}) = 1\} \quad \text{and} \quad R^c = \{\mathbf{x} : \delta(\mathbf{x}) = 0\}$$

where the set R is the *rejection region* (or *critical region*), since it corresponds to an observed sample $\mathbf{x}$ which results in H_0 being rejected (and hence R^c is the *non-rejection region*).

For a given statistical test, the decision function, δ , and rejection region, R , are equivalent since knowledge of one completely determines the other. We formalise as follows.

Decision function and rejection region

The **decision function** δ for the **rejection region** R is:

$$\delta(\mathbf{x}) = \begin{cases} 1 & \text{for } \mathbf{x} \in R \\ 0 & \text{for } \mathbf{x} \notin R. \end{cases}$$

Similarly, for a specific decision function δ , the rejection region is:

$$R = \{\mathbf{x} \in \mathbb{R}^n : \delta(\mathbf{x}) = 1\}.$$

5.4.2 Test statistics

The crucial part of a hypothesis test is to identify an appropriate test statistic, $T(\mathbf{X})$. One of the desired features of $T(\mathbf{X})$ is to have an interpretation such that large (or small) values of it provide evidence *against* H_0 . We also want to know the distribution of $T(\mathbf{X})$ under H_0 , that is when H_0 holds, i.e. we require the distribution of $T(\mathbf{X}) | H_0$.

We focus on test statistics which have tabulated distributions under H_0 (such as $N(0, 1)$, t , χ^2 and F), but this is only for convenience and it is not a strict requirement.

Example 5.1 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample of size n from $N(\mu, \sigma^2)$, with σ^2 known. Is μ equal to μ_0 or larger for a given value of μ_0 ?

The hypotheses of the test are:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu > \mu_0.$$

A statistical test may be based on the test statistic $\bar{X}$ with rejection region:

$$R = \left(\mu_0 + 1.96 \times \frac{\sigma}{\sqrt{n}}, \infty \right).$$

The decision rule is then to reject H_0 if:

$$\bar{x} > \mu_0 + 1.96 \times \frac{\sigma}{\sqrt{n}}.$$

5.4.3 Types of errors in tests, power function and p -value

Consider a test of $H_0 : \theta \in \Theta_0$ vs. $H_1 : \theta \in \Theta_1$. It is possible that the wrong decision is made, such that there are two types of error.

- **Type I error:** If $\theta \in \Theta_0$ (H_0 is true) but the test rejects H_0 .
- **Type II error:** If $\theta \in \Theta_1$ (H_1 is true) but the test does not reject H_0 .

The test outcome space is hence:

	Not reject H_0	Reject H_0
H_0 is true	Correct decision	Type I error
H_1 is true	Type II error	Correct decision

To assess the statistical performance of a hypothesis test procedure we can look at the *probability* of an error. The probability of a Type I error is associated with the **significance level** and the **size** of the test. Note that these terms are not synonymous due to subtle differences in their definitions.

Significance level versus size

Consider testing the null hypothesis of $H_0 : \theta \in \Theta_0$.

The test has **significance level** α if:

$$\sup_{\theta \in \Theta_0} P_{\theta}(\text{reject } H_0) \leq \alpha.$$

The test has **size** α if:

$$\sup_{\theta \in \Theta_0} P_{\theta}(\text{reject } H_0) = \alpha.$$

Some points to note are the following.

- i. A test of size α has significance level α . The converse is *not* true.
- ii. A test with significance level α will always be at least as conservative as a test of size α .
- iii. If the null hypothesis is a simple hypothesis, then the size of the test is the probability of a Type I error.

Point iii. relates to the fact that we can only specify a precise probability for Type I errors when H_0 is a simple hypothesis. Similarly, we can only specify a precise probability for Type II errors when H_1 is a simple hypothesis. However, often H_1 is a composite hypothesis which leads us to the definition of the power function.

Power function

For $\theta \in \Theta$, the **power function** is defined as:

$$\beta(\theta) = P_{\theta}(\text{reject } H_0)$$

which is the probability that the null hypothesis is rejected if the true parameter value is θ (note it may be a scalar parameter θ).

Note that the **power of the test** when testing $H_0 : \theta \in \Theta_0$ vs. $H_1 : \theta \in \Theta_1$ is:

$$\beta(\theta) = P_{\theta}(\text{reject } H_0) = \begin{cases} P_{\theta}(\text{Type I error}) & \text{for } \theta \in \Theta_0 \\ 1 - P_{\theta}(\text{Type II error}) & \text{for } \theta \in \Theta_1 \end{cases}$$

and hence:

$$P_{\theta}(\text{Type II error}) = P_{\theta}(\text{not reject } H_0) = 1 - \beta(\theta) \quad \text{for } \theta \in \Theta_1.$$

If a hypothesis test procedure has size α then, from the definition of size, we have that:

$$\beta(\theta_0) \leq \alpha$$

for $\theta_0 \in \Theta_0$.

We may hence define the significance level and the size of a test in terms of the power function.

Significance level and size in terms of the power function

If a hypothesis test procedure has significance level α , then:

$$\sup_{\theta_0 \in \Theta_0} \beta(\theta) \leq \alpha.$$

If a hypothesis test procedure has size α , then:

$$\sup_{\theta_0 \in \Theta_0} \beta(\theta) = \alpha.$$

5. Hypothesis testing

Ideally we would like the power function $\beta(\theta)$ to be 0 when $\theta \in \Theta_0$ and 1 when $\theta \in \Theta_1$, which would mean:

$$\beta(\theta) = \begin{cases} 0 & \text{for } \theta \in \Theta_0 \\ 1 & \text{for } \theta \notin \Theta_0 \end{cases}$$

such that the hypothesis test procedure has a size of 0 and a power of 1. However, this is not possible as there is uncertainty about the true values of the parameters being tested. This results in a trade-off between size and power.

In practice we fix the size α to a small value (usually 0.05, although depending on the severity of a Type I error in the context of the hypothesis testing procedure other values such as 0.10 or 0.01 may be chosen) and for a given size we try to maximise the power. Therefore:

- failure to reject the null hypothesis does not imply that it holds and we say that we ‘do not reject H_0 ’ rather than say we ‘accept H_0 ’
- we usually set the alternative hypothesis to contain the statement which we are interested in proving.

Given the binary nature of a statistical test (i.e. we either reject H_0 or we do not), it is desirable to quantify the *weight of evidence* against H_0 . The observed value of the test statistic, $T(\mathbf{x})$, could be used, although what an ‘extreme’ test statistic value is depends on the probability distribution of $T(\mathbf{X}) | H_0$.

Example 5.2 Consider an upper-tailed test with $\alpha = 0.05$. If $T(\mathbf{X}) | H_0 \sim N(0, 1)$, then rejection of H_0 would occur if the test statistic value $T(\mathbf{x})$ exceeded $z_{0.05} = 1.645$, but if $T(\mathbf{X}) | H_0 \sim \chi_1^2$ then rejection of H_0 would occur if the test statistic value exceeded $\chi_{0.05, 1}^2 = 3.841$. Hence a test statistic value of, say, $T(\mathbf{x}) = 2.345$ would be statistically significant if $T(\mathbf{X}) | H_0 \sim N(0, 1)$, but insignificant if $T(\mathbf{X}) | H_0 \sim \chi_1^2$.

Therefore, for reporting purposes, we would prefer a *normalised* version of a test statistic value which is easier to interpret, such that normalisation results in a value between 0 and 1. The *p-value* provides such a measure.

p-value

Let $T(\mathbf{X})$ be a test statistic used in a hypothesis testing procedure of a scalar parameter θ . Without loss of generality, suppose H_0 is rejected for large values of $T(\mathbf{X})$. For an observed sample $\mathbf{x}$ the corresponding **p-value** is:

$$p(\mathbf{x}) = \sup_{\theta \in \Theta_0} P(T(\mathbf{X}) \geq T(\mathbf{x}))$$

where $T(\mathbf{x})$ is the test statistic value.

Some points to note are the following.

- i. Clearly $0 \leq p(\mathbf{x}) \leq 1$. The closer the *p-value* is to 0, the more likely we are to reject H_0 .

- ii. In words, a p -value is the probability of observing the result of the sample or a more extreme result. ‘Extreme’ in the sense of evidence against H_0 . For a simple null hypothesis, we may omit the ‘sup’ term.
- iii. If we have a fixed significance level α , then we can describe the rejection region as:

$$R = \{\mathbf{x} : p(\mathbf{x}) \leq \alpha\}.$$

We reject H_0 if the probability of observing a more extreme result than that of the observed sample $\mathbf{x}$ is small (less than α).

- iv. A similar definition can be made if small values of $T(\mathbf{X})$ give evidence against H_0 .

Example 5.3 Suppose $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ is a random sample from $N(\mu, \sigma^2)$, with σ^2 known. We test $H_0 : \mu = \mu_0$ vs. $H_1 : \mu > \mu_0$.

A test of size α can be constructed with the rejection region:

$$R = \left\{ \mathbf{x} : \bar{x} > \mu_0 + z_\alpha \times \frac{\sigma}{\sqrt{n}} \right\}.$$

The power function of the test is:

$$\begin{aligned} \beta(\mu) &= P_\mu(\mathbf{X} \in R) = P\left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} > z_\alpha\right) \\ &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > z_\alpha - \frac{\mu - \mu_0}{\sigma/\sqrt{n}}\right) \\ &= 1 - \Phi\left(z_\alpha - \frac{\mu - \mu_0}{\sigma/\sqrt{n}}\right) \end{aligned}$$

where $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

As expected, the power function is a strictly increasing function of μ which takes the value of α when $\mu = \mu_0$.

The p -value associated with this test is:

$$p(\mathbf{x}) = 1 - \Phi\left(\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}\right)$$

such that small p -values arise when $\bar{x}$ is much larger than μ_0 .

Example 5.4 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, with σ^2 unknown. We want to test $H_0 : \mu = \mu_0$ vs. $H_1 : \mu > \mu_0$.

In order to construct a hypothesis test procedure we use the pivotal function:

$$\frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}.$$

The rejection region for a test of size α is:

$$R = \left\{ \mathbf{x} : \bar{x} > \mu_0 + t_{\alpha, n-1} \times \frac{s}{\sqrt{n}} \right\}.$$

5.4.4 Constructing statistical tests

The procedure for constructing a statistical test can be given by the following general directions.

1. Find an appropriate test statistic $T(\mathbf{X})$. Determine whether large or small values of $T(\mathbf{X})$ provide evidence against H_0 . Also, find the distribution of $T(\mathbf{X})$ under H_0 .
2. Given the significance level or size α , write (where R is unknown):

$$P_{\theta_0}(T(\mathbf{X}) \in R) \leq \alpha \quad \text{or} \quad P_{\theta_0}(T(\mathbf{X}) \in R) = \alpha.$$

3. Solve the equation to get R . The test rule is then ‘reject H_0 ’ if:

$$\{\mathbf{x} : T(\mathbf{x}) \in R\}.$$

Example 5.5 Consider again Example 5.3 with $H_0 : \mu = \mu_0$ vs. $H_1 : \mu > \mu_0$.

1. The sample mean is $\bar{X}$. Large values provide evidence against H_0 . Under H_0 , $\bar{X} \sim N(\mu_0, \sigma^2/n)$, so a proposed test statistic is:

$$T(\mathbf{X}) = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1).$$

2. Under H_0 we know that:

$$P_{\mu_0} \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > z_\alpha \right) = P \left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} > z_\alpha \right) = \alpha.$$

3. From:

$$P \left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} > z_\alpha \right) = \alpha$$

we can get to:

$$P \left(\bar{X} > \mu_0 + z_\alpha \times \frac{\sigma}{\sqrt{n}} \right) = \alpha$$

hence:

$$R = \left\{ \mathbf{x} : \bar{x} > \mu_0 + z_\alpha \times \frac{\sigma}{\sqrt{n}} \right\}.$$

Example 5.6 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, where both μ and σ^2 are unknown. Consider testing the hypotheses:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu < \mu_0.$$

We find the rejection region, the power function and the p -value of a test with size α . A sensible test statistic is the following:

$$T(\mathbf{X}) = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

where S is the sample standard deviation. Small values of $\bar{X}$ correspond to evidence against H_0 . Also, under $H_0 : \mu = \mu_0$, we get that $T(\mathbf{X}) \sim t_{n-1}$.

Let R denote the rejection region. We want:

$$P_{\mu_0}(\mathbf{X} \in R) = \alpha.$$

For $H_0 : \mu = \mu_0$, we can write:

$$P_{\mu} \left(\frac{\bar{X} - \mu}{S/\sqrt{n}} < -t_{\alpha, n-1} \right) = P \left(\frac{\bar{X} - \mu_0}{S/\sqrt{n}} < -t_{\alpha, n-1} \right) = \alpha.$$

We can rewrite as:

$$P \left(\bar{X} < \mu_0 - t_{\alpha, n-1} \times \frac{S}{\sqrt{n}} \right) = \alpha.$$

Hence the rejection region is:

$$R = \left\{ \mathbf{x} : \bar{x} < \mu_0 - t_{\alpha, n-1} \times \frac{s}{\sqrt{n}} \right\}.$$

The power function of the test is:

$$\beta(\mu) = P_{\mu} \left(\bar{X} < \mu_0 - t_{\alpha, n-1} \times \frac{S}{\sqrt{n}} \right) = P_{\mu} \left(\frac{\bar{X} - \mu}{S/\sqrt{n}} < \frac{\mu_0 - \mu}{S/\sqrt{n}} - t_{\alpha, n-1} \right)$$

where:

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

The p -value is:

$$\begin{aligned} p(\mathbf{x}) &= P_{\mu=\mu_0}(\bar{X} \leq \bar{x}) = P_{\mu=\mu_0} \left(\frac{\bar{X} - \mu}{S/\sqrt{n}} \leq \frac{\bar{x} - \mu}{S/\sqrt{n}} \right) = P \left(\frac{\bar{X} - \mu_0}{S/\sqrt{n}} \leq \frac{\bar{x} - \mu_0}{S/\sqrt{n}} \right) \\ &= P \left(T(\mathbf{X}) \leq \frac{\bar{x} - \mu_0}{S/\sqrt{n}} \right). \end{aligned}$$

Activity 5.1 Someone tells you that the p -value is the probability that the null hypothesis is true. How would you explain why this view is mistaken?

Activity 5.2 Consider the pivotal functions of Chapter 4. Explain how pivotal functions could be used in hypothesis testing.

5.5 Most powerful tests

The statistical tests we are interested in by construction control for the probability of a Type I error (it is at most α). A good test should also have a small probability of committing a Type II error. In other words it should also be a *powerful* test.

In this section we will be comparing tests. As such, it is helpful to refer to tests using a specific identifier. We characterise a test by its decision function, δ , or equivalently by its rejection region, R . For a specific test ' T ', we denote its decision function by δ_T and its rejection region by R_T , while β_T denotes its power function (for example, $\beta_T(\boldsymbol{\theta})$ provides the probability of H_0 being rejected by test T if $\boldsymbol{\theta}$ is the (vector of) true parameter value(s)).

The following lemma connects the power function and decision function which is useful in a subsequent proof in this section.

Relationship between power function and decision function

If δ_T is the decision function for a statistical test T , then the power function is:

$$\beta_T(\boldsymbol{\theta}) = E_{\boldsymbol{\theta}}(\delta_T(\mathbf{X})).$$

Proof: Noting that $\delta_T(\mathbf{x}) = 0$ for $\mathbf{x} \notin R_T$, we have:

$$\begin{aligned} \beta_T(\boldsymbol{\theta}) &= P_{\boldsymbol{\theta}}(H_0 \text{ is rejected by test } T) = P_{\boldsymbol{\theta}}(\mathbf{X} \in R_T) \\ &= \int_{R_T} f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) d\mathbf{x} \\ &= \int_{\mathbb{R}^n} \delta_T(\mathbf{x}) f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) d\mathbf{x} \\ &= E_{\boldsymbol{\theta}}(\delta_T(\mathbf{X})). \end{aligned}$$

■

In order to develop a theory for choosing between competing hypothesis testing procedures based on their respective power, consider the test of the following simple hypotheses:

$$H_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0 \quad \text{vs.} \quad H_1 : \boldsymbol{\theta} = \boldsymbol{\theta}_1.$$

The power of a test T is hence $\beta_T(\boldsymbol{\theta}_1)$. We now define a *most powerful test*.

Most powerful test

Consider the test of $H_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$ vs. $H_1 : \boldsymbol{\theta} = \boldsymbol{\theta}_1$, i.e. of two simple hypotheses.

The test T is the **most powerful test** if:

$$\beta_T(\boldsymbol{\theta}_1) \geq \beta_S(\boldsymbol{\theta}_1)$$

for all tests S such that $\beta_S(\boldsymbol{\theta}_0) = \beta_T(\boldsymbol{\theta}_0)$. Hence T is a most powerful test if its power is greater than or equal to any other test of the same size.

Given this definition of a most powerful test, it is instructive to refer to a most powerful test of size α , with the size being explicitly stated to emphasise its role in the definition of a most powerful test.

5.5.1 Neyman–Pearson lemma

We now state and prove the Neyman–Pearson lemma, which demonstrates that *likelihood ratio tests* are the most powerful tests, among all possible statistical tests.

Neyman–Pearson lemma

Consider a test procedure T of size α for $H_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$ vs. $H_1 : \boldsymbol{\theta} = \boldsymbol{\theta}_1$. If the rejection region for T is:

$$R_T = \{\mathbf{x} \in \mathbb{R}^n : L_{\mathbf{X}}(\boldsymbol{\theta}_1; \mathbf{x}) - k_{\alpha} L_{\mathbf{X}}(\boldsymbol{\theta}_0; \mathbf{x}) > 0\}$$

equivalently:

$$R_T = \left\{ \mathbf{x} \in \mathbb{R}^n : \frac{L_{\mathbf{X}}(\boldsymbol{\theta}_1; \mathbf{x})}{L_{\mathbf{X}}(\boldsymbol{\theta}_0; \mathbf{x})} > k_{\alpha} \right\}$$

then T is the most powerful test of size α .

Here $L_{\mathbf{X}}$ is the likelihood function and k_{α} is a constant dependent on α . Note that $\alpha = P_{\boldsymbol{\theta}_0}(\mathbf{X} \in R_T)$.

We give the proof for continuous random variables. For discrete random variables just replace integrals with sums.

Proof: Let S be a test of size α for $H_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$ vs. $H_1 : \boldsymbol{\theta} = \boldsymbol{\theta}_1$. We begin by considering the power of S minus k_{α} times its size:

$$\beta_S(\boldsymbol{\theta}_1) - k_{\alpha}\beta_S(\boldsymbol{\theta}_0) = \int_{\mathbb{R}^n} \delta_S(\mathbf{x})(f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}_1) - k_{\alpha}f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}_0)) d\mathbf{x} \quad (1)$$

$$= \int_{\mathbb{R}^n} \delta_S(\mathbf{x})(L_{\mathbf{X}}(\boldsymbol{\theta}_1; \mathbf{x}) - k_{\alpha}L_{\mathbf{X}}(\boldsymbol{\theta}_0; \mathbf{x})) d\mathbf{x} \quad (2)$$

$$\leq \int_{R_T} \delta_S(\mathbf{x})(L_{\mathbf{X}}(\boldsymbol{\theta}_1; \mathbf{x}) - k_{\alpha}L_{\mathbf{X}}(\boldsymbol{\theta}_0; \mathbf{x})) d\mathbf{x} \quad (3)$$

$$\leq \int_{R_T} \delta_T(\mathbf{x})(L_{\mathbf{X}}(\boldsymbol{\theta}_1; \mathbf{x}) - k_{\alpha}L_{\mathbf{X}}(\boldsymbol{\theta}_0; \mathbf{x})) d\mathbf{x} \quad (4)$$

$$= \int_{R_T} \delta_T(\mathbf{x})(f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}_1) - k_{\alpha}f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}_0)) d\mathbf{x} \quad (2)$$

$$= \int_{\mathbb{R}^n} \delta_T(\mathbf{x})(f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}_1) - k_{\alpha}f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}_0)) d\mathbf{x} \quad (5)$$

$$= \beta_T(\boldsymbol{\theta}_1) - k_{\alpha}\beta_T(\boldsymbol{\theta}_0). \quad (1)$$

Since both S and T are of size α , we know $\beta_S(\boldsymbol{\theta}_0) = \beta_T(\boldsymbol{\theta}_0) = \alpha$ and the above demonstrates that $\beta_S(\boldsymbol{\theta}_1) \leq \beta_T(\boldsymbol{\theta}_1)$. We conclude that T is a most powerful test of size α . ■

The equation references above relate to the following explanatory notes.

Explanatory notes for the proof of the Neyman–Pearson lemma

1. For any test S , we have:

$$\begin{aligned}\beta_S(\boldsymbol{\theta}) &= E_{\boldsymbol{\theta}}(\delta_S(\mathbf{X})) \\ &= \int_{\mathbb{R}^n} \delta_S(\mathbf{x}) f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) d\mathbf{x}.\end{aligned}$$

2. The density function $f_{\mathbf{X}}$ and the likelihood function $L_{\mathbf{X}}$ are functionally identical, hence for any $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ we have:

$$\int_{\mathbb{R}^n} \delta_S(\mathbf{x}) f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) d\mathbf{x} = \int_{\mathbb{R}^n} \delta_S(\mathbf{x}) L_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x}) d\mathbf{x}.$$

3. The statement of the Neyman–Pearson lemma defines the rejection region:

$$R_T = \{\mathbf{x} \in \mathbb{R}^n : L_{\mathbf{X}}(\boldsymbol{\theta}_1; \mathbf{x}) - k_{\alpha} L_{\mathbf{X}}(\boldsymbol{\theta}_0; \mathbf{x}) > 0\}.$$

The decision function $\delta_S(\mathbf{x})$ only takes the values 0 and 1, hence:

$$\delta_S(\mathbf{x})(L_{\mathbf{X}}(\boldsymbol{\theta}_1; \mathbf{x}) - k_{\alpha} L_{\mathbf{X}}(\boldsymbol{\theta}_0; \mathbf{x})) = \begin{cases} \geq 0 & \text{for any } \mathbf{x} \in R_T \\ \leq 0 & \text{for any } \mathbf{x} \notin R_T. \end{cases}$$

An integral over just the region R_T excludes all possible negative values of the function. Therefore, the result will be at least as large as that given by integrating over the whole of $\mathbb{R}^n$.

4. By definition of the decision function, $\delta_T(\mathbf{x}) = 1$ for all $\mathbf{x} \in R_T$. Hence since any decision function only takes the values 0 and 1, we have:

$$\delta_T(\mathbf{x}) \geq \delta_S(\mathbf{x}) \quad \text{for all } \mathbf{x} \in R_T.$$

5. By definition of the decision function, $\delta_T(\mathbf{x}) = 0$ for all $\mathbf{x} \notin R_T$. Hence integrating any product of $\delta_T(\mathbf{x})$ over R_T will yield the same result as integrating over the whole of $\mathbb{R}^n$.

6. The α index on k_{α} is to emphasise the dependence of the choice of k_{α} on the size of the test. Note that:

$$\begin{aligned}P_{\boldsymbol{\theta}_0}(\text{H}_0 \text{ is rejected by test } T) &= P_{\boldsymbol{\theta}_0}(\mathbf{X} \in R_T) \\ &= P_{\boldsymbol{\theta}_0}(L_{\mathbf{X}}(\boldsymbol{\theta}_1; \mathbf{X}) - k_{\alpha} L_{\mathbf{X}}(\boldsymbol{\theta}_0; \mathbf{X}) > 0).\end{aligned}$$

Hence k_{α} is chosen to satisfy:

$$P_{\boldsymbol{\theta}_0}\left(\frac{L_{\mathbf{X}}(\boldsymbol{\theta}_1; \mathbf{X})}{L_{\mathbf{X}}(\boldsymbol{\theta}_0; \mathbf{X})} > k_{\alpha}\right) = \alpha$$

such that we consider the *ratio* of the likelihood functions calculated at θ_1 and θ_0 , respectively.

So far we have considered tests of simple hypotheses. We now extend the concept of most powerful tests to *uniformly* most powerful tests to cover composite cases.

Uniformly most powerful test (UMPT)

Consider the test of $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \in \Theta_1$. The Neyman–Pearson lemma is no longer directly applicable, but in many cases the most powerful test of $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$ is the same for every $\theta_1 \in \Theta_1$.

The test T is the **uniformly most powerful test (UMPT)** of size α if:

$$\beta_T(\theta_1) \geq \beta_S(\theta_1)$$

for all tests S such that $\beta_S(\theta_0) = \beta_T(\theta_0)$.

Hence T is the uniformly most powerful test if its power is greater than or equal to any other test of the same size for any $\theta_1 \in \Theta_1$.

Composite null hypotheses require special consideration to ensure that the test is the appropriate size, as the next example demonstrates.

Example 5.7 Suppose T is the most powerful test of size α when testing:

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta = \theta_1$$

for $\theta_0 < \theta_1$. Now consider the composite hypotheses:

$$H_0 : \theta \leq \theta_0 \quad \text{vs.} \quad H_1 : \theta > \theta_0.$$

The test T is the uniformly most powerful test of size α for these hypotheses if:

- i. the rejection region R_T does not depend on the value of θ_1
- ii. the probability of rejecting H_0 is at most α for any $\theta \leq \theta_0$, i.e. we have:

$$\beta_T(\theta) \leq \beta_T(\theta_0) \quad \text{for } \theta \leq \theta_0.$$

Similar statements can be made for testing:

$$H_0 : \theta \geq \theta_0 \quad \text{vs.} \quad H_1 : \theta < \theta_1$$

as well as the vector parameter case.

Example 5.8 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, with σ^2 known. We want to test:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu = \mu_1.$$

The likelihood ratio from the Neyman–Pearson lemma is (noting Example 2.7):

$$\begin{aligned}
 \frac{L_{\mathbf{X}}(\mu_1; \mathbf{X})}{L_{\mathbf{X}}(\mu_0; \mathbf{X})} &= \frac{\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu_1)^2\right)}{\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu_0)^2\right)} \\
 &= \frac{(2\pi\sigma^2)^{-n/2} \exp(-(n(\bar{X} - \mu_1)^2 + (n-1)S^2)/2\sigma^2)}{(2\pi\sigma^2)^{-n/2} \exp(-(n(\bar{X} - \mu_0)^2 + (n-1)S^2)/2\sigma^2)} \\
 &= \exp\left(\frac{1}{2\sigma^2} (2n\bar{X}\mu_1 - n\mu_1^2 - 2n\bar{X}\mu_0 + n\mu_0^2)\right) \\
 &= \exp\left(\frac{n}{2\sigma^2} ((\mu_0 - \mu_1)^2 - 2\bar{X}(\mu_0 - \mu_1))\right) \\
 &= \exp\left(\frac{n}{2\sigma^2} ((\mu_0 - \mu_1)(\mu_0 + \mu_1 - 2\bar{X}))\right).
 \end{aligned}$$

If $\mu_0 < \mu_1$ the likelihood ratio is large when $\bar{X}$ is large. If $\mu_0 > \mu_1$ the likelihood ratio is large when $\bar{X}$ is small. Therefore, we may create a most powerful test based on $\bar{X}$. It is known that:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

In the case of $\mu_0 < \mu_1$, a most powerful test has rejection region:

$$R = \left\{ \mathbf{x} : \bar{x} > \mu_0 + z_\alpha \times \frac{\sigma}{\sqrt{n}} \right\}.$$

This rejection region does not depend on μ_1 , so the test is uniformly most powerful for testing:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu > \mu_0.$$

For the composite null hypothesis case of:

$$H_0 : \mu \leq \mu_0 \quad \text{vs.} \quad H_1 : \mu > \mu_0$$

to establish that the rejection region above provides a uniformly most powerful test we need to show that the probability of rejecting H_0 is at most α for any $\mu \leq \mu_0$.

This is true since:

$$P_\mu(\mathbf{X} \in R) = \Phi\left(z_{1-\alpha} + \frac{\mu - \mu_0}{\sigma/\sqrt{n}}\right)$$

and, if $\mu < \mu_0$, then the argument of $\Phi(\cdot)$ is less than z_α . Since $\Phi(\cdot)$ is a non-decreasing function, we conclude that the probability of rejecting H_0 is less than or equal to α .

Example 5.9 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $\text{Pois}(\lambda)$ and suppose we test:

$$H_0 : \lambda = \lambda_0 \quad \text{vs.} \quad H_1 : \lambda = \lambda_1.$$

The likelihood ratio from the Neyman–Pearson lemma is:

$$\frac{L_{\mathbf{X}}(\lambda_1; \mathbf{X})}{L_{\mathbf{X}}(\lambda_0; \mathbf{X})} = \frac{e^{-n\lambda_1} \lambda_1^{\sum_{i=1}^n X_i} / \prod_{i=1}^n X_i!}{e^{-n\lambda_0} \lambda_0^{\sum_{i=1}^n X_i} / \prod_{i=1}^n X_i!} = e^{n(\lambda_0 - \lambda_1)} \left(\frac{\lambda_1}{\lambda_0} \right)^{\sum_{i=1}^n X_i}.$$

A most powerful test has a rejection region such that:

$$P \left(\sum_{i=1}^n X_i > \frac{\log k - n(\lambda_0 - \lambda_1)}{\log \lambda_1 - \log \lambda_0} = k_1 \right) = \alpha$$

but we also know that $\sum_i X_i \sim \text{Pois}(n\lambda_0)$ under H_0 so we can find k_1 .

Let $n = 8$, $\lambda_0 = 2$ (hence $\sum_i X_i \sim \text{Pois}(16)$) and $\lambda_1 = 6$. The size is 0.058 with $k_1 = 22$ and 0.037 with $k_1 = 23$. A test with $\alpha = 0.05$ corresponds to $k_1 = 23$.

Example 5.10 Suppose that $X \sim \text{Bin}(n, \pi)$ and we wish to test:

$$H_0 : \pi = \pi_0 \quad \text{vs.} \quad H_1 : \pi > \pi_0.$$

We begin by considering the simple alternative hypothesis $H_1 : \pi = \pi_1$, for $\pi_1 > \pi_0$. The likelihood ratio from the Neyman–Pearson lemma is:

$$\frac{L_{\mathbf{X}}(\pi_1; \mathbf{X})}{L_{\mathbf{X}}(\pi_0; \mathbf{X})} = \frac{\binom{n}{X} \pi_1^X (1 - \pi_1)^{n-X}}{\binom{n}{X} \pi_0^X (1 - \pi_0)^{n-X}} = \left(\frac{\pi_1(1 - \pi_0)}{\pi_0(1 - \pi_1)} \right)^X \left(\frac{1 - \pi_1}{1 - \pi_0} \right)^n.$$

Since $\pi_1 > \pi_0$, we have that:

$$\frac{\pi_1(1 - \pi_0)}{\pi_0(1 - \pi_1)} > 1.$$

We conclude that the value of the likelihood ratio increases as X increases, hence we can construct a most powerful test of the form ‘reject H_0 for $X > k$ ’, for some k . The value of k (and hence the rejection region) will depend only on the $\text{Bin}(n, \pi_0)$ distribution. Therefore, the rejection region will not depend on π_1 so the test is uniformly most powerful.

Example 5.11 Suppose that we have a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ from $N(0, \sigma^2)$.

(a) Find the uniformly most powerful test for the testing problem:

$$H_0 : \sigma^2 = \sigma_0^2 \quad \text{vs.} \quad H_1 : \sigma^2 = \sigma_1^2$$

where $\sigma_0^2 < \sigma_1^2$ and where σ_0^2 and σ_1^2 are known constants.

(b) Given that $\sigma_0^2 = 1$, $\sigma_1^2 = 2$ and $n = 10$, find the power function of the testing procedure of size α .

(c) Is this a uniformly most powerful test for testing $H_0 : \sigma^2 = \sigma_0^2$ vs. $H_1 : \sigma^2 > \sigma_0^2$?
Hint: Check whether the rejection region of the previous test depends on σ_1^2 .

- (d) Is this a uniformly most powerful test for testing $H_0 : \sigma^2 \leq \sigma_0^2$ vs. $H_1 : \sigma^2 > \sigma_0^2$?

Hint: Show that:

$$\sup_{\sigma^2 \leq \sigma_0^2} P_{\sigma^2}(\mathbf{X} \in R) = P_{\sigma_0^2}(\mathbf{X} \in R)$$

where R is the rejection region of the previous test.

Solution

The likelihood function is:

$$L_{\mathbf{X}}(\sigma^2; \mathbf{x}) = (2\pi)^{-n/2} \sigma^{-n} \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2 \right).$$

- (a) For testing the simple hypotheses:

$$H_0 : \sigma^2 = \sigma_0^2 \quad \text{vs.} \quad H_1 : \sigma^2 = \sigma_1^2$$

the most powerful test is defined in terms of the likelihood ratio statistic:

$$\begin{aligned} LR(\mathbf{X}) &= \frac{L_{\mathbf{X}}(\sigma_1^2; \mathbf{X})}{L_{\mathbf{X}}(\sigma_0^2; \mathbf{X})} = \frac{\sigma_1^{-n} \exp \left(-\frac{1}{2\sigma_1^2} \sum_{i=1}^n X_i^2 \right)}{\sigma_0^{-n} \exp \left(-\frac{1}{2\sigma_0^2} \sum_{i=1}^n X_i^2 \right)} \\ &= \left(\frac{\sigma_0}{\sigma_1} \right)^n \exp \left(\frac{1}{2} \left(\frac{1}{\sigma_0^2} - \frac{1}{\sigma_1^2} \right) \sum_{i=1}^n X_i^2 \right). \end{aligned}$$

Since $0 < \sigma_0^2 < \sigma_1^2$, we get that:

$$\frac{1}{\sigma_0^2} - \frac{1}{\sigma_1^2} > 0$$

and hence the function above is increasing in $\sum_{i=1}^n X_i^2$. Therefore, $LR(\mathbf{X}) > k$ for

some suitable k is equivalent to $\sum_{i=1}^n X_i^2 > k_1$ for some suitable k_1 .

Under $H_0 : \sigma^2 = \sigma_0^2$, we have that $X_i/\sigma_0 \sim N(0, 1)$ for $i = 1, 2, \dots, n$, and hence:

$$\frac{X_i^2}{\sigma_0^2} \sim \chi_1^2 \quad \Rightarrow \quad \frac{\sum_{i=1}^n X_i^2}{\sigma_0^2} \sim \chi_n^2.$$

In order to have a test of size α the condition below should be satisfied under H_0 . We have:

$$P_{\sigma_0^2} \left(\sum_{i=1}^n X_i^2 > k_1 \right) = P_{\sigma_0^2} \left(\frac{1}{\sigma_0^2} \sum_{i=1}^n X_i^2 > \frac{k_1}{\sigma_0^2} \right) = \alpha.$$

If we set $k_1 = \sigma_0^2 \times \chi_{\alpha, n}^2$, the above is ensured. Hence we reject H_0 if:

$$\sum_{i=1}^n x_i^2 > \sigma_0^2 \times \chi_{\alpha, n}^2.$$

(b) For $n = 10$, under $H_0 : \sigma^2 = \sigma_0^2 = 1$ we have:

$$\sum_{i=1}^n X_i^2 \sim \chi_{10}^2.$$

To test against $H_1 : \sigma^2 = 2$, we reject H_0 if and only if:

$$\sum_{i=1}^n x_i^2 > \chi_{\alpha, 10}^2 = 18.30 \quad \text{for } \alpha = 0.05.$$

To get the power note that if $\sigma^2 = 2$, then:

$$\sum_{i=1}^n \frac{X_i^2}{2} \sim \chi_{10}^2.$$

Hence the power is:

$$\begin{aligned} P\left(\sum_{i=1}^n X_i^2 > 18.30 \mid \sigma^2 = 2\right) &= P\left(\frac{1}{2} \sum_{i=1}^n X_i^2 > 9.15\right) \\ &= P(Y > 9.15) \\ &= 0.518 \end{aligned}$$

where $Y \sim \chi_{10}^2$.

(c) Note this test does not depend on σ_1^2 , and hence it is the most powerful test for any $\sigma_1^2 > \sigma_0^2$. Therefore, it would be the most powerful test for testing:

$$H_0 : \sigma^2 = \sigma_0^2 \quad \text{vs.} \quad H_1 : \sigma^2 > \sigma_0^2.$$

(d) For any $\sigma^2 \leq \sigma_0^2$, we have:

$$\begin{aligned} P_{\sigma^2}\left(\sum_{i=1}^n X_i^2 > \sigma_0^2 \times \chi_{\alpha, n}^2\right) &= P_{\sigma^2}\left(\frac{1}{\sigma^2} \sum_{i=1}^n X_i^2 > \frac{\sigma_0^2}{\sigma^2} \times \chi_{\alpha, n}^2\right) \\ &\leq P_{\sigma^2}\left(\frac{1}{\sigma^2} \sum_{i=1}^n X_i^2 > \chi_{\alpha, n}^2\right) \\ &= \alpha \end{aligned}$$

since:

$$\frac{\sigma_0^2}{\sigma^2} \times \chi_{\alpha, n}^2 \geq \chi_{\alpha, n}^2.$$

Since:

$$\sup_{\sigma^2 \leq \sigma_0^2} P_{\sigma^2}\left(\sum_{i=1}^n X_i^2 > \sigma_0^2 \times \chi_{\alpha, n}^2\right) = \alpha$$

this test is the desired uniformly most powerful test for testing:

$$H_0 : \sigma^2 \leq \sigma_0^2 \quad \text{vs.} \quad H_1 : \sigma^2 > \sigma_0^2.$$

Activity 5.3 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(0, \theta)$. Construct a most powerful test of:

$$H_0 : \theta = 1 \quad \text{vs.} \quad H_1 : \theta = 2.$$

Find the power of the test when $n = 10$ for a 5% significance level.

Activity 5.4 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(0, \sigma^2)$. Find the uniformly most powerful test of:

$$H_0 : \sigma^2 \leq \sigma_0^2 \quad \text{vs.} \quad H_1 : \sigma^2 > \sigma_0^2$$

where $\sigma_0^2 > 0$ is a given constant.

5.6 Likelihood ratio test

Let $\Theta = \Theta_0 \cup \Theta_0^c$ and consider hypothesis testing problems with:

$$H_0 : \boldsymbol{\theta} \in \Theta_0 \quad \text{vs.} \quad H_1 : \boldsymbol{\theta} \in \Theta_0^c.$$

We can think of H_0 as imposing a constraint on the possible values of $\boldsymbol{\theta}$. In this set-up, since $\Theta = \Theta_0 \cup \Theta_0^c$ the pair of hypotheses cover the whole parameter space. The likelihood ratio provides us with an explicit test statistic for this situation based on the likelihood function (hence the name).

Likelihood ratio test statistic

Let $\mathbf{X}$ be a random sample and define the likelihood function by $L_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X})$. The **likelihood ratio test statistic** associated with testing $H_0 : \boldsymbol{\theta} \in \Theta_0$ is:

$$LR(\mathbf{X}) = \frac{\sup_{\boldsymbol{\theta} \in \Theta} L_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X})}{\sup_{\boldsymbol{\theta} \in \Theta_0} L_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X})}$$

where Θ is the entire parameter space.

The likelihood ratio test statistic is connected with maximum likelihood estimation. Consider an observed sample $\mathbf{x}$. Hence the observed value of the likelihood ratio test statistic is:

$$LR(\mathbf{x}) = \frac{\sup_{\boldsymbol{\theta} \in \Theta} L_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x})}{\sup_{\boldsymbol{\theta} \in \Theta_0} L_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x})}.$$

The numerator is the *global maximum* of the likelihood function, attained at the maximum likelihood estimate $\hat{\boldsymbol{\theta}}$. The denominator is the maximum of the likelihood function *under the constraint* that $\boldsymbol{\theta} \in \Theta_0$. Let $\hat{\boldsymbol{\theta}}_0$ denote the maximum value of this constrained likelihood function. Hence the observed value of the likelihood ratio test

statistic can be expressed as:

$$LR(\mathbf{x}) = \frac{L_{\mathbf{X}}(\hat{\boldsymbol{\theta}}; \mathbf{x})}{L_{\mathbf{X}}(\hat{\boldsymbol{\theta}}_0; \mathbf{x})}.$$

This value must be ≥ 1 (since the constrained parameter estimate, $\hat{\boldsymbol{\theta}}_0$, cannot be better than the unconstrained estimate, $\hat{\boldsymbol{\theta}}$, in the sense that the associated likelihood value cannot be larger), although rejection of H_0 only occurs if the value is significantly greater than 1, i.e. when $\hat{\boldsymbol{\theta}}_0$ is significantly worse than $\hat{\boldsymbol{\theta}}$.

Activity 5.5 Briefly explain why:

$$\frac{L_{\mathbf{X}}(\hat{\boldsymbol{\theta}}; \mathbf{x})}{L_{\mathbf{X}}(\hat{\boldsymbol{\theta}}_0; \mathbf{x})} \geq 1$$

for all observed samples $\mathbf{x}$.

To determine ‘significantly worse’, we control the size of the test such that we choose a critical value, c , such that:

$$P_{\boldsymbol{\theta}_0}(LR(\mathbf{X}) > c) = \alpha.$$

In practice, the values of the likelihood function (upon which the observed value of the likelihood ratio test statistic is calculated) are determined based on a numerical optimisation procedure. As a consequence, the exact sampling distribution of the likelihood ratio is usually unknown. However, under certain conditions, an *approximate* sampling distribution is given to:

$$2 \log LR(\mathbf{X}).$$

Details are discussed later in the chapter after we cover a common application of likelihood ratio tests.

Example 5.12 Suppose that $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ and $\mathbf{Y} = \{Y_1, Y_2, \dots, Y_n\}$ are two independent random samples from two exponential distributions with means $\mu_1 = 1/\lambda_1$ and $\mu_2 = 1/\lambda_2$, respectively. We want to test:

$$H_0 : \mu_1 = \mu_2 \quad \text{vs.} \quad H_1 : \mu_1 \neq \mu_2.$$

Note that under H_0 the parameters are the same, so we may write $\mu_1 = \mu_2 = \mu$, where μ is the common parameter value.

The likelihood function is:

$$L_{\mathbf{X}, \mathbf{Y}}(\mu_1, \mu_2; \mathbf{x}, \mathbf{y}) = \mu_1^{-n} \exp\left(-\sum_{i=1}^n \frac{x_i}{\mu_1}\right) \mu_2^{-n} \exp\left(-\sum_{i=1}^n \frac{y_i}{\mu_2}\right).$$

Under the unconstrained model of H_1 we have $\hat{\mu}_1 = \bar{X}$ and $\hat{\mu}_2 = \bar{Y}$ since the sample means are the maximum likelihood estimators of the respective population means.

Under the constrained model of H_0 we get:

$$\hat{\mu} = \frac{\bar{X} + \bar{Y}}{2}$$

which is the maximum likelihood estimator of μ . Note here that since the sample sizes of $\mathbf{X}$ and $\mathbf{Y}$ are both n , this is simply the arithmetic mean (i.e. unweighted mean) of $\bar{X}$ and $\bar{Y}$.

Hence the likelihood ratio test statistic is:

$$LR(\mathbf{X}, \mathbf{Y}) = \frac{L(\hat{\mu}_1, \hat{\mu}_2; \mathbf{x}, \mathbf{y})}{L(\hat{\mu}; \mathbf{x}, \mathbf{y})} = \frac{(\bar{X} + \bar{Y})^{2n}/2^{2n}}{\bar{X}^n \bar{Y}^n} = 2^{-2n} \left(\sqrt{\frac{\bar{X}}{\bar{Y}}} + \sqrt{\frac{\bar{Y}}{\bar{X}}} \right)^{2n}.$$

We do not know the distribution of the likelihood ratio test statistic. We may attempt to isolate $T = \bar{X}/\bar{Y}$, but since $LR(\mathbf{X}, \mathbf{Y})$ is not monotone in T we cannot construct a test.

We will see later in the chapter how to deal with such cases by constructing asymptotic tests.

Example 5.13 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $N(\mu, \sigma^2)$, with σ^2 known. Suppose we test:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu \neq \mu_0.$$

The maximum likelihood estimator of μ is $\hat{\mu} = \bar{X}$, hence the likelihood ratio test statistic is:

$$\begin{aligned} LR(\mathbf{X}) &= \frac{L_{\mathbf{X}}(\hat{\mu}; \mathbf{X})}{L_{\mathbf{X}}(\mu_0; \mathbf{X})} = \frac{(2\pi\sigma^2)^{-n/2} \exp(-(n(\bar{X} - \hat{\mu})^2 + (n-1)S^2)/(2\sigma^2))}{(2\pi\sigma^2)^{-n/2} \exp(-(n(\bar{X} - \mu_0)^2 + (n-1)S^2)/(2\sigma^2))} \\ &= \exp\left(\frac{n(\bar{X} - \mu_0)^2}{2\sigma^2}\right). \end{aligned}$$

The test $LR(\mathbf{X}) > k$ is equivalent to the test:

$$\left| \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \right| > k_1.$$

We can write the previous rejection region as:

$$R = \left\{ \mathbf{x} : \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} < -k_1 \right\} \cup \left\{ \mathbf{x} : \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} > k_1 \right\}$$

or:

$$R = \left\{ \mathbf{x} : \bar{x} < \mu_0 - k_1 \times \frac{\sigma}{\sqrt{n}} \right\} \cup \left\{ \mathbf{x} : \bar{x} \geq \mu_0 + k_1 \times \frac{\sigma}{\sqrt{n}} \right\}.$$

Since:

$$\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$$

if we set $k_1 = z_{\alpha/2}$ the size of the test will be α . Hence:

$$\begin{aligned} P_{\mu_0}(\mathbf{X} \in R) &= P_{\mu_0} \left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} < -z_{\alpha/2} \right) + P_{\mu_0} \left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} > z_{\alpha/2} \right) \\ &= \Phi(z_{1-\alpha/2}) + 1 - \Phi(z_{\alpha/2}) \\ &= \frac{\alpha}{2} + 1 - \left(1 - \frac{\alpha}{2} \right) \\ &= \alpha. \end{aligned}$$

Equivalently we can also use the fact that:

$$\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1) \Rightarrow \frac{(\bar{X} - \mu_0)^2}{\sigma^2/n} \sim \chi_1^2.$$

Note also that:

$$\frac{(\bar{X} - \mu_0)^2}{\sigma^2/n} = 2 \log \left(\exp \left(\frac{n(\bar{X} - \mu_0)^2}{2\sigma^2} \right) \right) = 2 \log LR(\mathbf{X}) \sim \chi_1^2.$$

5.7 Testing in the presence of nuisance parameters

Suppose that a random sample $\mathbf{X}$ is drawn from a distribution parameterised by the vector of parameters $\boldsymbol{\theta}$. Now suppose that $\boldsymbol{\theta}$ can be split into two groups of parameters: $\boldsymbol{\psi}$ being the main parameters of interest, and $\boldsymbol{\lambda}$ being the parameters which are of little interest – known as **nuisance parameters**. Hence we may write:

$$\boldsymbol{\theta} = \begin{pmatrix} \boldsymbol{\psi} \\ \boldsymbol{\lambda} \end{pmatrix} = (\boldsymbol{\psi}^T, \boldsymbol{\lambda}^T)^T$$

where the final transpose ensures that $\boldsymbol{\theta}$ is a column vector. So $\boldsymbol{\theta}$, $\boldsymbol{\psi}$ and $\boldsymbol{\lambda}$ are just lists of scalar parameters.

For brevity, we may omit the transposes and simply write:

$$\boldsymbol{\theta} = (\boldsymbol{\psi}, \boldsymbol{\lambda})$$

and note it could be that $\boldsymbol{\psi}$ and/or $\boldsymbol{\lambda}$ contain a single (i.e. scalar) parameter.

Since we have little-to-no interest in the values of the nuisance parameters (i.e. in $\boldsymbol{\lambda}$), we may wish to test the value(s) of the main parameter(s) by testing:

$$H_0 : \boldsymbol{\psi} = \boldsymbol{\psi}_0 \quad \text{vs.} \quad H_1 : \boldsymbol{\psi} \neq \boldsymbol{\psi}_0.$$

The alternative hypothesis is clearly a composite hypothesis, but what about H_0 ? Well, given we place no constraint on the value(s) of the nuisance parameter(s), H_0 is also a composite hypothesis which is more readily apparent if we write the hypotheses as:

$$H_0 : \boldsymbol{\psi} = \boldsymbol{\psi}_0, \boldsymbol{\lambda} \in \boldsymbol{\Lambda} \quad \text{vs.} \quad H_1 : \boldsymbol{\psi} \neq \boldsymbol{\psi}_0, \boldsymbol{\lambda} \in \boldsymbol{\Lambda}$$

where $\boldsymbol{\Lambda}$ is the parameter space of $\boldsymbol{\lambda}$.

Using this specification, we can think of H_0 as imposing a constraint on the model being fitted. The constraint is that $\boldsymbol{\psi}$ must take the values specified by $\boldsymbol{\psi}_0$. Hence the likelihood ratio test statistic compares two models, each with a relative advantage.

- The **unconstrained model** will provide a better fit since the associated likelihood will be larger. We denote the unconstrained maximum likelihood estimate by:

$$\hat{\boldsymbol{\theta}} = (\hat{\boldsymbol{\psi}}, \hat{\boldsymbol{\lambda}}).$$

- The **constrained model** has fewer parameters requiring estimation. If the dimension of $\boldsymbol{\psi}$ is k , then the values of the k parameters contained in $\boldsymbol{\psi}$ are fixed, such that we only need to estimate the nuisance parameters which are in $\boldsymbol{\lambda}$, where $\dim(\boldsymbol{\lambda}) < \dim(\boldsymbol{\theta})$. Under the constraint that $\boldsymbol{\psi} = \boldsymbol{\psi}_0$, the maximum likelihood estimate of $\boldsymbol{\lambda}$ is generated by maximising:

$$L_{\mathbf{X}}(\boldsymbol{\psi}_0, \boldsymbol{\lambda}; \mathbf{x})$$

with respect to $\boldsymbol{\lambda}$. We denote the resulting estimate as $\hat{\boldsymbol{\lambda}}_0$. Hence the constrained maximum likelihood estimate of $\boldsymbol{\theta}$ is:

$$\hat{\boldsymbol{\theta}}_0 = (\boldsymbol{\psi}_0, \hat{\boldsymbol{\lambda}}_0).$$

Likelihood ratio test in the presence of nuisance parameters

When testing:

$$H_0 : \boldsymbol{\psi} = \boldsymbol{\psi}_0, \boldsymbol{\lambda} \in \Lambda \quad \text{vs.} \quad H_1 : \boldsymbol{\psi} \neq \boldsymbol{\psi}_0, \boldsymbol{\lambda} \in \Lambda$$

the likelihood ratio test statistic is:

$$LR(\mathbf{x}) = \frac{\sup_{\boldsymbol{\psi}, \boldsymbol{\lambda}} L_{\mathbf{X}}(\boldsymbol{\psi}, \boldsymbol{\lambda}; \mathbf{x})}{\sup_{\boldsymbol{\lambda}} L_{\mathbf{X}}(\boldsymbol{\lambda}; \boldsymbol{\psi}_0, \mathbf{x})}.$$

The test may be viewed as a comparison between two models:

- the constrained model under H_0 with parameters $\boldsymbol{\psi}_0$ and $\boldsymbol{\lambda}$
- the unconstrained model under H_1 with parameters $\boldsymbol{\psi}$ and $\boldsymbol{\lambda}$.

Generally, in statistics, models with many parameters have better fit but do not always give better predictions. **Parsimonious models** achieve a good fit with not too many parameters and usually perform better in terms of prediction. The likelihood ratio tests provide a useful tool for finding parsimonious models.

In particular if we tested $H_0 : \boldsymbol{\psi} = \mathbf{0}$ then we effectively remove k parameters (where k is the number of parameters in $\boldsymbol{\psi}$) from the model. This produces a trade-off between the number of parameters and goodness of fit of the model. Goodness of fit can always be improved by adding more parameters to the model, but will likely suffer from less reliable predictions.

According to the **principle of parsimony** (also known as **Occam's razor**) in modelling we value simplicity over complexity, other things equal. A 'good' model is one which provides a sufficiently adequate fit with the smallest number of parameters. The likelihood ratio test is used in practical situations to help determine which features of a model are important, and which features can be ignored.

Example 5.14 Consider a random sample $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$, where $X_1 \sim N(\lambda, 1)$ and $X_i \sim N(\lambda + \psi_i, 1)$, for $i = 2, 3, \dots, n$.

We want to test for differences in the means, i.e. we test:

$$H_0 : \psi_i = 0 \quad \text{for all } i \quad \text{vs.} \quad H_1 : \psi_i \neq 0 \quad \text{for some } i.$$

Here H_0 means that all of the observations have the same mean, λ , while H_1 means that *at least one* differs from the others.

The likelihood function is:

$$L_{\mathbf{X}}(\psi_2, \psi_3, \dots, \psi_n, \lambda; \mathbf{X}) = \left(\frac{1}{\sqrt{2\pi}} \right)^n \exp \left(-\frac{1}{2}(X_1 - \lambda)^2 - \frac{1}{2} \sum_{i=2}^n (X_i - \lambda - \psi_i)^2 \right).$$

In the unconstrained case, we are estimating n means from n observations which will result in maximum likelihood estimators of:

$$\hat{\lambda} = X_1 \quad \text{and} \quad \hat{\lambda} + \hat{\psi}_i = X_i \quad \text{for } i = 2, 3, \dots, n.$$

Under H_0 , we have that $\hat{\lambda}_0 = \bar{X}$. Therefore, the likelihood ratio test statistic is:

$$\begin{aligned} LR(\mathbf{X}) &= \frac{L_{\mathbf{X}}(\hat{\psi}_2, \hat{\psi}_3, \dots, \hat{\psi}_n, \hat{\lambda}; \mathbf{X})}{L_{\mathbf{X}}(0, 0, \dots, 0, \hat{\lambda}_0; \mathbf{X})} \\ &= \exp \left(\frac{1}{2} \sum_{i=1}^n (X_i - \bar{X})^2 \right). \end{aligned}$$

We will reject H_0 for sufficiently large values of:

$$\sum_{i=1}^n (X_i - \bar{X})^2.$$

5.8 Properties of the likelihood ratio

We saw in Chapter 3 that maximum likelihood estimators have many desirable statistical properties. This motivates us in this section to establish statistical properties of likelihood ratio tests (given they make use of maximum likelihood estimators of unknown parameters, nuisance or otherwise).

5.8.1 Likelihood ratio tests and sufficiency

The likelihood ratio test has a strong connection with sufficiency. Indeed, we can show that the observed likelihood ratio test statistic calculated using sufficient statistics is identical to that calculated using the full random sample.

Likelihood ratio test and sufficiency

Let $\mathbf{X}$ be a random sample parameterised by $\boldsymbol{\theta}$ and suppose we are interested in testing:

$$H_0 : \boldsymbol{\theta} \in \Theta_0.$$

If $T(\mathbf{X})$ is a sufficient statistic for $\boldsymbol{\theta}$, then the likelihood ratio statistic based on $T(\mathbf{X})$ is identical to that based on $\mathbf{X}$.

Equivalently, let $LR(\mathbf{X})$ and $LR^*(T(\mathbf{X}))$ be the likelihood ratio tests based on $\mathbf{X}$ and $T(\mathbf{X})$, respectively. For every $\mathbf{x}$ in the sample space:

$$LR(\mathbf{x}) = LR^*(T(\mathbf{x})).$$

Proof: We first clarify what is meant by ‘the likelihood ratio statistic based on $T(\mathbf{X})$ ’. We define the associated test statistic as:

$$LR^*(T(\mathbf{x})) = \frac{\sup_{\boldsymbol{\theta} \in \Theta} L_{T(\mathbf{x})}(\boldsymbol{\theta}; T(\mathbf{x}))}{\sup_{\boldsymbol{\theta} \in \Theta_0} L_{T(\mathbf{x})}(\boldsymbol{\theta}; T(\mathbf{x}))}$$

where $L_{T(\mathbf{x})}$ is the likelihood function of $T(\mathbf{X})$. It remains to show that:

$$LR(\mathbf{x}) = LR^*(T(\mathbf{x})) \quad \text{for all } \mathbf{x}.$$

We use the fact that, since $T(\mathbf{X})$ is sufficient for $\boldsymbol{\theta}$, the conditional mass/density $f_{\mathbf{X}|T(\mathbf{X})}$ does not depend on $\boldsymbol{\theta}$. Hence, for any observed sample $\mathbf{x}$, we have:

$$\begin{aligned} LR(\mathbf{x}) &= \frac{\sup_{\boldsymbol{\theta} \in \Theta} L_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x})}{\sup_{\boldsymbol{\theta} \in \Theta_0} L_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x})} \\ &= \frac{\sup_{\boldsymbol{\theta} \in \Theta} f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta})}{\sup_{\boldsymbol{\theta} \in \Theta_0} f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta})} \\ &= \frac{\sup_{\boldsymbol{\theta} \in \Theta} f_{T(\mathbf{X})}(T(\mathbf{x}); \boldsymbol{\theta}) f_{\mathbf{X}|T(\mathbf{X})}(\mathbf{x} | T(\mathbf{x}))}{\sup_{\boldsymbol{\theta} \in \Theta_0} f_{T(\mathbf{X})}(T(\mathbf{x}); \boldsymbol{\theta}) f_{\mathbf{X}|T(\mathbf{X})}(\mathbf{x} | T(\mathbf{x}))} \\ &= \frac{\sup_{\boldsymbol{\theta} \in \Theta} f_{T(\mathbf{X})}(T(\mathbf{x}); \boldsymbol{\theta})}{\sup_{\boldsymbol{\theta} \in \Theta_0} f_{T(\mathbf{X})}(T(\mathbf{x}); \boldsymbol{\theta})} \\ &= \frac{\sup_{\boldsymbol{\theta} \in \Theta} L_{T(\mathbf{X})}(\boldsymbol{\theta}; T(\mathbf{x}))}{\sup_{\boldsymbol{\theta} \in \Theta_0} L_{T(\mathbf{X})}(\boldsymbol{\theta}; T(\mathbf{x}))} \\ &= LR^*(T(\mathbf{x})). \end{aligned}$$

■

5.8.2 Asymptotic sampling distributions in likelihood ratio testing

We have previously mentioned that the sampling distribution of the likelihood ratio test statistic is usually unknown. We now consider two results related to the *asymptotic* sampling distribution of $2 \log LR(\mathbf{X})$, without proof.

Asymptotic distribution of likelihood ratio (simple case)

Let $\mathbf{X}$ be a random sample parameterised by the scalar parameter θ , and let $LR(\mathbf{X})$ be the likelihood ratio test statistic associated with testing $H_0 : \theta = \theta_0$.

Under H_0 , then:

$$2 \log LR(\mathbf{X}) \xrightarrow{d} \chi_1^2$$

as $n \rightarrow \infty$, provided certain regulatory conditions hold.

The number of degrees of freedom of the asymptotic chi-squared distribution depends on the number of *restrictions* which we place on the model. In the above case there is a single restriction as θ is a scalar parameter, which can be thought of as a vector of dimension 1. We consider the case of nuisance parameters, which recall we denote by $\boldsymbol{\lambda}$, such that we are interested in testing k parameters of interest represented by $\boldsymbol{\psi}$.

Asymptotic distribution of likelihood ratio (nuisance parameter case)

Let $\mathbf{X}$ be a random sample parameterised by the vector parameter $\boldsymbol{\theta} = (\boldsymbol{\psi}, \boldsymbol{\lambda})$, where k is the dimension of $\boldsymbol{\psi}$. Let $LR(\mathbf{X})$ be the likelihood ratio test statistic associated with testing $H_0 : \boldsymbol{\psi} = \boldsymbol{\psi}_0$.

Under H_0 , then:

$$2 \log LR(\mathbf{X}) \xrightarrow{d} \chi_k^2$$

as $n \rightarrow \infty$, provided certain regulatory conditions hold.

Example 5.15 Continuing Example 5.14, it is clear that:

$$2 \log LR(\mathbf{X}) = \sum_{i=1}^n (X_i - \bar{X})^2.$$

We know that if $X_i \sim N(\mu, \sigma^2)$ then:

$$\sum_{i=1}^n (X_i - \bar{X})^2 \sim \chi_1^2.$$

Therefore, in this case the *asymptotic* sampling distribution of the test statistic is also the *exact* sampling distribution of the test statistic for any sample size n . The rejection region for a test of size α is hence:

$$R = \left\{ \mathbf{x} : \sum_{i=1}^n (x_i - \bar{x})^2 > \chi_{\alpha, 1}^2 \right\}.$$

If we were to reject H_0 in this test, we would conclude that there is evidence that the mean of at least one observation differs from the others.

Example 5.16 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $\text{Exp}(\mu)$, where $\mu = 1/\lambda$. We want to test:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu \neq \mu_0.$$

The maximum likelihood estimator of μ is $\hat{\mu} = \bar{X}$.

The likelihood ratio test statistic is:

$$LR(\mathbf{X}) = \frac{\sup_{\mu > 0} L_{\mathbf{X}}(\mu; \mathbf{X})}{L_{\mathbf{X}}(\mu_0; \mathbf{X})} = \frac{\hat{\mu}^{-n} \exp(-n\bar{X}/\hat{\mu})}{\mu_0^{-n} \exp(-n\bar{X}/\mu_0)} = \frac{\bar{X}^{-n} \exp(-n)}{\mu_0^{-n} \exp(-n\bar{X}/\mu_0)}.$$

We cannot construct an exact test since:

- the distribution of $LR(\mathbf{X})$ is unknown
- the distribution of $\bar{X}$ is known, but $LR(\mathbf{X})$ is non-monotone in $\bar{X}$.

Consider the quantity:

$$2 \log LR(\mathbf{X}) = 2n \left(\log(\mu_0) - \log(\bar{X}) - \left(1 - \frac{\bar{X}}{\mu_0} \right) \right).$$

Under H_0 , we have:

$$2 \log LR(\mathbf{X}) \xrightarrow{d} \chi_1^2$$

as $n \rightarrow \infty$. Hence the asymptotic likelihood ratio test of size α rejects H_0 if:

$$2 \log LR(\mathbf{x}) > \chi_{\alpha, 1}^2$$

where $\chi_{\alpha, 1}^2$ is top 100α th percentile of a χ_1^2 distribution.

Example 5.17 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from $\text{Pois}(\lambda)$. We want to test:

$$H_0 : \lambda = \lambda_0 \quad \text{vs.} \quad H_1 : \lambda \neq \lambda_0.$$

The maximum likelihood estimator of λ is $\hat{\lambda} = \bar{X}$.

The likelihood ratio test statistic is:

$$LR(\mathbf{X}) = \frac{\sup_{\lambda > 0} L_{\mathbf{X}}(\lambda; \mathbf{X})}{L_{\mathbf{X}}(\lambda_0; \mathbf{X})} = \frac{\exp(-n\hat{\lambda})\hat{\lambda}^{n\bar{X}}}{\exp(-n\lambda_0)\lambda_0^{n\bar{X}}} = \exp(n(\lambda_0 - \bar{X})) \left(\frac{\lambda_0}{\bar{X}} \right)^{-n\bar{X}}.$$

We cannot construct an exact test since:

- the distribution of $LR(\mathbf{X})$ is unknown
- the distribution of $\bar{X}$ is known, but $LR(\mathbf{X})$ is non-monotone in $\bar{X}$.

Consider the quantity:

$$2 \log LR(\mathbf{X}) = 2n \left((\lambda_0 - \bar{X}) - \bar{X} \log \left(\frac{\lambda_0}{\bar{X}} \right) \right).$$

Under H_0 , we have:

$$2 \log LR(\mathbf{X}) \xrightarrow{d} \chi_1^2$$

as $n \rightarrow \infty$. Hence the asymptotic likelihood ratio test of size α rejects H_0 if:

$$2 \log LR(\mathbf{x}) > \chi_{\alpha,1}^2.$$

Example 5.18 (A fuller version of Example 5.16.) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from an exponential distribution with $\mu = 1/\lambda$. In other words for each X_i , $i = 1, 2, \dots, n$, the density function is:

$$f_X(x) = \begin{cases} \mu^{-1} e^{-x/\mu} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where $\lambda > 0$ is an unknown parameter.

- (a) Obtain the test statistic for the likelihood ratio test for testing:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu \neq \mu_0.$$

- (b) State the asymptotic distribution of this likelihood ratio test statistic under H_0 .
 (c) If $n = 5$, $t = 9.6$ and $\mu_0 = 1$, will you reject H_0 at the 5% significance level?

Solution

- (a) The joint density function (and hence likelihood function) of the random sample is:

$$f_{\mathbf{X}}(\mathbf{x}; \mu) = L_{\mathbf{X}}(\mu; \mathbf{x}) = \mu^{-n} \exp \left(-\frac{1}{\mu} \sum_{i=1}^n x_i \right)$$

and the log-likelihood function for the observed sample $\mathbf{x}$ is:

$$l_{\mathbf{X}}(\mu; \mathbf{x}) = -n \log(\mu) - \frac{1}{\mu} \sum_{i=1}^n x_i$$

and the score function is:

$$s_{\mathbf{X}}(\mu; \mathbf{x}) = -\frac{n}{\mu} + \frac{1}{\mu^2} \sum_{i=1}^n x_i.$$

The score function is equal to 0 only at $\mu = \bar{X}$. The second derivative of the log-likelihood is:

$$\frac{\partial^2}{\partial \mu^2} l_{\mathbf{X}}(\mu; \mathbf{x}) = \frac{n}{\mu^2} - \frac{2n\bar{x}}{\mu^3}$$

which is negative at $\mu = \bar{x}$. Hence we conclude that the likelihood function is maximised at $\bar{X}$.

The likelihood ratio statistic is:

$$\begin{aligned} LR(\mathbf{X}) &= \frac{L_{\mathbf{X}}(\hat{\mu}; \mathbf{X})}{L_{\mathbf{X}}(\mu_0; \mathbf{X})} = \frac{\hat{\mu}^{-n} \exp\left(-\sum_{i=1}^n X_i / \hat{\mu}\right)}{\mu_0^{-n} \exp\left(-\sum_{i=1}^n X_i / \mu_0\right)} \\ &= \frac{\bar{X}^{-n} \exp(-n)}{\mu_0^{-n} \exp(-n\bar{X}/\mu_0)}. \end{aligned}$$

H_0 is rejected if $LR(\mathbf{x}) > k$, where k is a suitably-chosen constant.

(b) Under H_0 , we have:

$$2 \log LR(\mathbf{X}) = 2n \left(\log(\mu_0) - \log(\bar{X}) - \left(1 - \frac{\bar{X}}{\mu_0}\right) \right) \xrightarrow{d} \chi_1^2$$

as $n \rightarrow \infty$. We reject H_0 if:

$$2 \log LR(\mathbf{x}) > \chi_{\alpha,1}^2.$$

(c) Note that:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{9.6}{5} = 1.92.$$

Plugging into $2 \log LR(\mathbf{x})$ gives:

$$2 \log LR(\mathbf{x}) = 2.677.$$

Since this value is less than $\chi_{0.05,1}^2 = 3.841$, we do not reject H_0 .

Example 5.19 Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ and $\mathbf{Y} = \{Y_1, Y_2, \dots, Y_m\}$ be two independent random samples drawn from two Poisson distributions with means $\mu > 0$ and $\lambda > 0$, respectively.

(a) Write down the likelihood function for (μ, λ) . Find the maximum likelihood estimators of (μ, λ) . Also, find the maximum likelihood estimator of μ when $\mu = \lambda$. (No need to check second derivatives.)

(b) Find the likelihood ratio statistic for testing:

$$H_0 : \mu = \lambda \quad \text{vs.} \quad H_1 : \mu \neq \lambda.$$

(c) Under H_0 the distribution of $2 \log LR(\mathbf{X}, \mathbf{Y})$ is approximately χ_1^2 . Test H_0 at the 10% significance level when:

$$n = 20, \quad m = 20, \quad \sum_{i=1}^{20} x_i = 18.4 \quad \text{and} \quad \sum_{j=1}^{20} y_j = 25.6.$$

Solution

(a) The likelihood function is:

$$L_{\mathbf{X}, \mathbf{Y}}(\mu, \lambda; \mathbf{X}, \mathbf{Y}) = \prod_{i=1}^n \left(\frac{e^{-\mu} \mu^{X_i}}{X_i!} \right) \prod_{j=1}^m \left(\frac{e^{-\lambda} \lambda^{Y_j}}{Y_j!} \right) = \frac{\mu^{\sum_{i=1}^n X_i} \lambda^{\sum_{j=1}^m Y_j}}{\prod_{i=1}^n X_i! \prod_{j=1}^m Y_j!} e^{-n\mu - m\lambda}.$$

The log-likelihood function for (μ, λ) is:

$$l_{\mathbf{X}, \mathbf{Y}}(\mu, \lambda; \mathbf{X}, \mathbf{Y}) = \sum_{i=1}^n X_i \log(\mu) - n\mu + \sum_{j=1}^m Y_j \log(\lambda) - m\lambda + c$$

where c is independent of μ and λ .

The score functions for μ and λ are, respectively:

$$s_{\mathbf{X}}(\mu; \mathbf{X}) = \frac{1}{\mu} \sum_{i=1}^n X_i - n \quad \text{and} \quad s_{\mathbf{Y}}(\lambda; \mathbf{Y}) = \frac{1}{\lambda} \sum_{j=1}^m Y_j - m$$

which lead to the maximum likelihood estimators of μ and λ to be $\hat{\mu} = \bar{X}$ and $\hat{\lambda} = \bar{Y}$, respectively.

When $\mu = \lambda$, the log-likelihood function is:

$$l_{\mathbf{X}, \mathbf{Y}}(\mu, \mu; \mathbf{X}, \mathbf{Y}) = \log L_{\mathbf{X}, \mathbf{Y}}(\mu, \mu; \mathbf{X}, \mathbf{Y}) = \left(\sum_{i=1}^n X_i + \sum_{j=1}^m Y_j \right) \log \mu - (m+n)\mu$$

leading to the maximum likelihood estimator:

$$\tilde{\mu} = \frac{1}{m+n} \left(\sum_{i=1}^n X_i + \sum_{j=1}^m Y_j \right).$$

(b) The likelihood ratio statistic is:

$$\begin{aligned} LR(\mathbf{X}, \mathbf{Y}) &= \frac{L_{\mathbf{X}, \mathbf{Y}}(\hat{\mu}, \hat{\lambda}; \mathbf{X}, \mathbf{Y})}{L_{\mathbf{X}, \mathbf{Y}}(\tilde{\mu}, \tilde{\mu}; \mathbf{X}, \mathbf{Y})} \\ &= \frac{\hat{\mu}^{\sum_{i=1}^n X_i} \hat{\lambda}^{\sum_{j=1}^m Y_j} \exp \left(- \sum_{i=1}^n X_i \right) \exp \left(- \sum_{j=1}^m Y_j \right)}{\tilde{\mu}^{\sum_{i=1}^n X_i + \sum_{j=1}^m Y_j} \exp \left(- \left(\sum_{i=1}^n X_i + \sum_{j=1}^m Y_j \right) \right)} \\ &= \frac{\hat{\mu}^{\sum_{i=1}^n X_i} \hat{\lambda}^{\sum_{j=1}^m Y_j}}{\tilde{\mu}^{\sum_{i=1}^n X_i + \sum_{j=1}^m Y_j}}. \end{aligned}$$

(c) With the given data:

$$\tilde{\mu} = \frac{18.4 + 25.6}{40} = 1.1, \quad \hat{\mu} = \frac{18.4}{20} = 0.92 \quad \text{and} \quad \hat{\lambda} = \frac{25.6}{20} = 1.28$$

hence $LR(\mathbf{x}, \mathbf{y}) = 1.8071$. Under H_0 , then:

$$2 \log LR(\mathbf{X}, \mathbf{Y}) \xrightarrow{d} \chi_1^2.$$

Since $2 \log LR(\mathbf{x}, \mathbf{y}) = 1.1835$, which is smaller than $\chi_{0.10,1}^2 = 2.706$, we do not reject H_0 .

Activity 5.6 Suppose that $\{X_1, X_2, \dots, X_n\}$ and $\{Y_1, Y_2, \dots, Y_n\}$ are two independent random samples from $\text{Exp}(\lambda_1)$ and $\text{Exp}(\lambda_2)$, respectively. Find the likelihood ratio test of:

$$H_0 : \lambda_1 = \lambda_2 \quad \text{vs.} \quad H_1 : \lambda_1 \neq \lambda_2.$$

Specify the asymptotic distribution of the test statistic under H_0 .

Activity 5.7 A survey of the use of a particular product was conducted in four areas. In each area, 200 randomly selected potential users were interviewed. The number from area i who said that they would use the product was x_i . Construct a likelihood ratio test to determine whether the proportion of the population using the product is the same in each of the four areas. Perform the test at the 5% significance level for the case $x_1 = 76$, $x_2 = 53$, $x_3 = 59$ and $x_4 = 48$.

5.9 Other tests based on the likelihood

We briefly consider some other statistical tests based on the likelihood. You may meet these in future studies.

The Wald test

The **Wald test** is suitable for testing simple null hypotheses such as:

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta \neq \theta_0.$$

The test statistic is:

$$\frac{\hat{\theta} - \theta_0}{\text{S.E.}(\hat{\theta})} \xrightarrow{d} N(0, 1)$$

where $\hat{\theta}$ is the maximum likelihood estimator of θ , and a reasonable estimator of its standard error is:

$$\text{S.E.}(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})}$$

given by Fisher information. Asymptotically, the Wald test statistic is distributed as $N(0, 1)$.

The score test

Similar to the Wald test, when testing $H_0 : \theta = \theta_0$, the **score test** takes the form:

$$Z = \frac{S_{\mathbf{X}}(\theta_0; \mathbf{X})}{\sqrt{\mathcal{I}_{\mathbf{X}}(\theta_0)}} \xrightarrow{d} N(0, 1)$$

where $S_{\mathbf{X}}(\theta_0; \mathbf{X})$ is the score function and $\mathcal{I}_{\mathbf{X}}(\theta_0)$ is the Fisher information. Asymptotically, the score test statistic is distributed as $N(0, 1)$.

Multivariate versions of the above tests exist. These tests are similar to the likelihood ratio test, but not identical. As with the likelihood ratio test, their distributions are generally unknown. For ‘large’ sample sizes the likelihood ratio, score and Wald tests are equivalent.

5.10 A reminder of your learning outcomes

On completion of this chapter, you should be able to:

- explain the distinction between simple and composite hypotheses, and the distinction between the size and significance level of a statistical test
- define Type I and Type II errors and describe the association between these errors and the concepts of size and power
- summarise the use of p -values in hypothesis testing
- construct hypothesis tests using pivotal functions
- state and prove the Neyman–Pearson lemma
- construct most powerful tests for simple hypotheses
- determine whether a test is uniformly most powerful
- explain the rationale behind the likelihood ratio test
- describe how the likelihood ratio test is used in the presence of nuisance parameters
- construct likelihood ratio tests for standard cases.

5.11 Sample examination questions

Solutions can be found in Appendix C.

1. Let $X \sim \text{Bin}(n, \pi)$ with $n = 10$ and $\pi = 0.25$ or 0.50 . Suppose we test:

$$H_0 : \pi = 0.50 \quad \text{vs.} \quad H_1 : \pi = 0.25$$

and H_0 is rejected if the observed value x (i.e. a sample of size of $n = 1$) is less than or equal to 3.

5. Hypothesis testing

- (a) Find the size and the power of the test.
- (b) Suppose that we observe $x = 1$. Find the p -value of the test.

2. Let $\mathbf{X} = (X_1, X_2)$ be a random sample from an exponential distribution. We want to test the hypotheses:

$$H_0 : \mu = 2 \quad \text{vs.} \quad H_1 : \mu = 1.$$

Consider a particular test which rejects H_0 if the observed values x_1 and x_2 are such that:

$$\frac{f_X(x_1; \mu = 2) f_X(x_2; \mu = 2)}{f_X(x_1; \mu = 1) f_X(x_2; \mu = 1)} \leq \frac{1}{2}.$$

Find the size and the power of the test.

3. Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a distribution with the following density function:

$$f_X(x; \beta) = \beta x^{\beta-1}, \quad \text{for } 0 \leq x \leq 1, \beta > 0$$

and 0 otherwise.

- (a) Consider the test $H_0 : \beta = \beta_0$ vs. $H_1 : \beta \neq \beta_0$. Find the likelihood ratio test statistic.
- (b) Use the likelihood ratio test statistic to provide the rejection region of an asymptotic test at a $100\alpha\%$ significance level, where $\alpha \in [0, 1]$.
- (c) Describe how you would obtain an asymptotic p -value for the test in part (b).

Appendix A

Non-examinable proofs

A.1 Chapter 2 – Fundamentals of statistical inference

Minimal sufficiency

Consider the partition generated by the statistic $\mathbf{U} = \mathbf{T}(\mathbf{X})$ such that $\mathbf{T} : P \rightarrow Q$ for some sets P and Q . As with any function, $\mathbf{T}$ defines a partition on its domain P . We say $\mathbf{x}$ and $\mathbf{y}$ belong to the same element of the partition if $\mathbf{T}(\mathbf{x}) = \mathbf{T}(\mathbf{y})$. The elements of this partition can be indexed by the range Q , so for $\mathbf{q} \in Q$ we have:

$$A_{\mathbf{q}} = \{\mathbf{x} \in P : \mathbf{T}(\mathbf{x}) = \mathbf{q}\}.$$

Proof: In order to establish that a collection of sets form a partition, we have to establish that the members are mutually exclusive and collectively exhaustive. Mutual exclusivity comes from the fact that T is a function. Suppose $\mathbf{x} \in A_i$ and $\mathbf{x} \in A_j$. By definition this means $T(\mathbf{x}) = i$ and $T(\mathbf{x}) = j$. Since T is a function, we must have $i = j$ and, hence, $A_i = A_j$. Exhaustiveness simply comes from the fact that T is defined on P . We might also want to insist that for every $q \in Q$ we can find an $\mathbf{x} \in P$, such that $T(\mathbf{x}) = q$. This then guarantees that $A_q \neq \emptyset$ for any q . ■

Characterisation of minimal sufficiency

Proof: We prove sufficiency first. Let $\mathbf{T} : P \rightarrow Q$ and consider the partition generated by $\mathbf{T}$, such that:

$$A_{\mathbf{q}} = \{\mathbf{x} \in P : \mathbf{T}(\mathbf{x}) = \mathbf{q}\}.$$

Let $\mathbf{y}(\mathbf{q}) \in P$ be a fixed point in $A_{\mathbf{q}}$, hence:

$$\mathbf{y}(\mathbf{q}) = \mathbf{y}(\mathbf{T}(\mathbf{x})) \quad \text{for all } \mathbf{x} \in A_{\mathbf{q}}.$$

We now define the following functions:

$$\mathbf{d} : P \rightarrow P \quad \text{by } \mathbf{d}(\mathbf{x}) = \mathbf{y}(\mathbf{T}(\mathbf{x}))$$

$$c : P \rightarrow \mathbb{R} \quad \text{by } c(\mathbf{x}) = k(\mathbf{x}, \mathbf{d}(\mathbf{x}))$$

$$b : Q \rightarrow \mathbb{R} \quad \text{by } b(\mathbf{q}, \boldsymbol{\theta}) = f_{\mathbf{x}}(\mathbf{y}(\mathbf{q}); \boldsymbol{\theta}).$$

For any point $\mathbf{x} \in P$ we have $\mathbf{x} \in A_{\mathbf{q}}$ where $\mathbf{T}(\mathbf{x}) = \mathbf{T}(\mathbf{y}(\mathbf{q}))$ and hence:

$$\begin{aligned} f_{\mathbf{x}}(\mathbf{x}; \boldsymbol{\theta}) &= k(\mathbf{x}, \mathbf{y}(\mathbf{q})) f_{\mathbf{x}}(\mathbf{y}(\mathbf{q}); \boldsymbol{\theta}) \\ &= k(\mathbf{x}, \mathbf{y}(\mathbf{T}(\mathbf{x}))) f_{\mathbf{x}}(\mathbf{y}(\mathbf{T}(\mathbf{x})); \boldsymbol{\theta}) \\ &= k(\mathbf{x}, \mathbf{d}(\mathbf{x})) b(\mathbf{T}(\mathbf{x}), \boldsymbol{\theta}) \\ &= c(\mathbf{x}) b(\mathbf{T}(\mathbf{x}), \boldsymbol{\theta}). \end{aligned}$$

To prove minimal sufficiency, we consider a sufficient statistic $\mathbf{V} = \mathbf{S}(\mathbf{X})$. By the factorisation theorem, we can find functions g and h such that:

$$f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) = h(\mathbf{x}) g(\mathbf{S}(\mathbf{x}), \boldsymbol{\theta}).$$

Therefore, if $\mathbf{S}(\mathbf{x}) = \mathbf{S}(\mathbf{y})$, then:

$$f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) = h(\mathbf{x}) g(\mathbf{S}(\mathbf{y}), \boldsymbol{\theta}) = \frac{h(\mathbf{x})}{h(\mathbf{y})} f_{\mathbf{X}}(\mathbf{y}; \boldsymbol{\theta}) = k(\mathbf{x}, \mathbf{y}) f_{\mathbf{X}}(\mathbf{y}; \boldsymbol{\theta}).$$

By the conditions in the proposition, this implies that $\mathbf{T}(\mathbf{x}) = \mathbf{T}(\mathbf{y})$. Hence we can find a function $\mathbf{R}$ such that $\mathbf{T}(\mathbf{x}) = \mathbf{R}(\mathbf{S}(\mathbf{x}))$ for all $\mathbf{x}$ and hence $\mathbf{U} = \mathbf{R}(\mathbf{V})$. Therefore, $\mathbf{U}$ is minimal sufficient. ■

A.2 Chapter 3 – Point estimation

Proof of the central limit theorem.

Proof: For the proof we assume that the moment generating function of the X_i s exists. If we define:

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$$

then:

$$Z_n = \frac{S_n - n\mu}{\sqrt{n}\sigma}$$

where $S_n = \sum_{i=1}^n X_i$. We can obtain an expression for the moment generating function of Z in terms of $M_X(t)$ by noting that:

$$M_{S_n}(t) = E(e^{tS_n}) = E\left(e^{t \sum_{i=1}^n X_i}\right) = E\left(\prod_{i=1}^n e^{tX_i}\right) = \prod_{i=1}^n E(e^{tX_i}) = (M_X(t))^n$$

since the X_i s are independent and identically distributed. Hence:

$$\begin{aligned} M_{Z_n}(t) &= E(e^{tZ_n}) = E\left(\exp\left(t\left(\frac{S_n - n\mu}{\sqrt{n}\sigma}\right)\right)\right) \\ &= \exp\left(-\frac{\sqrt{n}\mu t}{\sigma}\right) E\left(\exp\left(\frac{t}{\sqrt{n}\sigma} S_n\right)\right) \\ &= \exp\left(-\frac{\sqrt{n}\mu t}{\sigma}\right) M_{S_n}\left(\frac{t}{\sqrt{n}\sigma}\right) \\ &= \exp\left(-\frac{\sqrt{n}\mu t}{\sigma}\right) \left(M_X\left(\frac{t}{\sqrt{n}\sigma}\right)\right)^n. \end{aligned}$$

Taking logs produces the cumulant generating function:

$$K_{Z_n}(t) = -\frac{\sqrt{n}\mu t}{\sigma} + nK_X\left(\frac{t}{\sqrt{n}\sigma}\right).$$

By the properties of cumulants, we have:

$$K_X(t) = \mu t + \frac{\sigma^2 t^2}{2} + \text{terms in } t^3 \text{ and higher order.}$$

Hence:

$$\begin{aligned} K_{Z_n}(t) &= -\frac{\sqrt{n}\mu t}{\sigma} + n \left(\mu \frac{t}{\sqrt{n}\sigma^2} + \sigma^2 \frac{t^2}{2n\sigma^2} + \text{terms involving } \left(\frac{1}{n}\right)^{3/2} \text{ and higher order} \right) \\ &= \frac{t^2}{2} + \text{terms involving } \left(\frac{1}{n}\right)^{1/2} \text{ and higher order.} \end{aligned}$$

As $n \rightarrow \infty$, the terms involving $(1/n)^{1/2}$ and higher order will tend to zero. Therefore:

$$K_{Z_n}(t) \rightarrow \frac{t^2}{2} \quad \text{as } n \rightarrow \infty.$$

Hence the cumulant generating function of Z_n converges to that of a standard normal distribution. Since there is a one-to-one mapping between cumulant generating function and distribution of a random variable, we have that:

$$Z_n \xrightarrow{d} N(0, 1) \quad \text{as } n \rightarrow \infty.$$

■

Four representations of the sample variance

i. We will make use of the fact that:

$$\sum_{i=1}^n (X_i - \bar{X}_n) = \sum_{i=1}^n X_i - n\bar{X}_n = 0.$$

Expanding the expression for $(n-1)S_n^2$ and using the above we have:

$$\begin{aligned} (n-1)S_n^2 &= \sum_{i=1}^n (X_i - \bar{X}_n)^2 \\ &= \sum_{i=1}^n X_i(X_i - \bar{X}_n) - \bar{X}_n \sum_{i=1}^n (X_i - \bar{X}_n) \\ &= \sum_{i=1}^n X_i(X_i - \bar{X}_n). \end{aligned}$$

ii. Following on from part i., multiplying out the remaining bracket on the right-hand side gives the result:

$$\begin{aligned} (n-1)S_n^2 &= \sum_{i=1}^n (X_i^2 - X_i\bar{X}_n) \\ &= \sum_{i=1}^n X_i^2 - \bar{X}_n \sum_{i=1}^n X_i \\ &= \sum_{i=1}^n X_i^2 - n\bar{X}_n^2. \end{aligned}$$

iii. Note that:

$$\sum_{i=1}^n (X_i - \bar{X}_n) = 0 \quad \Rightarrow \quad (X_1 - \bar{X}_n) = - \sum_{i=2}^n (X_i - \bar{X}_n)$$

hence:

$$(n-1)S_n^2 = (X_1 - \bar{X}_n)^2 + \sum_{i=2}^n (X_i - \bar{X}_n)^2 = \left(\sum_{i=2}^n (X_i - \bar{X}_n) \right)^2 + \sum_{i=2}^n (X_i - \bar{X}_n)^2.$$

iv. We prove directly:

$$\begin{aligned} (n-1)S_n^2 &= \sum_{i=1}^n (X_i - \bar{X}_n)^2 \\ &= \sum_{i=1}^n \left((X_i - \bar{X}_{n-1}) - \frac{1}{n}(X_n - \bar{X}_{n-1}) \right)^2 \\ &= \sum_{i=1}^n \left((X_i - \bar{X}_{n-1})^2 - \frac{2}{n}(X_i - \bar{X}_{n-1})(X_n - \bar{X}_{n-1}) + \frac{1}{n^2}(X_n - \bar{X}_{n-1})^2 \right). \end{aligned}$$

Using the fact that:

$$\sum_{i=1}^{n-1} (X_i - \bar{X}_{n-1}) = \sum_{i=1}^{n-1} X_i - (n-1)\bar{X}_{n-1} = 0$$

then:

$$\begin{aligned} (n-1)S_n^2 &= \sum_{i=1}^{n-1} (X_i - \bar{X}_{n-1})^2 + (X_n - \bar{X}_{n-1})^2 - \frac{2}{n}(X_n - \bar{X}_{n-1})^2 + \frac{1}{n}(X_n - \bar{X}_{n-1})^2 \\ &= (n-2)S_{n-1}^2 + \frac{n-1}{n}(X_n - \bar{X}_{n-1})^2. \end{aligned}$$

■

Statistical properties of the sample variance

Throughout this proof we will use the fact that for central sample moments we may assume a population mean of zero and assume, without loss of generality, that $E(X) = 0$ and hence $E(X^2) = \text{Var}(X) = \sigma^2$. We will also use the technique of splitting $(X_i - \bar{X})$ into independent terms as follows:

$$(X_i - \bar{X}) = \left(X_i - \frac{1}{n} \sum_{j=1}^n X_j \right) = \left(\left(X_i - \frac{1}{n} X_i \right) - \frac{1}{n} \sum_{j \neq i} X_j \right) = (V_i - W_i)$$

where:

$$V_i = X_i - \frac{1}{n} X_i = \frac{n-1}{n} X_i \quad \text{and} \quad W_i = \frac{1}{n} \sum_{j \neq i} X_j.$$

Hence we may write:

$$(n-1)S^2 = \sum_{i=1}^n X_i(V_i - W_i).$$

Using the fact that $E(X) = 0$ and hence $E(X_i) = 0$ for all $i = 1, 2, \dots, n$, we have $E(V_i) = 0$ and $E(W_i) = 0$. The V_i s and W_i s are identically distributed because the X_i s are identically distributed. Since $X_i \perp\!\!\!\perp X_j$ for $i \neq j$, we have that $X_i \perp\!\!\!\perp W_i$, $V_i \perp\!\!\!\perp W_i$ and, for $i \neq j$, $X_i \perp\!\!\!\perp V_j$.

i. We have:

$$\begin{aligned} E((n-1)S^2) &= E\left(\sum_{i=1}^n X_i(V_i - W_i)\right) \\ &= \sum_{i=1}^n (E(X_i V_i) - E(X_i W_i)) \\ &= \sum_{i=1}^n E(X_i V_i) \quad (\text{since } X_i \perp\!\!\!\perp W_i \text{ and } E(X_i) = E(W_i) = 0) \\ &= \sum_{i=1}^n \frac{n-1}{n} E(X_i^2) \\ &= \frac{n-1}{n} n\sigma^2 \\ &= (n-1)\sigma^2 \end{aligned}$$

and so $E(S^2) = \sigma^2$.

ii. Using standard results for variance and unbiasedness, we have:

$$\text{Var}(S^2) = E(S^4) - (E(S^2))^2 = E(S^4) - \sigma^4.$$

So we need to calculate $E(S^4)$. We have:

$$\begin{aligned} E((n-1)^2 S^4) &= E\left(\left(\sum_{i=1}^n X_i(V_i - W_i)\right)^2\right) \\ &= \sum_{i=1}^n \sum_{j=1}^n E(X_i(V_i - W_i)X_j(V_j - W_j)) \\ &= \sum_{i=1}^n \sum_{j=1}^n (E(X_i V_i X_j V_j) - E(X_i V_i X_j W_j) - E(X_i W_i X_j V_j) + E(X_i W_i X_j W_j)). \end{aligned}$$

When $i = j$, the second and third terms of the summand are zero because $W_i \perp\!\!\!\perp X_i$, $W_i \perp\!\!\!\perp V_i$ and $E(W_i) = 0$. When $i \neq j$, the second term is zero because $X_j \perp\!\!\!\perp X_i$, $X_j \perp\!\!\!\perp V_i$, $X_j \perp\!\!\!\perp W_j$ and $E(X_j) = 0$. By swapping i and j an identical

argument shows that the third term is also zero when $i \neq j$. Since the sequences are identically distributed:

$$\begin{aligned} E((n-1)^2 S^4) &= \sum_{i=1}^n (E(X_i^2 V_i^2) + E(X_i^2) E(W_i^2)) + \sum_{i \neq j} \sum_{j=1}^n ((E(X_i V_i))^2 + (E(X_i W_j))^2) \\ &= n \left(\frac{(n-1)^2}{n^2} \mu'_4 + \frac{n-1}{n^2} (\mu'_2)^2 \right) + n(n-1) \left(\frac{(n-1)^2}{n^2} (\mu'_2)^2 + \frac{1}{n^2} (\mu'_2)^2 \right). \end{aligned}$$

Therefore:

$$E(S^4) = \frac{1}{n} \mu'_4 + \frac{n^2 - 2n + 3}{n(n-1)} (\mu'_2)^2$$

and so:

$$\text{Var}(S^2) = \frac{1}{n} \mu'_4 + \frac{n^2 - 2n + 3}{n(n-1)} (\mu'_2)^2 - (\mu'_2)^2 = \frac{1}{n} \left(\mu'_4 - \frac{n-3}{n-1} \sigma^4 \right).$$

■

Association between the sample mean and sample variance

We proceed using the approach and terminology introduced above.

- i. We know that $E(\bar{X}) = 0$ and $E(X_i - \bar{X}) = 0$, so to prove that $\bar{X}$ and $(X_i - \bar{X})$ are uncorrelated, we just need to verify that $E(\bar{X}(X_i - \bar{X})) = 0$. We have:

$$\begin{aligned} E(\bar{X}(X_i - \bar{X})) &= E \left(\frac{1}{n} \sum_{j=1}^n X_j (V_i - W_i) \right) \\ &= \frac{1}{n} \sum_{j=1}^n (E(X_j V_i) - E(X_j W_i)) \\ &= \frac{1}{n} \left(E(X_i V_i) - \sum_{j \neq i} E(X_j W_i) \right) \\ &= \frac{1}{n} \left(\frac{n-1}{n} \sigma^2 - (n-1) \frac{1}{n} \sigma^2 \right) \\ &= 0. \end{aligned}$$

- ii. The covariance result can now be shown directly:

$$\begin{aligned} \text{Cov}(\bar{X}, S^2) &= E(\bar{X} S^2) - E(\bar{X}) E(S^2) \\ &= E(\bar{X} S^2) \\ &= E \left(\left(\frac{1}{n} \sum_{i=1}^n X_i \right) \left(\frac{1}{n-1} \sum_{j=1}^n X_j (V_j - W_j) \right) \right) \\ &= \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j=1}^n (E(X_i X_j V_j) - E(X_i X_j W_j)). \end{aligned}$$

When $i \neq j$, the first term in the summand is zero because $X_j \perp\!\!\!\perp X_i$ and $X_j \perp\!\!\!\perp V_j$. The second term is also zero because $X_j \perp\!\!\!\perp X_i$ and $X_j \perp\!\!\!\perp W_j$. When $i = j$, the second term is zero because $W_i \perp\!\!\!\perp X_i$. Hence:

$$\text{Cov}(\bar{X}, S^2) = \frac{1}{n(n-1)} \sum_{i=1}^n \mathbb{E}(X_i^2 V_i) = \frac{1}{n(n-1)} n \frac{n-1}{n} \mu'_3 = \frac{1}{n} \mu'_3.$$

■

Density function of the i th order statistic

Proof: We begin by considering the definition of the density function as a derivative:

$$f_{X_{(i)}}(x) = \frac{d}{dx} F_{X_{(i)}}(x) = \lim_{h \rightarrow 0} \frac{F_{X_{(i)}}(x+h) - F_{X_{(i)}}(x)}{h} = \lim_{h \rightarrow 0} \frac{P(x < X_{(i)} \leq x+h)}{h}.$$

Consider the event $x < X_{(i)} \leq x+h$, i.e. the event that the i th smallest value falls in the interval $(x, x+h]$. Partitioning the real line, $\mathbb{R}$, into three intervals:

$$(-\infty, x], \quad (x, x+h] \quad \text{and} \quad (x+h, \infty)$$

then the probabilities that an individual sample value falls into these intervals are, respectively:

$$F_X(x), \quad F_X(x+h) - F_X(x) \quad \text{and} \quad 1 - F_X(x+h).$$

For $x < X_{(i)} \leq x+h$ to occur, we require at least one observation in $(x, x+h]$, at most $i-1$ observations in $(-\infty, x]$ and at most $n-i$ observations in $(x+h, \infty)$. Hence $P(x < X_{(i)} \leq x+h)$ is the sum of values from a trinomial mass function:

$$\begin{aligned} P(x < X_{(i)} \leq x+h) \\ = \sum_{j=1}^n \sum_{k=\min(i-j, 0)}^{\max(i-1, n-j)} \frac{n!}{k! j! (n-k-j)!} (F_X(x))^k (F_X(x+h) - F_X(x))^j (1 - F_X(x+h))^{n-k-j}. \end{aligned} \tag{1}$$

Consider the term $(F_X(x+h) - F_X(x))^j$. Since f_X is the derivative of F_X , we have:

$$F_X(x+h) - F_X(x) = h f_X(x) + o(h)$$

and hence:

$$(F_X(x+h) - F_X(x))^j = o(h) \quad \text{for } j > 1.$$

Dividing by h and taking the limit as $h \rightarrow 0$ in (1), the only term which does not go to zero is the one corresponding to $j = 1$, hence:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{P(x < X_{(i)} \leq x+h)}{h} \\ = \frac{n!}{(i-1)! 1! (n-i)!} (F_X(x))^{i-1} \frac{F_X(x+h) - F_X(x)}{h} (1 - F_X(x+h))^{n-i}. \end{aligned}$$

The result then follows from the properties of limits of functions since:

$$\lim_{h \rightarrow 0} \frac{F_X(x+h) - F_X(x)}{h} = f_X(x)$$

by definition, and:

$$\lim_{h \rightarrow 0} F_X(x+h) = F_X(x)$$

by right continuity. ■

Joint density function of the i th and j th order statistics

We begin from the definition of $f_{X_{(i)}, X_{(j)}}$ as the derivative of $F_{X_{(i)}, X_{(j)}}$, such that:

$$\begin{aligned} f_{X_{(i)}, X_{(j)}}(u, v) &= \frac{\partial}{\partial v} \frac{\partial}{\partial u} F_{X_{(i)}, X_{(j)}}(u, v) \\ &= \lim_{h \rightarrow 0} \frac{F_{X_{(i)}, X_{(j)}}(u+h, v+h) - F_{X_{(i)}, X_{(j)}}(u+h, v) - F_{X_{(i)}, X_{(j)}}(u, v+h) + F_{X_{(i)}, X_{(j)}}(u, v)}{h^2} \\ &= \lim_{h \rightarrow 0} \frac{P(u < X_{(i)} \leq u+h, v < X_{(j)} \leq v+h)}{h^2}. \end{aligned}$$

Consider $P(u < X_{(i)} \leq u+h, v < X_{(j)} \leq v+h)$, i.e. the probability that the i th order statistic is in $(u, u+h]$ and the j th order statistic is in $(v, v+h]$. Partitioning the real line, $\mathbb{R}$, into five intervals:

$$(-\infty, u], \quad (u, u+h], \quad (u+h, v], \quad (v, v+h] \quad \text{and} \quad (v+h, \infty)$$

then the probabilities that an individual sample value falls into these intervals are, respectively:

$$F_X(u), \quad F_X(u+h) - F_X(u), \quad F_X(v) - F_X(u+h), \quad F_X(v+h) - F_X(v) \quad \text{and} \quad 1 - F_X(v+h).$$

Hence $P(u < X_{(i)} \leq u+h, v < X_{(j)} \leq v+h)$ is the sum of values from a multinomial mass function (with n trials and five possible outcomes for each trial).

Dividing by h^2 and taking the limit as $h \rightarrow 0$, the only terms which do not go to zero are the ones corresponding to having one observation in $(u, u+h]$ and one observation in $(v, v+h]$. Therefore:

$$\begin{aligned} &\lim_{h \rightarrow 0} \frac{P(u < X_{(i)} \leq u+h, v < X_{(j)} \leq v+h)}{h^2} \\ &= \lim_{h \rightarrow 0} \left(\frac{n!}{(i-1)! 1! (j-i-1)! 1! (n-j)!} (F_X(u))^{i-1} \frac{F_X(u+h) - F_X(u)}{h} \right. \\ &\quad \left. (F_X(v) - F_X(u+h))^{j-i-1} \frac{F_X(v+h) - F_X(v)}{h} (1 - F_X(v+h))^{n-j} \right). \end{aligned}$$

The result follows by the properties of limits of functions. ■

Asymptotic normality of maximum likelihood estimators

Assuming a random sample $\mathbf{X}$ we begin by considering the score function evaluated at the maximum likelihood estimator. The Taylor series expansion of $s_{\mathbf{X}}(\hat{\theta}; \mathbf{X})$ around the true value θ gives us:

$$s_{\mathbf{X}}(\hat{\theta}; \mathbf{X}) = s_{\mathbf{X}}(\theta; \mathbf{X}) + (\hat{\theta} - \theta) \frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}) + o((\hat{\theta} - \theta)^2) + \dots$$

Since $\hat{\theta}$ is a consistent estimator of θ , we would hope that the term in $(\hat{\theta} - \theta)^2$ and higher-order terms would contribute very little to the probability distribution associated with $s_{\mathbf{X}}(\hat{\theta}; \mathbf{X})$, hence it is reasonable to assume that:

$$s_{\mathbf{X}}(\hat{\theta}; \mathbf{X}) \approx s_{\mathbf{X}}(\theta; \mathbf{X}) + (\hat{\theta} - \theta) \frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}).$$

In regular cases, the score function evaluated at $\hat{\theta}$ is zero, that is:

$$s_{\mathbf{X}}(\hat{\theta}; \mathbf{X}) = 0$$

hence:

$$s_{\mathbf{X}}(\theta; \mathbf{X}) + (\hat{\theta} - \theta) \frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X}) \approx 0.$$

Rearranging and multiplying both sides by $\sqrt{n}$ gives us:

$$\sqrt{n}(\hat{\theta} - \theta) \approx \frac{\sqrt{n} \frac{s_{\mathbf{X}}(\theta; \mathbf{X})}{n}}{-\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X})}. \quad (1)$$

Since we have assumed a random sample, we know that the score function can be expressed as the sum of the marginal score functions:

$$s_{\mathbf{X}}(\theta; \mathbf{X}) = \sum_{i=1}^n s_X(\theta; X_i).$$

The X_i s are independent and identically distributed, hence the random variables $s_X(\theta; X_i)$ must also be independent and identically distributed. We have that:

$$E(s_X(\theta; X_i)) = 0 \quad \text{and} \quad \text{Var}(s_X(\theta; X_i)) = \mathcal{I}_X(\theta).$$

With some additional weak assumptions, the conditions for the central limit theorem are met, hence:

$$\frac{s_{\mathbf{X}}(\theta; \mathbf{X})}{\sqrt{n}} = \sqrt{n} \frac{\sum_{i=1}^n s_X(\theta; X_i)}{n} \xrightarrow{d} N(0, \mathcal{I}_{\mathbf{X}}(\theta)).$$

Therefore, the numerator of (1) converges in distribution to $N(0, \mathcal{I}_{\mathbf{X}}(\theta))$. The denominator of (1) is the sample mean of a sequence of independent and identically distributed random variables of the form:

$$-\frac{\partial}{\partial \theta} s_X(\theta; X)$$

and we know that:

$$E\left(-\frac{\partial}{\partial \theta} s_X(\theta; X)\right) = \mathcal{I}_X(\theta).$$

By the law of large numbers, the denominator of (1) converges to $\mathcal{I}_{\mathbf{X}}(\theta)$. The result then follows immediately. ■

Appendix B

Solutions to Activities

B.1 Chapter 2 – Fundamentals of statistical inference

1. Begin with the left-hand side and proceed as follows:

$$\sum_{i=1}^n \sum_{j=1}^n (x_i - x_j)^2 = \sum_{i=1}^n \left[\sum_{j=1}^n (x_i - x_j)^2 \right].$$

Now, expand the square:

$$= \sum_{i=1}^n \left[\sum_{j=1}^n (x_i^2 - 2x_i x_j + x_j^2) \right].$$

Next, sum separately inside [] so we have:

$$= \sum_{i=1}^n \left[\sum_{j=1}^n x_i^2 + \sum_{j=1}^n (-2x_i x_j) + \sum_{j=1}^n x_j^2 \right].$$

Now, factor out x_i terms inside [] to give:

$$= \sum_{i=1}^n \left[x_i^2 \sum_{j=1}^n 1 - 2x_i \sum_{j=1}^n x_j + \sum_{j=1}^n x_j^2 \right].$$

Now, recall that $\bar{x} = \sum_{i=1}^n x_i / n$, so re-write as:

$$= \sum_{i=1}^n \left[nx_i^2 - 2x_i n\bar{x} + \sum_{j=1}^n x_j^2 \right].$$

Next, expand the bracket:

$$= \sum_{i=1}^n nx_i^2 + \sum_{i=1}^n (-2x_i n\bar{x}) + \sum_{i=1}^n \left(\sum_{j=1}^n x_j^2 \right).$$

Re-arrange again:

$$= n \sum_{i=1}^n x_i^2 - 2n\bar{x} \sum_{i=1}^n x_i + \left(\sum_{j=1}^n x_j^2 \right) \sum_{i=1}^n 1.$$

Apply the ' $\bar{x}$ trick' once more:

$$= n \sum_{i=1}^n x_i^2 - 2n\bar{x} \times n\bar{x} + \left(\sum_{j=1}^n x_j^2 \right) n.$$

Factor out the common n to give:

$$= n \left[\sum_{i=1}^n x_i^2 - 2n\bar{x}^2 + \sum_{j=1}^n x_j^2 \right].$$

Without loss of generality, we can re-define the index j as index i so:

$$= n \left[\sum_{i=1}^n x_i^2 - 2n\bar{x}^2 + \sum_{i=1}^n x_i^2 \right].$$

Finally, add terms, factor out $2n$, apply the ' $\bar{x}$ trick' ... and you're done!

$$= 2n \left[\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right] = 2n \left[\sum_{i=1}^n (x_i - \bar{x})^2 \right].$$

2. (a) For $x_i = 0, 1, 2, \dots$, the joint density function is:

$$f_{\mathbf{X}}(\mathbf{x}; \lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = \frac{\lambda^{\sum_{i=1}^n x_i} e^{-n\lambda}}{\prod_{i=1}^n x_i!}.$$

By the factorisation theorem, we can set:

$$g\left(\sum_{i=1}^n x_i, \lambda\right) = \lambda^{\sum_{i=1}^n x_i} e^{-n\lambda} \quad \text{and} \quad h(\mathbf{x}) = \frac{1}{\prod_{i=1}^n x_i!}$$

hence $\sum_{i=1}^n X_i$ is a sufficient statistic for λ .

- (b) For $x_i = 1, 2, \dots$, the joint density function is:

$$f_{\mathbf{X}}(\mathbf{x}; \pi) = \prod_{i=1}^n (1 - \pi)^{x_i-1} \pi = (1 - \pi)^{\sum_{i=1}^n x_i - n} \pi^n.$$

By the factorisation theorem, we can set:

$$g\left(\sum_{i=1}^n x_i, \pi\right) = (1 - \pi)^{\sum_{i=1}^n x_i - n} \pi^n \quad \text{and} \quad h(\mathbf{x}) = 1$$

hence $\sum_{i=1}^n X_i$ is a sufficient statistic for π .

3. The intuition here is that any points which are indistinguishable in terms of the output of R (that is, $R(\mathbf{x}) = R(\mathbf{y})$) are also indistinguishable in terms of the output of T (that is, $T(\mathbf{x}) = T(\mathbf{y})$). Hence the output of R will preserve sufficient resolution for us to reconstruct T . A more rigorous argument follows.

Our approach is to define a mapping b by $T(\mathbf{x}) = b(R(\mathbf{x}))$ and show that it is a function. We know that, whenever $R(\mathbf{x}) = R(\mathbf{y})$, we have $T(\mathbf{x}) = T(\mathbf{y})$. Therefore, for any $p \in P$, if $b(p) = z_1$ and $b(p) = z_2$, then $z_1 = z_2$. We conclude that b is a function.

4. This result comes from direct calculation. Since $\mathbf{X}$ is a random sample we have:

$$f_{\mathbf{X}}(\mathbf{x}; \mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x_i - \mu)^2\right)$$

and:

$$f_{\mathbf{X}}(\mathbf{y}; \mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - \mu)^2\right).$$

Dividing these two terms yields:

$$\frac{f_{\mathbf{X}}(\mathbf{x}; \mu, \sigma^2)}{f_{\mathbf{X}}(\mathbf{y}; \mu, \sigma^2)} = \exp\left(-\frac{1}{2\sigma^2} \left(\sum_{i=1}^n (x_i - \mu)^2 - \sum_{i=1}^n (y_i - \mu)^2\right)\right).$$

Multiplying out the brackets and cancelling some terms then gives the result.

5. Using the original definition of Fisher information is considerably more tedious. Noting that $E(X_i) = 1/\lambda$ and $E(X_i^2) = 2/\lambda^2$ for $i = 1, 2, \dots, n$, we have:

$$\begin{aligned} \mathcal{I}_{\mathbf{X}}(\lambda) &= E\left(\left(\frac{\partial}{\partial \lambda} l_{\mathbf{X}}(\lambda; \mathbf{X})\right)^2\right) \\ &= E\left(\left(\frac{n}{\lambda} - \sum_{i=1}^n X_i\right)^2\right) \\ &= E\left(\frac{n^2}{\lambda^2} - \frac{2n}{\lambda} \sum_{i=1}^n X_i + \left(\sum_{i=1}^n X_i\right)^2\right) \\ &= \frac{n^2}{\lambda^2} - \frac{2n}{\lambda} \sum_{i=1}^n E(X_i) + \sum_{i=1}^n \sum_{j=1}^n E(X_i X_j) \\ &= \frac{n^2}{\lambda^2} - \frac{2n}{\lambda} \frac{n}{\lambda} + n E(X^2) + (n^2 - n)(E(X))^2 \\ &= \frac{n^2}{\lambda^2} - \frac{2n^2}{\lambda^2} + \frac{2n}{\lambda^2} + \frac{n^2 - n}{\lambda^2} \\ &= \frac{n}{\lambda^2}. \end{aligned}$$

In the above we are exploiting the fact that $\{X_1, X_2, \dots, X_n\}$ is a sequence of independent and identically distributed random variables. Hence:

$$E(X_i X_j) = \begin{cases} E(X^2) & \text{if } i = j \\ (E(X))^2 & \text{if } i \neq j. \end{cases}$$

When $n = 1$, the individual Fisher information is:

$$\mathcal{I}_X(\lambda) = \frac{1}{\lambda^2}.$$

6. For the normal distribution with σ^2 known, we have:

$$s_{\mathbf{X}}(\mu; \mathbf{X}) = \frac{\partial}{\partial \mu} l(\mu; \mathbf{X}) = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

and hence:

$$\mathcal{I}_{\mathbf{X}}(\mu) = -E \left(\frac{\partial^2}{\partial \mu^2} l(\mu; \mathbf{X}) \right) = -E \left(\frac{1}{\sigma^2} \sum_{i=1}^n (-1) \right) = -E \left(-\frac{n}{\sigma^2} \right) = \frac{n}{\sigma^2}.$$

When $n = 1$, the individual Fisher information is:

$$\mathcal{I}_X(\mu) = \frac{1}{\sigma^2}.$$

7. Throughout this question we will use the following:

$$l_X(\theta; x) = a(\theta) t(x) + b(\theta) + c(x)$$

$$s_X(\theta; x) = \frac{\partial}{\partial \theta} l_X(\theta; x) = a'(\theta) t(x) + b'(\theta)$$

$$\frac{\partial^2}{\partial \theta^2} l_X(\theta; x) = a''(\theta) t(x) + b''(\theta).$$

- (a) We know that $E(s_X(\theta; X)) = 0$, so:

$$E(a'(\theta) t(X) + b'(\theta)) = 0 \quad \Rightarrow \quad E(t(X)) = -\frac{b'(\theta)}{a'(\theta)}.$$

- (b) We use the original definition of Fisher information:

$$\begin{aligned} \mathcal{I}_X(\theta) &= -E \left(\frac{\partial^2}{\partial \theta^2} l_X(\theta; X) \right) = -E(a''(\theta) t(X) + b''(\theta)) \\ &= -a''(\theta) E(t(X)) - b''(\theta) \\ &= \frac{a''(\theta) b'(\theta) - a'(\theta) b''(\theta)}{a'(\theta)}. \end{aligned}$$

- (c) We use the definition of Fisher information as the variance of the score function:

$$\mathcal{I}_X(\theta) = \text{Var}(s_X(\theta; X)) = (a'(\theta))^2 \text{Var}(t(X))$$

hence:

$$\text{Var}(t(X)) = \frac{a''(\theta) b'(\theta) - a'(\theta) b''(\theta)}{(a'(\theta))^3}.$$

We apply these results to the binomial distribution. The log-likelihood for a single observation can be written as:

$$l_X(\pi; x) = (\log \pi - \log(1 - \pi))x + n \log(1 - \pi) + \log \binom{n}{x}.$$

In this application:

$$a(\pi) = (\log \pi - \log(1 - \pi))$$

$$a'(\pi) = \frac{1}{\pi} + \frac{1}{1 - \pi} = \frac{1}{\pi(1 - \pi)}$$

$$a''(\pi) = -\frac{1}{\pi^2} + \frac{1}{(1 - \pi)^2} = \frac{2\pi - 1}{\pi^2(1 - \pi)^2}$$

and:

$$b(\pi) = n \log(1 - \pi)$$

$$b'(\pi) = -\frac{n}{1 - \pi}$$

$$b''(\pi) = -\frac{n}{(1 - \pi)^2}.$$

Here t is just the identity function $t(X) = X$. Therefore:

$$\text{E}(X) = -\frac{b'(\pi)}{a'(\pi)} = \frac{n}{1 - \pi} \pi(1 - \pi) = n\pi$$

and:

$$\text{Var}(X) = \frac{a''(\pi) b'(\pi) - a'(\pi) b''(\pi)}{(a'(\pi))^3} = \dots = n\pi(1 - \pi).$$

8. We just need to work through the definitions and perform the calculus. We will use the second derivative of the log-likelihood function to calculate the Fisher information matrix. The log-likelihood function is:

$$l_{\mathbf{X}}(\boldsymbol{\pi}; \mathbf{x}) = c + x_1 \log \pi_1 + x_2 \log \pi_2 + (n - x_1 - x_2) \log(1 - \pi_1 - \pi_2)$$

where c does not depend on $\boldsymbol{\pi} = (\pi_1, \pi_2)$. The score vector function is:

$$s_{\mathbf{X}}(\boldsymbol{\pi}; \mathbf{x}) = \begin{pmatrix} \frac{x_1}{\pi_1} - \frac{n - x_1 - x_2}{1 - \pi_1 - \pi_2} \\ \frac{x_2}{\pi_2} - \frac{n - x_1 - x_2}{1 - \pi_1 - \pi_2} \end{pmatrix}.$$

The Fisher information matrix is then:

$$\begin{aligned}
 \mathcal{I}_{\mathbf{X}}(\boldsymbol{\pi}) &= -\mathbb{E}(\nabla_{\boldsymbol{\pi}} s_{\mathbf{X}}(\boldsymbol{\pi}; \mathbf{X})) \\
 &= -\mathbb{E} \left(\begin{pmatrix} \left(\frac{X_1}{\pi_1^2} - \frac{n - X_1 - X_2}{(1 - \pi_1 - \pi_2)^2} \right) & -\frac{n - X_1 - X_2}{(1 - \pi_1 - \pi_2)^2} \\ -\frac{n - X_1 - X_2}{(1 - \pi_1 - \pi_2)^2} & \frac{X_2}{\pi_2^2} - \frac{n - X_1 - X_2}{(1 - \pi_1 - \pi_2)^2} \end{pmatrix} \right) \\
 &\quad \vdots \\
 &= \begin{pmatrix} \frac{n}{\pi_1} + \frac{n}{1 - \pi_1 - \pi_2} & \frac{n}{1 - \pi_1 - \pi_2} \\ \frac{n}{1 - \pi_1 - \pi_2} & \frac{n}{\pi_2} + \frac{n}{1 - \pi_1 - \pi_2} \end{pmatrix}.
 \end{aligned}$$

9. The proof exploits the results for the univariate case.

i. The score vector function is defined as:

$$\mathbf{s}_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x}) = \left(\frac{\partial}{\partial \theta_1} l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x}), \frac{\partial}{\partial \theta_2} l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x}), \dots, \frac{\partial}{\partial \theta_r} l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x}) \right)^T.$$

Every element of this vector has zero expectation since:

$$\begin{aligned}
 \mathbb{E} \left(\frac{\partial}{\partial \theta_j} l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X}) \right) &= \int_{\mathbb{R}^n} \frac{1}{L_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x})} \frac{\partial}{\partial \theta_j} L_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{x}) f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) d\mathbf{x} \\
 &= \frac{\partial}{\partial \theta_j} \int_{\mathbb{R}^n} f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) d\mathbf{x} \\
 &= 0.
 \end{aligned}$$

Hence $\mathbb{E}(\mathbf{s}_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X})) = \mathbf{0}$.

ii. We start from the right-hand side:

$$\begin{aligned}
 \text{Var}(\mathbf{s}_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X})) &= \mathbb{E}(\mathbf{s}_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X}) \mathbf{s}_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X})^T) - \mathbb{E}(\mathbf{s}_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X})) \mathbb{E}(\mathbf{s}_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X}))^T \\
 &= \mathbb{E}(\mathbf{s}_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X}) \mathbf{s}_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X})^T) \\
 &= \mathcal{I}_{\mathbf{X}}(\boldsymbol{\theta}).
 \end{aligned}$$

iii. In what follows we will use the fact that:

$$\frac{\partial}{\partial \theta_i} f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) = \left(\frac{\partial}{\partial \theta_i} \log f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) \right) f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}).$$

We use a slight variant of the argument used in the univariate case.

$$\begin{aligned}
& \frac{\partial}{\partial \theta_i} \mathbb{E} \left(\frac{\partial}{\partial \theta_j} \log f_{\mathbf{X}}(\mathbf{X}; \boldsymbol{\theta}) \right) \\
&= \frac{\partial}{\partial \theta_i} \int_{\mathbb{R}^n} \left(\frac{\partial}{\partial \theta_j} \log f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) \right) f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) \, d\mathbf{x} \\
&= \int_{\mathbb{R}^n} \left(\left(\frac{\partial^2}{\partial \theta_i \partial \theta_j} \log f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) \right) f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) + \frac{\partial}{\partial \theta_j} \log f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) \frac{\partial}{\partial \theta_i} f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) \right) d\mathbf{x} \\
&= \int_{\mathbb{R}^n} \left(\frac{\partial^2}{\partial \theta_i \partial \theta_j} \log f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) + \frac{\partial}{\partial \theta_i} \log f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) \frac{\partial}{\partial \theta_j} \log f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) \right) f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) \, d\mathbf{x} \\
&= \mathbb{E} \left(\frac{\partial^2}{\partial \theta_i \partial \theta_j} \log f_{\mathbf{X}}(\mathbf{X}; \boldsymbol{\theta}) \right) + \mathbb{E} \left(\frac{\partial}{\partial \theta_i} \log f_{\mathbf{X}}(\mathbf{X}; \boldsymbol{\theta}) \frac{\partial}{\partial \theta_j} \log f_{\mathbf{X}}(\mathbf{X}; \boldsymbol{\theta}) \right) \\
&= \mathbb{E} \left(\frac{\partial^2}{\partial \theta_i \partial \theta_j} l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X}) \right) + \mathbb{E} \left(\frac{\partial}{\partial \theta_i} l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X}) \frac{\partial}{\partial \theta_j} l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X}) \right).
\end{aligned}$$

By part i., the left-hand side of the equation above is zero. By definition, the second term on the right-hand side is the (i, j) th element of $\mathcal{I}_{\mathbf{X}}(\boldsymbol{\theta})$. Therefore:

$$\mathcal{I}_{\mathbf{X}}(\boldsymbol{\theta})_{i,j} = -\mathbb{E} \left(\frac{\partial^2}{\partial \theta_i \partial \theta_j} l_{\mathbf{X}}(\boldsymbol{\theta}; \mathbf{X}) \right).$$

The result follows by definition of $\nabla_{\boldsymbol{\theta}}$. ■

B.2 Chapter 3 – Point estimation

1. To derive the condition for consistency, we require an expression for the expected value of $\hat{\mu}$, which is:

$$\begin{aligned}
\mathbb{E}(\hat{\mu}) &= \mathbb{E} \left(\sum_{i=1}^n a_i X_i \right) \\
&= \sum_{i=1}^n a_i \mathbb{E}(X_i) \\
&= \mu \sum_{i=1}^n a_i.
\end{aligned}$$

Hence for unbiasedness we require:

$$\sum_{i=1}^n a_i = 1.$$

Given that the condition for unbiasedness is met, the mean squared error is equal to the variance. We have:

$$\begin{aligned}
 \text{MSE}(\hat{\mu}) &= \text{Var}(\hat{\mu}) = \text{Var}\left(\sum_{i=1}^n a_i X_i\right) \\
 &= \text{E}\left(\sum_{i=1}^n a_i (X_i - \mu) \sum_{j=1}^n a_j (X_j - \mu)\right) \\
 &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j) \\
 &= \sum_{i=1}^n a_i^2 \text{Var}(X_i) + \sum_{j=1}^n \sum_{i \neq j} a_i a_j \text{Cov}(X_i, X_j) \\
 &= \sigma^2 \sum_{i=1}^n a_i^2 + \rho \sigma^2 \sum_{j=1}^n \sum_{i \neq j} a_i a_j.
 \end{aligned}$$

2. We have that $S \sim \text{Bin}(n, \pi)$. Therefore:

$$\text{E}(S) = n\pi, \quad \text{Var}(S) = n\pi(1 - \pi)$$

and:

$$\text{E}(S^2) = \text{Var}(S) + (\text{E}(S))^2 = n\pi(1 - \pi) + n^2\pi^2.$$

To check unbiasedness, we have:

$$\begin{aligned}
 \text{E}\left(\frac{S(S-1)}{n(n-1)}\right) &= \frac{1}{n(n-1)}(\text{E}(S^2) - \text{E}(S)) \\
 &= \frac{1}{n(n-1)}(n\pi(1 - \pi) + n^2\pi^2 - n\pi) \\
 &= \frac{1}{n(n-1)}(n(n-1)\pi^2) \\
 &= \pi^2.
 \end{aligned}$$

The weak law of large numbers tells us that $S/n \xrightarrow{p} \pi$. Hence:

$$\frac{S-1}{n-1} = \left(\frac{S}{n} - \frac{1}{n}\right) \left(1 - \frac{1}{n}\right)^{-1} \xrightarrow{p} \pi.$$

Using the result in Example 3.2 we have:

$$\frac{S(S-1)}{n(n-1)} \xrightarrow{p} \pi^2$$

and hence we have a consistent estimator of π^2 .

3. We use the definition of convergence in probability:

$$\begin{aligned} P(|\hat{\mu}_n - \mu| \geq \epsilon) &= P(|\bar{X} + Y_n - \mu| \geq \epsilon) \\ &\leq P\left(\left\{|\bar{X} - \mu| \geq \frac{\epsilon}{2}\right\} \cup \left\{|Y_n| \geq \frac{\epsilon}{2}\right\}\right) \\ &\leq P\left(|\bar{X} - \mu| \geq \frac{\epsilon}{2}\right) + P\left(|Y_n| \geq \frac{\epsilon}{2}\right). \end{aligned}$$

We know that $\bar{X} \xrightarrow{p} \mu$ and so:

$$P\left(|\bar{X} - \mu| \geq \frac{\epsilon}{2}\right) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

By the definition of Y_n , we have:

$$P\left(|Y_n| \geq \frac{\epsilon}{2}\right) = \frac{1}{n} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Therefore, $P(|\hat{\mu}_n - \mu| \geq \epsilon)$ is a positive sequence which is bounded above by a sequence which converges to zero. We conclude that:

$$\lim_{n \rightarrow \infty} P(|\hat{\mu}_n - \mu| \geq \epsilon) \rightarrow 0$$

and so:

$$\hat{\mu}_n \xrightarrow{p} \mu.$$

So this unappealing estimator is consistent. However, the point is that while consistency may be considered a desirable attribute for a ‘good’ estimator, it is certainly not a sufficient condition. Hence we cannot assume that a consistent estimator is a ‘good’ estimator.

4. Recall from **ST2133 Advanced statistics: distribution theory** that if $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$ then the joint moment generating function of X and Y is:

$$M_{X,Y}(t, u) = \exp\left(t E(X) + u E(Y) + \frac{1}{2}(t^2 \text{Var}(X) + 2tu \text{Cov}(X, Y) + u^2 \text{Var}(Y))\right). \quad (1)$$

Now consider the joint moment generating function of U and S , which is:

$$\begin{aligned} M_{U,S}(r, t) &= E(e^{rU+tS}) = E\left(\exp\left(rX_1 + t \sum_{i=1}^n X_i\right)\right) \\ &= E\left(\exp\left((r+t)X_1 + t \sum_{i=2}^n X_i\right)\right) \\ &= M_X(r+t)(M_X(t))^{n-1} \\ &= \exp\left((r+t)\mu + \frac{(r+t)^2}{2}\right) \left(\exp\left(\mu t + \frac{t^2}{2}\right)\right)^{n-1} \\ &= \exp\left(r\mu + t\mu + \frac{1}{2}(r^2 + 2rt + t^2) + (n-1)t\mu + \frac{(n-1)t^2}{2}\right) \\ &= \exp\left(r\mu + nt\mu + \frac{1}{2}(r^2 + 2rt + t^2n)\right). \end{aligned}$$

Comparison with (1) yields the required result.

5. We know that $\sum_{i=1}^n X_i$ is sufficient for π . By the Rao–Blackwell theorem:

$$E \left(X_1 \mid \sum_{i=1}^n X_i \right)$$

is an estimator of π with a lower mean squared error. Consider:

$$\sum_{j=1}^n E \left(X_j \mid \sum_{i=1}^n X_i \right) = E \left(\sum_{j=1}^n X_j \mid \sum_{i=1}^n X_i \right) = \sum_{i=1}^n X_i = n\bar{X}. \quad (1)$$

Note the symmetry here:

$$E \left(X_j \mid \sum_{i=1}^n X_i \right) = E \left(X_1 \mid \sum_{i=1}^n X_i \right)$$

for any $j = 1, 2, \dots, n$. Therefore:

$$\sum_{j=1}^n E \left(X_j \mid \sum_{i=1}^n X_i \right) = n E \left(X_1 \mid \sum_{i=1}^n X_i \right).$$

Comparison with (1) yields:

$$E \left(X_1 \mid \sum_{i=1}^n X_i \right) = \bar{X}.$$

We conclude, of course, that $\bar{X}$ is a better estimator of π than X_1 .

6. We begin with the moment generating function:

$$\begin{aligned} M_{\bar{X}_n}(t) &= E(e^{t\bar{X}_n}) = E \left(\exp \left(\frac{t}{n} \sum_{i=1}^n X_i \right) \right) \\ &= E \left(\prod_{i=1}^n \exp \left(\frac{t}{n} X_i \right) \right) && \text{(simplifying)} \\ &= \prod_{i=1}^n E \left(\exp \left(\frac{t}{n} X_i \right) \right) && \text{(by independence)} \\ &= \left(M_X \left(\frac{t}{n} \right) \right)^n. && \text{(as identically distributed)} \end{aligned}$$

Taking logarithms gives the cumulant generating function:

$$\begin{aligned} K_{\bar{X}_n}(t) &= nK_X \left(\frac{t}{n} \right) \\ &= n \left(\mu \frac{t}{n} + \frac{\sigma^2}{2} \frac{t^2}{n^2} + \text{terms in } \left(\frac{1}{n} \right)^3 \text{ and higher order} \right) \\ &= \mu t + \frac{\sigma^2}{2} \frac{t^2}{n} + \text{terms in } \left(\frac{1}{n} \right)^2 \text{ and higher order.} \end{aligned}$$

We conclude that $K_{\bar{X}_n}(t) \rightarrow \mu t$ as $n \rightarrow \infty$. This is the cumulant generating function of a degenerate random variable with all of the probability mass at μ . We conclude that $\bar{X}_n \xrightarrow{d} \mu$ as $n \rightarrow \infty$.

7. Letting $Y = X_1 + X_2$, due to independence we have:

$$M_Y(t) = E(e^{t(X_1+X_2)}) = M_{X_1}(t) M_{X_2}(t).$$

Hence:

$$K_Y(t) = K_{X_1}(t) + K_{X_2}(t) = (\mu_1 + \mu_2)t + \frac{(\sigma_1^2 + \sigma_2^2)t^2}{2}.$$

Uniqueness of cumulant generating functions gives the result.

8. If $Z \sim N(0, 1)$ and $X = Z^2 \sim \chi_1^2$, then the moment generating function of X is:

$$M_X(t) = (1 - 2t)^{-1/2}.$$

Proceeding by induction, suppose that $X_k \sim \chi_k^2$ and that the given form of the moment generating function holds for X_k . By definition of a chi-squared random variable:

$$X_{k+1} = X_k + V_{k+1}$$

where $V_{k+1} \sim \chi_1^2$, and $V_{k+1} \perp\!\!\!\perp X_k$. Standard moment generating function results give:

$$M_{X_{k+1}}(t) = M_{X_k}(t) M_{V_{k+1}}(t) = (1 - 2t)^{-k/2} (1 - 2t)^{-1/2} = (1 - 2t)^{-(k+1)/2}.$$

Hence if the given form of moment generating function holds for k , it holds for $k + 1$. We have shown that this is true for $k = 1$, and hence the result holds by induction.

9. Using the definition of the chi-squared distribution, we could write:

$$X_1 = \sum_{i=1}^m Y_i \quad \text{and} \quad X_2 = \sum_{i=m+1}^{m+n} Y_i$$

where $Y_1, Y_2, \dots, Y_m$ and $Y_{m+1}, Y_{m+2}, \dots, Y_{m+n}$ are sequences of independent and identically distributed random variables. Since $X_1 \perp\!\!\!\perp X_2$ we must have $Y_i \perp\!\!\!\perp Y_j$ for $i \neq j$. We conclude that:

$$X_1 + X_2 = \sum_{i=1}^{m+n} Y_i \sim \chi_{m+n}^2.$$

10. We know that S^2 is an unbiased estimator of σ^2 , so:

$$E(\hat{\sigma}^2) = E\left(\frac{n-1}{n} S^2\right) = \frac{n-1}{n} E(S^2) = \frac{n-1}{n} \sigma^2 = \sigma^2 - \frac{\sigma^2}{n}.$$

Hence the bias is:

$$\text{Bias}(\hat{\sigma}^2) = E(\hat{\sigma}^2) - \sigma^2 = \sigma^2 - \frac{\sigma^2}{n} - \sigma^2 = -\frac{\sigma^2}{n}.$$

For the variance, we have:

$$\begin{aligned}
 \text{Var}(\hat{\sigma}^2) &= \text{Var}\left(\frac{n-1}{n}S^2\right) = \frac{(n-1)^2}{n^2}\text{Var}(S^2) \\
 &= \frac{(n-1)^2}{n^2} \frac{1}{n} \left(\mu'_4 - \frac{n-3}{n-1}\sigma^4\right) \\
 &= \frac{(n-1)^2}{n^2} \frac{1}{n} \frac{2n\sigma^4}{n-1} \\
 &= \frac{2(n-1)}{n^2}\sigma^4.
 \end{aligned}$$

Therefore, the mean squared error is:

$$\text{MSE}_{\sigma^2}(\hat{\sigma}^2) = \text{Var}(\hat{\sigma}^2) + (\text{Bias}(\hat{\sigma}^2))^2 = \frac{2(n-1)}{n^2}\sigma^4 + \frac{\sigma^4}{n^2} = \frac{2\sigma^4}{n}.$$

Note that this estimator is consistent since:

$$\lim_{n \rightarrow \infty} \text{MSE}_{\sigma^2}(\hat{\sigma}^2) = \lim_{n \rightarrow \infty} \frac{2\sigma^4}{n} = 0.$$

11. Suppose that:

$$\frac{1}{2n} < \alpha < 0.50$$

then:

$$0.50 < 1 - \alpha < 1 - \frac{1}{2n}.$$

By definition:

$$Q_{1-\alpha} = X_{(n+1-\{n(1-(1-\alpha))\})} = X_{(n+1-\{n\alpha\})}.$$

Similar reasoning holds if $0.50 < \alpha < 1 - 1/(2n)$.

12. In general, the density function of the sample minimum is:

$$nf_X(x)(\bar{F}_X(x))^{n-1}.$$

(a) We have:

$$F_X(x) = 1 - (1+x)^{-a} \quad \text{and} \quad \bar{F}_X(x) = (1+x)^{-a}$$

hence:

$$\begin{aligned}
 f_{X_{(1)}}(x) &= nf_X(x)(\bar{F}_X(x))^{n-1} \\
 &= n \frac{a}{(1+x)^{a+1}} ((1+x)^{-a})^{n-1} \\
 &= \frac{na}{(1+x)^{na+1}}
 \end{aligned}$$

and so $X_{(1)} \sim \text{Pareto}(na)$.

(b) We have:

$$F_X(x) = 1 - \exp(-cx^\tau) \quad \text{and} \quad \bar{F}_X(x) = \exp(-cx^\tau)$$

hence:

$$\begin{aligned} f_{X_{(1)}}(x) &= n f_X(x) (\bar{F}_X(x))^{n-1} \\ &= nc\tau x^{\tau-1} \exp(-cx^\tau) (\exp(-cx^\tau))^{n-1} \\ &= nc\tau x^{\tau-1} \exp(-ncx^\tau) \end{aligned}$$

and so $X_{(1)} \sim \text{Weibull}(nc, \tau)$.

(c) We have:

$$F_X(x) = 1 - \exp(-\lambda e^x) \quad \text{and} \quad \bar{F}_X(x) = \exp(-\lambda e^x)$$

hence:

$$\begin{aligned} f_{X_{(1)}}(x) &= n f_X(x) (\bar{F}_X(x))^{n-1} \\ &= n\lambda e^x \exp(-\lambda e^x) (\exp(-\lambda e^x))^{n-1} \\ &= n\lambda e^x \exp(-n\lambda e^x) \end{aligned}$$

and so $X_{(1)} \sim \text{Extreme value}(n\lambda)$.

13. In order to calculate the Cramér–Rao lower bound we just need information which is based on the log-likelihood function. For a single observation, the log-likelihood function is:

$$l_X(\pi; x) = \log \binom{n}{x} + x \log \pi + (n - x) \log(1 - \pi)$$

and the score function is:

$$s_X(\pi; x) = \frac{\partial}{\partial \pi} l_X(\pi; x) = \frac{x}{\pi} - \frac{n - x}{1 - \pi} = \frac{x - \pi n}{\pi(1 - \pi)}.$$

Hence Fisher information is:

$$\mathcal{I}_X(\pi) = \text{Var}(s_X(\pi; X)) = \frac{1}{\pi^2(1 - \pi)^2} n\pi(1 - \pi) = \frac{n}{\pi(1 - \pi)}.$$

In this binomial case where $g(\pi) = \pi^2$, we only have one observation so the Cramér–Rao lower bound for the variance of an unbiased estimator is given by:

$$\frac{\left(\frac{d}{d\pi} g(\pi)\right)^2}{\mathcal{I}_X(\pi)} = \frac{4\pi^2}{n/(\pi(1 - \pi))} = \frac{4\pi^3(1 - \pi)}{n}.$$

If n is known, we can see from the form of the score function:

$$s_X(\pi; x) = \frac{n}{\pi(1 - \pi)} \left(\frac{x}{n} - \pi \right)$$

that only x/n (which is an unbiased estimator of π) can attain the Cramér–Rao lower bound.

14. The likelihood function is:

$$L_{\mathbf{X}}(\theta; \mathbf{x}) = \frac{2^n \prod_{i=1}^n x_i}{\theta^{2n}}$$

for $0 \leq x_i \leq \theta$, for $i = 1, 2, \dots, n$, and 0 otherwise. The log-likelihood function is:

$$l_{\mathbf{X}}(\theta; \mathbf{x}) = n \log 2 + \sum_{i=1}^n \log x_i - 2n \log \theta$$

for $\max_{1 \leq i \leq n} x_i \leq \theta$.

The support of this function is particularly important, since it is defined by θ . A quick sketch would indicate that the function is strictly decreasing for $\theta > \max_{1 \leq i \leq n} x_i$, and zero for $\theta < \max_{1 \leq i \leq n} x_i$, so the maximum is attained at $\max_{1 \leq i \leq n} x_i$. Hence this is the maximum likelihood estimate.

15. We have that $R \sim \text{Bin}(n, \pi)$, so that $E(R) = n\pi$ and hence $E(R/n) = \pi$. It follows that:

$$E\left(1 - \frac{R}{n}\right) = 1 - E\left(\frac{R}{n}\right) = 1 - \pi.$$

So the obvious plug-in estimator of $\pi(1 - \pi)$ is:

$$\frac{R}{n} \left(1 - \frac{R}{n}\right) = \frac{R}{n} - \left(\frac{R}{n}\right)^2.$$

Now:

$$n\pi(1 - \pi) = \text{Var}(R) = E(R^2) - (E(R))^2 = E(R^2) - (n\pi)^2.$$

So:

$$E\left(\left(\frac{R}{n}\right)^2\right) = \frac{1}{n^2} E(R^2) = \frac{1}{n^2} (n\pi(1 - \pi) + n^2\pi^2)$$

hence:

$$\begin{aligned} E\left(\frac{R}{n} - \left(\frac{R}{n}\right)^2\right) &= \frac{1}{n} E(R) - \frac{1}{n^2} (n\pi(1 - \pi) + n^2\pi^2) \\ &= \frac{n\pi}{n} - \frac{n^2\pi^2}{n^2} - \frac{\pi(1 - \pi)}{n} \\ &= \pi - \pi^2 - \frac{\pi(1 - \pi)}{n}. \end{aligned}$$

However, $\pi(1 - \pi) = \pi - \pi^2$, so:

$$E\left(\frac{R}{n} - \left(\frac{R}{n}\right)^2\right) = \pi(1 - \pi) - \frac{\pi(1 - \pi)}{n} = \pi(1 - \pi) \times \frac{n-1}{n}.$$

It follows that:

$$\pi(1 - \pi) = \frac{n}{n-1} \times E\left(\frac{R}{n} - \left(\frac{R}{n}\right)^2\right) = E\left(\frac{R}{n-1} - \frac{R^2}{n(n-1)}\right).$$

So we have found an unbiased estimator of $\pi(1 - \pi)$, but it could do with tidying up! When this is done, we see that:

$$\frac{R(n - R)}{n(n - 1)}$$

is an unbiased estimator of $\pi(1 - \pi)$, not our initial plug-in estimator. We see that the unbiasedness of R/n when estimating π is *not* an invariant property.

16. The log-likelihood function for a single observation is:

$$l_X(\mu; x) = \log L_X(\mu; x) = \log f_X(x; \mu) = (x - \mu) - 2 \log(1 + \exp(x - \mu)).$$

Differentiating yields:

$$s_X(\mu; x) = -1 + 2 \frac{\exp(x - \mu)}{1 + \exp(x - \mu)} = -1 + 2F_X(x; \mu).$$

Hence the score function for the whole sample is:

$$s_{\mathbf{X}}(\mu; \mathbf{x}) = \sum_{i=1}^n (-1 + 2F_X(x_i; \mu)).$$

By noting that $F_X(X; \mu) \sim \text{Uniform}[0, 1]$ we can calculate the Fisher information associated with a single observation:

$$\mathcal{I}_X(\mu) = \text{Var}(s_X(\mu; X)) = \text{Var}(-1 + 2F_X(X; \mu)) = 4 \times \frac{1}{12} = \frac{1}{3}$$

since the variance of the $\text{Uniform}[0, 1]$ distribution is $1/12$. Hence the total Fisher information is:

$$\mathcal{I}_{\mathbf{X}}(\mu) = \frac{n}{3}.$$

Putting this together produces the method of scoring iterations:

$$\begin{aligned} \mu_{k+1}^* &= \mu_k^* + (\mathcal{I}_{\mathbf{X}}(\mu_k^*))^{-1} s_{\mathbf{X}}(\mu_k^*; \mathbf{x}) \\ &= \mu_k^* + \frac{3}{n} \sum_{i=1}^n (-1 + 2F_X(x_i; \mu_k^*)). \end{aligned}$$

B.3 Chapter 4 – Interval estimation

1. In classical statistics parameters are treated as unknown constants, i.e. parameters are not random variables. As such it does not make sense to talk about the probability that a parameter falls into a particular interval. It either does, or it does not!
2. The probability calculation is straightforward, i.e. if $Z \sim N(0, 1)$ then:

$$P(Z \geq |0.126|) = 0.95.$$

The interval is not desirable in practice as it does not contain $\bar{X}$, which is the obvious estimator of μ .

3. We first note that:

$$\sum_{i=1}^{10} \frac{(X_i - \bar{X})^2}{\sigma^2} \sim \chi_9^2 \quad \text{and} \quad \sum_{i=1}^{10} \frac{(X_i - \mu)^2}{\sigma^2} \sim \chi_{10}^2.$$

If we take an interval which excludes the upper and lower 5%, the relevant chi-squared values are:

$$\chi_{0.95,9}^2 = 3.3251, \quad \chi_{0.05,9}^2 = 16.9190, \quad \chi_{0.95,10}^2 = 3.9403 \quad \text{and} \quad \chi_{0.05,10}^2 = 18.3070.$$

We conclude that:

$$P \left(3.3251 \leq \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{\sigma^2} \leq 16.9190 \right) = 0.90$$

hence:

$$P \left(\frac{1}{16.9190} \sum_{i=1}^n (X_i - \bar{X})^2 \leq \sigma^2 \leq \frac{1}{3.3251} \sum_{i=1}^n (X_i - \bar{X})^2 \right) = 0.90.$$

Therefore:

$$a_1 = 16.9190 \quad \text{and} \quad a_2 = 3.3251.$$

By similar reasoning:

$$b_1 = 18.3070 \quad \text{and} \quad b_2 = 3.9403.$$

In order to calculate the expected length, we first note that if $X \sim \chi_k^2$, then:

$$E(\sigma^2 X) = \sigma^2 E(X) = k\sigma^2$$

since $E(X) = k$. Using the fact that:

$$\sum_{i=1}^{10} \frac{(X_i - \bar{X})^2}{\sigma^2} \sim \chi_9^2$$

the expected length of the first interval is given by:

$$\left(\frac{1}{a_2} - \frac{1}{a_1} \right) E \left(\sum_{i=1}^n (X_i - \bar{X})^2 \right) = \left(\frac{1}{a_2} - \frac{1}{a_1} \right) 9\sigma^2 = 2.175\sigma^2.$$

By similar reasoning, the expected length of the second interval is $1.992\sigma^2$. We conclude that the second interval is shorter than the first. This is what we would expect as the second interval exploits the fact that μ is known.

4. When μ is known and we define:

$$Y = \sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma^2}$$

then Y is a pivotal function. The only unknown parameter is σ^2 and, since $Y \sim \chi_n^2$, the distribution of Y does not depend on σ^2 . Therefore:

$$P\left(\chi_{1-\alpha/2,n}^2 \leq \sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma^2} \leq \chi_{\alpha/2,n}^2\right) = 1 - \alpha.$$

We conclude that:

$$\left[\frac{1}{\chi_{\alpha/2,n}^2} \sum_{i=1}^n (X_i - \mu)^2, \frac{1}{\chi_{1-\alpha/2,n}^2} \sum_{i=1}^n (X_i - \mu)^2 \right]$$

is an interval estimator of σ^2 with a $100(1 - \alpha)\%$ confidence level. Hence using the fact that the expected value of a χ_n^2 -distributed random variable is n , the expected length of the confidence interval is:

$$\left(\frac{1}{\chi_{1-\alpha/2,n}^2} - \frac{1}{\chi_{\alpha/2,n}^2} \right) n\sigma^2.$$

When μ is unknown, the fact that:

$$\sum_{i=1}^n \frac{(X_i - \bar{X})^2}{\sigma^2} \sim \chi_{n-1}^2$$

provides a pivotal function. Therefore:

$$P\left(\chi_{1-\alpha/2,n-1}^2 \leq \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{\sigma^2} \leq \chi_{\alpha/2,n-1}^2\right) = 1 - \alpha.$$

We conclude that:

$$\left[\frac{1}{\chi_{\alpha/2,n-1}^2} \sum_{i=1}^n (X_i - \bar{X})^2, \frac{1}{\chi_{1-\alpha/2,n-1}^2} \sum_{i=1}^n (X_i - \bar{X})^2 \right]$$

is an interval estimator of σ^2 with a $100(1 - \alpha)\%$ confidence level. Hence using the fact that the expected value of a χ_{n-1}^2 -distributed random variable is $n - 1$, the expected length of the confidence interval is:

$$\left(\frac{1}{\chi_{1-\alpha/2,n-1}^2} - \frac{1}{\chi_{\alpha/2,n-1}^2} \right) (n - 1)\sigma^2.$$

5. In order to construct a pivotal function we first note that:

$$\frac{\sum_{i=1}^n (X_i - \mu_X)^2}{\sigma_X^2} \sim \chi_n^2 \quad \text{and} \quad \frac{\sum_{i=1}^m (Y_i - \mu_Y)^2}{\sigma_Y^2} \sim \chi_m^2.$$

Note that the F distribution was introduced in **ST104b Statistics 2**. If V and W are two independent random variables, where $V \sim \chi_p^2$ and $W \sim \chi_k^2$, then:

$$\frac{V/p}{W/k} \sim F_{p,k}.$$

Hence from the definition of the F distribution, if we let:

$$U = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \mu_X)^2}{\frac{1}{m} \sum_{i=1}^m (Y_i - \mu_Y)^2} \frac{\sigma_Y^2}{\sigma_X^2}$$

then U is a pivotal function for the ratio σ_X^2/σ_Y^2 . This ratio is the only unknown parameter and $U \sim F_{n,m}$, i.e. the distribution does not depend on the unknown parameter (ratio).

Hence:

$$P \left(F_{n,m,1-\alpha/2} \leq \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \mu_X)^2}{\frac{1}{m} \sum_{i=1}^m (Y_i - \mu_Y)^2} \frac{\sigma_Y^2}{\sigma_X^2} \leq F_{n,m,\alpha/2} \right) = 1 - \alpha.$$

We conclude that:

$$\left[\frac{1}{F_{n,m,\alpha/2}} \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \mu_X)^2}{\frac{1}{m} \sum_{i=1}^m (Y_i - \mu_Y)^2}, \frac{1}{F_{n,m,1-\alpha/2}} \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \mu_X)^2}{\frac{1}{m} \sum_{i=1}^m (Y_i - \mu_Y)^2} \right]$$

is an interval estimator of σ_X^2/σ_Y^2 with a $100(1 - \alpha)\%$ confidence level.

6. This involves two levels of approximation. In the first instance we use the normal approximation of the binomial distribution. For the situation described in the question:

$$\sum_{i=1}^n X_i \sim \text{Bin}(n, \pi).$$

The distribution of $\sum_{i=1}^n X_i$ is approximated by $N(n\pi, n\pi(1 - \pi))$. If we define:

$$Z = \frac{\sum_{i=1}^n X_i - n\pi}{\sqrt{n\pi(1 - \pi)}}$$

then $Z \sim N(0, 1)$, approximately for large n . Note we omit the continuity correction since we are assuming n is large. Hence Z is approximately a pivotal function for π . In order to use this function we make a further approximation. Defining the usual estimator of the Bernoulli parameter π , that is:

$$\hat{\pi} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

we approximate the denominator of the pivotal statistic by $n\hat{\pi}(1 - \hat{\pi})$. After rearrangement, we have:

$$\frac{\hat{\pi} - \pi}{\sqrt{\hat{\pi}(1 - \hat{\pi})/n}} \sim N(0, 1)$$

approximately for large n . We conclude that:

$$\left[\hat{\pi} - z_{\alpha/2} \times \sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{n}}, \hat{\pi} + z_{\alpha/2} \times \sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{n}} \right]$$

is an interval estimator of π with an approximate $100(1 - \alpha)\%$ confidence level.

7. We have that for a Poisson distribution with parameter λ , the total Fisher information is given by (see Example 2.30):

$$\mathcal{I}_{\mathbf{X}}(\lambda) = \frac{n}{\lambda}.$$

If $\hat{\lambda}$ is the maximum likelihood estimator of λ , then:

$$\frac{\hat{\lambda} - \lambda}{\sqrt{\lambda/n}} \xrightarrow{d} N(0, 1).$$

Approximating Fisher information with $\sqrt{n/\hat{\lambda}}$, we conclude that:

$$\left[\hat{\lambda} - z_{\alpha/2} \times \sqrt{\frac{\hat{\lambda}}{n}}, \hat{\lambda} + z_{\alpha/2} \times \sqrt{\frac{\hat{\lambda}}{n}} \right]$$

is an interval estimator of λ with an approximate $100(1 - \alpha)\%$ confidence level.

8. We need to find i and j such that:

$$F_Y(j - 1) - F_Y(i - 1) \approx 0.95$$

where $Y \sim \text{Bin}(100, 0.75)$. Since:

$$P(Y \leq 66) = 0.0276 \quad \text{and} \quad P(Y \leq 83) = 0.9789$$

we conclude that $[X_{(67)}, X_{(84)}]$ is an interval estimator for the upper quartile with a confidence level of approximately 95%.

B.4 Chapter 5 – Hypothesis testing

1. The null hypothesis is a statement about population parameters. In classical statistical inference, these are fixed unknown constants, not random variables. Hence it does not make sense to talk about the probability that the null hypothesis is true. The p -value gives an indication of the probability of a particular sample result or something more extreme, conditional on the null hypothesis being true.
2. We can assume (without losing too much generality) that we are interested in testing:

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta < \theta_0.$$

In addition, suppose that we have a pivotal function $g(\mathbf{X}, \theta)$. We know that the distribution of $g(\mathbf{X}, \theta)$ does not depend on θ . Suppose that this distribution is known so we can calculate k_α such that:

$$P(g(\mathbf{X}, \theta) < k_\alpha) = \alpha.$$

We can now construct a test at the $100\alpha\%$ significance level by taking the rejection region to be:

$$R = \{\mathbf{x} : g(\mathbf{x}, \theta_0) < k_\alpha\}.$$

3. Suppose that we are testing:

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta = \theta_1.$$

The likelihood ratio from the Neyman–Pearson lemma is:

$$\frac{L_{\mathbf{X}}(\theta_1; \mathbf{X})}{L_{\mathbf{X}}(\theta_0; \mathbf{X})} = \left(\frac{\theta_0}{\theta_1}\right)^{n/2} \exp\left(\frac{1}{2} \left(\frac{\theta_1 - \theta_0}{\theta_0 \theta_1}\right) \sum_{i=1}^n X_i^2\right).$$

A most powerful test will reject H_0 when this ratio is large. If $\theta_1 > \theta_0$, then $L_{\mathbf{X}}(\theta_1; \mathbf{X})/L_{\mathbf{X}}(\theta_0; \mathbf{X})$ is large when $\sum_{i=1}^n X_i^2$ is large. Under H_0 we have:

$$\frac{\sum_{i=1}^n X_i^2}{\theta_0} \sim \chi_n^2.$$

This pivotal quantity is the basis for a test of size α with rejection region:

$$R = \left\{ \mathbf{x} : \sum_{i=1}^n x_i^2 > \theta_0 \times \chi_{\alpha, n}^2 \right\}.$$

The power of this test is:

$$\begin{aligned} \beta(\theta_1) &= P_{\theta_1} \left(\sum_{i=1}^n X_i^2 > \theta_0 \times \chi_{\alpha, n}^2 \right) \\ &= P_{\theta_1} \left(\frac{1}{\theta_1} \sum_{i=1}^n X_i^2 > \frac{\theta_0}{\theta_1} \times \chi_{\alpha, n}^2 \right) \\ &= 1 - F_Y \left(\frac{\theta_0}{\theta_1} \times \chi_{\alpha, n}^2 \right) \end{aligned}$$

where $Y \sim \chi_n^2$. This is a result of the fact that under H_1 we have:

$$\frac{\sum_{i=1}^n X_i^2}{\theta_1} \sim \chi_n^2.$$

Here we have $\theta_0 = 1$, $n = 10$ and $\alpha = 0.05$. Hence the rejection region is:

$$R = \left\{ \mathbf{x} : \sum_{i=1}^n x_i^2 > 18.3070 \right\}.$$

We also have that $\theta_1 = 2$, hence the power is:

$$\beta(2) = 1 - F_Y \left(\frac{18.3070}{2} \right) = 0.518.$$

4. The rejection region for the test is:

$$R = \left\{ \mathbf{x} : \sum_{i=1}^n x_i^2 > \sigma_0^2 \times \chi_{\alpha, n}^2 \right\}.$$

This region does not depend on σ_1^2 , so the test is uniformly most powerful for testing:

$$H_0 : \sigma^2 = \sigma_0^2 \quad \text{vs.} \quad H_1 : \sigma^2 > \sigma_0^2.$$

Now consider the composite null hypothesis $H_0 : \sigma^2 \leq \sigma_0^2$. In order to establish that the test is a uniformly most powerful test of size α in this case, we need to show that:

$$P_{\sigma^2}(\text{reject } H_0) \leq \alpha$$

for $\sigma^2 \leq \sigma_0^2$. This is straightforward since:

$$P_{\sigma^2}(\text{reject } H_0) = \beta(\sigma^2) = 1 - F_Y \left(\frac{\sigma_0^2}{\sigma^2} \times \chi_{\alpha, n}^2 \right)$$

where $Y \sim \chi_n^2$. For $\sigma^2 \leq \sigma_0^2$, we know that $\sigma_0^2/\sigma^2 \geq 1$. As F_Y is a non-decreasing function:

$$F_Y \left(\frac{\sigma_0^2}{\sigma^2} \times \chi_{\alpha, n}^2 \right) \geq F_Y(\chi_{\alpha, n}^2) = \alpha \quad \Rightarrow \quad 1 - F_Y \left(\frac{\sigma_0^2}{\sigma^2} \times \chi_{\alpha, n}^2 \right) \leq \alpha$$

which gives the result.

5. $L_{\mathbf{X}}(\hat{\theta}; \mathbf{x})$ is the maximum value of the likelihood function over all parameters. This includes parameters which satisfy the restrictions imposed by the null hypothesis. Hence $L_{\mathbf{X}}(\hat{\theta}; \mathbf{x}) \geq L_{\mathbf{X}}(\hat{\theta}_0; \mathbf{x})$ and the result follows immediately.
6. The likelihood function is:

$$L_{\mathbf{X}, \mathbf{Y}}(\lambda_1, \lambda_2; \mathbf{X}, \mathbf{Y}) = \lambda_1^n \lambda_2^n \exp \left(-\lambda_1 \sum_{i=1}^n X_i - \lambda_2 \sum_{i=1}^n Y_i \right).$$

Under H_0 we have that $\lambda_1 = \lambda_2 = \lambda$, so the likelihood becomes:

$$L_{\mathbf{X}, \mathbf{Y}}(\lambda, \lambda; \mathbf{X}, \mathbf{Y}) = \lambda^{2n} \exp \left(-\lambda \left(\sum_{i=1}^n X_i + \sum_{i=1}^n Y_i \right) \right).$$

We can calculate the restricted maximum likelihood estimator by differentiating the likelihood:

$$\frac{\partial}{\partial \lambda} L_{\mathbf{X}, \mathbf{Y}}(\lambda, \lambda; \mathbf{X}, \mathbf{Y}) = \lambda^{2n-1} \exp \left(-\lambda \left(\sum_{i=1}^n X_i + \sum_{i=1}^n Y_i \right) \right) \left(2n - \lambda \left(\sum_{i=1}^n X_i + \sum_{i=1}^n Y_i \right) \right).$$

Setting this derivative equal to zero and solving for λ yields:

$$\hat{\lambda} = \frac{2n}{\sum_{i=1}^n X_i - \sum_{i=1}^n Y_i} = \frac{2}{\bar{X} + \bar{Y}}.$$

Similar reasoning under $H_1 : \lambda_1 \neq \lambda_2$ yields:

$$\hat{\lambda}_1 = \frac{1}{\bar{X}} \quad \text{and} \quad \hat{\lambda}_2 = \frac{1}{\bar{Y}}.$$

Therefore, the likelihood ratio test statistic is:

$$LR(\mathbf{X}, \mathbf{Y}) = \frac{(1/\bar{X})^n (1/\bar{Y})^n \exp(-2n)}{(2/(\bar{X} + \bar{Y}))^{2n} \exp(-2n)} = \left(\frac{1}{2}\right)^{2n} \left(\frac{\bar{X} + \bar{Y}}{\sqrt{\bar{X}\bar{Y}}}\right)$$

and so:

$$2 \log LR(\mathbf{X}, \mathbf{Y}) = 4n \log \left(\sqrt{\frac{\bar{X}}{\bar{Y}}} + \sqrt{\frac{\bar{Y}}{\bar{X}}} \right) - 4n \log 2.$$

Under H_0 , the asymptotic distribution of $2 \log LR(\mathbf{X}, \mathbf{Y})$ is χ_1^2 . Therefore, a test which rejects H_0 when:

$$2 \log LR(\mathbf{x}, \mathbf{y}) > \chi_{\alpha, 1}^2$$

will have approximate size α .

7. We test:

$$H_0 : \pi_1 = \pi_2 = \pi_3 = \pi_4 \quad \text{vs.} \quad H_1 : \pi_i \neq \pi_j \text{ for some } i \text{ and } j.$$

For notational convenience we use $\boldsymbol{\pi} = (\pi_1, \pi_2, \pi_3, \pi_4)$ and $\mathbf{x} = (x_1, x_2, x_3, x_4)$. The likelihood function is then:

$$L(\boldsymbol{\pi}; \mathbf{x}) = \prod_{i=1}^n \binom{200}{x_i} \pi_i^{x_i} (1 - \pi_i)^{200 - x_i}.$$

We can readily show that the constrained maximum likelihood estimate is

$\hat{\boldsymbol{\pi}} = (\hat{\pi}_0, \hat{\pi}_0, \hat{\pi}_0, \hat{\pi}_0)$, where:

$$\hat{\pi}_0 = \frac{\sum_{i=1}^n x_i}{4 \times 200}$$

and the unconstrained maximum likelihood estimates are:

$$\hat{\pi}_i = \frac{x_i}{200} \quad \text{for } i = 1, 2, 3, 4.$$

We conclude that the likelihood ratio test statistic will have the form:

$$LR(\mathbf{X}) = \frac{\prod_{i=1}^n \hat{\pi}_i^{X_i} (1 - \hat{\pi}_i)^{200 - X_i}}{\hat{\pi}_0^{\sum_{i=1}^n X_i} (1 - \hat{\pi}_0)^{800 - \sum_{i=1}^n X_i}}$$

and hence:

$$2 \log LR(\mathbf{X}) = 2 \sum_{i=1}^4 (X_i \log \hat{\pi}_i + (200 - X_i) \log(1 - \hat{\pi}_i)) \\ - 2 \sum_{i=1}^n X_i \log \hat{\pi}_0 - 2 \left(800 - \sum_{i=1}^n X_i \right) \log(1 - \hat{\pi}_0).$$

In order to be able to specify the asymptotic distribution of the test statistic, we can view this problem in terms of a nuisance parameter. Consider setting:

$$\pi_i = \pi_1 + \psi_i \quad \text{for } i = 2, 3, 4.$$

We then test:

$$H_0 : \psi_2 = \psi_3 = \psi_4 = 0 \quad \text{vs.} \quad H_1 : \psi_i \neq 0 \text{ for some } i.$$

The parameters are then $(\pi_1, \psi_2, \psi_3, \psi_4)$, with π_1 the nuisance parameter. We have three non-nuisance parameters restricted by H_0 , so $2 \log LR(\mathbf{X})$ has an approximate χ^2_3 distribution. Hence a test which rejects H_0 when:

$$2 \log LR(\mathbf{x}) > \chi^2_{\alpha, 3}$$

will have approximate size α .

Here we have $2 \log LR(\mathbf{x}) = 10.535$, and since $\chi^2_{0.05, 3} = 7.815$ we reject H_0 and conclude that there is evidence to suggest that the proportions using the product differ between areas.



Appendix C

Solutions to Sample examination questions

C.1 Chapter 3 – Point estimation

1. (a) The random variable is the sample minimum, i.e. the first order statistic $X_{(1)}$. The density function of the i th order statistic $X_{(i)}$ is:

$$f_{X_{(i)}}(x) = \frac{n!}{(i-1)!(n-i)!} f_X(x) (F_X(x))^{i-1} (1 - F_X(x))^{n-i}.$$

Substituting $i = 1$ in the above we get:

$$f_{X_{(1)}}(x) = n f_X(x) (1 - F_X(x))^{n-1}.$$

We know (or can calculate) that the distribution function of the exponential distribution is:

$$F_X(x) = \begin{cases} 1 - \exp\left(-\frac{x}{\mu}\right) & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

hence:

$$f_{X_{(1)}}(x) = n \frac{1}{\mu} \exp\left(-\frac{x}{\mu}\right) \left(1 - \left(1 - \exp\left(-\frac{x}{\mu}\right)\right)\right)^{n-1} = \frac{n}{\mu} \exp\left(-\frac{nx}{\mu}\right)$$

for $x \geq 0$, and 0 otherwise. The distribution of $X_{(1)}$ is an exponential with mean μ/n .

- (b) The first population moment is:

$$E(X) = \mu.$$

For the method of moments estimator we just need to equate μ with $M_1 = \bar{X}$ (the first sample moment), which immediately gives us the estimator:

$$\hat{\mu} = \bar{X}.$$

- (c) The likelihood function is, due to independence:

$$L_{\mathbf{X}}(\mathbf{x}; \mu) = \prod_{i=1}^n \frac{1}{\mu} \exp\left(-\frac{x_i}{\mu}\right) = \mu^{-n} \exp\left(-\frac{1}{\mu} \sum_{i=1}^n x_i\right).$$

Hence the log-likelihood function is:

$$l_{\mathbf{X}}(\mathbf{x}; \mu) = -n \log \mu - \frac{1}{\mu} \sum_{i=1}^n x_i.$$

The score function is:

$$s_{\mathbf{X}}(\mu; \mathbf{x}) = \frac{\partial}{\partial \mu} l_{\mathbf{X}}(\mu; \mathbf{x}) = -\frac{n}{\mu} + \frac{1}{\mu^2} \sum_{i=1}^n x_i.$$

The maximum likelihood estimator is:

$$s_{\mathbf{X}}(\mu; \mathbf{x}) \Big|_{\mu=\hat{\mu}} = -\frac{n}{\hat{\mu}} + \frac{1}{\hat{\mu}^2} \sum_{i=1}^n x_i = 0$$

which solves for $\hat{\mu} = \bar{X}$.

The derivative of the score function (equivalently, the second derivative of the log-likelihood function) is:

$$\frac{\partial}{\partial \mu} s_{\mathbf{X}}(\mu; \mathbf{x}) = \frac{\partial^2}{\partial \mu^2} l_{\mathbf{X}}(\mu; \mathbf{x}) = \frac{n}{\mu^2} - \frac{2}{\mu^3} \sum_{i=1}^n x_i.$$

Hence Fisher information is (noting that $E(X_i) = \mu$ for $i = 1, 2, \dots, n$):

$$\mathcal{I}_{\mathbf{X}}(\mu) = -E \left(\frac{n}{\mu^2} - \frac{2}{\mu^3} \sum_{i=1}^n X_i \right) = -\frac{n}{\mu^2} + \frac{2}{\mu^3} \sum_{i=1}^n E(X_i) = -\frac{n}{\mu^2} + \frac{2n\mu}{\mu^3} = \frac{n}{\mu^2}.$$

2. (a) Since the X_i s are independent and identically distributed, the joint density function can be written as:

$$\begin{aligned} f_{\mathbf{X}}(\mathbf{x}; \theta) &= \prod_{i=1}^n f_X(x_i; \theta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\theta}} \exp \left(-\frac{x_i^2}{2\theta} \right) \\ &= (2\pi\theta)^{-n/2} \exp \left(-\frac{1}{2\theta} \sum_{i=1}^n x_i^2 \right). \end{aligned}$$

If we set:

$$g \left(\sum_{i=1}^n x_i^2, \theta \right) = f_{\mathbf{X}}(\mathbf{x}; \theta) \quad \text{and} \quad h(\mathbf{x}) = 1$$

we can write:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = g \left(\sum_{i=1}^n x_i^2, \theta \right) h(\mathbf{x}).$$

According to the factorisation theorem, this implies that the one-dimensional statistic $\sum_{i=1}^n X_i^2$ is a sufficient statistic for θ .

The log-likelihood function is:

$$l_{\mathbf{X}}(\theta; \mathbf{x}) = \log L_{\mathbf{X}}(\theta; \mathbf{x}) = \log f_{\mathbf{X}}(\mathbf{x}; \theta) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \theta - \frac{1}{2\theta} \sum_{i=1}^n x_i^2.$$

(b) The score function for θ is:

$$s_{\mathbf{X}}(\theta; \mathbf{x}) = \frac{\partial}{\partial \theta} l_{\mathbf{X}}(\theta; \mathbf{x}) = -\frac{n}{2\theta} + \frac{1}{2\theta^2} \sum_{i=1}^n x_i^2.$$

Solving $s_{\mathbf{X}}(\hat{\theta}; \mathbf{x}) = 0$ yields a potential maximum likelihood estimator. Hence:

$$\frac{1}{2\hat{\theta}^2} \sum_{i=1}^n x_i^2 = \frac{n}{2\hat{\theta}} \Rightarrow \hat{\theta} = \frac{1}{n} \sum_{i=1}^n X_i^2 = \frac{1}{n} S(\mathbf{X}).$$

where the sufficient statistic is denoted as $S(\mathbf{X})$. The second derivative of the log-likelihood function with respect to θ is:

$$\frac{\partial^2}{\partial \theta^2} l_{\mathbf{X}}(\theta; \mathbf{x}) = \frac{n}{2\theta^2} - \frac{S(\mathbf{x})}{\theta^3}.$$

If we evaluate the above function at $\theta = \hat{\theta}$, we get:

$$\left. \frac{\partial^2}{\partial \theta^2} l_{\mathbf{X}}(\theta; \mathbf{x}) \right|_{\theta=\hat{\theta}} = \frac{n^3}{2(S(\mathbf{x}))^2} - \frac{n^3 S(\mathbf{x})}{(S(\mathbf{x}))^3} = -\frac{n^3}{2(S(\mathbf{x}))^2} < 0.$$

As this is negative, we conclude that the maximum likelihood estimator of θ is:

$$\hat{\theta} = T(\mathbf{X}) = \frac{S(\mathbf{X})}{n} = \frac{1}{n} \sum_{i=1}^n X_i^2.$$

We can show that the maximum likelihood estimator is a minimum variance unbiased estimator (MVUE) in different ways. However, first we have to verify that it is an unbiased estimator of θ . Indeed:

$$E(\hat{\theta}) = E\left(\frac{1}{n} \sum_{i=1}^n X_i^2\right) = \frac{1}{n} \sum_{i=1}^n E(X_i^2) = \frac{1}{n} n E(X_i^2) = \text{Var}(X_i) = \theta.$$

Method 1: The score function may be written as:

$$\begin{aligned} s_{\mathbf{X}}(\theta; \mathbf{x}) &= \frac{n}{2\theta^2} \left(\frac{1}{n} \sum_{i=1}^n x_i^2 - \theta \right) \\ &= b(\theta)(T(\mathbf{x}) - g(\theta)) \end{aligned}$$

where $b(\theta) = n/(2\theta^2)$, $g(\theta) = \theta$ and $T(\mathbf{X})$ is an unbiased estimator of θ . The factorisation criterion means that the maximum likelihood estimator attains the Cramér–Rao lower bound and hence is a minimum variance unbiased estimator.

Method 2: Since the Fisher information is equal to:

$$\mathcal{I}_{\mathbf{X}}(\theta) = -E\left(\frac{\partial}{\partial \theta} s_{\mathbf{X}}(\theta; \mathbf{X})\right) = -E\left(\frac{n}{2\theta^2} - \frac{S(\mathbf{X})}{\theta^3}\right) = \frac{n}{2\theta^2}$$

the Cramér–Rao lower bound for an unbiased estimator of θ is:

$$\frac{1}{\mathcal{I}_{\mathbf{X}}(\theta)} = \frac{2\theta^2}{n}.$$

Since:

$$X_i \sim N(0, \theta) \Rightarrow \frac{X_i}{\sqrt{\theta}} \sim N(0, 1) \Rightarrow \frac{X_i^2}{\theta} \sim \chi_1^2$$

and noting that the variance of a chi-squared-distributed random variable is twice its degrees of freedom (or using the hint in the question!), we get that:

$$\text{Var}\left(\frac{X_i^2}{\theta}\right) = 2 \Rightarrow \text{Var}(X_i^2) = 2\theta^2.$$

Therefore, by independence of the X_i s, we have:

$$\text{Var}(\hat{\theta}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i^2\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i^2) = \frac{1}{n^2} n \text{Var}(X_i^2) = \frac{2\theta^2}{n}.$$

Since the maximum likelihood estimator is unbiased and its variance attains the Cramér–Rao lower bound, it is also a minimum variance unbiased estimator of θ .

3. (a) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ and $\mathbf{x} = \{x_1, x_2, \dots, x_n\}$, hence the likelihood function is:

$$\begin{aligned} L_{\mathbf{X}}(\alpha, \theta; \mathbf{x}) &= f_{\mathbf{X}}(\mathbf{x}; \alpha, \theta) = \prod_{i=1}^n \alpha e^{-\alpha(x_i - \theta)} I(x_i \geq \theta) \\ &= \alpha^n \exp\left(-\alpha \sum_{i=1}^n x_i + n\alpha\theta\right) I\left(\min_{1 \leq i \leq n} x_i \geq \theta\right) \end{aligned}$$

and 0 otherwise.

- i. Applying the factorisation theorem to the above $f_{\mathbf{X}}$, we have that $\sum_{i=1}^n X_i$ is a sufficient statistic for α when θ is known, as we can write:

$$f_{\mathbf{X}}(\mathbf{x}; \alpha) = g\left(\sum_{i=1}^n X_i, \alpha\right) h(\mathbf{x})$$

if we set:

$$g\left(\sum_{i=1}^n X_i, \alpha\right) = \alpha^n \exp\left(-\alpha \sum_{i=1}^n x_i + n\alpha\theta\right)$$

and:

$$h(\mathbf{x}) = I\left(\min_{1 \leq i \leq n} x_i \geq \theta\right).$$

Since the statistic is of dimension 1, it is also a minimal sufficient statistic.

Furthermore, the log-likelihood function is:

$$l_{\mathbf{X}}(\alpha; \mathbf{x}) = n \log \alpha - \alpha \sum_{i=1}^n x_i + n\alpha\theta.$$

Setting:

$$\frac{\partial}{\partial \alpha} l_{\mathbf{X}}(\alpha; \mathbf{x}) \Big|_{\alpha=\hat{\alpha}} = \frac{n}{\hat{\alpha}} - \sum_{i=1}^n x_i + n\theta = 0$$

and solving for $\hat{\alpha}$ leads us to the candidate maximum likelihood estimator:

$$\hat{\alpha} = \frac{n}{\sum_{i=1}^n X_i - n\theta}.$$

After checking that:

$$\frac{\partial^2}{\partial \alpha^2} l_{\mathbf{X}}(\alpha; \mathbf{x}) = -\frac{n}{\alpha^2} < 0$$

for all α , we conclude that the maximum likelihood estimator is indeed $\hat{\alpha}$.

- ii. When α is known, $\min_{1 \leq i \leq n} X_i$ is a sufficient statistic for θ , as we can write:

$$f_{\mathbf{X}}(\mathbf{x}; \theta) = g\left(\min_{1 \leq i \leq n} X_i, \theta\right) h(\mathbf{x})$$

if we set:

$$g\left(\min_{1 \leq i \leq n} X_i, \theta\right) = e^{n\alpha\theta} I\left(\min_{1 \leq i \leq n} x_i \geq \theta\right)$$

and:

$$h(\mathbf{x}) = \alpha^n \exp\left(-\alpha \sum_{i=1}^n x_i\right).$$

Since the statistic is of dimension 1, it is also a minimal sufficient statistic.

Now the likelihood function is of the form:

$$L_{\mathbf{X}}(\theta; \mathbf{x}) \propto e^{n\alpha\theta} I\left(\min_{1 \leq i \leq n} x_i \geq \theta\right)$$

which is an increasing (positive) function in θ for $\theta \leq \min_{1 \leq i \leq n} X_i$. Hence it obtains the maximum at the largest possible value of θ which is equal to $\min_{1 \leq i \leq n} X_i$. Hence the maximum likelihood estimator of θ is the sample minimum:

$$\hat{\theta} = \min_{1 \leq i \leq n} X_i = X_{(1)}.$$

- (b) When θ is known, by the invariance property of maximum likelihood estimators part (a) implies that the maximum likelihood estimator of $\mu = 1/\alpha$ is:

$$\hat{\mu} = \frac{1}{\hat{\alpha}} = \bar{X} - \theta.$$

We now check for unbiasedness. Let $U_i = X_i - \theta$. If we now apply the transformation $X_i = U_i + \theta$ we get that:

$$f_{U_i}(u; \alpha) = f_{X_i}(u_i + \theta; \alpha, \theta) = \alpha e^{-\alpha u + \theta - \theta} \times 1 = \alpha e^{-\alpha u}$$

for $u \geq 0$. Hence $U_i \sim \text{Exp}(\alpha)$ and so $E(U_i) = 1/\alpha = \mu$. This implies that:

$$E(X_i) = E(U_i + \theta) = E(U_i) + \theta = \mu + \theta = E(\bar{X}).$$

Hence:

$$E(\hat{\mu}) = E(\bar{X} - \theta) = E(\bar{X}) - \theta = \mu + \theta - \theta = \mu$$

and so $\hat{\mu}$ is an unbiased estimator of μ .

We now check attainment of the Cramér–Rao lower bound. Note the log-likelihood function is:

$$l_{\mathbf{X}}(\mu; \mathbf{x}) = -n \log \mu - \frac{1}{\mu} \sum_{i=1}^n x_i + \frac{n\theta}{\mu}.$$

Hence the score function is:

$$\begin{aligned} -\frac{n}{\mu} + \frac{1}{\mu^2} \sum_{i=1}^n x_i - \frac{n\theta}{\mu^2} &= \frac{n}{\mu^2} \left(\left(\frac{1}{n} \sum_{i=1}^n x_i - \theta \right) - \mu \right) \\ &= b(\mu)(T(\mathbf{x}) - g(\mu)) \end{aligned}$$

where:

$$b(\mu) = \frac{n}{\mu^2}, \quad g(\mu) = \mu \quad \text{and} \quad T(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n x_i - \theta.$$

Hence the maximum likelihood estimator:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i - \theta = \bar{X} - \mu$$

is an unbiased estimator of μ which attains the Cramér–Rao lower bound for the variance, hence it is a minimum variance unbiased estimator.

C.2 Chapter 4 – Interval estimation

1. (a) i. If μ is known, we have:

$$\frac{X_i - \mu}{\sigma} \sim N(0, 1) \quad \Rightarrow \quad \frac{(X_i - \mu)^2}{\sigma^2} \sim \chi_1^2.$$

Since the X_i s are independent, we have:

$$\frac{\sum_{i=1}^n (X_i - \mu)^2}{\sigma^2} \sim \chi_n^2$$

which is a pivotal function for σ^2 .

- ii. If μ is unknown, we know that:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

provides a pivotal function for σ^2 for this case, where:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

denotes the sample variance.

- (b) If μ is known, since $\sum_{i=1}^{10} (X_i - \mu)^2 / \sigma^2 \sim \chi_{10}^2$. Leaving out 5% at both ends of the distribution, we have:

$$\begin{aligned} 0.90 &= P \left(3.94 \leq \frac{\sum_{i=1}^{10} (X_i - \mu)^2}{\sigma^2} \leq 18.31 \right) \\ &= P \left(\frac{\sum_{i=1}^{10} (X_i - \mu)^2}{18.31} \leq \sigma^2 \leq \frac{\sum_{i=1}^{10} (X_i - \mu)^2}{3.94} \right). \end{aligned}$$

Let $a_1 = 3.94$ and $a_2 = 18.31$.

Similarly, if μ is unknown, since $\sum_{i=1}^{10} (X_i - \bar{X})^2 / \sigma^2 \sim \chi_9^2$. Leaving out 5% at both ends of the distribution, we have:

$$\begin{aligned} 0.90 &= P \left(3.325 \leq \frac{\sum_{i=1}^{10} (X_i - \bar{X})^2}{\sigma^2} \leq 16.92 \right) \\ &= P \left(\frac{\sum_{i=1}^{10} (X_i - \bar{X})^2}{16.92} \leq \sigma^2 \leq \frac{\sum_{i=1}^{10} (X_i - \bar{X})^2}{3.325} \right). \end{aligned}$$

Let $b_1 = 3.325$ and $b_2 = 16.92$.

The average lengths of the confidence intervals are:

$$\left(\frac{1}{a_1} - \frac{1}{a_2} \right) E \left(\sum_{i=1}^{10} (X_i - \mu)^2 \right) = \left(\frac{1}{3.94} - \frac{1}{18.31} \right) \times 10\sigma^2 = 1.99\sigma^2$$

and:

$$\left(\frac{1}{b_1} - \frac{1}{b_2} \right) E \left(\sum_{i=1}^{10} (X_i - \bar{X})^2 \right) = \left(\frac{1}{3.325} - \frac{1}{16.92} \right) \times 9\sigma^2 = 2.18\sigma^2.$$

- (c) The first interval is shorter since it makes use of the known mean μ , as expected.

2. (a) The likelihood function is, due to independence:

$$\begin{aligned}
 L_{\mathbf{X}}(\mu; \mathbf{x}) &= f_{\mathbf{X}}(\mathbf{x}; \mu) = \prod_{i=1}^n f_X(x_i; \mu) \\
 &= \prod_{i=1}^n \left(\frac{1}{2\pi x_i^3} \right)^{1/2} \exp \left(\frac{1}{\mu} - \frac{x_i}{2\mu^2} - \frac{1}{2x_i} \right) \\
 &= \left(2\pi \prod_{i=1}^n x_i^3 \right)^{-n/2} \exp \left(\frac{n}{\mu} - \frac{\sum_{i=1}^n x_i}{2\mu^2} - \frac{1}{2} \sum_{i=1}^n \frac{1}{x_i} \right) \\
 &= \exp \left(\frac{n}{\mu} - \frac{\sum_{i=1}^n x_i}{2\mu^2} \right) \exp \left(-\frac{1}{2} \sum_{i=1}^n \frac{1}{x_i} \right) \left(2\pi \prod_{i=1}^n x_i^3 \right)^{-n/2}.
 \end{aligned}$$

We can factorise $f_{\mathbf{X}}(\mu; \mathbf{x})$ as $g(T(\mathbf{x}), \mu) h(\mathbf{x})$ by setting:

$$g \left(\sum_{i=1}^n x_i, \mu \right) = \exp \left(\frac{n}{\mu} - \frac{\sum_{i=1}^n x_i}{2\mu^2} \right)$$

and:

$$h(\mathbf{x}) = \exp \left(-\frac{1}{2} \sum_{i=1}^n \frac{1}{x_i} \right) \left(2\pi \prod_{i=1}^n x_i^3 \right)^{-n/2}.$$

Hence $\sum_{i=1}^n X_i$ is a one-dimensional sufficient statistic for μ .

- (b) Note that the likelihood function is proportional to:

$$L_{\mathbf{X}}(\mu; \mathbf{x}) \propto \exp \left(\frac{n}{\mu} - \frac{\sum_{i=1}^n x_i}{2\mu^2} \right).$$

Therefore, the log-likelihood function can be written as:

$$l_{\mathbf{X}}(\mu; \mathbf{x}) = \log L_{\mathbf{X}}(\mu; \mathbf{x}) \propto \frac{n}{\mu} - \frac{\sum_{i=1}^n x_i}{2\mu^2}.$$

The score function is:

$$s_{\mathbf{X}}(\mu; \mathbf{x}) = \frac{\partial}{\partial \mu} l_{\mathbf{X}}(\mu; \mathbf{x}) = -\frac{n}{\mu^2} + \frac{\sum_{i=1}^n x_i}{\mu^3}.$$

Setting $s_{\mathbf{X}}(\hat{\mu}; \mathbf{x}) = 0$ gives:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

as a candidate maximum likelihood estimator. Since:

$$\left. \frac{\partial^2}{\partial \mu^2} l_{\mathbf{X}}(\mu; \mathbf{x}) \right|_{\mu=\bar{x}} = \frac{2n}{\mu^3} - \frac{3n\bar{x}}{\mu^4} \Big|_{\mu=\bar{x}} = \frac{2n}{\bar{x}^3} - \frac{3n\bar{x}}{\bar{x}^4} = -\frac{n}{\bar{x}^3} < 0$$

we conclude that this is indeed the maximum likelihood estimator.

- (c) We know that for large sample sizes, approximately:

$$\hat{\mu} \sim N\left(\mu, \frac{1}{\mathcal{I}_{\mathbf{X}}(\mu)}\right) = N\left(\mu, \frac{\mu^3}{n}\right)$$

where the total Fisher information is:

$$\mathcal{I}_{\mathbf{X}}(\mu) = -E\left(\frac{\partial^2}{\partial \mu^2} l_{\mathbf{X}}(\mu; \mathbf{X})\right) = -E\left(\frac{2n}{\mu^3} - \frac{3n\bar{X}}{\mu^4}\right) = -\frac{2n}{\mu^3} + \frac{3n}{\mu^3} = \frac{n}{\mu^3}$$

noting that $E(\bar{X}) = \mu$. In an extra level of approximation we may use:

$$\hat{\mu} \sim N\left(\mu, \frac{1}{\mathcal{I}_{\mathbf{X}}(\hat{\mu})}\right) = N\left(\mu, \frac{\hat{\mu}^3}{n}\right).$$

Let $S_{\mu} = \hat{\mu}^3/n = \bar{X}^3/n$. We can write:

$$\frac{\hat{\mu} - \mu}{\sqrt{S_{\mu}}} \sim N(0, 1)$$

approximately, for large n . Using the pivotal function technique for constructing interval estimators, we get the following asymptotic interval estimator with a $100(1 - \alpha)\%$ confidence level:

$$\left[\hat{\mu} - z_{\alpha/2} \times \sqrt{S_{\mu}}, \hat{\mu} + z_{\alpha/2} \times \sqrt{S_{\mu}} \right].$$

3. (a) We have that the maximum likelihood estimator of π is $\hat{\pi} = \bar{X}$, and the total Fisher information is:

$$\mathcal{I}_{\mathbf{X}}(\pi) = \frac{n}{\pi(1 - \pi)}.$$

We seek an estimator of $g(\pi) = \pi/(1 - \pi)$. We have that:

$$\frac{\partial}{\partial \pi} g(\pi) = \frac{1 - \pi + \pi}{(1 - \pi)^2} = \frac{1}{(1 - \pi)^2}.$$

Hence:

$$\left(\frac{\partial}{\partial \pi} g(\pi) \right)^2 = \frac{1}{(1 - \pi)^4}.$$

Since the maximum likelihood estimator of π is $\hat{\pi} = \bar{X}$, by the invariance property the maximum likelihood estimator of $\pi/(1 - \pi)$ is $\bar{X}/(1 - \bar{X})$, which is a consistent, asymptotically unbiased and efficient estimator of $\pi/(1 - \pi)$.

- (b) The asymptotic sampling distribution of this estimator of the odds of success is:

$$\frac{\bar{X}}{1 - \bar{X}} \sim N\left(\frac{\pi}{1 - \pi}, \frac{1/(\pi(1 - \pi))^4}{n/(\pi(1 - \pi))}\right) = N\left(\frac{\pi}{1 - \pi}, \frac{\pi}{n(1 - \pi)^3}\right)$$

approximately, for large n .

If we further approximate the variance of this sampling distribution by:

$$\frac{\bar{X}}{n(1 - \bar{X})^3}$$

we have:

$$\frac{\bar{X}}{1 - \bar{X}} \sim N\left(\frac{\pi}{1 - \pi}, \frac{\bar{X}}{n(1 - \bar{X})^3}\right)$$

approximately for large n . Standardising:

$$\frac{\bar{X}/(1 - \bar{X}) - \pi/(1 - \pi)}{\sqrt{\bar{X}/(n(1 - \bar{X})^3)}} \sim N(0, 1)$$

approximately, for large n . This leads to the following asymptotic $100(1 - \alpha)\%$ interval estimator of $\pi/(1 - \pi)$ of:

$$\left[\frac{\bar{X}}{1 - \bar{X}} - z_{\alpha/2} \times \sqrt{\frac{\bar{X}}{n(1 - \bar{X})^3}}, \frac{\bar{X}}{1 - \bar{X}} + z_{\alpha/2} \times \sqrt{\frac{\bar{X}}{n(1 - \bar{X})^3}} \right].$$

C.3 Chapter 5 – Hypothesis testing

1. (a) We want to test:

$$H_0 : \pi = 0.50 \quad \text{vs.} \quad H_1 : \pi = 0.25.$$

To find the size of the test we need to find the probability:

$$P(\text{reject } H_0 \mid \pi = 0.50)$$

and for the power we need to calculate the probability:

$$P(\text{reject } H_0 \mid \pi = 0.25).$$

Since we reject H_0 if $x \leq 3$ and $X \sim \text{Bin}(10, \pi)$, we get the size:

$$P(\text{reject } H_0 \mid \pi = 0.50) = \sum_{x=0}^3 P(X = x \mid \pi = 0.50) \approx 0.1719$$

and the power is:

$$P(\text{reject } H_0 \mid \pi = 0.25) = \sum_{x=0}^3 P(X = x \mid \pi = 0.25) \approx 0.7759.$$

- (b) The p -value is:

$$P(X \leq 1 \mid \pi = 0.50) = \sum_{x=0}^1 P(X = x \mid \pi = 0.50) \approx 0.0107.$$

2. Parameterised in terms of μ , since:

$$f_X(x; \lambda) = \frac{1}{\mu} \exp\left(-\frac{x}{\mu}\right)$$

then we can write the rejection region as:

$$R = \left\{ \frac{\frac{1}{2} \exp\left(-\frac{x_1}{2}\right) \frac{1}{2} \exp\left(-\frac{x_2}{2}\right)}{\exp(-x_1) \exp(-x_2)} \leq \frac{1}{2} \right\}$$

which is equivalent to:

$$x_1 + x_2 \leq 2 \log 2.$$

Since both X_1 and X_2 are $\text{Exp}(\mu)$, $T = X_1 + X_2$ is distributed as $\text{Gamma}(2, 1/\mu)$. Since H_0 is a simple hypothesis, the significance level of the test is equal to the probability:

$$P(T \leq 2 \log 2 \mid \mu = 2) \approx 0.1534.$$

The power of the test is equal to the probability:

$$P(T \leq 2 \log 2 \mid \mu = 1) \approx 0.4034.$$

Parameterised in terms of λ , we would have:

$$f_X(x; \lambda) = \lambda e^{-\lambda x}$$

then we can write the rejection region as:

$$R = \left\{ \frac{2 \exp(-2x_1) 2 \exp(-2x_2)}{\exp(x_1) \exp(-x_2)} \leq \frac{1}{2} \right\}$$

which is equivalent to:

$$x_1 + x_2 \geq 3 \log 2.$$

Since both X_1 and X_2 are $\text{Exp}(\lambda)$, $T = X_1 + X_2$ is distributed as $\text{Gamma}(2, \lambda)$. Since H_0 is a simple hypothesis, the significance level of the test is equal to the probability:

$$P(T \geq 3 \log 2 \mid \lambda = 2) \approx 0.0806.$$

The power of the test is equal to the probability:

$$P(T \geq 3 \log 2 \mid \lambda = 1) \approx 0.3849.$$

3. (a) The likelihood function is:

$$L_{\mathbf{X}}(\beta; \mathbf{x}) = f_{\mathbf{X}}(\mathbf{x}; \beta) = \prod_{i=1}^n \beta x_i^{\beta-1} = \beta^n \left(\prod_{i=1}^n x_i \right)^{\beta-1}.$$

The likelihood ratio test statistic, LR , for this test can be written as:

$$LR(\mathbf{X}) = \frac{L_{\mathbf{X}}(\hat{\beta}; \mathbf{x})}{L_{\mathbf{X}}(\beta_0; \mathbf{x})}$$

where $\hat{\beta}$ is the maximum likelihood estimator of β . To find $\hat{\beta}$ consider the log-likelihood function which is:

$$l_{\mathbf{X}}(\beta; \mathbf{x}) = \log L_{\mathbf{X}}(\beta; \mathbf{x}) = n \log \beta + (\beta - 1) \sum_{i=1}^n \log x_i$$

and the score function:

$$s_{\mathbf{X}}(\beta; \mathbf{x}) = \frac{\partial}{\partial \beta} l_{\mathbf{X}}(\beta; \mathbf{x}) = \frac{n}{\beta} + \sum_{i=1}^n \log x_i.$$

Setting $s_{\mathbf{X}}(\beta; \mathbf{x}) = 0$ gives:

$$\hat{\beta} = -\frac{n}{\sum_{i=1}^n \log x_i}$$

as a candidate maximum likelihood estimator. Since:

$$\frac{\partial^2}{\partial \beta^2} l_{\mathbf{X}}(\beta; \mathbf{x}) = -\frac{n}{\beta^2} < 0$$

for all β , we conclude that this is indeed the maximum likelihood estimator.

Hence we can write the likelihood ratio test statistic as:

$$LR(\mathbf{X}) = \frac{\hat{\beta}^n \left(\prod_{i=1}^n X_i \right)^{\hat{\beta}-1}}{\beta_0^n \left(\prod_{i=1}^n X_i \right)^{\beta_0-1}}$$

where:

$$\hat{\beta} = -\frac{n}{\sum_{i=1}^n \log x_i}.$$

- (b) The distribution of $2 \log LR(\mathbf{X})$ converges to χ_1^2 under H_0 for large n . Large values of $2 \log LR(\mathbf{X})$ provide evidence against H_0 . Hence a test of size $100\alpha\%$ will reject H_0 if:

$$2 \log LR(\mathbf{x}) > \chi_{\alpha, 1}^2.$$

- (c) For the asymptotic test in part (b) we can use the observed value of $LR(\mathbf{x})$ and the fact that $2 \log LR(\mathbf{X})$ is asymptotically χ_1^2 under H_0 . The p -value is then:

$$P(Y \geq LR(\mathbf{x}))$$

where $Y \sim \chi_1^2$.

Appendix D

Sample examination paper

Important note: This Sample examination paper reflects the examination and assessment arrangements for this course in the academic year 2020–21. The format and structure of the examination may have changed since the publication of this subject guide. You can find the most recent examination papers on the VLE where all changes to the format of the examination are posted.

Time allowed: 2 hours.

Candidates should answer all **FOUR** questions: **QUESTION 1** of Section A (40 marks) and all **THREE** questions from Section B (60 marks in total). **Candidates are strongly advised to divide their time accordingly.**

A handheld calculator may be used when answering questions on this paper and it must comply in all respects with the specification given with your Admission Notice. The make and type of machine must be clearly stated on the front cover of the answer book.

SECTION A

Answer Question 1 from this section.

1. (a) Define the mean squared error of an estimator of the parameter θ . State and prove the relationship between the mean squared error, bias and variance of an estimator.

(7 marks)

- (b) Suppose that we draw a random sample of size 5 from the probability distribution:

$$p_X(x; \theta) = \begin{cases} \theta^x(1 - \theta)^{1-x} & \text{for } x = 0, 1 \\ 0 & \text{otherwise.} \end{cases}$$

Calculate the method of moments estimate of the parameter θ if the sample is the set of numbers 0, 0, 1, 0 and 1.

(7 marks)

- (c) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a normal distribution with the parameters μ and σ^2 unknown. Consider the sample mean and sample variance, respectively:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{and} \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

It is known that in this case:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

where χ_{n-1}^2 denotes a chi-squared distribution with $n-1$ degrees of freedom. It is also known that if $X \sim \chi_{n-1}^2$, then:

$$E(X) = n-1 \quad \text{and} \quad \text{Var}(X) = 2(n-1).$$

- i. Find the mean squared error associated with the estimator S^2 of the parameter σ^2 .

(6 marks)

- ii. Consider the alternative estimator of σ^2 , $(n-1)S^2/n$. Find the mean squared error of this alternative estimator and compare it with that of S^2 .

(10 marks)

- (d) i. Provide the definition of a sufficient statistic.

(4 marks)

- ii. State the factorisation theorem.

(6 marks)

SECTION B

Answer all three questions from this section.

2. (a) Describe how a maximum likelihood estimator may be used to form an approximate pivotal function for large sample sizes.

(6 marks)

- (b) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a distribution with density function:

$$f_X(x; \alpha, \theta) = \begin{cases} \alpha e^{-\alpha(x-\theta)} & \text{for } x > \theta \\ 0 & \text{otherwise} \end{cases}$$

where $\alpha > 0$, $\theta > 0$ are unknown parameters. Use the factorisation theorem to determine:

- i. a 1-dimensional sufficient statistic for α when θ is known.

(3 marks)

- ii. a 1-dimensional sufficient statistic for θ when α is known.

(4 marks)

- iii. a 2-dimensional sufficient statistic for α and θ .

(3 marks)

- (c) Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from the (continuous) uniform distribution such that $X \sim \text{Uniform}[0, \theta]$, where $\theta > 0$. Find the method of moments estimator of θ .

(4 marks)

3. Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample from a distribution with probability density function:

$$f_X(x; \alpha) = \begin{cases} \frac{1}{(k-1)!} \alpha^k x^{k-1} e^{-\alpha x} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

where $k \geq 1$ is a known integer, and $\alpha > 0$ is an unknown parameter.

- (a) Find the maximum likelihood estimator of α , and also of $1/\alpha$. You should show that the stationary point of the log-likelihood about α is a maximum.

(7 marks)

- (b) Do the variances of the maximum likelihood estimators of α and $1/\alpha$ obtained in (a) attain the Cramér–Rao lower bound?

(6 marks)

- (c) Construct an asymptotic interval estimator of α with a confidence level of 95%.

Hint: It is known that if $Z \sim N(0, 1)$, then $z_{0.025} = 1.96$.

(7 marks)

4. (a) State the Neyman–Pearson lemma. Explain how it may be used to derive the uniformly most powerful test (UMPT) for a one-sided null hypothesis against a one-sided alternative hypothesis.

(7 marks)

- (b) Let $X \sim \text{Bin}(12, \pi)$. In other words:

$$p_X(x; \pi) = \binom{12}{x} \pi^x (1 - \pi)^{12-x} \quad \text{for } x = 0, 1, 2, \dots, 12$$

and 0 otherwise.

- i. Consider the test of:

$$H_0 : \pi = 0.50 \quad \text{vs.} \quad H_1 : \pi > 0.50.$$

What values of X provide evidence against H_0 ? Find the p -value for $X = 9$. Would you reject H_0 at the 5% significance level?

(5 marks)

- ii. Now consider the test of:

$$H_0 : \pi = 0.50 \quad \text{vs.} \quad H_1 : \pi \neq 0.50.$$

Obtain the test statistic for the likelihood ratio test.

(5 marks)

- iii. Describe how you would construct an asymptotic test based on the likelihood ratio test statistic. Briefly comment on the suitability of this test.

(3 marks)

END OF PAPER

Appendix E

Solutions to Sample examination paper

SECTION A

1. (a) If $\hat{\theta}$ is an estimator of the parameter θ , then its mean squared error is:

$$\text{MSE}_{\theta}(\hat{\theta}) = \text{Var}_{\theta}(\hat{\theta}) + (\text{Bias}_{\theta}(\hat{\theta}))^2.$$

Proof: We have (dropping the θ subscript):

$$\begin{aligned} \text{MSE}_{\theta}(\hat{\theta}) &= \text{E}((\hat{\theta} - \theta)^2) \\ &= \text{E}((\underbrace{\hat{\theta} - \text{E}(\hat{\theta}) + \text{E}(\hat{\theta})}_{=0} - \theta)^2) \\ &= \text{E}((\hat{\theta} - \text{E}(\hat{\theta}))^2 + (\text{E}(\hat{\theta}) - \theta)^2 + 2(\hat{\theta} - \text{E}(\hat{\theta}))(\text{E}(\hat{\theta}) - \theta)) \\ &= \text{E}((\hat{\theta} - \text{E}(\hat{\theta}))^2) + \text{E}((\text{Bias}(\hat{\theta}))^2) + 2 \underbrace{(\text{E}(\hat{\theta}) - \text{E}(\hat{\theta}))(\text{E}(\hat{\theta}) - \text{E}(\theta))}_{=0} \\ &= \text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2. \end{aligned}$$

- (b) The first population moment is:

$$\mu_1 = \text{E}(X) = 0 \times \theta^0(1 - \theta)^{1-0} + 1 \times \theta^1(1 - \theta)^{1-1} = \theta.$$

Method of moments equates corresponding population and sample moments, which then leads to:

$$\hat{\theta}(\mathbf{X}) = \bar{X} = M_1.$$

Hence the method of moments estimate of θ is:

$$\hat{\theta}(\mathbf{x}) = \frac{0 + 0 + 1 + 0 + 1}{5} = 0.4.$$

- (c) i. Since the expected value of χ_{n-1}^2 is $n - 1$, we have:

$$\text{E}\left(\frac{(n-1)S^2}{\sigma^2}\right) = n - 1 \quad \Rightarrow \quad \frac{n-1}{\sigma^2} \text{E}(S^2) = n - 1$$

and so $\text{E}(S^2) = \sigma^2$. The estimator S^2 is hence an unbiased estimator of σ^2 .

A similar calculation provides the variance:

$$\text{Var}\left(\frac{(n-1)S^2}{\sigma^2}\right) = 2(n-1) \quad \Rightarrow \quad \frac{(n-1)^2}{\sigma^4} \text{Var}(S^2) = 2(n-1)$$

hence:

$$\text{Var}(S^2) = \frac{2\sigma^4}{n-1}.$$

Therefore, the mean squared error of S^2 is:

$$\text{MSE}(S^2) = \text{Var}(S^2) + (\text{Bias}(S^2))^2 = \frac{2\sigma^4}{n-1}.$$

- ii. The bias of the estimator $(n-1)S^2/n$ is given by:

$$\text{Bias}\left(\frac{n-1}{n}S^2\right) = \text{E}\left(\frac{n-1}{n}S^2\right) - \sigma^2 = \frac{n-1}{n}\text{E}(S^2) - \sigma^2 = -\frac{\sigma^2}{n}.$$

The variance of the estimator $(n-1)S^2/n$ is given by:

$$\text{Var}\left(\frac{n-1}{n}S^2\right) = \left(\frac{n-1}{n}\right)^2 \text{Var}(S^2) = \frac{(n-1)^2}{n^2} \frac{2\sigma^4}{n-1} = \frac{2(n-1)}{n^2} \sigma^4.$$

Hence the mean squared error of the estimator $(n-1)S^2/n$ is given by:

$$\text{MSE}\left(\frac{n-1}{n}S^2\right) = \text{Var}(S^2) + (\text{Bias}(S^2))^2 = \frac{2(n-1)}{n^2} \sigma^4 + \frac{\sigma^4}{n^2} = \frac{2n-1}{n^2} \sigma^4.$$

Note that the mean squared error of the estimator $(n-1)S^2/n$ is smaller than that of the estimator S^2 for $n > 1$.

- (d) i. Let $\mathbf{X}$ be a random sample parameterised by $\boldsymbol{\theta}$. A statistic $\mathbf{U} = \mathbf{T}(\mathbf{X})$ is a sufficient statistic for $\boldsymbol{\theta}$ if the conditional distribution of $\mathbf{X}$ given $\mathbf{U}$ does not depend on $\boldsymbol{\theta}$.
- ii. Let $\mathbf{X} = \{X_1, X_2, \dots, X_n\}$ be a random sample with joint density function $f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta})$. The statistic $\mathbf{U} = \mathbf{T}(\mathbf{X})$ is sufficient for the vector of parameters $\boldsymbol{\theta}$ if and only if we can find functions g and h such that:

$$f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) = g(\mathbf{T}(\mathbf{x}), \boldsymbol{\theta}) h(\mathbf{x})$$

for all $\mathbf{x} \in \mathbb{R}^n$ and $\boldsymbol{\theta} \in \boldsymbol{\Theta}$. Note $\mathbf{T}(\mathbf{X})$ and $\boldsymbol{\theta}$ can be scalars, denoted $T(\mathbf{X})$ and θ , respectively.

SECTION B

2. (a) As explained in Section 4.7.

(b) The joint density function is:

$$f_{\mathbf{X}}(\mathbf{x}; \alpha, \theta) = \alpha^n \exp \left(-\alpha \sum_{i=1}^n x_i + n\alpha\theta \right) I(x_{(1)} > \theta)$$

and 0 otherwise.

i. If θ is known, then we may set:

$$g \left(\sum_{i=1}^n x_i, \alpha \right) = \alpha^n \exp \left(-\alpha \sum_{i=1}^n x_i + n\alpha\theta \right) \quad \text{and} \quad h(\mathbf{x}) = I(x_{(1)} > \theta).$$

Since:

$$f_{\mathbf{X}}(\mathbf{x}; \alpha, \theta) = g \left(\sum_{i=1}^n x_i, \alpha \right) h(\mathbf{x})$$

the factorisation theorem implies that the statistic $\sum_{i=1}^n X_i$ is sufficient for α .

ii. If α is known, then we may set:

$$g(x_{(1)}, \theta) = e^{n\alpha\theta} I(x_{(1)} > \theta) \quad \text{and} \quad h(\mathbf{x}) = \alpha^n \exp \left(-\alpha \sum_{i=1}^n x_i \right).$$

Since:

$$f_{\mathbf{X}}(\mathbf{x}; \alpha, \theta) = g(x_{(1)}, \theta) h(\mathbf{x})$$

the factorisation theorem implies that the statistic $X_{(1)}$ is sufficient for θ .

iii. If neither θ nor α are known, we may set:

$$g \left(\sum_{i=1}^n x_i, x_{(1)}, \alpha, \theta \right) = f_{\mathbf{X}}(\mathbf{x}; \alpha, \theta) \quad \text{and} \quad h(\mathbf{x}) = 1.$$

The factorisation theorem implies that the statistic $\left(\sum_{i=1}^n X_i, X_{(1)} \right)$ is sufficient for (α, θ) .

(c) The density function of X is:

$$f_X(x; \theta) = \begin{cases} \theta^{-1} & \text{for } 0 \leq x \leq \theta \\ 0 & \text{otherwise.} \end{cases}$$

Therefore, the first population moment is:

$$\mu_1 = E(X) = \frac{1}{\theta} \int_0^\theta x \, dx = \frac{1}{\theta} \left[\frac{x^2}{2} \right]_0^\theta = \frac{\theta}{2}.$$

Hence setting $\hat{\mu}_1 = M_1$, we have:

$$\frac{\hat{\theta}}{2} = \bar{X} \quad \Rightarrow \quad \hat{\theta} = 2\bar{X}.$$

3. (a) The log-likelihood function is:

$$l_{\mathbf{X}}(\alpha; \mathbf{x}) = nk \log \alpha - \alpha \sum_{i=1}^n x_i + c$$

where c is independent of α . Hence:

$$s_{\mathbf{X}}(\alpha; \mathbf{x}) = \frac{\partial}{\partial \alpha} l_{\mathbf{X}}(\alpha; \mathbf{x}) = \frac{nk}{\alpha} - \sum_{i=1}^n x_i.$$

Setting $s_{\mathbf{X}}(\alpha; \mathbf{x}) = 0$ leads to the maximum likelihood estimator of α , which is:

$$\hat{\alpha} = \frac{nk}{\sum_{i=1}^n X_i} = \frac{k}{\bar{X}}.$$

Note that:

$$\frac{\partial^2}{\partial \alpha^2} l_{\mathbf{X}}(\alpha; \mathbf{x}) = -\frac{nk}{\alpha^2} < 0$$

for all α , hence this is indeed the maximum likelihood estimator.

By the invariance property, the maximum likelihood estimator of $1/\alpha$ is:

$$\frac{1}{\hat{\alpha}} = \frac{\sum_{i=1}^n X_i}{nk} = \frac{\bar{X}}{k}.$$

- (b) The score function may be written as:

$$s_{\mathbf{X}}(\alpha; \mathbf{x}) = nk \left(\frac{1}{\alpha} - \frac{\bar{X}}{k} \right)$$

which indicates that only the parameter $\theta = 1/\alpha$ has an unbiased estimator which attains the Cramér–Rao lower bound. Hence the maximum likelihood estimator of α does not attain the Cramér–Rao lower bound whereas the maximum likelihood estimator of $1/\alpha$ does.

- (c) Fisher information is equal to:

$$\mathcal{I}_{\mathbf{X}}(\alpha) = -E \left(\frac{\partial^2}{\partial \alpha^2} l_{\mathbf{X}}(\alpha; \mathbf{x}) \right) = \frac{nk}{\alpha^2}$$

hence the asymptotic distribution of $\hat{\alpha}$ is:

$$\hat{\alpha} \sim N \left(\alpha, \frac{\alpha^2}{nk} \right)$$

approximately, for large n . This is the first level of approximation. The second level sets the asymptotic distribution of $\hat{\alpha}$ to:

$$\hat{\alpha} \sim N \left(\alpha, \frac{\hat{\alpha}^2}{nk} \right)$$

approximately, for large n . Now we can construct the pivotal function:

$$\frac{\hat{\alpha} - \alpha}{\hat{\alpha}/\sqrt{nk}} \sim N(0, 1)$$

approximately, for large n , which leads to the following asymptotic interval estimator:

$$\left[\hat{\alpha} - 1.96 \times \frac{\hat{\alpha}}{\sqrt{nk}}, \hat{\alpha} + 1.96 \times \frac{\hat{\alpha}}{\sqrt{nk}} \right]$$

with a confidence level of approximately 95%.

4. (a) As explained in Section 5.5.

(b) i. Large values of X provide evidence against H_0 . Hence the p -value for $X = 9$ is:

$$\begin{aligned} P(X \geq 9 | \pi = 0.50) &= \sum_{x=9}^{12} \binom{12}{x} (0.50)^x (0.50)^{12-x} \\ &= \sum_{x=9}^{12} \binom{12}{x} (0.50)^{12} \\ &= 0.073. \end{aligned}$$

Since the p -value is larger than $\alpha = 0.05$, we do not reject H_0 at the 5% significance level.

ii. In order to find the likelihood ratio test statistic we first need to find the maximum likelihood estimator $\hat{\pi}$. The likelihood function is:

$$L_X(\pi; x) = \binom{12}{x} \pi^x (1 - \pi)^{12-x}.$$

The log-likelihood function is:

$$l_X(\pi; x) = \log L_X(\pi; x) = \log \binom{12}{x} + x \log \pi + (12 - x) \log(1 - \pi).$$

The score function is:

$$s_X(\pi; x) = \frac{\partial}{\partial \pi} l_X(\pi; x) = \frac{x}{\pi} - \frac{12 - x}{1 - \pi}.$$

Setting $s_X(\pi; x) = 0$ leads to:

$$\hat{\pi} = \frac{X}{12}$$

as a candidate maximum likelihood estimator. Since:

$$\frac{\partial^2}{\partial \pi^2} l_X(\pi; x) = -\frac{x}{\pi^2} - \frac{12 - x}{(1 - \pi)^2} < 0$$

we conclude that this is indeed the maximum likelihood estimator.

The likelihood ratio statistic, $LR(X)$, is the following:

$$LR(X) = \frac{L_X(\hat{\pi}; X)}{L_X(0.50; X)} = \frac{\hat{\pi}^X (1 - \hat{\pi})^{12-X}}{(0.5)^{12}}.$$

iii. For large n we note that:

$$2 \log LR(X) \xrightarrow{d} \chi_1^2$$

and, since large values of $2 \log LR(X)$ are evidence against H_0 , a size α test will reject H_0 if:

$$2 \log LR(x) > \chi_{\alpha, 1}^2.$$

For small n this will be a poor approximation.